MACHINE LEARNING

Lecture Notes

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LIST OF TABLES

NOTATION

This table collects the symbols used throughout the book for quick reference. Chapter 2 states the underlying conventions (parentheses vs. brackets, uppercase vs. lowercase, etc.) in prose; this page is only a lookup table.

Sets and spaces

Symbol	Meaning
$\mathcal{X}$	Feature (input) space
$\mathcal{Y}$	Label (output) space
$\mathcal{H}$	Hypothesis class
$\mathcal{D}$	Data distribution over $\mathcal{X} \times \mathcal{Y}$
$\mathcal{D}_{\mathcal{X}}$	Marginal of $\mathcal{D}$ on $\mathcal{X}$
$[m]$	$\{1, \ldots, m\}$
$\mathrm{dist}(\cdot, \cdot)$	Distance function of a metric space (Chapter 3)

Data and samples

Symbol	Meaning
S	A data set / sample
m	Size of S
d	Dimension of the feature space
C	Number of classes
x, x_i	An input (feature vector)
y, y_i	A true label
$\widehat{y}$	A predicted label
z, z_i	A score / pre-activation (also a training example in generic contexts)

Probability

Symbol	Meaning
$P(\cdot)$	Probability of an event or of a discrete random variable's value

Symbol	Meaning
$p(\cdot)$	Probability **density** of a continuous random variable
$\mathbb{E}[\cdot]$, $\mathrm{Var}[\cdot]$, $\mathrm{Cov}[\cdot, \cdot]$	Expectation, variance, covariance
$\mathbb{1}(\cdot)$	Indicator function
log	Natural logarithm
σ	Standard deviation (bandwidth of an RBF kernel in Chapter 11 only)
$\sigma_i \in \{-1, +1\}$	A Rademacher sign (from Chapter 7 onwards)
$\mathrm{kl}(z \,\|\, \mu)$	KL divergence between Bernoulli(z) and Bernoulli(μ) (Chapter 5)
$\mathrm{KL}(P \,\|\, Q)$	Kullback-Leibler divergence $\mathbb{E}_P[\log \frac{dP}{dQ}]$ (Chapter 7)
$\mathrm{TV}(P, Q)$, $\chi^2(P \,\|\, Q)$	Total variation distance, chi-squared divergence (Chapter 7)
$P^{\otimes m}$, $\mathcal{D}^m$	Law of m i.i.d. draws
$I(j, S)$	Mutual information between an index j and a sample S (Chapter 7)

Losses and risks

Symbol	Meaning
$\ell(y, \widehat{y})$	Pointwise loss — true label first, prediction second
$R(h)$	True risk (generalization error) of h
$\widehat{R}_S(h)$	Empirical risk of h on S — the subscript always names the sample (e.g. $\widehat{R}_{S_{train}}$ on the training split, $\widehat{R}_{S'}$ on a ghost sample, $\widehat{R}_B$ on a mini-batch)
$\mathcal{L}(\cdot)$	Training objective (empirical risk plus a regularizer)
$R^*_{\mathcal{D}}$	Bayes error of $\mathcal{D}$

Learning-theoretic quantities

Symbol	Meaning
h	A hypothesis
h_S	A hypothesis learned from S (the subscript names the sample the algorithm was run on)
f	The true labeling function
f^*	The Bayes predictor
A	A learning algorithm
ϵ	An accuracy parameter
δ	A failure probability

Symbol	Meaning
$d_{VC}(\mathcal{H})$	VC dimension of $\mathcal{H}$
$\tau_{\mathcal{H}}(m)$	Growth function of $\mathcal{H}$
$\mathfrak{R}, \widehat{\mathfrak{R}}_S$	Rademacher complexity (population, empirical)
$N(\epsilon, \mathcal{H}, \text{dist})$	Covering number of $\mathcal{H}$ at scale ϵ
$\langle a, b\rangle_S$, $\|g\|_S$	Empirical inner product $\frac{1}{m}\sum_i a_i b_i$ and norm on the sample; $L_p(S)$ the empirical L_p distance (Chapter 7)
$\text{conv}(\mathcal{H})$	Convex hull: all finite convex combinations of members of $\mathcal{H}$
$\mathcal{F}_{A,L_1}(\Psi)$	Combinations of members of Ψ with total weight $\sum_j \|w_j\| \leq A$ (Chapter 7)
ϵ-uniformly stable	Stability parameter of an algorithm (Definition 7.8.1); here ϵ is a sensitivity, not an accuracy
$r_m(\mathcal{P}, Q)$	Minimax risk (Definition 7.9.1)

Model and optimization symbols

Symbol	Meaning
w	A weight vector
λ	A regularization coefficient
α	A learning rate
β	Smoothness constant: ∇f is β-Lipschitz (Chapters 7, 10, 12); inverse temperature in PAC-Bayes (Chapter 7)
λ-strongly convex	See Theorem 7.8.3
γ	Weak-learning edge (Chapter 12)
η_t	Step size of gradient boosting (Chapter 12)

Neural networks (Chapter 10)

Symbol	Meaning
$g(\cdot)$	Activation function
z_j^l	Pre-activation of neuron j in layer l
h_j^l	Activation (output) of neuron j in layer l
W_l, b^l	Weight matrix and bias vector of layer l
δ_j^l	Backpropagated error of neuron j in layer l

Kernel methods (Chapter 11)

Symbol	Meaning
$k(\cdot, \cdot)$	A kernel function
$\phi(\cdot)$	A feature map
$\mathcal{H}_k$	The RKHS of kernel k
K	A Gram matrix

CHAPTER 1

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CHAPTER 2

BASIC CONCEPTS

2.1 What is machine learning?

Notational conventions used throughout these notes.

Probability. $P(\cdot)$ denotes a probability — of an event, or of a discrete random variable taking a value, so that $P(X = x)$ is the probability mass function of a discrete X. Lowercase $p(\cdot)$ is reserved for the probability **density** of a continuous random variable, which is not itself a probability. Where a joint object mixes the two (a continuous feature vector with a discrete label, say) we write p for the joint and P for its discrete factors. Subscripts name the distribution being integrated against, e.g. $P_{S\sim\mathcal{D}^m}(\cdot)$. Probabilities always take parentheses; expectations, variances and covariances take brackets: $\mathbb{E}[\cdot]$, $\mathrm{Var}[\cdot]$, $\mathrm{Cov}[\cdot,\cdot]$, with subscripts naming the variable averaged over ($\mathbb{E}_S$, $\mathbb{E}_{x\sim\mathcal{D}}$). $\mathbb{1}(\cdot)$ is the indicator function and log always the **natural** logarithm.

Data. m is the size of a data set S, d the dimension of the feature space, C the number of classes, $[m] := \{1, \dots, m\}$. Calligraphic letters denote spaces: $\mathcal{X}$ the feature space, $\mathcal{Y}$ the label space, $\mathcal{H}$ a hypothesis class, $\mathcal{D}$ a data distribution over $\mathcal{X} \times \mathcal{Y}$, and $\mathcal{D}_{\mathcal{X}}$ its marginal on $\mathcal{X}$.

Losses and risks. The pointwise loss is always written $\ell(y, \widehat{y})$ — **true label first, prediction second**. $R(h)$ is the true risk and $\widehat{R}_S(h)$ the empirical risk on S: **a subscript on a risk always names the sample it is computed on** ($\widehat{R}_S$ on S, $\widehat{R}_{S'}$ on a second sample S', $\widehat{R}_B$ on a mini-batch B), never a purpose (training vs. testing) — when a risk is computed on a training or test split, we say so in words. A calligraphic $\mathcal{L}$ denotes a training objective (an empirical risk plus a regularizer).

Recurring symbols. h is a hypothesis, f the labeling function and f^* the Bayes predictor, A a learning algorithm, w a weight vector, ϵ an accuracy and δ a failure probability, α a learning rate, λ a regularization coefficient. From Chapter 7 onwards, $\sigma_i \in \{-1, +1\}$ are Rademacher signs; the activation function of a neural network is therefore written $g(\cdot)$, not $\sigma(\cdot)$, to keep the two apart. σ without a subscript is a standard deviation (and, in Chapter 11 only, the bandwidth of an RBF kernel, which is its universal name there).

Definition 2.1.1

"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E". (Mitchell, 1997)

(Mitchell's P for "performance measure" is the only place in these notes where the letter P does not denote a probability; from the next section on we call it the **loss** ℓ and the **risk** R.)

Purpose: Designing algorithms to solve T with maximum P and minimum:

1. time complexity (same as in any application)
2. space complexity (same as in any application)
3. sample complexity (new!)

2.1.1 Formal definitions

Experience (E)

Our computer program observes the environment by means of a **data set**, a set of tuples called **data points**:

$$S := \{(x_1, y_1), \ldots, (x_m, y_m)\}.$$

Each tuple comprises an **input** observation $x_i \in \mathcal{X}$, also called a **feature vector**, and a corresponding output observation $y_i \in \mathcal{Y}$.

The set $\mathcal{X}$ is called the **feature space** and the set $\mathcal{Y}$ a **label space**. In some contexts, S is called a **sample**.

We assume that there exists a **labeling function** $f : \mathcal{X} \to \mathcal{Y}$ such that $y_i = f(x_i)$, which is **unknown** to us.

We further assume that each observation in the data set is an **independent and identically distributed (i.i.d.)** sample from an unknown data distribution $\mathcal{D}$. An i.i.d. data set is denoted as $S := \{(x_i, y_i) \overset{i.i.d.}{\sim} \mathcal{D} : i \in [m]\}$.

Imagine playing backgammon. Each roll of the dice is a sample. These are independent samples if consecutive rolls do not influence each other. The samples are identically distributed if we use the same dice throughout the game. The i.i.d. assumption is at the heart of machine learning. The same process needs to be repeated for an agent to be able to learn it, just like us humans.

Task (T)

Our task is to predict the outputs y_i from input observations x_i as accurately as possible. In mathematical terms, we look for a **hypothesis** $h : \mathcal{X} \to \mathcal{Y}$ that approximates the unknown labeling function, i.e. it should satisfy $f(x) \approx h(x), \forall x \in \mathcal{X}$.

- If $\mathcal{Y}$ is a set with discrete elements, then our task is called **classification**, because it boils down to assigning an input x_i to an appropriate class. Imagine x_i is a picture expressed as an array of pixel intensity values, $\mathcal{Y}$ is then the type of the object in the foreground.

- If $\mathcal{Y}$ is a continuous set such as $\mathbb{R}$, then our task is called **regression**. Imagine x_i is the properties of a house (size, age, coordinates), $\mathcal{Y}$ can be its price.

We do not search over all conceivable maps $\mathcal{X} \to \mathcal{Y}$. We commit in advance to a **hypothesis class** $\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$ and search inside it. This commitment is the **inductive bias** of the learner. Chapter 6 shows that learning is impossible without one.

Performance Measure (P)

We define what we mean by an **accurate prediction** via a **loss function**: $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}^+$ which maps a pair (y, y') comprising a prediction y' and a corresponding true label y to a score inversely proportional to the quality of the prediction. Commonsense design choices for supervised learning problems are:

1. the **zero-one loss** defined as $\ell(y, y') := \mathbb{1}(y \neq y')$ for classification, and
2. the **squared error** defined as $\ell(y, y') := (y - y')^2$ for regression.

Here $\mathbb{1}$ is the indicator function that returns 1 if the predicate in its argument holds (e.g. if y differs from y') and 0 otherwise. Put together, we are interested in a **learning algorithm** $A(\cdot)$ that takes a data set S as input and returns a hypothesis. Let us denote this hypothesis as h_S, where the subscript S is to highlight its dependence on the data set. Then we can express the learning process as $h_S \leftarrow A(S)$. We expect from this algorithm to minimize the **generalization error (true risk)**, which is defined for zero-one loss as:

$$R(h) := \mathbb{E}_{x \sim \mathcal{D}_{\mathcal{X}}}\big[\ell(f(x), h(x))\big] = P_{x \sim \mathcal{D}_{\mathcal{X}}}\big(f(x) \neq h(x)\big),$$

where the second equality is specific to the zero-one loss.

Some intuition before we dissect the notation. A new application of our program will keep receiving input observations after we ship it, and we care about how often it will be wrong on *those*, not on the ones we have already collected. Since future inputs arrive according to the same chances as the ones in S (which is exactly what the i.i.d. assumption ensures), the fair way to score a hypothesis is: *average its loss over the chances of seeing each input*. That is all $R(h)$ is — a weighted average of the pointwise losses $\ell(f(x), h(x))$, one weight per input x, with inputs we are likely to see counted more heavily than rare ones. If the loss is the zero-one loss, this average counts each input where h and f disagree, weighted in the same way; an average of zeros and ones weighted by non-negative chances that sum to one is precisely a probability, which is the second equality above.

Two remarks unpack the expectation symbol. First, $\mathcal{D}_{\mathcal{X}}$, read "$\mathcal{D}$ marginalized on $\mathcal{X}$", is the distribution of input observations alone — the chances of encountering each x, with the label y **integrated out**. In our current setup the label is a deterministic function of the input, $y = f(x)$, so knowing x already pins down y; carrying the label along would tell us nothing new, and $R(h)$ averages only over x. (When we later encounter noisy data, labels will have a randomness of their own, and we will need the full joint distribution of the pair (x, y) — a preview of that treatment appears in Chapter 5.) Second, you can read $\mathbb{E}_{x \sim \mathcal{D}_{\mathcal{X}}}$ as *"the long-run average of $\ell(f(x), h(x))$ over inputs x drawn repeatedly from $\mathcal{D}_{\mathcal{X}}$"*. Probability theory will be covered properly in Chapter 5; for now, this long-run-frequency reading is all you need. In words, $R(h)$ is the true chance that our hypothesis will make a different prediction than the true labeling function on a freshly drawn input.

2.1.2 The Empirical Risk Minimization (ERM) Paradigm

The goal of a learning algorithm A is to find the hypothesis that minimizes the generalization error:

$$h_* := \arg\min_{h \in \mathcal{H}} R(h).$$

We cannot solve this optimization problem because we do not know $\mathcal{D}$ and f. We only have an idea about these unknowns via the data set S. Let us then use the data set to curate a quantity that approximates the generalization error:

$$\widehat{R}_S(h) := \frac{1}{m} \sum_{i \in [m]} \ell(y_i, h(x_i)) = \frac{|\{i \in [m] : h(x_i) \neq y_i\}|}{m}$$

where $[m] := \{1, \ldots, m\}$ and $|\cdot|$ counts the number of elements a set contains, e.g. $|\{1, 2\}| = 2$. The quantity $\widehat{R}_S(h)$ is called the **training error** or **empirical risk**. We can then find a suitable hypothesis by minimizing the empirical risk:

$$h_S := \arg\min_{h \in \mathcal{H}} \widehat{R}_S(h).$$

This paradigm, called **Empirical Risk Minimization (ERM)**, is at the center of machine learning. Yet, it has certain weaknesses. For instance, consider the following solution:

$$h_S(x) := \begin{cases} y_i, & \text{if } \exists i \in [m] \text{ s.t. } x_i = x, \\ 0, & \text{otherwise.} \end{cases}$$

This solution will provide us zero $\widehat{R}_S(h_S)$. It may look as if the problem is solved, but actually this is the moment where all problems start.

2.1.3 Example: Polynomial curve fitting

We observe data like below for 50 input observations x and the corresponding outputs y. We aim to find a function $y = f(x)$ that maps inputs to outputs.

```
import matplotlib.pyplot as plt
import torch as th

th.manual_seed(2530)  # fix the random draws so that the figures are reproducible

# This is the true data generating process.
# We do not have access to this information in a typical machine learning problem
inputs = th.linspace(0, 1, 50, dtype=th.float64)
outputs = th.sin(2 * th.pi * inputs)
labels = outputs + th.randn(inputs.shape[0], dtype=th.float64) * 0.25

plt.plot(inputs, labels, 'bo')
plt.xlabel("inputs (x)")
```

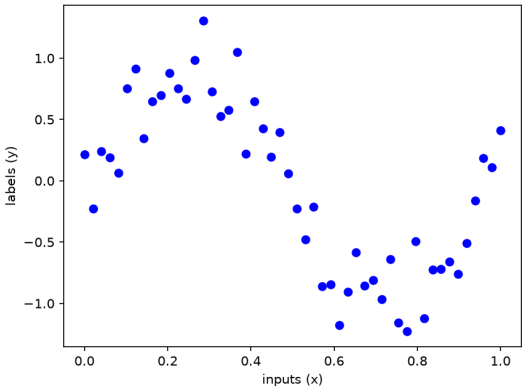


Figure 2.1: Synthetic noisy observations of a sinusoidal labeling function.

```
plt.ylabel("labels (y)")
plt.show()
```

Express our observations as a set of input-output tuples: $S = \{(x_1, y_1), \cdots, (x_{50}, y_{50})\}$. We would like to use them for two purposes:

- to **learn** the unknown function f,
- to **evaluate** the success of the learning process (i.e. how well we expect the learned hypothesis to predict future observations).

Let us then split our data into two equal partitions randomly, one per purpose: $S = S_{train} \cup S_{test}$.

S_{train} is called the **training set**. Our learning algorithm will use it to find a hypothesis that minimizes the empirical risk.

S_{test} is called the **test set**. We will use it to check how well the empirical risk represents the true risk. In formal terms, our goal is to approximate $R(h_{S_{train}})$.

Both splits are data sets in their own right, so every quantity we have defined for S applies to them verbatim. In particular, the empirical risk of a hypothesis h computed on the training set is $\widehat{R}_{S_{train}}(h)$, and on the test set $\widehat{R}_{S_{test}}(h)$: the subscript names the sample the average is taken over, which is the whole content of the symbol. We will call these two quantities the **training error** and the **test error** of h. Since our hypothesis $h_{S_{train}}$ is chosen by minimizing $\widehat{R}_{S_{train}}$, the test error $\widehat{R}_{S_{test}}(h_{S_{train}})$ is the honest estimate of $R(h_{S_{train}})$: S_{test} plays no role whatsoever in training, so $\widehat{R}_{S_{test}}(h_{S_{train}})$ behaves like a fresh measurement of how $h_{S_{train}}$ fares on new data.

```
num_samples = inputs.shape[0]
num_train_samples = num_samples // 2
# draw a random permutation of indices
idx = th.randperm(num_samples)
inputs_train = inputs[idx[:num_train_samples]]
labels_train = labels[idx[:num_train_samples]]
inputs_test = inputs[idx[num_train_samples:]]
labels_test = labels[idx[num_train_samples:]]
```

By visual inspection, we find that the input and the output have a nonlinear and periodic relationship. Let us then choose a hypothesis space suitable to this approximation, for instance a polynomial of degree M:

$$y(x) := \sum_{m=0}^{M} w_m x^m.$$

If $M = 0$, then our model learns to fit a constant: $y(x) = w_0$.
If $M = 1$, then it fits a line: $y(x) = w_0 + w_1 x$.
If $M = 2$, it fits a parabola: $y(x) = w_0 + w_1 x + w_2 x^2$, and so on.

```
class PolynomialModel:
    def __init__(self, num_polynomial_degrees=1, regularizer=0.0):
        self.M = num_polynomial_degrees
        self.regularization_factor = regularizer

    def extract_features(self, inputs):
        # Column m of the feature matrix Z holds x^m, for m = 0, ..., M
        powers = th.arange(self.M + 1, dtype=inputs.dtype)
        return inputs.unsqueeze(1) ** powers

    def learn(self, inputs, labels):
        Z = self.extract_features(inputs)
        ridge = self.regularization_factor * th.eye(Z.shape[1], dtype=Z.dtype)
        A = Z.T @ Z + ridge
        # w_S = (Z^T Z + lambda I)^{-1} Z^T y, obtained by solving the linear
        # system A w = Z^T y rather than by forming A^{-1} explicitly
        self.weights = th.linalg.solve(A, Z.T @ labels)

    def predict(self, inputs):
        return self.extract_features(inputs) @ self.weights
```

The **extract_features** function converts each one-dimensional observation to an $M-$ dimensional **feature vector**. This operation is called **feature extraction**.

The **learn** function takes a set of inputs and their corresponding labels in its arguments and fits the unknown model parameters $w_0, \cdots, w_M$ to data. We will cover later how exactly it does that.

The **predict** function takes only inputs in its arguments. It predicts their corresponding labels itself, hence the name. The source code of advanced machine learning models we will see later on follows the same template.

```
model = PolynomialModel(num_polynomial_degrees=3) # Create the model instance

# Training
model.learn(inputs_train, labels_train)

# Testing
```

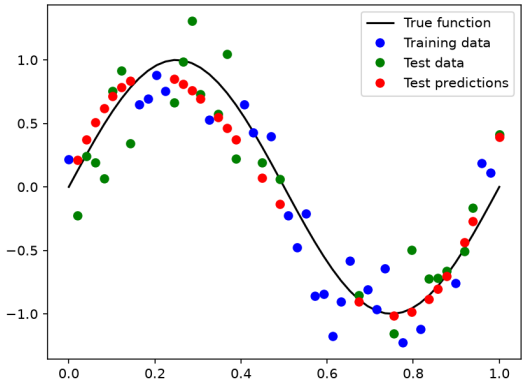


Figure 2.2: The true labeling function together with the training set, the test set, and the model's predictions on the test set.

```
predictions_test = model.predict(inputs_test)
root_mean_squared_error = ((labels_test - predictions_test) ** 2).mean().sqrt()

print(root_mean_squared_error.item())
```

```
0.25155540073010746
```

Let us visualize all we have. In the plot below, the training set is shown as blue circles, the test set as green circles, and the predictions of our machine learning model as red circles. The black curve is the labeling function we are trying to find out. Note that we can observe only noisy evaluations of it, just as in real life. Data collection processes are affected by unintended factors, which we call **noise**.

```
plt.plot(inputs,outputs,'k-', label="True function")
plt.plot(inputs_train,labels_train,'bo', label="Training data")
plt.plot(inputs_test, labels_test,'go', label="Test data")
plt.plot(inputs_test,predictions_test,'ro', label="Test predictions")
plt.legend(loc="upper right")
plt.show()
```

The properties of the model we have to choose before running the learning algorithm are called **hyperparameters**. For instance, the polynomial degree M is a hyperparameter. Let us simply try all options from 1 to 10 and see how they affect the model's behavior.

Since we have a regression problem, we follow common sense and choose the squared error loss function. We take its square root to bring it to the same scale as the label space:

$$\mathrm{RMSE}_S(h) := \sqrt{\widehat{R}_S(h)} = \sqrt{\frac{1}{m}\sum_{i\in[m]}(h(x_i)-y_i)^2}.$$

This loss function is called the **Root Mean Squared Error (RMSE)**. As with any subscripted risk, the sample named in RMSE_S is the one the average is taken over — in the experiment below, $\mathrm{RMSE}_{S_{train}}$ and $\mathrm{RMSE}_{S_{test}}$.

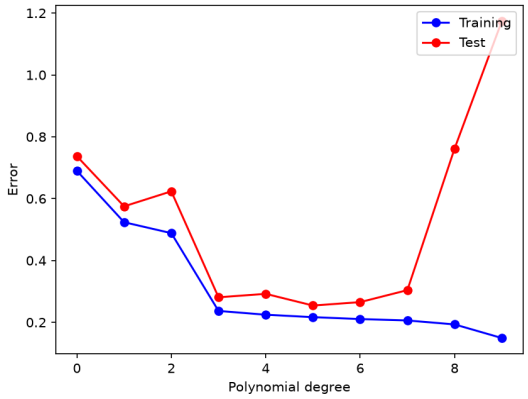


Figure 2.3: Training and test error as a function of the polynomial degree M.

```
# The performance score
def root_mean_squared_error(labels_true, predictions):
    return ((labels_true - predictions) ** 2).mean().sqrt()

Mmax = 10
train_errors = th.zeros(Mmax, dtype=th.float64)
test_errors = th.zeros(Mmax, dtype=th.float64)

for M in range(Mmax):
    # Create the model instance
    model = PolynomialModel(num_polynomial_degrees=M)

    # Training: fit parameters to data on the training split.
    model.learn(inputs_train, labels_train)
    predictions_train = model.predict(inputs_train)
    train_err = root_mean_squared_error(labels_train, predictions_train)

    # Testing: predict on the test split and evaluate performance.
    predictions_test = model.predict(inputs_test)
    test_err = root_mean_squared_error(labels_test, predictions_test)

    train_errors[M] = train_err
    test_errors[M] = test_err

plt.plot(th.arange(Mmax), train_errors, "bo-", label="Training")
plt.plot(th.arange(Mmax), test_errors, "ro-", label="Test")
plt.xlabel("Polynomial degree")
plt.ylabel("Error")
plt.legend(loc="upper right")
plt.show()
```

Expected: As M increases, the training error $\widehat{R}_{S_{train}}(h_{S_{train}})$ monotonically decreases.
Surprise: However, the test error $\widehat{R}_{S_{test}}(h_{S_{train}})$ decreases up to $M = 5$, and then increases — slowly

at first, then sharply: by $M = 9$ it is almost five times its minimum.

What may have gone wrong? Let us take a closer look into individual cases.

```
model0 = PolynomialModel(num_polynomial_degrees=0)
model0.learn(inputs_train, labels_train)
pred_0 = model0.predict(inputs)

model1 = PolynomialModel(num_polynomial_degrees=1)
model1.learn(inputs_train, labels_train)
pred_1 = model1.predict(inputs)

model4 = PolynomialModel(num_polynomial_degrees=4)
model4.learn(inputs_train, labels_train)
pred_4 = model4.predict(inputs)

model9 = PolynomialModel(num_polynomial_degrees=9)
model9.learn(inputs_train, labels_train)
pred_9 = model9.predict(inputs)

fig, axs = plt.subplots(2, 2)

for ax in axs.flat:
    ax.plot(inputs, outputs,'k-', label="True f")
    ax.plot(inputs_train, labels_train,'ro', label="Tr data")

axs[0, 0].plot(inputs, pred_0,'b-', label="M=0")
axs[0, 1].plot(inputs, pred_1,'g-', label="M=1")
axs[1, 0].plot(inputs, pred_4,'c-', label="M=4")
axs[1, 1].plot(inputs, pred_9,'m-', label="M=9")

for ax in axs.flat:
    ax.legend(loc="upper right")
    ax.label_outer()
```

The model had poor performance because it could not express a periodic relationship with a constant $M = 0$ or a line $M = 1$. Poor model fit due to limited model capacity is called **underfitting**. An underfitted model performs poorly on both training and test data, i.e. it has high $\widehat{R}_{S_{train}}(h_{S_{train}})$ and high $\widehat{R}_{S_{test}}(h_{S_{train}})$.

What goes on for $M = 9$ is interesting. A polynomial with such high degree is flexible enough to make sharp turns. The model learned to use this flexibility to pass through every training point, chasing even the noise in the observed labels and turning sharply wherever a training point demands it; but then it missed the trend and diverged from the true labeling function, swinging far from it in the gaps between training points. It is called **overfitting** when a model fits to training data extremely well, i.e. low $\widehat{R}_{S_{train}}(h_{S_{train}})$, but delivers poor performance on test data, i.e. high $\widehat{R}_{S_{test}}(h_{S_{train}})$.

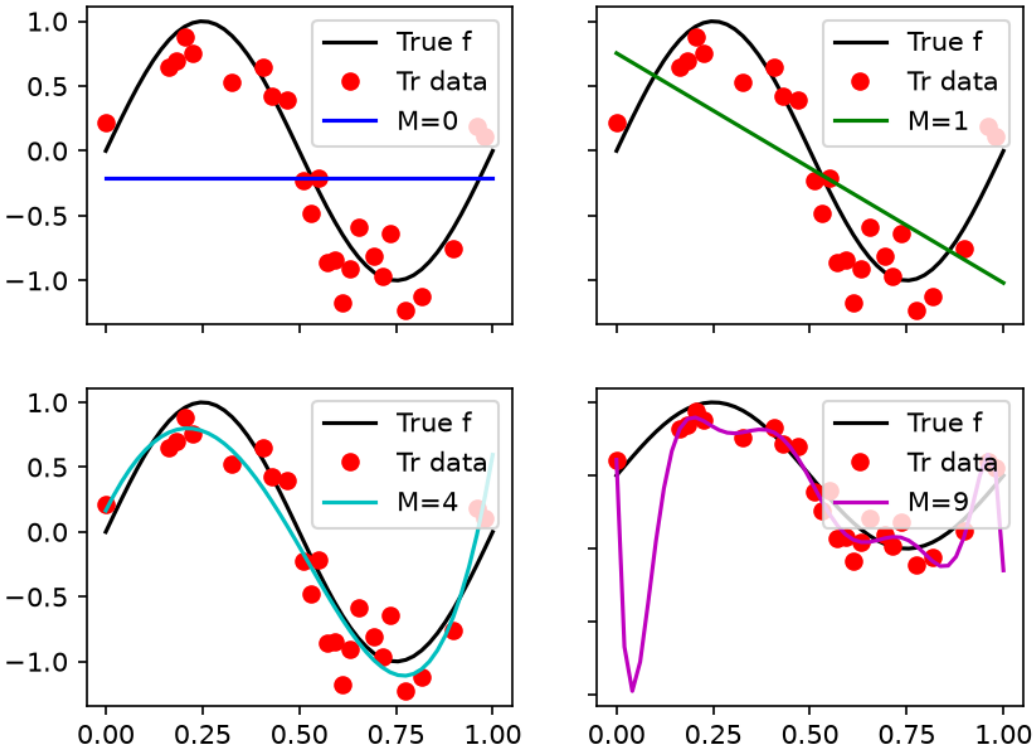


Figure 2.4: Polynomial fits of degree $M \in \{0, 1, 4, 9\}$ against the true function and training data, illustrating underfitting and overfitting.

Remark 2.1.2

Designing highly-expressive models while keeping them immune to overfitting is one of the fundamental problems of machine learning.

To mitigate overfitting, we need to devise an algorithm that encourages minimum empirical risk and penalizes the increase of model capacity. One choice could be as follows:

$$h_{S_{train}} := \arg\min_{h \in \mathcal{H}} \widehat{R}_{S_{train}}(h) + \lambda \sum_{m=0}^{M} w_m^2.$$

Here the term $\lambda \sum_{m=0}^{M} w_m^2$ is called a **regularizer** and λ a **regularization coefficient**. Let us see what happens then.

```
for M in range(Mmax):
    # Create the model instance
    model = PolynomialModel(num_polynomial_degrees=M, regularizer=0.01)

    # Training: fit parameters to data on the training split.
    model.learn(inputs_train, labels_train)
    predictions_train = model.predict(inputs_train)
    train_err = root_mean_squared_error(labels_train, predictions_train)

    # Testing: predict on the test split and evaluate performance.
    predictions_test = model.predict(inputs_test)
    test_err = root_mean_squared_error(labels_test, predictions_test)

    train_errors[M] = train_err
    test_errors[M] = test_err

plt.plot(th.arange(Mmax), train_errors, "bo-", label="Training")
plt.plot(th.arange(Mmax), test_errors, "ro-", label="Test")
plt.xlabel("Polynomial degree")
plt.ylabel("Error")
plt.legend(loc="upper right")
plt.show()
```

As seen, having a regularizer term significantly mitigates overfitting when the model capacity, i.e. the polynomial degree, increases: the test error now stays essentially flat beyond the optimal degree instead of exploding.

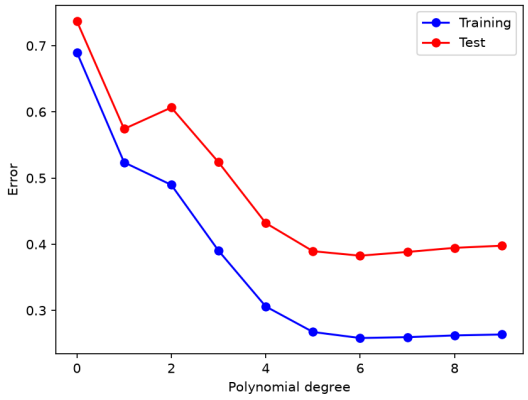


Figure 2.5: Training and test error versus polynomial degree with L_2 regularization, showing reduced overfitting compared to the unregularized fit.

CHAPTER 3

LINEAR PREDICTORS

Assume we observe data that look as below.

```
import matplotlib.pyplot as plt
import torch as th

th.manual_seed(0)  # fix the random draws so that the figures are reproducible

# This is the true data generating process.
# We do not have access to this information in a typical machine learning problem
inputs = th.linspace(-5, 5, 50, dtype=th.float64)
outputs = inputs * 0.5 + 3
labels = outputs + th.randn(inputs.shape[0], dtype=th.float64)

grid = th.linspace(-5, 5, 500, dtype=th.float64)
plt.plot(inputs, labels, 'bo')
plt.plot(grid, grid * 0.5 + 3, 'k-')
plt.xlabel("inputs (x)")
plt.ylabel("labels (y)")
plt.show()
```

Our visual inspection tells us that there is a strong linear correlation between the inputs and the outputs. Then let us choose our hypothesis set accordingly:

$$\mathcal{H} := \{h : h(x) = w_0 + w_1 x,\ (w_0, w_1) \in \mathbb{R}^2\}.$$

Denoting $w := (w_0, w_1)$, we can re-express the hypothesis as $h(x) := w^\top x$, where $\top$ indicates a matrix transpose.

Assume we are given a training set $S = \{(x_i, y_i) : i \in [m]\}$. Its mean squared-error empirical risk for our chosen hypothesis h is:

$$\widehat{R}_S(w) := \frac{1}{m} \sum_{i \in [m]} (w^\top x_i - y_i)^2.$$

Let us perform empirical risk minimization:

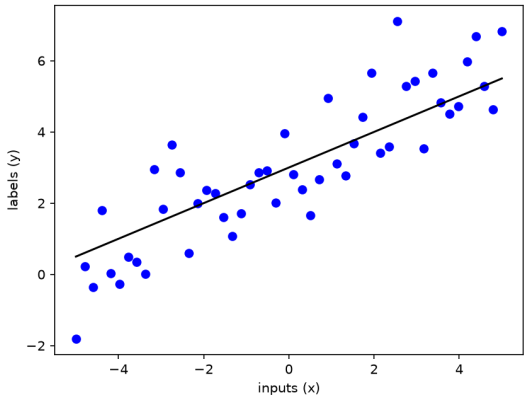


Figure 3.1: Synthetic data with a strong linear relationship between input and label, together with the underlying linear generating function.

$$w_S := \arg\min_w \widehat{R}_S(w).$$

The squared error loss gets its minimum value at the point where its derivative is zero. Hence, we can find the solution at w_S that satisfies $\nabla_w \widehat{R}_S(w)|_{w:=w_S} = 0$, that is

$$\begin{aligned}
\widehat{R}_S(w) &= \frac{1}{m}\sum_{i\in[m]}\left(w^\top x_i x_i^\top w - 2w^\top x_i y_i + y_i^2\right) \\
\Rightarrow \nabla_w \widehat{R}_S(w) &= \frac{1}{m}\sum_{i\in[m]}\left(2x_i x_i^\top w - 2x_i y_i\right) \triangleq 0 \\
&\Rightarrow \sum_{i\in[m]} x_i x_i^\top w_S = \sum_{i\in[m]} x_i y_i \\
&\Rightarrow w_S = \left(\sum_{i\in[m]} x_i x_i^\top\right)^{-1} \sum_{i\in[m]} x_i y_i
\end{aligned}$$

Fitting a linear model to data is often referred to as **least-squares regression** and w_S the **least-squares solution**.

Denote each data point by a column vector $z_i := (x_i, 1)$. Expressing the linear model as $w^\top z_i$ and using the definition of the dot product we have

$$w^\top z_i = x_i w_1 + w_0$$

where the term w_0 is referred to as the **bias**. The bias gives the model the opportunity to shift the predictor to an appropriate position before fitting the line. As we will see later, this flexibility sometimes brings substantial improvement in model fit.

Let us collect all input samples into a $\mathbb{R}^{m\times 2}$ matrix that contains data points on its rows:

$$Z = \begin{bmatrix} z_1^\top \\ \vdots \\ z_m^\top \end{bmatrix}$$

Let us also collect the corresponding labels in a vector $y := (y_1, \ldots, y_m)$. We can re-express the least-squares solution in vector notation as follows

$$w_S := (Z^\top Z)^{-1} Z^\top y.$$

Let us implement it and solve the task.

```
num_samples = inputs.shape[0]
num_train_samples = num_samples
idx = th.randperm(num_samples)
inputs_train = inputs[idx[:num_train_samples]]
labels_train = labels[idx[:num_train_samples]]

class LinearRegression:
    def extract_features(self, inputs):
        # Append a constant 1 to every input, so that w^T z = w_1 x + w_0
        ones = th.ones(inputs.shape[0], 1, dtype=inputs.dtype)
        return th.cat((inputs.unsqueeze(1), ones), dim=1)

    def learn(self, inputs, labels):
        Z = self.extract_features(inputs)
        # This is where the least-squares solution is implemented!
        # w_S = (Z^T Z)^{-1} Z^T y, obtained by solving the linear system
        # (Z^T Z) w = Z^T y, which is numerically safer than inverting Z^T Z
        self.weights = th.linalg.solve(Z.T @ Z, Z.T @ labels)

    def predict(self, inputs):
        return self.extract_features(inputs) @ self.weights

model = LinearRegression()

model.learn(inputs_train, labels_train)
predictions = model.predict(grid)

plt.plot(inputs, labels, 'bo')
plt.plot(grid, grid * 0.5 + 3, 'k-', label='true labeling function')
plt.plot(grid, predictions, 'r-', label='learned hypothesis')
plt.legend(loc="upper left")
plt.xlabel("inputs (x)")
plt.ylabel("labels (y)")
plt.show()
```

3.1 Metric spaces

We would like the spaces used in machine learning — feature spaces, parameter spaces, and later even spaces of hypotheses — to carry a notion of *how far apart* two of their elements are. The standard way to

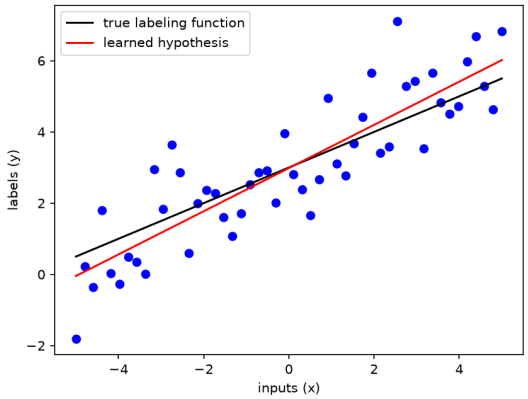


Figure 3.2: The learned least-squares line together with the true labeling function and the training data.

formalize this is not to build the notion into the space itself, but to equip a set with a distance function from the outside:

Definition 3.1.1: Metric space

A **metric space** is a pair $(\mathcal{X}, \text{dist})$ consisting of a non-empty set $\mathcal{X}$ and a function $\text{dist} : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ satisfying the following axioms, $\forall a, b, c \in \mathcal{X}$:

1. **(M1) identity of indiscernibles:** $\text{dist}(a, b) = 0 \iff a = b$,
2. **(M2) symmetry:** $\text{dist}(a, b) = \text{dist}(b, a)$,
3. **(M3) triangle inequality:** $\text{dist}(a, c) \leq \text{dist}(a, b) + \text{dist}(b, c)$.

Note what the definition does *not* assume: that $\mathcal{X}$ is a vector space, or that dist returns non-negative values. Neither is needed. Non-negativity is a consequence, not an axiom:

Remark 3.1.2

Non-negativity, $\text{dist}(a, b) \geq 0$, follows directly from M1, M2, and M3 by setting $c := a$ in M3: $0 = \text{dist}(a, a) \leq \text{dist}(a, b) + \text{dist}(b, a) = 2\,\text{dist}(a, b)$, hence $\text{dist}(a, b) \geq 0$.

This is the level of generality we need: in this chapter $\mathcal{X}$ will be a space of real vectors, but in Chapter 7 the very same definition will be applied to $\mathcal{X} = \mathcal{H}$, a set of hypotheses, and it is precisely the distance *between hypotheses* that the covering-number generalization bounds there are built on.

For $p \geq 1$, the L_p-norm of a vector $u \in \mathbb{R}^d$ is defined as follows

$$||u||_p := \left(\sum_{j=1}^{d} |u_j|^p \right)^{1/p}.$$

If $p = 2$, we get the well-known **Euclidean norm**.
If $p = 1$, we get the **Manhattan Norm**.
As $p \to \infty$, we tend to get the **Maximum Norm** (L_∞), i.e. $||u||_\infty := \max\{|u_1|, \ldots, |u_d|\}$.
Every such norm turns $\mathbb{R}^d$ into a metric space: the pair $\left(\mathbb{R}^d, \text{dist}\right)$ with

$$\text{dist}(a, b) := ||a - b||_p$$

satisfies M1–M3. Symmetry and the identity of indiscernibles are immediate from the properties of a norm; the triangle inequality for dist is inherited from the subadditivity of the norm, $||u + v||_p \leq ||u||_p + ||v||_p$.

For $0 < p < 1$ the same expression is still well defined and still traces the iso-contours plotted below, but it is no longer a norm: it violates subadditivity, so the pair it defines violates M3 and is not a metric space.

Let us plot the behavior of these norms as a function of p. The red lines below are iso-contours of $||u||_p = 1$ for $u = (a_1, a_2)$ and different choices of p. The unit ball shrinks as p decreases: a small p charges a vector heavily for spreading its mass over many coordinates. A vector can stay inside a small-p ball only by concentrating all of its mass on a few coordinates and setting the rest exactly to zero. We call such a vector — one with only a few non-zero entries — **sparse**. The converse, a vector with many small non-zero entries, is called **dense**. Sparsity will matter to us a great deal: recall that in a linear model each coordinate of w multiplies one feature of the input, so a zero coordinate of w switches the corresponding feature *off* — a sparse w is a hypothesis that bases its predictions on a small subset of the available features, ignoring the rest.

```
p_values = [0.5, 1, 2, 3, 7, float('inf')]
a1, a2 = th.meshgrid(th.linspace(-1.2, 1.2, 101),
                     th.linspace(-1.2, 1.2, 101), indexing='xy')
fig, axes = plt.subplots(ncols=(len(p_values) + 1) // 2,
                         nrows=2, figsize=(14, 7))
for p, ax in zip(p_values, axes.flat):
    if p == float('inf'):
        zz = th.maximum(a1.abs(), a2.abs())
    else:
        zz = (a1.abs()**p + a2.abs()**p)**(1./p)
    ax.contour(a1, a2, zz, [1], colors='red', linewidths=2)
    ax.set_title("p= {0}".format(p))

plt.show()
```

3.2 Regularized least squares

We have seen in Chapter 2 that $\widehat{R}_S(h_S) = 0$ can be achieved by memorizing the training set, resulting in overfitting. Let us then write the hypothesis space as a nested chain of subclasses of growing capacity, $\mathcal{H}_1 \subset \mathcal{H}_2 \subset \cdots$ with $\mathcal{H} = \bigcup_i \mathcal{H}_i$, and devise an objective that pursues the dual goals of fitting the data as well as possible and constraining model complexity concurrently:

$$h_S := \arg\min_{h \in \mathcal{H}} \widehat{R}_S(h) + \lambda \cdot \text{pen}(i(h)),$$
$$i(h) := \min\{i : h \in \mathcal{H}_i\}.$$

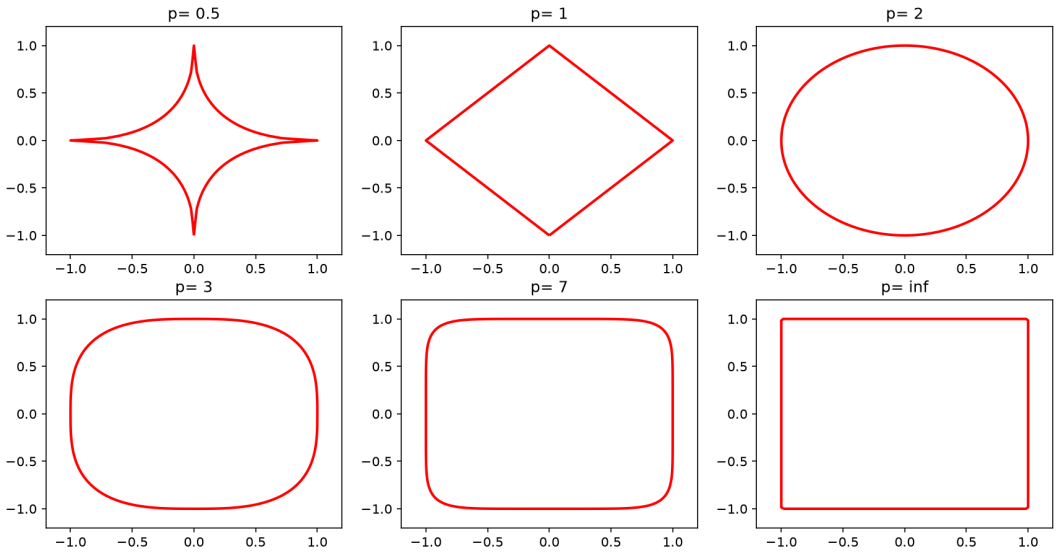


Figure 3.3: Unit-ball contours of the ℓ_p norm for $p \in \{0.5, 1, 2, 3, 7, \infty\}$, shrinking towards the axes as p decreases.

Here λ is called a **regularization coefficient** and $\text{pen}(\cdot)$, an increasing function of the subclass index, a **regularizer**. As we will see in more detail in Chapter 6, this paradigm is called **Structural Risk Minimization (SRM)**, one of the biggest achievements of machine learning research in the pre-deep-learning era.

Let us next apply the idea of constraining the hypothesis space to the least squares problem:

$$w_S := \arg\min_w \frac{1}{m} \sum_{i \in [m]} (w^\top x_i - y_i)^2$$

$$\text{s.t. } ||w||_p^p \leq \eta$$

where the abbreviation s.t. stands for **subject to**, meaning that we search for a solution within a feasible set of parameters $\{w : ||w||_p \leq \eta\}$. This constrained optimization problem can be expressed equivalently as:

$$w_S := \arg\min_w \max_{\lambda \geq 0} \frac{1}{m} \sum_{i \in [m]} (w^\top x_i - y_i)^2 + \lambda(||w||_p^p - \eta)$$

where $\lambda \geq 0$ is the **Lagrange multiplier** of the constraint — a standard device from constrained optimization: instead of enforcing the budget η explicitly, we allow the constraint to be violated but charge a price λ per unit of violation, and let the minimization itself decide how much violation is worth paying for. For every budget η there is a λ whose unconstrained problem has the same solution. Fixing λ instead of η and dropping the constant $\lambda\eta$ gives the **regularized least squares** objective:

$$w_S := \arg\min_w \underbrace{\frac{1}{m} \sum_{i \in [m]} (w^\top x_i - y_i)^2}_{\widehat{R}_S(w)} + \lambda ||w||_p^p.$$

The special case of $p = 2$ deserves detailed investigation:

$$w_S := \arg\min_w \frac{1}{m} \sum_{i \in [m]} (w^\top x_i - y_i)^2 + \lambda ||w||_2^2.$$

We can rewrite the loss of this optimization in vector form as:

$$\mathcal{L}_S(w) := \frac{1}{m} \|Zw - y\|_2^2 + \lambda w^\top w.$$

Let us find the optimal weights that minimize the loss by setting its gradient to zero once again:

$$\begin{aligned}
\mathcal{L}_S(w) &= \frac{1}{m} w^\top Z^\top Z w - \frac{2}{m} w^\top Z^\top y + \frac{1}{m} y^\top y + \lambda w^\top w \\
&= w^\top \left(\frac{1}{m} Z^\top Z + \lambda I \right) w - \frac{2}{m} w^\top Z^\top y + \frac{1}{m} y^\top y \\
\Rightarrow \nabla_w \mathcal{L}_S(w) &= 2 \left(\frac{1}{m} Z^\top Z + \lambda I \right) w - \frac{2}{m} Z^\top y \triangleq 0 \\
\Rightarrow w_S &= \left(\frac{1}{m} Z^\top Z + \lambda I \right)^{-1} \frac{1}{m} Z^\top y = \left(Z^\top Z + m\lambda I \right)^{-1} Z^\top y.
\end{aligned}$$

The result, known as **ridge regression** (Hoerl and Kennard, 1970), differs from the least-squares solution $w_S = (Z^\top Z)^{-1} Z^\top y$ by only the added term $m\lambda I$. This term makes the matrix invertible even when $Z^\top Z$ is singular, and it pulls every coordinate of w_S towards zero. Methods that pull parameters towards zero instead of fitting them freely are known in statistics as **parameter shrinkage** methods. Note that ridge shrinks but does not sparsify: it drives all coefficients *close* to zero, but (as we will see in the weight plots at the end of this chapter) none of them *exactly* to zero, so every feature keeps a small say in the prediction. We will see next that $p = 1$ does sparsify. The regularizer of ridge regression is referred to as **weight decay**, a name still in active use by deep learning libraries.

Let us next see ridge regression in action.

```
class RidgeRegression:
    def extract_features(self, inputs):
        ones = th.ones(inputs.shape[0], 1, dtype=inputs.dtype)
        return th.cat((inputs, ones), dim=1)

    def learn(self, inputs, labels, lambda_coef=0.1):
        Z = self.extract_features(inputs)
        # w_S = (Z^T Z + lambda I)^{-1} Z^T y: the least-squares system above
        # with lambda I added to the matrix being inverted
        ridge = lambda_coef * th.eye(Z.shape[1], dtype=Z.dtype)
        A = Z.T @ Z + ridge
        self.weights = th.linalg.solve(A, Z.T @ labels)

    def predict(self, inputs):
        return self.extract_features(inputs) @ self.weights
```

This time we work on the Diabetes data set, which consists of 442 data points. Each data point represents a patient with the following 10 features:

1. age
2. sex
3. body-mass index
4. average blood pressure
5. total serum cholesterol
6. low-density lipoprotein level
7. high-density lipoprotein level
8. total cholesterol level
9. possibly log of serum triglycerides level
10. blood sugar level

The goal is to predict a quantitative measure of disease progression, the higher the more severe.

```
from sklearn.datasets import load_diabetes

# The only library-provided piece is the raw data set itself; the split,
# the normalization, and both estimators are implemented below.
d = load_diabetes()
X = th.tensor(d.data, dtype=th.float64)
y = th.tensor(d.target, dtype=th.float64)

# hold out 20% of the data as the test split
perm = th.randperm(X.shape[0])
n_test = int(0.20 * X.shape[0])
X_test, y_test = X[perm[:n_test]], y[perm[:n_test]]
X_train, y_train = X[perm[n_test:]], y[perm[n_test:]]

print("Means:")
print(X_train.mean(dim=0))
print("Variances:")
print(X_train.var(dim=0, unbiased=False))
```

```
Means:
tensor([-2.3438e-03,  5.7266e-05,  6.4150e-04, -1.2388e-03,
        -2.0432e-03, -2.1458e-03, -1.7280e-04, -1.1079e-03,
        -6.1292e-04, -1.4638e-03],
       dtype=torch.float64)
Variances:
tensor([0.0023, 0.0023, 0.0024, 0.0023, 0.0022,
        0.0021, 0.0022, 0.0022, 0.0023, 0.0023],
       dtype=torch.float64)
```

Notably, different features have different scales. This may cause artifacts in model fitting, such as prioritization of features according to their scales instead of their relevance to the predicted quantity of interest.

We can mitigate these artifacts by **normalizing** the data. One commonplace approach is **z-score normalization**, which assumes that each feature is normally distributed with some mean μ and variance σ^2. That is, each coordinate x_j of a d-dimensional observation x comes from a sampling process as below:

$$\epsilon \sim \mathcal{N}(0, 1),$$
$$x_j = \mu_j + \sigma_j \epsilon.$$

This operation makes the assumption that all the observed variation in a feature's values (i.e. the differences across individual observations, stemming from whatever factors generated them) is already inside the first step, the standard normal sample; the second step merely scales and shifts this sample. Z-score normalization reverses the second operation to bring all features back to the first step:

$$x'_j := \frac{x_j - \mu_j}{\sigma_j},$$
$$\forall j \in [d].$$

For μ_j and σ_j, we use the sample mean and standard deviation of the corresponding feature, computed on the **training split only** so that no information leaks from the test split. This preprocessing step is also referred to as **standardization**. We will revisit its probability-theoretic justification later.

```
# z-score normalization, with mu and sigma taken from the training split only
mu = X_train.mean(dim=0)
sd = X_train.std(dim=0, unbiased=False)
X_train = (X_train - mu) / sd
X_test = (X_test - mu) / sd

print("Means:")
print(X_train.mean(dim=0))
print("Variances:")
print(X_train.var(dim=0, unbiased=False))
```

```
Means:
tensor([ 5.0180e-18,  3.6380e-17,  1.5054e-17,  3.7635e-18,
         2.8226e-17,  1.7563e-17, -4.0144e-17,  4.5162e-17,
        -5.0180e-18,  1.7563e-17],
       dtype=torch.float64)
Variances:
tensor([1.0000, 1.0000, 1.0000, 1.0000, 1.0000,
        1.0000, 1.0000, 1.0000, 1.0000, 1.0000],
       dtype=torch.float64)
```

Now we are ready to train and test ridge regression on the Diabetes data set.

```
model_ridge = RidgeRegression()

model_ridge.learn(X_train, y_train, lambda_coef=1)
```

```
predictions = model_ridge.predict(X_train)
train_error = ((predictions - y_train)**2).mean().sqrt()
print("Train RMSE: {:.2f}".format(train_error))

predictions = model_ridge.predict(X_test)
test_error = ((predictions - y_test)**2).mean().sqrt()
print("Test RMSE: {:.2f}".format(test_error))
```

```
Train RMSE: 54.35
Test RMSE: 50.92
```

We have a decent result. But can we do even better? Remember that the coordinates of w pair with the features of the input: a zero coordinate switches a feature off. Can we therefore learn a hypothesis that relies only on a small subset of the features — that is, can we make our solution **sparse**, in the sense of having only a few non-zero coordinates? Inspired by the shape of the L_p balls illustrated above — recall that small p keeps a vector inside the ball only by zeroing out most of its coordinates — we can next try out $p = 1$. For d-dimensional input vectors x_i, the resulting objective reads:

$$\mathcal{L}_S(w) := \frac{1}{m} \sum_{i \in [m]} (w^\top x_i - y_i)^2 + \lambda \sum_{j=1}^{d} |w_j|.$$

This approach is known as **Least Absolute Shrinkage Selection Operator (LASSO) Regression** (Tibshirani, 1996).

Unlike ridge regression and least squares, the Lasso objective does not have an analytical solution. One reason is that the L_1 term is not differentiable at $w_j = 0$ — which, as it will turn out, is exactly where many coordinates of its solution like to sit. There is no closed-form formula for a w that would satisfy

$$\nabla_w \Big(\frac{1}{m} \sum_{i \in [m]} (w^\top x_i - y_i)^2 + \lambda \sum_{j=1}^{d} |w_j| \Big) \triangleq 0.$$

If we cannot go straight to the value that minimizes the loss, we can instead find its direction and take a step towards there. Remember that the derivative of a function points to the direction towards which a function grows. Then negating it should point us towards a neighboring position with lower loss than our current position. It is reasonable to hope that repetitively taking small steps towards a neighboring point with lower loss will bring us to the point where the loss is minimum. Consider a series $w_0, w_1, \ldots$ created by the following succession rule and some arbitrary initialization w_0:

$$w_{t+1} := w_t - \alpha \nabla_w \mathcal{L}_S(w)|_{w:=w_t}$$

This approach is called **gradient descent**. It is in use with nearly all modern machine learning approaches. The coefficient $\alpha > 0$ is called a **learning rate**. It determines how fast the model parameters will change in a single iteration. Too large a learning rate may result in missing the optimal solution due to large oscillations around it. Too small a learning rate may result in infeasibly long training time.

Gradient descent requires repetitive evaluation of the gradient of the loss with respect to the parameters at every iteration: $\nabla_w \mathcal{L}_S(w)|_{w:=w_t}$. Hence its implementation on the computer requires an analytical

calculation of this gradient. This may be time consuming for complex loss functions. Deep learning libraries such as PyTorch and TensorFlow allow us to automate this process, a mechanism called **automatic differentiation**: flagging a tensor with `requires_grad_()` makes the library record every operation it takes part in, and a later call to `backward()` replays that record in reverse, accumulating $\nabla_w \mathcal{L}_S(w)$ into the tensor's `.grad` field. Note that only the *gradient computation* is automated — the update rule $w_{t+1} := w_t - \alpha \nabla_w \mathcal{L}_S(w)$ is still ours to write, and we write it out explicitly below rather than delegating it to a library optimizer, so that nothing about the algorithm is hidden.

```
class LassoRegression:
    def __init__(self, n_dims, lambda_coef=1.0):
        self.lambda_coef = lambda_coef
        # The parameters are plain tensors flagged for gradient tracking:
        # requires_grad_() tells torch to record every operation they take
        # part in, so that backward() can later replay the record in reverse.
        self.w = th.randn(n_dims, 1).requires_grad_()
        self.b = th.randn(1).requires_grad_()

    def predict(self, inputs):
        return inputs @ self.w + self.b

    def learn(self, inputs, labels, alpha=0.01, num_steps=1):
        for _ in range(num_steps):
            # Forward pass: evaluate the objective at the current parameters
            loss = ((self.predict(inputs) - labels)**2).mean() \
                   + self.lambda_coef * self.w.abs().sum()
            # Backward pass: automatic differentiation fills w.grad and b.grad
            loss.backward()
            # The gradient descent step itself, written out explicitly. The
            # no_grad() block switches the recording off, since the update is
            # not part of the function we are differentiating.
            with th.no_grad():
                for param in (self.w, self.b):
                    param -= alpha * param.grad
                    param.grad.zero_()   # clear it for the next iteration


# Single precision is enough for an iterative solver, and it is what the
# deep learning libraries use by default
X_train = X_train.float()
X_test = X_test.float()
y_train = y_train.float().reshape(-1, 1)
y_test = y_test.float().reshape(-1, 1)

# Train our model
model_lasso = LassoRegression(n_dims=X_train.shape[1], lambda_coef=1)
# Number of gradient descent iterations
num_iterations = 250
```

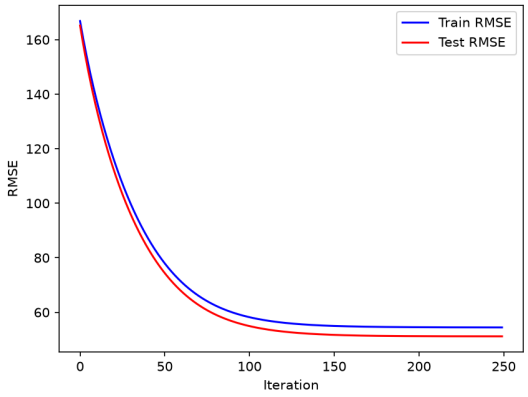


Figure 3.4: Learning curve: training and test RMSE of Lasso regression across gradient descent iterations.

```
# Collect the train and test errors here.
train_errors = th.zeros(num_iterations)
test_errors = th.zeros(num_iterations)

for ii in range(num_iterations):
    model_lasso.learn(X_train, y_train)

    with th.no_grad():   # evaluation needs no gradient record
        predictions = model_lasso.predict(X_train)
        train_errors[ii] = ((predictions - y_train)**2).mean().sqrt()

        # Test our model
        predictions = model_lasso.predict(X_test)
        test_errors[ii] = ((predictions - y_test)**2).mean().sqrt()

# Plot the learning curve
plt.plot(th.arange(num_iterations), train_errors, 'b-', label="Train RMSE")
plt.plot(th.arange(num_iterations), test_errors, 'r-', label="Test RMSE")
plt.xlabel("Iteration")
plt.ylabel("RMSE")
plt.legend(loc="upper right")
plt.show()
```

The figure above is called the **learning curve**. It depicts the evolution of model performance throughout the learning process.

Remark 3.2.1

Learning curves give plenty of information about the model behavior. Hence, plotting and visually inspecting them facilitates debugging.

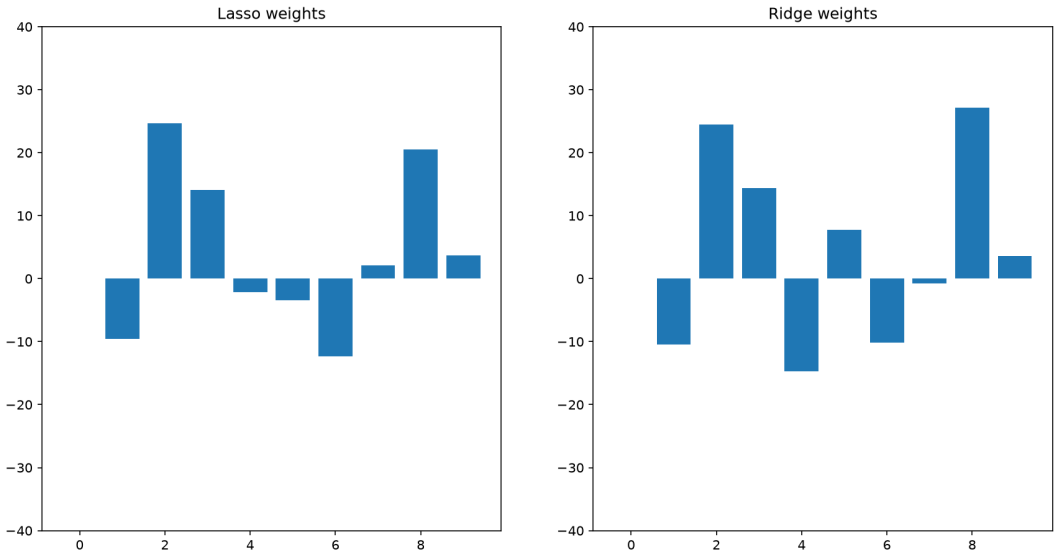


Figure 3.5: Learned weight coefficients under Lasso versus ridge regularization, showing that Lasso drives more coefficients to zero.

Let us next see how much shrinkage we gained from the Lasso and ridge regression regularizers. As can be seen below, Lasso sparsifies the parameters more than ridge: most of its weight coordinates sit exactly at zero, i.e. the learned hypothesis ignores the corresponding features altogether, while ridge keeps every coordinate slightly non-zero, i.e. every feature keeps a small influence.

```
fig, axes = plt.subplots(ncols=2, nrows=1, figsize=(14, 7))

axes[0].bar(th.arange(X_train.shape[1]),
            model_lasso.w.view(-1).detach())
axes[0].set_ylim(-40,40)
axes[0].set_title("Lasso weights")
axes[1].bar(th.arange(X_train.shape[1]),
            model_ridge.weights[:10])
axes[1].set_ylim(-40,40)
axes[1].set_title("Ridge weights")
plt.show()
```

CHAPTER 4

CLASSIFICATION

4.1 Logistic Regression

Assume we have a binary classification problem. We aim to fit an h to a data set $S = \{(x_i, y_i)|i \in [m]\}$ that consists of (x, y) pairs with $y \in \{0, 1\}$.

As in the previous lecture, we choose the hypothesis to be a linear function of the input:

$$\mathcal{H} := \{h : h(x) = w^\top x,\ w \in \mathbb{R}^d\}.$$

For notational convenience, let us ignore the bias term as one can always append a constant one to the input vector x to account for the bias.

The difference of the current setting is that the output space is discrete. Let us interpret the output of our hypothesis, the **score** $z_i := w^\top x_i$, as follows:

- Its sign indicates the class label. If $z_i > 0$, then the hypothesis predicts that x_i belongs to class 1. If $z_i < 0$, then the hypothesis predicts that x_i belongs to class 0.
- Its magnitude $|z_i|$ indicates the confidence of the hypothesis about the class assignment.

When interpreted this way, z_i is called the **discriminant function** (in Chapter 10 the same quantity reappears, under the same symbol, as the pre-activation of a neuron). While readily usable in its current shape, such a model is not easy to train: the prediction depends on the *sign* of z_i, which is not differentiable with respect to the parameters w, so no gradient-based method can tune them. The repair is to re-express the hypothesis as a probability — a differentiable quantity — while keeping the score's sign-and-magnitude reading intact. Let $P_i := P(y_i{=}1 \mid x_i)$ denote the probability that x_i belongs to class 1.

The question is *which* function of P_i the score should equal. Equating z_i with P_i itself cannot work: a probability lives in $(0, 1)$, while the score ranges over all of $\mathbb{R}$. What the score's reading does suggest is a continuous quantity that measures how probable class 1 is *relative to* class 0. The natural candidate is the **odds** $\dfrac{P_i}{1 - P_i}$: it equals 1 when the two classes are equally likely, grows above 1 as class 1 becomes the more probable, and shrinks below 1 as class 0 does. Taken as the score's scale, however, the odds treat the two sides of $P_i = \frac{1}{2}$ unevenly, without any reason to prefer one class over the other: for $P_i < \frac{1}{2}$ they are squeezed into the bounded interval $(0, 1)$, while for $P_i > \frac{1}{2}$ they stretch over all of $(1, \infty)$, so mirrored probabilities yield *reciprocal* odds rather than *opposite* ones — $P_i = 0.9$ gives odds 9, whereas

$P_i = 0.1$ gives $\frac{1}{9}$. The score, by contrast, is symmetric about zero: exchanging the roles of the two classes should merely negate it. The logarithm repairs exactly this asymmetry: the **log odds** $\log \frac{P_i}{1-P_i}$ is 0 at $P_i = \frac{1}{2}$, takes mirrored values at mirrored probabilities ($\log 9 = -\log \frac{1}{9}$), and ranges over all of $\mathbb{R}$ — a scale symmetric about zero, on a par with the score. Equating the score with the log odds yields:

$$\log \frac{P_i}{1-P_i} = w^\top x_i.$$

Let us solve for P_i stepwise:

$$\begin{aligned}
&\frac{P_i}{1-P_i} = e^{w^\top x_i} \\
&\Rightarrow P_i = e^{w^\top x_i} - P_i e^{w^\top x_i} \\
&\Rightarrow P_i + P_i e^{w^\top x_i} = e^{w^\top x_i} \\
&\Rightarrow P_i(1 + e^{w^\top x_i}) = e^{w^\top x_i} \\
&\Rightarrow P_i = \frac{e^{w^\top x_i}}{1 + e^{w^\top x_i}} \\
&\Rightarrow P_i = \frac{e^{w^\top x_i} e^{-w^\top x_i}}{(1 + e^{w^\top x_i}) e^{-w^\top x_i}} \\
&\Rightarrow P_i = \frac{1}{1 + e^{-w^\top x_i}}.
\end{aligned}$$

The result $g(u) = \frac{1}{1+e^{-u}}$ has the form of a function known as the **sigmoid function**. The expression $\log(P_i/(1-P_i))$ is sometimes called the **log odds** or the **logit function**. It is the inverse of the sigmoid function: $g^{-1}(P) = \log(P/(1-P))$ for some probability P. When $u := w^\top x_i$, the model is called **logistic regression** (Cox, 1958). Note what the construction achieved: the sign of the score still carries the prediction — $g(u) \geq \frac{1}{2}$ exactly when $u \geq 0$, since g is strictly increasing with $g(0) = \frac{1}{2}$ — and the magnitude of the score still carries confidence, now as distance from the decision boundary $\frac{1}{2}$ on the probability scale, and every bit of it is differentiable in w.

We would like our hypothesis to maximize the true class probability of all data points in the data set. Since we assume our data points to be independent, the probability of the data set is the product of the probabilities of each data point, and maximizing it point by point maximizes it jointly:

$$\prod_{i=1}^{m} P_i^{y_i}(1-P_i)^{1-y_i}.$$

Maximizing this product is mathematically equivalent to maximizing its logarithm, since log is strictly increasing. The logarithm is also what makes the objective usable, for two reasons. First, *numerical precision*: the product of many probabilities, each below 1, shrinks towards zero exponentially fast in m — a million data points of probability around 0.5 multiply to about $10^{-301030}$, far below what any computer can represent — and once a product underflows, comparing candidate parameters becomes impossible. The logarithm converts the product into a *sum*, which grows linearly in m instead of shrinking exponentially, and stays within representable range. Second, *optimization*: sums split over data points, so the objective and its gradient decompose into per-point contributions — a structure gradient descent (and

the mini-batching of Chapter 10) relies on — and the exponent in each factor, $P_i^{y_i}(1-P_i)^{1-y_i}$, comes down from the power to a multiplier, where $P_i = g(w^\top x_i)$ and g itself already involves an exponential. Hence, we can define the **binary cross-entropy loss** as the negated log-probability of the data set:

$$\mathcal{L}_{CE}(w) = -\sum_{i=1}^{m} \log g(w^\top x_i)^{y_i}(1-g(w^\top x_i))^{1-y_i} = -\sum_{i=1}^{m} \Big[y_i \log g(w^\top x_i) + (1-y_i)\log(1-g(w^\top x_i))\Big].$$

4.1.1 Extension to multi-class classification

Let us quickly extend the above formulation to multi-class classification. Assume we have C classes, that is, we aim to fit an h to a data set $S = \{(x_i, y_i) | i \in [m]\}$ that consists of (x, y) pairs with $y \in \{1, \ldots, C\}$. We will then need to model the class probabilities of C different classes, $P_1, \ldots, P_C$, from C per-class scores $u_c := w_c^\top x_i$. The binary construction generalizes directly: the log odds $\log \frac{P}{1-P}$ measures how probable a class is relative to *one specific other class*, so with C classes we pick one of them — call it class $j \in [C]$ — as the **baseline**, and let each score u_c measure class c against that baseline:

$$\log \frac{P_c}{P_j} = u_c = w_c^\top x_i, \qquad c \neq j.$$

Let us solve for the class probabilities step by step.

Step 1: exponentiate each equation. For every class $c \neq j$,

$$\frac{P_c}{P_j} = e^{u_c} \quad \Longrightarrow \quad P_c = P_j\, e^{u_c}.$$

The baseline's own equation is trivially $\log \frac{P_j}{P_j} = 0$, so class j has no score of its own — it is the reference against which the others are measured, and its "exponent" is $e^0 = 1$.

Step 2: determine P_j from normalization. The probabilities must sum to one. Substituting $P_c = P_j e^{u_c}$ for every $c \neq j$:

$$1 = \sum_{c=1}^{C} P_c = P_j + \sum_{c \neq j} P_j\, e^{u_c} = P_j\Big(1 + \sum_{c \neq j} e^{u_c}\Big) = P_j \sum_{c=1}^{C} e^{u_c},$$

where the last equality uses $1 = e^{u_j}$ with $u_j := 0$. Hence

$$P_j = \frac{1}{\sum_{c'=1}^{C} e^{u_{c'}}}.$$

Step 3: back-substitute. Combining $P_c = P_j e^{u_c}$ with the expression for P_j gives every class probability at once:

$$P_c = \frac{e^{u_c}}{\sum_{c'=1}^{C} e^{u_{c'}}} =: \mathrm{softmax}(u)_c,$$

which is called the **softmax** function. Note that the choice of the baseline class has effectively disappeared from the result: picking a different class j' as the baseline shifts every score by a constant ($u'_c = u_c - u_{j'}$, leaving the ratios $e^{u_c}/e^{u_{c'}}$ untouched), so the softmax depends only on the *differences* between

the scores — with $C = 2$ it reduces to the sigmoid of exactly that difference, $P_{\bar{j}} = g(u_{\bar{j}} - u_j)$ for the one non-baseline class $\bar{j}$. For the very same reason we may drop the bookkeeping convention $u_j = 0$ altogether and simply give **every** class its own score $u_c = w_c^\top x_i$, one weight vector per class: a common shift of all scores changes no probability, so the redundancy is harmless, and this is the convention used in the loss below. The related loss function is then:

$$\mathcal{L}_{CE}(W) = -\sum_{i=1}^{m} \log \text{softmax}(u_i)_{y_i} = \sum_{i=1}^{m} \Big\{ -w_{y_i}^\top x_i + \log\Big(\sum_{c=1}^{C} e^{w_c^\top x_i}\Big)\Big\},$$

where $u_i := (w_1^\top x_i, \ldots, w_C^\top x_i)$ and $W = [w_1 \ldots w_C]$ is a matrix of the weight vectors for each class. This loss function is called the **cross-entropy loss**. The name is inherited from information theory, where the very same expression appears as a measure of the disagreement between two probability distributions; anticipating the probability chapter, the intuitive reading is enough for now: since $\text{softmax}(u_i)_{y_i}$ is the probability the model assigns to the correct label of x_i, the loss is the *negative logarithm* of the model's probability of the observed data — minimizing it means making the observed labels as probable as possible. (Its information-theoretic content will resurface in Chapter 9, where the closely related notion of entropy is used to grow decision trees.)

Let us take the well-known **Iris** data set as an example. Our task is to classify the iris flowers into three species. The data set is available in the sklearn library. The data set consists of 150 data points. Each data point has four features: sepal length, sepal width, petal length, and petal width. The data points are labeled as one of the three species: i) setosa, ii) versicolor, and iii) virginica. The data set is balanced: There are 50 data points for each species.

```
from sklearn.datasets import load_iris
import matplotlib.pyplot as plt
import torch as th

th.manual_seed(0)

# The only library-provided piece is the raw data set itself; the split and
# every step of the classifier are implemented below.
X_all, y_all = load_iris(return_X_y=True)
X_all = th.tensor(X_all).float()
y_all = th.tensor(y_all).long()

perm = th.randperm(X_all.shape[0])
n_train = X_all.shape[0] // 2
X_train, y_train = X_all[perm[:n_train]], y_all[perm[:n_train]]
X_test, y_test = X_all[perm[n_train:]], y_all[perm[n_train:]]
```

Let us put together what we learned thus far to make up an algorithm that is as general as possible. Concretely, let us use the cross-entropy loss for empirical risk minimization and an L_p regularizer with tunable p to control the complexity of the model. The resulting optimization problem is:

$$\mathcal{L}(W) := \mathcal{L}_{CE}(W) + \lambda \sum_{c=1}^{C} ||w_c||_p^p.$$

We minimize it by gradient descent, exactly as for the Lasso in Chapter 3: the softmax cross-entropy is written out in the form derived above — a difference of one log-sum and a second log-sum over the exponentials of the per-class scores, hence its usual name, the **log-sum-exp** form — automatic differentiation supplies $\nabla_W \mathcal{L}(W)$, and the update rule is coded by hand.

```
class GeneralizedLinearClassifier:
    def __init__(self, n_dims, n_classes=3, lambda_coef=1.0, p=2):
        self.lambda_coef, self.p = lambda_coef, p
        self.W = th.randn(n_dims, n_classes).requires_grad_()
        self.b = th.randn(n_classes).requires_grad_()

    def predict(self, inputs):
        return inputs @ self.W + self.b        # the C per-class logits u_c

    def cross_entropy(self, logits, labels):
        # -log softmax(u)_y = -u_y + log sum_c exp(u_c), exactly the loss
        # derived above. The row maximum is subtracted before exponentiating
        # so that exp() cannot overflow; it cancels out of the difference.
        shift = logits.max(dim=1, keepdim=True).values
        log_norm = shift.squeeze(1) + (logits - shift).exp().sum(dim=1).log()
        return (log_norm - logits.gather(1, labels.view(-1, 1)).squeeze(1)).mean()

    def learn(self, inputs, labels, alpha=0.1, num_steps=1):
        for _ in range(num_steps):
            # Forward pass: cross-entropy risk plus the L_p regularizer
            loss = self.cross_entropy(self.predict(inputs), labels) \
                   + self.lambda_coef * (self.W.abs()**self.p).sum()
            # Backward pass: automatic differentiation fills W.grad and b.grad
            loss.backward()
            # The gradient descent step, written out explicitly
            with th.no_grad():
                for param in (self.W, self.b):
                    param -= alpha * param.grad
                    param.grad.zero_()
```

Let us train our model next and plot its learning curve, i.e. the evolution of its error across iterations.

```
# z-score normalization, with mu and sigma taken from the training split only
mu = X_train.mean(dim=0)
sd = X_train.std(dim=0, unbiased=False)
X_train = (X_train - mu) / sd
X_test = (X_test - mu) / sd

model_glc = GeneralizedLinearClassifier(n_dims=X_train.shape[1],
                                        lambda_coef=0.01, p=2)
num_iterations = 1000
train_errors = th.zeros(num_iterations)
```

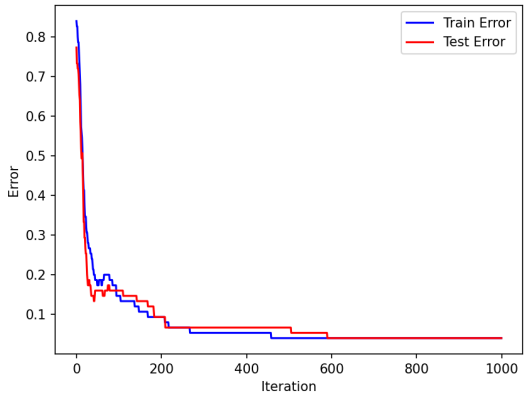


Figure 4.1: Training and test error of the generative linear classifier across gradient descent iterations.

```
test_errors = th.zeros(num_iterations)

for ii in range(num_iterations):
    model_glc.learn(X_train, y_train)
    with th.no_grad():
        pred_class = model_glc.predict(X_train).argmax(dim=1)
        train_errors[ii] = (pred_class != y_train).float().mean()
        pred_class = model_glc.predict(X_test).argmax(dim=1)
        test_errors[ii] = (pred_class != y_test).float().mean()

plt.plot(th.arange(num_iterations), train_errors, 'b-', label="Train Error")
plt.plot(th.arange(num_iterations), test_errors, 'r-', label="Test Error")
plt.xlabel("Iteration"); plt.ylabel("Error"); plt.legend(loc="upper right")
plt.show()
```

4.2 Performance Metrics for Classification

The goal of classification is to recognize a pattern, e.g. an object. From the viewpoint of a single object type, there can be four different outcomes of a classification task:

- True Positive (TP): The object is correctly classified as positive.
- False Positive (FP): The object is incorrectly classified as positive.
- True Negative (TN): The object is correctly classified as negative.
- False Negative (FN): The object is incorrectly classified as negative.

Let us make a matrix of these possible outcomes as a latex table:

	Predicted Positive	**Predicted Negative**
Actual Positive	TP	FN

	Predicted Positive	**Predicted Negative**
Actual Negative	FP	TN

The version of this table where its entries are filled with the number of data points that fall into each category is called a **confusion matrix**. We can compute many performance metrics from the confusion matrix:

1. **Accuracy**: (TP + TN) / (TP + FP + TN + FN), the ratio of the number of correctly classified objects to the total number of objects.
2. **Precision**: TP / (TP + FP), the ratio of the number of correctly classified objects to the number of objects classified as the object type.
3. **Recall** (a.k.a. **True Positive Rate**, **Sensitivity**): TP / (TP + FN), the ratio of the number of correctly classified objects to the number of objects that are actually of the object type.
4. **F1 score**: 2 × Precision × Recall / (Precision + Recall), the harmonic mean of the precision and the recall.
5. **False Positive Rate**: FP / (FP + TN), the ratio of the number of objects that are not of the object type but classified as the object type to the total number of objects that are not of the object type.

The rationale behind the F1 score is that precision and recall are rates. Harmonic mean is a more sensible score to take the average of multiple rates. Consider a car that travels a distance d with speed x and returns with speed y. The average speed of the whole travel is

$$\frac{2d}{\frac{d}{x}+\frac{d}{y}} = \frac{2}{\frac{1}{x}+\frac{1}{y}}$$

which is the harmonic mean of x and y. In words, harmonic mean is the reciprocal of the average of the reciprocals of a set of quantities.

One can calculate the confusion matrix also for more than two classes. Let us see what it looks like for the Iris data set.

```
with th.no_grad():
    pred_class = model_glc.predict(X_test).argmax(dim=1)

C = 3
# cm[a, p] counts the test points with actual label a and prediction p
cm = th.bincount(y_test * C + pred_class, minlength=C * C).reshape(C, C)
print(cm)

names = ['setosa', 'versicolor', 'virginica']
fig, ax = plt.subplots()
ax.imshow(cm, cmap='Blues')
for a in range(C):
    for p in range(C):
        ax.text(p, a, cm[a, p].item(), ha='center', va='center')
ax.set_xticks(range(C), names); ax.set_yticks(range(C), names)
ax.set_xlabel("Prediction"); ax.set_ylabel("Actual")
plt.show()
```

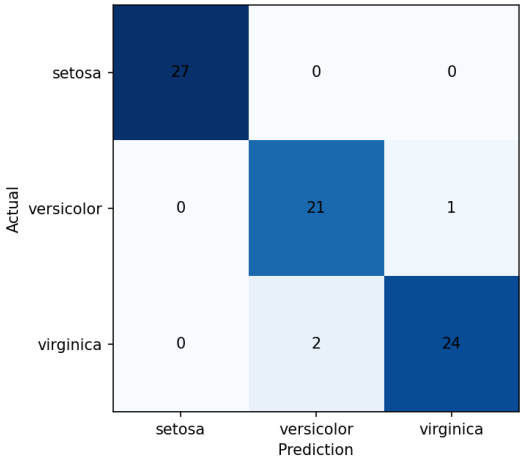


Figure 4.2: Confusion matrix of the classifier's predictions on the Iris test set.

```
tensor([[27,  0,  0],
        [ 0, 21,  1],
        [ 0,  2, 24]])
```

Except for accuracy, the other four performance metrics need to be calculated separately for each class. For instance, let us calculate them for versicolor, the class the model finds hardest.

```
c = 1                                                  # versicolor
precision = cm[c, c] / cm[:, c].sum()                  # sum towards the column
recall = cm[c, c] / cm[c, :].sum()                     # sum towards the row
print("Precision: ", precision.item())
print("Recall: ", recall.item())
print("F1: ", (2 * precision * recall / (precision + recall)).item())
```

```
Precision:  0.9130434989929199
Recall:  0.9545454382896423
F1:  0.9333332777023315
```

Another important performance metric is **Receiver Operating Characteristics (ROC)**. It is a plot of the true positive rate (TPR) against the false positive rate (FPR) for all possible thresholds applicable on the discriminant function. The **Area Under the ROC curve (AUC)** is a measure of the performance of the classifier. The higher the AUC, the better the classifier. A perfect classifier has an AUC of 1. A classifier that performs no better than random guessing has an AUC of 0.5.

Let us plot the ROC curve for the Iris data set.

```
with th.no_grad():
    logits = model_glc.predict(X_test)
    # The class posteriors P_c = softmax(u)_c of the model
    probs = (logits - logits.max(dim=1, keepdim=True).values).exp()
    probs = probs / probs.sum(dim=1, keepdim=True)
```

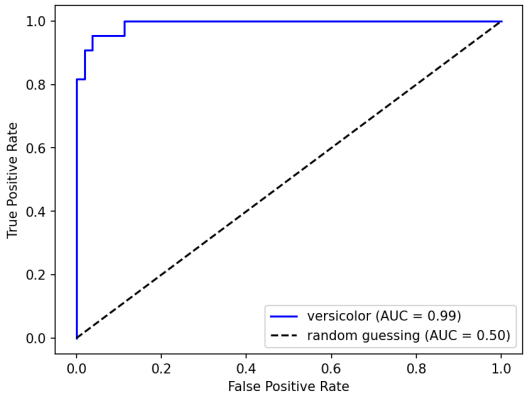


Figure 4.3: ROC curve for the versicolor-vs-rest classifier, with the corresponding AUC compared against random guessing.

```
# One-vs-rest score for versicolor (class 1) and the matching binary labels
scores = probs[:, 1]
positives = (y_test == 1)

# Sweep the threshold down through the observed scores, in decreasing order,
# and read off TPR and FPR at every position.
order = th.argsort(scores, descending=True)
hits = positives[order].float()
tp = th.cat([th.zeros(1), hits.cumsum(0)])
fp = th.cat([th.zeros(1), (1 - hits).cumsum(0)])
tpr, fpr = tp / tp[-1], fp / fp[-1]
auc = th.trapz(tpr, fpr)                # area under the curve, by trapezoids

plt.figure()
plt.plot(fpr, tpr, 'b-', label="versicolor (AUC = {:.2f})".format(auc))
plt.plot([0, 1], [0, 1], 'k--', label="random guessing (AUC = 0.50)")
plt.xlabel("False Positive Rate"); plt.ylabel("True Positive Rate")
plt.legend(loc="lower right")
plt.show()
```

Remark 4.2.1

Accuracy is a meaningful performance metric only when the classes are evenly distributed. The cases when the classes are not evenly distributed are called **imbalanced classification problems**. In such cases, we need to use other metrics such as precision, recall, and F1 score.

4.3 K-Fold Cross Validation

As is noticeable in the above example, the observed performance may depend greatly on the particular train-test split. To mitigate this problem, we can use **K-fold cross validation**. The idea is to split the data set into K folds. Then, we train the model on $K-1$ folds and test it on the remaining fold. We repeat this process K times, each time using a different fold as the test set. The final performance metric is the average of the performance metrics obtained in each iteration.

4.4 K-Nearest Neighbors (kNN) Classifier

The **k-nearest neighbors** (Fix and Hodges, 1951) classifier assigns a query point to the class of the majority of its k nearest neighbors. It is a **non-parametric classifier**, that is, it does not represent the model with a fixed number of parameters. Instead, it memorizes the whole training data and uses it to make predictions. The number of neighbors k is a hyperparameter of the model. While it is a strong classifier, its prediction-time complexity is unacceptable: $O(m)$, where m is the number of training data points.

One can implement a kNN in various ways. Given a query point x, one can find its k nearest neighbors $Ne(x;k) = \{(x_i, y_i) : [k]\}$ in the training data set and assign x to the class of the majority of its k nearest neighbors:

$$\widehat{y} = \arg\max_{c \in [C]} \sum_{i=1}^{k} \mathbb{1}(y_i = c).$$

One can also assign a weight to each neighbor inversely proportional to its distance to the query point and assign x to the class of the majority of its k nearest neighbors weighted by the inverse of their distances to x.

$$\widehat{y} = \arg\max_{c \in [C]} \sum_{i=1}^{k} \mathbb{1}(y_i = c) \frac{1}{\text{dist}(x, x_i)}$$

for some distance function $\text{dist}(\cdot, \cdot)$ on $\mathcal{X}$, i.e. the second component of a metric space $(\mathcal{X}, \text{dist})$ in the sense of Chapter 3 — any such choice makes "nearest" well defined.

The **Voronoi cell** of a point x is the set of points whose nearest neighbor is x. The Voronoi cells of the training data points form a partition of the input space. The decision boundary of a 1-nearest neighbor classifier is the set of points that are equidistant to two or more training data points. For general k, the decision boundary instead sits where the *majority label* among the k nearest neighbors changes — a subset of the points equidistant between two training points, but not all of them, since moving across such a point need not flip the majority vote. See an illustration from the Iris data set below.

```
from matplotlib.colors import ListedColormap

# A tiny subsample, so that the individual Voronoi cells stay visible
sub = th.randperm(X_all.shape[0])[:10]
X_knn, y_knn = X_all[sub][:, :2], y_all[sub]

def knn_predict(queries, X, y, k=1, weights="uniform", n_classes=3):
```

```
    """Assign each query to the class with the largest (weighted) vote among
    its k nearest training points -- the two rules stated above."""
    d = th.cdist(queries, X)                        # pairwise distances
    dist, idx = d.topk(k, dim=1, largest=False)     # the k nearest neighbours
    w = th.ones_like(dist) if weights == "uniform" else 1.0 / (dist + 1e-12)
    votes = th.zeros(queries.shape[0], n_classes)
    votes.scatter_add_(1, y[idx], w)                # accumulate 1(y_i = c) * w
    return votes.argmax(dim=1)

cmap_light = ListedColormap(["orange", "cyan", "cornflowerblue"])
cmap_bold = ["darkorange", "c", "darkblue"]
names = ['setosa', 'versicolor', 'virginica']

# The mesh on which the decision regions are painted
pad = 0.5
g0 = th.linspace(X_knn[:, 0].min() - pad, X_knn[:, 0].max() + pad, 200)
g1 = th.linspace(X_knn[:, 1].min() - pad, X_knn[:, 1].max() + pad, 200)
gg0, gg1 = th.meshgrid(g0, g1, indexing='xy')
mesh = th.stack([gg0.reshape(-1), gg1.reshape(-1)], dim=1)

def plot_regions(zz, k, weights):
    fig, ax = plt.subplots()
    # vmin/vmax are pinned to the class range: with auto-scaling, a class absent
    # from the predictions (e.g. a setosa point that never wins a 3-vote) would
    # shift the color mapping and paint the remaining classes in the wrong colors.
    ax.pcolormesh(gg0, gg1, zz, cmap=cmap_light, shading="auto", vmin=0, vmax=2)
    for c in range(3):
        pts = X_knn[y_knn == c]
        ax.scatter(pts[:, 0], pts[:, 1], c=cmap_bold[c],
                   edgecolor="black", label=names[c])
    ax.set_xlabel("sepal length (cm)"); ax.set_ylabel("sepal width (cm)")
    ax.legend()
    ax.set_title("kNN decision regions (k = %d, weights = '%s')" % (k, weights))

# With k = 1 the two weighting schemes coincide -- a single neighbor takes
# the whole vote whatever its weight -- and the decision regions are
# exactly the Voronoi cells of the training points.
plot_regions(knn_predict(mesh, X_knn, y_knn, k=1,
↪   weights="uniform").reshape(gg0.shape),
             k=1, weights="uniform")

# The weightings disagree only once several neighbors share the vote:
# with uniform weights each of the k neighbors counts equally, with
# distance weights a close neighbor can outvote several farther ones.
for weights in ["uniform", "distance"]:
    plot_regions(knn_predict(mesh, X_knn, y_knn, k=3,
↪   weights=weights).reshape(gg0.shape),
```

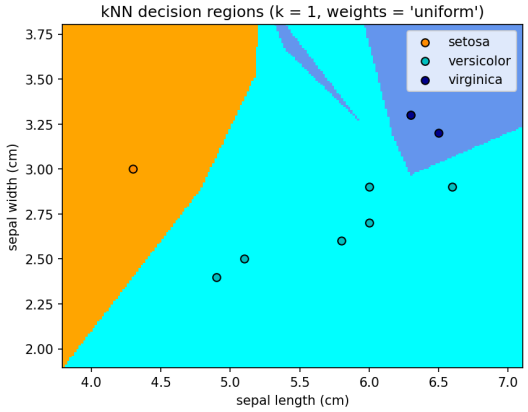


Figure 4.4: 1-nearest-neighbor decision regions on the Iris data: the decision regions are the Voronoi cells of the training points.

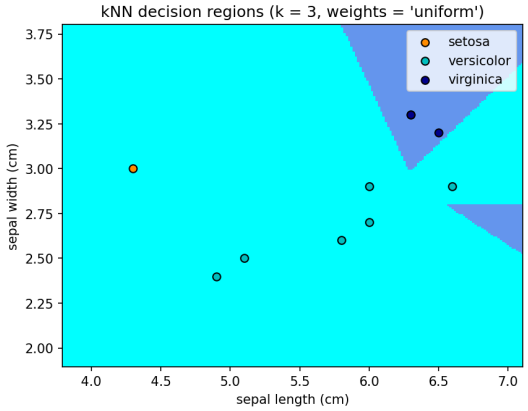


Figure 4.5: 3-nearest-neighbor decision regions with uniform neighbor weighting.

```
                k=3, weights=weights)

plt.show()
```

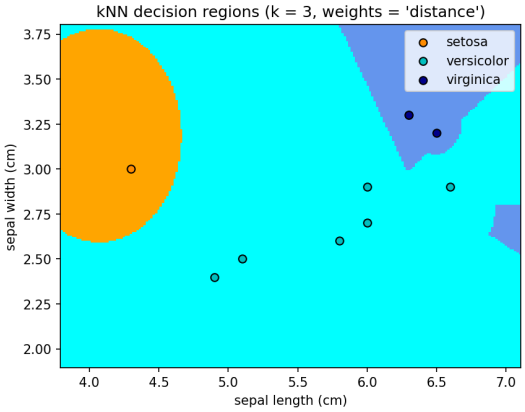


Figure 4.6: 3-nearest-neighbor decision regions with distance-based neighbor weighting, to be compared against the uniform-weighting figure above.

CHAPTER 5

PROBABILITY THEORY

This chapter builds the probabilistic machinery the rest of the course runs on. The notational conventions it uses were fixed in Chapter 2 and are assumed throughout.

5.1 Measure-Theoretic Foundations

Sample Space (Ω): The set of all possible outcomes of a random process.

Event (A): A subset of the sample space, $A \subset \Omega$.

2^Ω is the **powerset** of Ω. Let $\mathcal{F} \subseteq 2^\Omega$ be the collection of events we are willing to assign a probability to. Then $\mathcal{F}$ is called the **event space**. Technically speaking, it is a **σ-algebra**: a collection of subsets of Ω that contains Ω itself and is closed under complementation and countable unions. That is, $\Omega \in \mathcal{F}$, and if $A_1, A_2, \cdots \in \mathcal{F}$, then:

1. $A_1^c \in \mathcal{F}$,
2. $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

Closure under countable intersections, $\bigcap_{i=1}^{\infty} A_i \in \mathcal{F}$, follows from these two axioms via De Morgan's law: $\bigcap_i A_i = \left(\bigcup_i A_i^c\right)^c$.

Why we need this at all. For discrete sample spaces, such as the outcome of the roll of a pair of dice, it is safe to take $\mathcal{F} = 2^\Omega$: every subset is an event, and we can assign a probability to it. For continuous sample spaces, such as $\Omega = \mathbb{R}$, the powerset $2^{\mathbb{R}}$ is too large — it contains pathological sets to which no consistent notion of "length" (and hence probability) can be assigned. The fix is to only ever ask for the probability of sets built from intervals. The **Borel σ-algebra** $\mathcal{B}(\mathbb{R})$ is the smallest σ-algebra containing all open intervals $(a, b) \subset \mathbb{R}$; it contains every set we will ever need in practice (all open sets, closed sets, countable unions and intersections thereof) while excluding the pathological ones. When $\Omega = \mathbb{R}^d$, we use $\mathcal{F} = \mathcal{B}(\mathbb{R}^d)$, the σ-algebra generated analogously by open rectangles.

A **probability measure** is a function $P : \mathcal{F} \to [0, 1]$ such that:

1. $P(\Omega) = 1$,
2. $P(A) \geq 0$ for all $A \in \mathcal{F}$,
3. **(σ-additivity)** if $A_1, A_2, \ldots$ are pairwise disjoint events (i.e. $A_i \cap A_j = \emptyset$ for all $i \neq j$), then $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$.

Probability Space: The triple $(\Omega, \mathcal{F}, P)$. This is the complete mathematical object a random experiment is modeled by. Everything else in this chapter — random variables, expectations, concentration inequalities — is built on top of it.

Intuitive meaning of probability. While P is defined axiomatically above, the frequentist intuition behind it is the long-run frequency of an event across repeated, independent trials:

$$P(A) = \lim_{n \to \infty} \frac{n_A}{n},$$

where n_A is the number of times event A occurs in n trials. This intuition is useful for building understanding, but it is not the definition; the axioms above are.

5.2 Basic Rules

The rules below follow directly from the axioms and are standard fare; we state them compactly since they will be used freely throughout the rest of the notes without re-derivation.

- **Union bound.** From the inclusion-exclusion identity $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ and $P(A \cap B) \geq 0$, we get $P(A \cup B) \leq P(A) + P(B)$. By induction, for any (not necessarily disjoint) events $A_1, \ldots, A_n$,

  $$P\Big(\bigcup_{i=1}^{n} A_i\Big) \leq \sum_{i=1}^{n} P(A_i).$$

  This inequality — despite its triviality — is the single most used tool in this course: it lets us bound the probability that *any one* of many bad events happens by summing the probabilities of each bad event individually.

- **Conditional probability.** $P(A \mid B) := \dfrac{P(A \cap B)}{P(B)}$ for $P(B) > 0$, read as "the probability of A given that B has occurred." Rearranged, this gives the **product rule**: $P(A \cap B) = P(A \mid B)P(B)$.

- **Independence.** Events A, B are **independent** if $P(A \mid B) = P(A)$, equivalently $P(A \cap B) = P(A)P(B)$.

- **Law of total probability.** If $B_1, \ldots, B_n$ partition Ω (pairwise disjoint, $\bigcup_i B_i = \Omega$), then $P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i)$.

- **Bayes' rule** (Bayes and Price, 1763). From the product rule applied both ways, $P(A \mid B)P(B) = P(A \cap B) = P(B \mid A)P(A)$, hence

  $$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}.$$

While each of these facts is elementary on its own, Bayes' rule has outsized implications for statistical inference and machine learning. Assume $H_1, \ldots, H_n$ are mutually exclusive hypotheses partitioning a hypothesis space $\mathcal{H}$, and $\mathcal{D}$ is observed data. Combining Bayes' rule with the law of total probability (using $H_1, \ldots, H_n$ as the partition) gives

$$P(H_i \mid \mathcal{D}) = \frac{P(\mathcal{D} \mid H_i)P(H_i)}{P(\mathcal{D})} = \frac{P(\mathcal{D} \mid H_i)P(H_i)}{\sum_{j=1}^{n} P(\mathcal{D} \mid H_j)P(H_j)}.$$

Here:

- $P(H_i)$ is our **prior belief** in hypothesis H_i,
- $P(H_i \mid \mathcal{D})$ is our **posterior belief** in H_i after observing the data $\mathcal{D}$,
- $P(\mathcal{D} \mid H_i)$ is the **likelihood** of H_i, and $P(\mathcal{D})$ is the **evidence**.

This gives a general procedure for statistical inference: start with a prior belief, collect observations, update the belief. Chapter 8 builds an entire class of learning algorithms on exactly this procedure.

5.3 Random Variables

The probability measure P maps *sets* to numbers. It is not always convenient to work directly with sets; we would rather describe outcomes numerically and delegate the set-theoretic bookkeeping to the machinery below. This is what a random variable does.

Definition 5.3.1: Random variable

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $(\Lambda, \mathcal{G})$ a measurable space (e.g. $\Lambda = \mathbb{R}^k$ with its Borel σ-algebra $\mathcal{G} = \mathcal{B}(\mathbb{R}^k)$). A **random variable** is a function $X : \Omega \to \Lambda$ that is **measurable**, meaning

$$X^{-1}(A) := \{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{F}, \quad \forall A \in \mathcal{G}.$$

Measurability is exactly the condition needed for the expression "$P(X \in A)$" to make sense: it guarantees that the set of outcomes ω mapping into A is itself an event, i.e. an event to which P can assign a probability. Given this, we define the **induced probability measure** (a.k.a. the **law** or **distribution**) of X as

$$P_X(A) := P(X^{-1}(A)) = P(\{\omega \in \Omega : X(\omega) \in A\}), \quad A \in \mathcal{G}.$$

$(\Lambda, \mathcal{G}, P_X)$ is itself a probability space. This is the sense in which a random variable "transports" a probability space we may not care about the internals of (e.g. the space of all coin-toss sequences) onto a space we do care about (e.g. $\{0, 1, \ldots, n\}$, the count of heads).

Discrete random variables. When Λ is countable, e.g. $\Lambda = \{0, 1, \ldots, C-1\}$, every subset of Λ is measurable and it suffices to specify P_X on singletons:

$$P(X = x) := P_X(\{x\}) = P(\{\omega \in \Omega : X(\omega) = x\}), \quad x \in \Lambda.$$

This is called the **probability mass function (pmf)** of X, and $P_X(A) = \sum_{x \in A} P(X = x)$ recovers the measure on any event A. Note that we write a pmf with a **capital** P, because it *is* the probability of the

event $\{X = x\}$ — no new object is being introduced. Lowercase p is reserved, throughout these notes, for the densities of continuous random variables defined next, which are **not** probabilities of any event.

Continuous random variables. When $\Lambda = \mathbb{R}^k$, singletons typically carry zero probability ($P_X(\{x\}) = 0$), so a pmf is useless; instead we ask for P_X to be absolutely continuous with respect to the Lebesgue measure μ_{Leb} (ordinary k-dimensional volume). This means there exists a function $p_X \geq 0$, called the **probability density function (pdf)**, such that

$$P_X(A) = \int_A p_X(x)\, d\mu_{\text{Leb}}(x), \qquad \forall A \in \mathcal{B}(\mathbb{R}^k).$$

For $k = 1$ this is the familiar $P_X(A) = \int_A p_X(x)\, dx$. Not every random variable admits a density (some are discrete, some are neither), but every case relevant to this course is either discrete or admits one.

Example : three coin tosses

Toss a fair coin three times; write H for heads and T for tails. The sample space is

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\},$$

and since the coin is fair and the tosses are independent, $P(\omega) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$ for every $\omega \in \Omega$. Let $X : \Omega \to \{0, 1, 2, 3\}$ count the number of heads. Then X is a (measurable, since Ω is finite and $\mathcal{F} = 2^\Omega$) random variable with pmf

1. $P(X = 0) = P(\{TTT\}) = \frac{1}{8}$,
2. $P(X = 1) = P(\{TTH, THT, HTT\}) = \frac{3}{8}$,
3. $P(X = 2) = P(\{HTH, HHT, THH\}) = \frac{3}{8}$,
4. $P(X = 3) = P(\{HHH\}) = \frac{1}{8}$.

Random variables also let us query compound events easily, e.g.

$$P(X \geq 2) = P(\{HTH, HHT, THH, HHH\}) = \frac{1}{2}.$$

Joint distributions, marginals, independence. Given two random variables X, Y on the same probability space, their **joint distribution** is $P(X = x, Y = y) := P(\{\omega : X(\omega) = x, Y(\omega) = y\})$ (with the analogous density definition in the continuous case). We say X and Y are **independent** if $P(X = x, Y = y) = P(X = x)P(Y = y)$ for all x, y — the random-variable counterpart of event independence above.

When a variable in a joint distribution is not of direct interest, it is called a **nuisance variable**, and we can eliminate it by the law of total probability. This operation is called **marginalization**. For instance, if X is the outcome of a fair die roll and $Y \in \{\text{even}, \text{odd}\}$ its parity,

$$\begin{aligned} P(X = x) &= \sum_{y \in \{\text{even},\text{odd}\}} P(X = x, Y = y) \\ &= \sum_y P(X = x \mid Y = y) P(Y = y). \end{aligned}$$

Using $P(X = x)$ lets us query events concerning X alone, with Y's influence already averaged out proportionally to its own probability of occurrence. (As a sanity check on the definition of independence above: for a fair die, $X \leq 4$ and $Y =$ even are independent — $P(X \leq 4, Y{=}\text{even}) = \frac{2}{6}$ equals $P(X \leq 4)P(Y{=}\text{even}) = \frac{4}{6} \cdot \frac{3}{6}$ — while $X \leq 3$ and $Y =$ even are not, since low rolls are disproportionately odd.)

5.4 Expectation as an Integral

Definition 5.4.1: Expectation

The **expectation** of a random variable $X : \Omega \to \mathbb{R}$ is the Lebesgue integral of X against P:

$$\mathbb{E}[X] := \int_{\Omega} X(\omega)\, dP(\omega).$$

This single definition specializes to the two formulas you have likely seen before, depending on the law of X:

- if X is discrete with pmf $P(X = \cdot)$, the abstract integral reduces to a sum: $\mathbb{E}[X] = \sum_{x \in \Lambda} x\, P(X = x)$;
- if X is continuous with pdf p_X, it reduces to a Riemann-style integral: $\mathbb{E}[X] = \int_{\mathbb{R}} x\, p_X(x)\, dx$.

Framing expectation as a single integral against P, rather than as two unrelated formulas, is precisely the payoff of the measure-theoretic viewpoint: every fact we prove about $\mathbb{E}[\cdot]$ below (linearity, monotonicity) holds simultaneously for discrete and continuous random variables — and for mixtures of the two — because it is proven once, at the level of the integral, rather than twice.

In the three coin tosses example above, $\mathbb{E}[X] = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = 1.5$.

The expectation is also called the **mean** or **first moment** of X; it summarizes the **central tendency** of X with a single number, but says nothing about its **spread**. Spread is measured by the **variance**:

$$\mathrm{Var}[X] := \mathbb{E}\big[(X - \mathbb{E}[X])^2\big] = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

In the example above, $\mathrm{Var}[X] = 0^2 \cdot \frac{1}{8} + 1^2 \cdot \frac{3}{8} + 2^2 \cdot \frac{3}{8} + 3^2 \cdot \frac{1}{8} - 1.5^2 = 0.75$. The square root of the variance is the **standard deviation**, commonly denoted σ; here $\sigma = \sqrt{0.75} \approx 0.866$.

Variance is also called the **second (central) moment** of X. In general, the k**-th central moment** is $\mathbb{E}[(X - \mathbb{E}[X])^k]$ for a positive integer k: the first is (trivially) zero and the second is the variance. The third and fourth become scale-free, and acquire their usual names, only after being standardized by the appropriate power of σ: the **skewness** $\mathbb{E}[(X - \mathbb{E}[X])^3]/\sigma^3$ measures the asymmetry of the distribution, and the **kurtosis** $\mathbb{E}[(X - \mathbb{E}[X])^4]/\sigma^4$ the heaviness of its tails.

For two jointly distributed random variables X, Y, the **covariance** quantifies how their deviations from their own means co-vary:

$$\mathrm{Cov}[X, Y] := \mathbb{E}\big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\big] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Setting $Y = X$ recovers the variance, $\mathrm{Cov}[X, X] = \mathrm{Var}[X]$.

5.5 Estimators and Bias

Estimator. A function that maps a sample S to a value $\widehat{\theta}(S)$ is called an **estimator** of a parameter θ. We write the estimator and the value it produces with the same symbol $\widehat{\theta}$, suppressing the argument S when it is clear from context. The running example throughout this chapter is the **sample mean** of an i.i.d. sample $X_1, \dots, X_m \sim X$,

$$\widehat{\mu}_m := \frac{1}{m} \sum_{i=1}^{m} X_i,$$

which estimates the mean $\mu := \mathbb{E}[X]$.

Bias. The bias of an estimator is $\mathrm{Bias}(\widehat{\theta}) := \mathbb{E}[\widehat{\theta}(S)] - \theta$. An estimator is **unbiased** if $\mathrm{Bias}(\widehat{\theta}) = 0$. For example, $\widehat{\mu}_m$ is unbiased for μ:

$$\mathrm{Bias}(\widehat{\mu}_m) = \mathbb{E}[\widehat{\mu}_m] - \mu = \mathbb{E}\Big[\frac{1}{m}\sum_{i=1}^{m} X_i\Big] - \mu = \frac{1}{m}\sum_{i=1}^{m} \mathbb{E}[X_i] - \mu = \frac{1}{m}\sum_{i=1}^{m} \mu - \mu = 0.$$

By contrast, the sample variance $\widehat{\sigma}^2_m := \frac{1}{m}\sum_{i=1}^{m}(X_i - \widehat{\mu}_m)^2$ is a *biased* estimator of σ^2; the **unbiased sample variance** $\frac{1}{m-1}\sum_{i=1}^{m}(X_i - \widehat{\mu}_m)^2$ corrects for this via **Bessel's correction**. (We reserve S for a data set throughout these notes, which is why the sample variance is not written S^2 as it often is in statistics texts.)

Consistency. An estimator is **consistent** if $\lim_{m\to\infty} P(|\widehat{\theta}(S) - \theta| \geq \epsilon) = 0$ for every $\epsilon > 0$: the probability of a large error vanishes as the sample grows. The next section builds the tools to establish (and quantify) consistency.

5.6 Concentration of Measure Inequalities

Theorem 5.6.1: Markov's inequality

Let X be a non-negative random variable. Then for any $\epsilon > 0$,

$$P(X \geq \epsilon) \leq \frac{\mathbb{E}[X]}{\epsilon}.$$

Proof. Since $X \geq 0$, the pointwise inequality $X \geq \epsilon\, \mathbb{1}(X \geq \epsilon)$ holds for every outcome: if $X < \epsilon$ the right side is $0 \leq X$, and if $X \geq \epsilon$ the right side is $\epsilon \leq X$. Taking expectations and using monotonicity and linearity of the integral,

$$\mathbb{E}[X] \geq \epsilon\, \mathbb{E}\big[\mathbb{1}(X \geq \epsilon)\big] = \epsilon\, P(X \geq \epsilon),$$

where the last equality is the defining property of the indicator: $\mathbb{E}[\mathbb{1}(A)] = P(A)$. Dividing by $\epsilon > 0$ gives the claim □

Note how little the proof uses: only non-negativity and linearity. This is why Markov's inequality is the weakest of the three bounds in this section — and why every sharper bound below is obtained by first *transforming* X into a quantity to which Markov can be applied more profitably.

Example :
Consider ten Bernoulli random variables $X_1, \dots, X_{10}$ with $P(X_i = 1) = P(X_i = 0) = 0.5$. Since $\mathbb{E}[\sum_{i=1}^{10} X_i] = 5$, Markov's inequality only gives $P(\sum_{i=1}^{10} X_i \geq 3) \leq 5/3$, which exceeds 1 and is thus **vacuous**.

Theorem 5.6.2: Chebyshev's inequality

Let X have finite mean μ and variance σ^2. Then for any $\epsilon > 0$,

$$P(|X - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}.$$

Proof. The events $\{|X - \mu| \geq \epsilon\}$ and $\{(X - \mu)^2 \geq \epsilon^2\}$ are identical, since $u \mapsto u^2$ is increasing on $[0, \infty)$. The random variable $(X - \mu)^2$ is non-negative, so Theorem 5.6.1 applies to it with threshold ϵ^2:

$$P(|X - \mu| \geq \epsilon) = P\big((X - \mu)^2 \geq \epsilon^2\big) \leq \frac{\mathbb{E}[(X - \mu)^2]}{\epsilon^2} = \frac{\mathrm{Var}[X]}{\epsilon^2} = \frac{\sigma^2}{\epsilon^2}. \qquad \square$$

This is the transformation trick announced above, in its simplest form: squaring turns a two-sided deviation into a non-negative quantity, and improves on Markov by a factor $1/\epsilon$, at the cost of assuming a finite second moment. Applying Markov to e^{sX} instead of $(X - \mu)^2$ — the **Chernoff trick** — yields an *exponential* improvement, at the price of assuming a finite moment generating function; this is exactly the route Theorem 5.6.5 takes below.

Consider $S = \{X_1, \dots, X_m\}$ i.i.d. Bernoulli with sample average $\widehat{\mu}_m$ as defined above. How much can $\widehat{\mu}_m$ deviate from μ? Markov's inequality alone gives us

$$P(\widehat{\mu}_m - \mathbb{E}[\widehat{\mu}_m] \geq \epsilon) = P(\widehat{\mu}_m \geq \mu + \epsilon) \leq \frac{\mathbb{E}[\widehat{\mu}_m]}{\mu + \epsilon} = \frac{\mu}{\mu + \epsilon},$$

a bound that does not shrink no matter how large m is — useless for arguing that more data helps. Chebyshev's inequality does better:

$$\begin{aligned} P(|\widehat{\mu}_m - \mathbb{E}[\widehat{\mu}_m]| \geq \epsilon) &\leq \frac{\mathrm{Var}[\widehat{\mu}_m]}{\epsilon^2} = \frac{\mathrm{Var}\big[\frac{1}{m}\sum_{i=1}^{m} X_i\big]}{\epsilon^2} = \frac{1}{m^2}\frac{\sum_{i=1}^{m}\mathrm{Var}[X_i]}{\epsilon^2} \\ &= \frac{1}{m^2}\frac{m\,\mathrm{Var}[X_i]}{\epsilon^2} = \frac{\sigma^2}{m\epsilon^2}, \end{aligned}$$

using $\mathrm{Var}[\sum_i X_i] = \sum_i \mathrm{Var}[X_i]$ for independent X_i (second step) and $\mathrm{Var}[aX] = a^2\mathrm{Var}[X]$ for a constant a (third step). This bound shrinks as $\Theta(1/m)$, giving the **weak law of large numbers**: $\square$

Theorem 5.6.3: Weak law of large numbers

Let $X_1, \dots, X_m$ be i.i.d. with finite mean μ and variance σ^2. Then for any $\epsilon > 0$, $\lim_{m\to\infty} P(|\widehat{\mu}_m - \mu| \geq \epsilon) = 0$, where $\widehat{\mu}_m = \frac{1}{m}\sum_{i=1}^{m} X_i$.

Proof. $\widehat{\mu}_m$ is unbiased, $\mathbb{E}[\widehat{\mu}_m] = \mu$, and the display preceding this theorem computed $\mathrm{Var}[\widehat{\mu}_m] = \sigma^2/m$. Chebyshev's inequality (Theorem 5.6.2) applied to $\widehat{\mu}_m$ therefore gives

$$0 \leq P\big(|\widehat{\mu}_m - \mu| \geq \epsilon\big) \leq \frac{\sigma^2}{m\epsilon^2} \xrightarrow[m\to\infty]{} 0$$

for every fixed $\epsilon > 0$, and the claim follows by squeezing $\square$

Hence $\widehat{\mu}_m$ is a consistent estimator of μ. Note what the proof needed and what it did not: finiteness of σ^2 was used, but no boundedness and no distributional shape. In fact finiteness of the mean alone suffices, by a **truncation** argument. Let $\mathbb{E}|X| < \infty$, fix $\epsilon > 0$ and a level $t > 0$, and split $X_i = Y_i + Z_i$ with $Y_i := X_i \mathbb{1}(|X_i| \le t)$ and $Z_i := X_i \mathbb{1}(|X_i| > t)$, so that $\widehat{\mu}_m - \mu = (\overline{Y}_m - \mathbb{E}[Y_1]) + (\overline{Z}_m - \mathbb{E}[Z_1])$ with bars denoting sample means. If $|\widehat{\mu}_m - \mu| \ge \epsilon$, one of the two brackets is at least $\epsilon/2$ in absolute value. The bounded part has $\mathrm{Var}[Y_1] \le \mathbb{E}[Y_1^2] \le t^2$, so Chebyshev's inequality bounds the first probability by $\frac{4t^2}{m\epsilon^2}$. For the tail part, $\mathbb{E}\big|\overline{Z}_m - \mathbb{E}[Z_1]\big| \le 2\,\mathbb{E}|Z_1| = 2r(t)$ with $r(t) := \mathbb{E}\big[|X|\mathbb{1}(|X| > t)\big]$, so Markov's inequality bounds the second by $\frac{4r(t)}{\epsilon}$. Altogether

$$P\big(|\widehat{\mu}_m - \mu| \ge \epsilon\big) \;\le\; \frac{4t^2}{m\epsilon^2} + \frac{4\,r(t)}{\epsilon}.$$

As $t \to \infty$, $|X|\mathbb{1}(|X| > t)$ decreases to 0 pointwise and is dominated by the integrable $|X|$, so $r(t) \to 0$ by the dominated convergence theorem of Lebesgue integration. Given $\eta > 0$, first choose t with $4r(t)/\epsilon \le \eta/2$, then every $m \ge 8t^2/(\eta\epsilon^2)$ makes the right-hand side at most η. This is the weak law of large numbers in the generality due to Khinchin.

5.6.1 Hoeffding's inequality

Chebyshev's inequality gets tighter at the polynomial rate $1/m$. Boundedness lets us do much better — exponentially in m, with a bound due to (Hoeffding, 1963) that we derive below.

The device is the **Chernoff trick**: instead of applying Markov's inequality to X or to $(X-\mu)^2$, apply it to e^{sX} for a free parameter $s > 0$, then optimize over s. What makes this profitable is that the exponential turns a sum of independent variables into a *product* of expectations, and that each factor can be bounded using boundedness alone. The latter is the content of the following lemma.

Lemma 5.6.4: Hoeffding's lemma

Let X be a random variable with $\mathbb{E}[X] = 0$ and $P(a \le X \le b) = 1$. Then for every $s \in \mathbb{R}$,

$$\mathbb{E}\big[e^{sX}\big] \le \exp\Big(\frac{s^2(b-a)^2}{8}\Big).$$

Proof. Note $a \le 0 \le b$, since $\mathbb{E}[X] = 0$. Because $u \mapsto e^{su}$ is convex, every $x \in [a,b]$, written as the convex combination $x = \lambda b + (1-\lambda)a$ with $\lambda = \frac{x-a}{b-a}$, satisfies

$$e^{sx} \le \frac{x-a}{b-a}e^{sb} + \frac{b-x}{b-a}e^{sa}.$$

Taking expectations and using $\mathbb{E}[X] = 0$ to kill the x-dependent parts,

$$\mathbb{E}\big[e^{sX}\big] \le \frac{-a}{b-a}e^{sb} + \frac{b}{b-a}e^{sa} = p\,e^{sb} + (1-p)e^{sa},$$
$$p := \frac{-a}{b-a} \in [0,1].$$

Substitute $u := s(b-a)$, so that $sa = -pu$, and factor out e^{sa}:

$$p\,e^{sb} + (1-p)e^{sa} = e^{-pu}\big(1 - p + p\,e^{u}\big) = e^{\varphi(u)},$$
$$\varphi(u) := -pu + \log\big(1 - p + pe^{u}\big).$$

It remains to show $\varphi(u) \le u^2/8$. Differentiating,

$$\varphi'(u) = -p + \frac{pe^{u}}{1-p+pe^{u}},$$
$$\varphi''(u) = q(u)\big(1-q(u)\big) \ \text{ with } \ q(u) := \frac{pe^{u}}{1-p+pe^{u}} \in [0,1].$$

Hence $\varphi(0) = 0$, $\varphi'(0) = 0$, and $\varphi''(u) \le \tfrac{1}{4}$ everywhere, since $q(1-q) \le \tfrac{1}{4}$ for $q \in [0,1]$. Taylor's theorem with Lagrange remainder around 0 therefore gives $\varphi(u) = \tfrac{1}{2}\varphi''(\xi)u^2 \le u^2/8$ for some ξ between 0 and u. Substituting back $u = s(b-a)$ completes the proof □

Theorem 5.6.5: Hoeffding's inequality

Let $X_1, \dots, X_m$ be independent random variables such that $P(a_i \le X_i \le b_i) = 1$ for all i. Then for any $\epsilon > 0$,

$$P\Big(\frac{1}{m}\sum_{i=1}^{m} X_i - \mathbb{E}\Big[\frac{1}{m}\sum_{i=1}^{m} X_i\Big] \ge \epsilon\Big) \le \exp\Big(\frac{-2m^2\epsilon^2}{\sum_{i=1}^{m}(b_i-a_i)^2}\Big).$$

Proof. Center the variables, $Y_i := X_i - \mathbb{E}[X_i]$, so that $\mathbb{E}[Y_i] = 0$ and Y_i lies in an interval of the same width $b_i - a_i$. For any $s > 0$, the map $u \mapsto e^{su}$ is increasing, so Markov's inequality (Theorem 5.6.1) applied to the non-negative variable $e^{s\sum_i Y_i}$ gives

$$\begin{aligned} P\Big(\sum_{i=1}^{m} Y_i \ge m\epsilon\Big) = P\Big(e^{s\sum_i Y_i} \ge e^{sm\epsilon}\Big) &\le e^{-sm\epsilon}\,\mathbb{E}\Big[e^{s\sum_i Y_i}\Big] \\ &= e^{-sm\epsilon}\prod_{i=1}^{m}\mathbb{E}\big[e^{sY_i}\big] \\ &\le e^{-sm\epsilon}\exp\Big(\frac{s^2}{8}\sum_{i=1}^{m}(b_i-a_i)^2\Big), \end{aligned}$$

where the equality on the second line uses independence (the expectation of a product of independent variables factorizes) and the last step applies Lemma 5.6.4 to each Y_i separately. The bound holds for every $s > 0$, so we may take the best one. Writing $V := \sum_i (b_i-a_i)^2$, the exponent $-sm\epsilon + s^2V/8$ is a convex quadratic in s, minimized at $s^* = 4m\epsilon/V > 0$, where its value is

$$-\frac{4m^2\epsilon^2}{V} + \frac{2m^2\epsilon^2}{V} = -\frac{2m^2\epsilon^2}{V}.$$

Substituting and dividing the event by m gives the claim □

Applying Theorem 5.6.5 to the variables $-X_1, \dots, -X_m$ (which lie in $[-b_i, -a_i]$, of the same width) bounds the opposite tail by the same quantity:

$$P\Big(\mathbb{E}\Big[\frac{1}{m}\sum_{i=1}^{m} X_i\Big] - \frac{1}{m}\sum_{i=1}^{m} X_i \geq \epsilon\Big) \leq \exp\Big(\frac{-2m^2\epsilon^2}{\sum_{i=1}^{m}(b_i - a_i)^2}\Big),$$

and adding the two gives the two-sided version with an extra factor of 2.

We will use these inequalities to bound the generalization performance of learning algorithms. Let X_i be the loss of the i-th training example; then $\widehat{\mu}_m = \frac{1}{m}\sum_{i=1}^{m} X_i$ is the empirical risk and $\mu = \mathbb{E}[\widehat{\mu}_m]$ the expected risk. It is the lower tail that we need — the event that the true risk exceeds what training led us to believe. If $X_i \in [0, 1]$ for all i (so $b_i - a_i = 1$), it simplifies to

$$P(\mu \geq \widehat{\mu}_m + \epsilon) \leq e^{-2m\epsilon^2}.$$

Setting $\delta := e^{-2m\epsilon^2}$ and solving for ϵ gives $\epsilon = \sqrt{\log(1/\delta)/(2m)}$, hence

$$P\Big(\mu \geq \widehat{\mu}_m + \sqrt{\tfrac{\log(1/\delta)}{2m}}\Big) \leq \delta,$$
$$\text{equivalently} \quad P\Big(\mu \leq \widehat{\mu}_m + \sqrt{\tfrac{\log(1/\delta)}{2m}}\Big) \geq 1 - \delta.$$

Since we will later allocate μ to generalization performance and $\widehat{\mu}_m$ to training performance, it is this second, complementary form that we will reuse directly. Concentration inequalities of this form are called **generalization bounds**, and are the basis of statistical learning theory (Chapter 6).

5.6.2 Uniform Hoeffding bounds over several estimators

A single Hoeffding bound controls the deviation of *one* empirical average from its mean. In machine learning we rarely evaluate a single quantity: we compare many hypotheses' training errors, or track several estimators built from ranges (loss scales) that differ from one another. We now extend Theorem 5.6.5 to this setting, using only the union bound.

Setup. Suppose we have K estimators $\widehat{\mu}_m^{(1)}, \dots, \widehat{\mu}_m^{(K)}$, where $\widehat{\mu}_m^{(k)} := \frac{1}{m}\sum_{i=1}^{m} X_i^{(k)}$ is built from m independent observations $X_1^{(k)}, \dots, X_m^{(k)}$ (independent across i; the K estimators themselves need not be independent of each other) with each $X_i^{(k)} \in [a_k, b_k]$, i.e. the k-th estimator's underlying random variable lives in its *own* interval, of its own width $b_k - a_k$. Write $\mu^{(k)} := \mathbb{E}[\widehat{\mu}_m^{(k)}]$.

Theorem 5.6.6: Uniform Hoeffding bound

For any $\epsilon_1, \dots, \epsilon_K > 0$,

$$P\Big(\exists k \in [K] : \mu^{(k)} - \widehat{\mu}_m^{(k)} \geq \epsilon_k\Big) \leq \sum_{k=1}^{K} \exp\Big(\frac{-2m\epsilon_k^2}{(b_k - a_k)^2}\Big).$$

In the special case $\epsilon_k \equiv \epsilon$ and $b_k - a_k \equiv (b-a)$ for all k (a common range for every estimator), this simplifies to

$$P\Big(\exists k \in [K] : \mu^{(k)} - \widehat{\mu}_m^{(k)} \geq \epsilon\Big) \leq K \exp\Big(\frac{-2m\epsilon^2}{(b-a)^2}\Big).$$

Proof. Let B_k denote the "bad" event $\{\mu^{(k)} - \widehat{\mu}_m^{(k)} \geq \epsilon_k\}$. Then

$$\begin{aligned} P\Big(\exists k \in [K] : \mu^{(k)} - \widehat{\mu}_m^{(k)} \geq \epsilon_k\Big) &= P\Big(\bigcup_{k=1}^{K} B_k\Big) \\ &\leq \sum_{k=1}^{K} P(B_k) \\ &\leq \sum_{k=1}^{K} \exp\Big(\frac{-2m\epsilon_k^2}{(b_k - a_k)^2}\Big), \end{aligned}$$

where the first inequality is the union bound and the second applies Theorem 5.6.5 (in its lower-tail form) to each B_k separately, using only the independence of $X_1^{(k)}, \dots, X_m^{(k)}$ *within* estimator k — no relationship between different estimators $k \neq k'$ is required. Note that the m variables of estimator k all lie in the same interval, so $\sum_{i=1}^{m}(b_k - a_k)^2 = m(b_k - a_k)^2$, which is what turns the m^2 in Theorem 5.6.5 into the m above. □

What this yields. Setting the right-hand side to a target confidence δ and solving for a common ϵ (in the equal-range case) gives $\epsilon = (b-a)\sqrt{\log(K/\delta)/(2m)}$, so that simultaneously, for every one of the K estimators,

$$P\Big(\mu^{(k)} \leq \widehat{\mu}_m^{(k)} + (b-a)\sqrt{\tfrac{\log(K/\delta)}{2m}},\ \forall k \in [K]\Big) \geq 1 - \delta.$$

Two consequences matter directly for what follows in Chapter 6:

- **Effect of the hypothesis class.** If each estimator k corresponds to the training loss of a distinct hypothesis h_k in a finite hypothesis class $\mathcal{H} = \{h_1, \dots, h_K\}$, the bound degrades only logarithmically in $K = |\mathcal{H}|$ — a much larger hypothesis class costs comparatively little in confidence, but every additional hypothesis we simultaneously monitor must be paid for somewhere, and it is paid for here, inside the square root. This is the seed of the finite-hypothesis-class generalization bound proved in Chapter 6.
- **Effect of loss scale.** If different hypotheses (or different loss functions) produce losses on different scales $[a_k, b_k]$, the *per-estimator* range $(b_k - a_k)$ enters the bound directly and linearly. A

hypothesis whose loss can swing across a wide range is intrinsically harder to certify than one confined to a narrow range, even before we ask anything about the hypothesis class's size or structure — boundedness of the *loss*, not just of the sample, plays an essential role.

5.6.3 Bernstein's inequality

Chebyshev's inequality and Hoeffding's inequality bound the deviation of an empirical average using two different properties of the summands: Chebyshev pays with the variance, Hoeffding with the range. That these are not the same price matters. A Bernoulli(p) variable with a small p, say $p = 10^{-6}$, lies in $[0, 1]$, so Hoeffding treats it as a full-width $[0, 1]$ variable, but its variance $p(1-p) \approx 10^{-6}$ is six orders of magnitude smaller than the range would suggest: the variable is almost always (nearly) 0. The inequality due to (Bernstein, 1924) pays a variance-dependent price instead: the exponent is $\frac{m^2\epsilon^2}{2\sum_i \sigma_i^2 + \frac{2}{3}mb\epsilon}$, a variance term with a range-dependent correction that takes over only in the large-deviation regime. The proof runs the Chernoff trick of Theorem 5.6.5 once more, replacing Hoeffding's lemma (Lemma 5.6.4) by a moment bound that keeps track of the variance and charges the range only for the high-order part of the exponential series.

Lemma 5.6.7: Moment growth under boundedness

Let X be a random variable with $\mathbb{E}[X] = 0$, $\mathrm{Var}[X] = \sigma^2$, and $P(|X| \le b) = 1$ for some $b > 0$. Then for every integer $k \ge 2$,

$$\mathbb{E}\big[X^k\big] \le b^{k-2}\sigma^2.$$

Proof. Since $k \ge 2$ and $|X| \le b$ almost surely, we have $|X|^{k-2} \le b^{k-2}$ almost surely, and

$$\mathbb{E}\big[X^k\big] \le \mathbb{E}\big[|X|^k\big] = \mathbb{E}\big[|X|^{k-2}X^2\big] \le b^{k-2}\,\mathbb{E}\big[X^2\big] = b^{k-2}\sigma^2. \square$$

□

Lemma 5.6.8: Bennett's mgf bound

Let X be a random variable with $\mathbb{E}[X] = 0$, $\mathrm{Var}[X] = \sigma^2$, and $P(X \le b) = 1$ for some $b > 0$. Then for every $s > 0$,

$$\mathbb{E}\big[e^{sX}\big] \le \exp\left(\frac{\sigma^2}{b^2}\big(e^{sb} - 1 - sb\big)\right).$$

Proof. Expand the moment generating function into its power series and apply Lemma 5.6.7 to each

term:

$$\begin{aligned}\mathbb{E}\big[e^{sX}\big] = 1 + \sum_{k=2}^{\infty} \frac{s^k\,\mathbb{E}[X^k]}{k!} &\le 1 + \sum_{k=2}^{\infty} \frac{s^k\, b^{k-2}\sigma^2}{k!} \\ &= 1 + \frac{\sigma^2}{b^2}\sum_{k=2}^{\infty}\frac{(sb)^k}{k!} \\ &= 1 + \frac{\sigma^2}{b^2}\big(e^{sb} - 1 - sb\big) \\ &\le \exp\Big(\frac{\sigma^2}{b^2}\big(e^{sb} - 1 - sb\big)\Big),\end{aligned}$$

where the last step uses $1 + u \le e^u$ □

The contrast with Hoeffding's lemma (Lemma 5.6.4) is instructive: that lemma charges the full range, $\mathbb{E}[e^{sX}] \le e^{s^2(b-a)^2/8}$, while Bennett's bound keeps the *variance* σ^2 in front and replaces the quadratic-in-s charge by the exact Taylor remainder $e^{sb} - 1 - sb$ of a single summand hitting its range bound b. For small sb this remainder is $\approx \frac{1}{2}(sb)^2$, recovering a Hoeffding-like quadratic with the variance in front; for large sb it grows only like $e^{sb}/b^2 \cdot \sigma^2$, which is where the range eventually has to be paid for.

Theorem 5.6.9: Bernstein's inequality

Let $X_1, \dots, X_m$ be independent random variables with $\mathbb{E}[X_i] = 0$, $\mathrm{Var}[X_i] = \sigma_i^2$, and $P(|X_i| \le b) = 1$ for a common $b > 0$. Then for any $\epsilon > 0$,

$$P\Big(\frac{1}{m}\sum_{i=1}^{m} X_i \ge \epsilon\Big) \le \exp\Big(-\frac{m^2\epsilon^2}{2\sum_{i=1}^{m}\sigma_i^2 + \frac{2}{3}mb\epsilon}\Big).$$

Proof. For any $s > 0$, Markov's inequality (Theorem 5.6.1) applied to the increasing transformation $u \mapsto e^{su}$ of the sum, and independence, give

$$\begin{aligned}P\Big(\sum_{i=1}^{m} X_i \ge m\epsilon\Big) \le e^{-sm\epsilon}\prod_{i=1}^{m}\mathbb{E}\big[e^{sX_i}\big] &\le \exp\Big(-sm\epsilon + \frac{V}{b^2}\big(e^{sb} - 1 - sb\big)\Big) \\ &=: e^{h(s)},\end{aligned}$$

where the second inequality applies Lemma 5.6.8 to each X_i (valid since $P(X_i \le b) = 1$ follows from $P(|X_i| \le b) = 1$) and $V := \sum_i \sigma_i^2$. The bound holds for every $s > 0$, so it remains to minimize $h(s) = -sm\epsilon + \frac{V}{b^2}(e^{sb} - 1 - sb)$ over $s > 0$. Since $h''(s) = Ve^{sb} > 0$, h is strictly convex, $h(0) = 0$, and $h'(0) = -m\epsilon < 0$, so the infimum over $s > 0$ is attained at the unique stationary point $s^* > 0$ solving

$$h'(s^*) = -m\epsilon + \frac{V}{b}\big(e^{s^*b} - 1\big) = 0 \iff e^{s^*b} = 1 + \frac{m\epsilon b}{V}.$$

Define $u := \frac{m\epsilon b}{V} > 0$, so $s^*b = \log(1+u)$, $e^{s^*b} - 1 = u$, and $m\epsilon = Vu/b$. Substituting into h,

$$\begin{aligned}h(s^*) = -\frac{Vu}{b}\cdot\frac{\log(1+u)}{u} + \frac{V}{b^2}\big(u - \log(1+u)\big) &= -\frac{V}{b^2}\Big[(1+u)\log(1+u) - u\Big] \\ &= -\frac{V}{b^2}\,\psi(u),\end{aligned}$$

where $\psi(u) := (1+u)\log(1+u) - u$. The elementary inequality $\psi(u) \geq \frac{u^2/2}{1+u/3}$ for all $u \geq 0$ (Exercise) therefore gives

$$h(s^*) \leq -\frac{V}{b^2} \cdot \frac{u^2}{2 + \frac{2}{3}u} = -\frac{m^2\epsilon^2}{2V + \frac{2}{3}m\epsilon b},$$

where the last step substitutes $u = m\epsilon b/V$ and clears denominators:

$$\frac{V}{b^2} \cdot \frac{u^2}{2 + \frac{2}{3}u} = \frac{V}{b^2} \cdot \frac{m^2\epsilon^2 b^2/V^2}{2 + \frac{2}{3}\frac{m\epsilon b}{V}} = \frac{m^2\epsilon^2}{2V + \frac{2}{3}m\epsilon b}.$$

Since $e^{h(s)} \leq e^{h(s^*)}$ for every $s > 0$, the claim follows □

Exercise. Prove that $\psi(u) := (1+u)\log(1+u) - u \geq \frac{u^2/2}{1+u/3}$ for all $u \geq 0$, as follows. Let $f(u) := \frac{u^2/2}{1+u/3}$ and show that $\psi(0) = f(0) = 0$ and $\psi'(0) = f'(0) = 0$, and that the difference of the second derivatives is

$$\psi''(u) - f''(u) = \frac{u^2(9+u)}{(1+u)(3+u)^3} \geq 0,$$

using $(3+u)^3 - 27(1+u) = u^2(9+u)$. Conclude by two integrations from 0 to u.

Applying Theorem 5.6.9 to $-X_1, \ldots, -X_m$ (which satisfy the same hypotheses: $|-X_i| = |X_i| \leq b$ and $\mathrm{Var}[-X_i] = \sigma_i^2$) bounds the opposite tail identically, and adding the two gives the two-sided version with an extra factor of 2.

The two regimes. Specialize to the i.i.d. case $\sigma_i^2 \equiv \sigma^2$ and read the bound as a function of the deviation size ϵ:

- **Small-deviation (Gaussian) regime, $\epsilon \lesssim \sigma^2/b$.** The variance term $2m\sigma^2$ dominates the denominator and the bound reads $e^{-m\epsilon^2/2\sigma^2}$ — a Gaussian tail with the *true* variance in the denominator. This is where Bernstein beats Hoeffding: for variables confined to $[-b, b]$, Hoeffding's exponent is $\frac{m\epsilon^2}{2b^2}$, so the ratio of the exponents is b^2/σ^2 — six orders of magnitude in the Bernoulli(10^{-6}) example above. Since the deviations that matter for generalization bounds are typically of order $\sqrt{\sigma^2/m}$ — exactly the boundary of this regime — the Gaussian regime is the one that actually gets used.
- **Large-deviation (Poissonian) regime, $\epsilon \gg \sigma^2/b$.** The correction term $\frac{2}{3}mb\epsilon$ dominates the denominator and the bound reads $\exp(-\frac{3}{2}\frac{m\epsilon}{b})$ — linear in the deviation, the shape of a Poisson tail. No variance information can help this far out: the extreme tail of a bounded variable is controlled by how far it can reach, not by how much it typically fluctuates.

The variance-adaptivity of the Gaussian regime is what makes Bernstein's inequality the standard tool behind **empirical-Bernstein generalization bounds**, importance-weighted risk estimation, and the confidence bonuses of bandit algorithms — wherever the noise is heteroscedastic or the range is a pessimistic proxy for the variance.

5.6.4 Multiplicative Chernoff Bounds

Hoeffding's inequality treats the deviation ϵ as an *absolute* quantity: the same ϵ is charged whether the mean is $\mu = \frac{1}{2}$ or $\mu = 10^{-6}$. Bernstein's inequality repairs this through the variance; for $[0,1]$-valued

variables, whose variance is at most the mean, $\mathrm{Var}[X_i] \le \mathbb{E}[X_i^2] \le \mu$, the same repair can be carried out directly in terms of μ. For the small-mean case — which is exactly the regime of a well-trained classifier, whose error is close to zero — a *relative* deviation is the more natural scale, and a sharper bound is available. It is obtained by optimizing the Chernoff parameter exactly rather than crudely, at the cost of replacing the quadratic exponent by a binary version of the KL divergence. For $z, \mu \in (0,1)$ define

$$\mathrm{kl}(z \,\|\, \mu) := z \log \frac{z}{\mu} + (1-z) \log \frac{1-z}{1-\mu},$$

the **KL divergence between two Bernoulli laws** with parameters z and μ: it is zero iff $z = \mu$ and grows as z moves away from μ. (This is the two-outcome special case of the Kullback-Leibler divergence that reappears, in its general form, in the PAC-Bayes bounds of Chapter 7.)

Theorem 5.6.10: KL-form Chernoff bound

Let $X_1, \dots, X_m$ be i.i.d. with $P(X_i \in [0,1]) = 1$ and $\mathbb{E}[X_i] = \mu$. Then for every $z \in (\mu, 1)$,

$$P\Big(\frac{1}{m}\sum_{i=1}^{m} X_i \ge z\Big) \le e^{-m\,\mathrm{kl}(z\|\mu)},$$

and for every $z \in (0, \mu)$,

$$P\Big(\frac{1}{m}\sum_{i=1}^{m} X_i \le z\Big) \le e^{-m\,\mathrm{kl}(z\|\mu)}.$$

Proof. We start from the *chord trick* already used in Hoeffding's lemma (Lemma 5.6.4): for every $s > 0$, the function $x \mapsto e^{sx}$ is convex, so its graph on $[0,1]$ lies below the chord through the endpoints,

$$e^{sx} \;\le\; (1-x)\,e^{s\cdot 0} + x\,e^{s\cdot 1} \;=\; (1-x) + x\,e^{s} \qquad (x \in [0,1]).$$

Applying this to X_1 and taking expectations,

$$\mathbb{E}\big[e^{sX_1}\big] \le 1 - \mu + \mu\, e^{s},$$

and by independence,

$$P\Big(\sum_{i=1}^{m} X_i \ge mz\Big) \le e^{-smz}\,\mathbb{E}\big[e^{s\sum_i X_i}\big] \le \exp\big(-smz + m\log(1-\mu+\mu e^{s})\big),$$

using Markov's inequality (Theorem 5.6.1) with the increasing transformation $u \mapsto e^{su}$. The exponent $h(s) := -smz + m\log(1-\mu+\mu e^{s})$ is a convex function of s with $h(0) = 0$, and $h(s) \to +\infty$ as $s \to \infty$ (the linear term $ms(1-z)$ dominates), so its infimum over $s > 0$ is attained at the unique stationary point: setting $h'(s) = -mz + m\frac{\mu e^{s}}{1-\mu+\mu e^{s}} = 0$ and solving,

$$e^{s^*} = \frac{z(1-\mu)}{\mu(1-z)}$$

(note $s^* > 0$ since $z > \mu$). At this point $1 - \mu + \mu e^{s^*} = \frac{1-\mu}{1-z}$, so

$$h(s^*) = m\Big[- z \log \frac{z(1-\mu)}{\mu(1-z)} + \log \frac{1-\mu}{1-z} \Big] = -m \,\mathrm{kl}(z \| \mu),$$

where the last equality expands the logarithms and collects the z- and $(1-z)$-weighted terms. This proves the upper tail. For the lower tail, apply the upper-tail result to $Y_i := 1 - X_i$, whose mean is $1 - \mu$: the event $\{\widehat{\mu}_m \le z\}$ is $\{\overline{Y}_m \ge 1 - z\}$, and $\mathrm{kl}(1 - z \,\|\, 1 - \mu) = \mathrm{kl}(z \| \mu)$ by exchanging the roles of the two outcomes □

The gain over Hoeffding's $e^{-2m\epsilon^2}$ shows up when μ is small. At the extreme point $z = 0$ the exponent is *exact*: since $\mathrm{kl}(0 \| \mu) = -\log(1 - \mu) \ge \mu$,

$$P\big(\widehat{\mu}_m = 0\big) = (1 - \mu)^m = e^{-m \,\mathrm{kl}(0 \| \mu)} \le e^{-m\mu}.$$

Each of the m independent draws must miss the μ-rare event, which costs a factor $(1 - \mu)$ per draw — the exponent is **linear in the mean**, not quadratic in the deviation. The following corollary extracts from the KL form the explicit quadratic lower bounds on kl that make this usable.

Corollary 5.6.11

Multiplicative Chernoff bounds. Under the assumptions of Theorem 5.6.10, for every $\epsilon \in (0, 1 - \mu)$,

$$P\big(\widehat{\mu}_m \ge \mu + \epsilon\big) \le \exp\Big(- \frac{m\epsilon^2}{2\mu + \frac{2}{3}\epsilon} \Big),$$

and for every $\epsilon \in (0, \mu)$,

$$P\big(\widehat{\mu}_m \le \mu - \epsilon\big) \le \exp\Big(- \frac{m\epsilon^2}{2\mu} \Big).$$

Equivalently, with $\epsilon = \gamma\mu$,

$$P\big(\widehat{\mu}_m \ge (1+\gamma)\mu\big) \le \exp\Big(- \frac{m\mu\gamma^2}{2 + \frac{2}{3}\gamma} \Big), \qquad P\big(\widehat{\mu}_m \le (1-\gamma)\mu\big) \le \exp\Big(- \frac{m\mu\gamma^2}{2} \Big).$$

Proof. By Theorem 5.6.10 it suffices to bound $\mathrm{kl}(z \| \mu)$ from below at $z = \mu \pm \epsilon$. Since $\log v \ge 1 - \frac{1}{v}$ for $v > 0$, the second term of kl satisfies $(1 - z) \log \frac{1-z}{1-\mu} \ge (1 - z)\big(1 - \frac{1-\mu}{1-z}\big) = \mu - z$, so, writing $z = (1 + u)\mu$ with $u > -1$,

$$\mathrm{kl}(z \| \mu) \;\ge\; z \log \frac{z}{\mu} - (z - \mu) = \mu\big[(1+u)\log(1+u) - u\big] = \mu\,\psi(u),$$

with ψ the function of the proof of Bernstein's inequality (Theorem 5.6.9). For the upper tail, $u = \epsilon/\mu > 0$, and the inequality $\psi(u) \ge \frac{u^2/2}{1 + u/3}$ of the Exercise following Theorem 5.6.9 gives

$$\mathrm{kl}(\mu + \epsilon \| \mu) \;\ge\; \mu \,\frac{\epsilon^2/(2\mu^2)}{1 + \epsilon/(3\mu)} = \frac{\epsilon^2}{2\mu + \frac{2}{3}\epsilon}.$$

For the lower tail, $u = -\epsilon/\mu \in (-1, 0)$. There $\psi(0) = \psi'(0) = 0$ and $\psi''(v) = \frac{1}{1+v} \geq 1$ for $v \in (-1, 0]$, so Taylor's theorem with Lagrange remainder gives $\psi(u) \geq \frac{u^2}{2}$, and $\mathrm{kl}(\mu - \epsilon \| \mu) \geq \mu \frac{\epsilon^2}{2\mu^2} = \frac{\epsilon^2}{2\mu}$ □

The two tails are not symmetric. The upper-tail denominator cannot be replaced by 2μ: at $\mu = 0.1$, $\epsilon = 0.1$ that would claim the exponent $\frac{\epsilon^2}{2\mu} = 0.05$, but the exact exponent of Theorem 5.6.10 is only $\mathrm{kl}(0.2 \| 0.1) \approx 0.044$ (the corollary's exponent is $\frac{0.01}{0.2667} \approx 0.0375$). The lower tail — the empirical average falling below the mean — is the one that drives the generalization bounds of Chapter 6, and there the clean 2μ holds. Comparing with Bernstein's inequality is instructive: a $[0, 1]$-valued variable has $\mathrm{Var}[X_i] \leq \mathbb{E}[X_i^2] \leq \mu$, and Theorem 5.6.9 applied to $X_i - \mu$ (with $|X_i - \mu| \leq 1$) gives exactly the upper-tail bound above, while for the lower tail it gives the weaker denominator $2\mu + \frac{2}{3}\epsilon$. The KL form of Theorem 5.6.10 is sharper than both displays of the corollary, whose exponents are lower bounds on kl.

CHAPTER 6

STATISTICAL LEARNING THEORY

Solving hard real-world problems with machine learning demands advanced algorithmic designs. These designs should be based on a solid understanding of the principles that determine the behavior of learning algorithms. Statistical learning theory provides mathematical tools to develop such an understanding. It seeks answers to fundamental questions about the nature of learning from data such as

- Under which conditions can a learning algorithm generalize from the training data to new data?
- How can we measure the complexity of a learning task?
- How can we control the complexity of a learning algorithm?
- How can we estimate the generalization performance of a learning algorithm?

Every generalization bound in this course, no matter how it is derived, has the same shape: with high probability, the gap $R(h) - \widehat{R}_S(h)$ between true and empirical risk, uniformly over $h \in \mathcal{H}$, is controlled by a single number that measures **how complex $\mathcal{H}$ is**. What changes between different theories of generalization is only *how that number is defined* — as a raw count ($|\mathcal{H}|$, this chapter), a combinatorial shattering capacity ($d_{VC}(\mathcal{H})$, this chapter), an average best-case correlation with random noise (**Rademacher complexity**), a description length in a fixed metric (**covering numbers**), or a KL-divergence budget between a data-dependent posterior and a fixed prior over hypotheses (**PAC-Bayes bounds**) — the last three being the subject of Chapter 7. This chapter builds the foundational machinery — the finite-hypothesis-class bound, PAC learnability, the No Free Lunch theorem, VC dimension, and the Fundamental Theorem of Statistical Learning — using the coarsest and most concrete of these complexity measures, $|\mathcal{H}|$ and $d_{VC}(\mathcal{H})$. Chapter 7 then shows that these are two special cases of one abstract "complexity of a hypothesis class" idea, of which Rademacher complexity, covering numbers, and PAC-Bayes bounds are three more.

Building on what we covered in the probability theory chapter, let us start by investigating the last of the four questions above.

6.1 Generalization Bounds

Remember the basic definitions below.

Definition 6.1.1: Generalization error

Given a hypothesis $h \in \mathcal{H}$, a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, 1]$, and a data distribution $(x, y) \sim \mathcal{D}$, the generalization error of h is defined as

$$R(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(y, h(x))].$$

Definition 6.1.2: Empirical error

Given a hypothesis $h \in \mathcal{H}$, a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, 1]$, and a data set $S = \{(x_1, y_1), ..., (x_m, y_m)\}$, the empirical error of h is defined as

$$\widehat{R}_S(h) = \frac{1}{m}\sum_{i=1}^{m} \ell(y_i, h(x_i)).$$

Intuitively, $R(h)$ is the error we actually care about but can never observe directly, since it averages over the whole (unknown) data distribution $\mathcal{D}$; $\widehat{R}_S(h)$ is the only quantity we can compute, since it averages over the finite sample S we happen to have. The rest of this chapter is about how well the second quantity stands in for the first.

Using the definitions above and the concentration inequalities we covered in the probability theory chapter, we can derive the following theorem.

Theorem 6.1.3: Generalization bound

Let $\mathcal{H}$ be a finite hypothesis set. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the choice of $S \sim \mathcal{D}^m$, for all $h \in \mathcal{H}$

$$R(h) \leq \widehat{R}_S(h) + \sqrt{\frac{\log|\mathcal{H}| + \log\frac{1}{\delta}}{2m}}.$$

Proof. Let $\mathcal{H} = \{h_1, ..., h_{|\mathcal{H}|}\}$. Then we have

$$\begin{aligned} P_{S\sim\mathcal{D}^m}\left(\exists h_i \in \mathcal{H} : R(h_i) - \widehat{R}_S(h_i) > \epsilon\right) &= P_{S\sim\mathcal{D}^m}\left(\bigcup_{i=1}^{|\mathcal{H}|}\left\{R(h_i) - \widehat{R}_S(h_i) > \epsilon\right\}\right) \\ &\leq \sum_{i=1}^{|\mathcal{H}|} P_{S\sim\mathcal{D}^m}\left(R(h_i) - \widehat{R}_S(h_i) > \epsilon\right) \\ &\leq |\mathcal{H}|\exp(-2m\epsilon^2). \end{aligned}$$

where the first inequality follows from the union bound and the second from Hoeffding's inequality (Theorem 5.6.5) applied to each h_i separately, with the loss bounded in $[0, 1]$. Setting the right hand side to δ and solving for ϵ yields the final result □

This inequality is called a generalization bound, because it bounds the generalization error of any hypothesis $h \in \mathcal{H}$ in terms of its empirical error. We can derive a number of interesting consequences from this bound:

- The bound is uniform in the sense that it holds simultaneously for all hypotheses in $\mathcal{H}$.
- The bound is independent of the data distribution $\mathcal{D}$ and the loss function ℓ.
- The bound increases logarithmically with the size of the hypothesis set $|\mathcal{H}|$, that is, a richer hypothesis set is more likely to overfit.
- The bound decreases with the size of the training set m, that is, more data is less likely to overfit.
- For a fixed training set size m and two different hypothesis sets $\mathcal{H}_1$ and $\mathcal{H}_2$, the bound prefers the smaller hypothesis set. This is known as the **Occam's razor principle**. The principle was introduced by the 14th century theologian William of Ockham. It states that among competing hypotheses, the one with the fewest assumptions should be selected.
- The bound gives a **Probably Approximately Correct (PAC)** performance guarantee. The event that any hypothesis in $\mathcal{H}$ is **approximately correct** in the sense that its generalization error is at most ϵ with probability at least $1 - \delta$.

6.2 PAC Learnability

The formal notion below is due to (Valiant, 1984), who founded the **Probably Approximately Correct (PAC)** framework this whole chapter is built on.

Definition 6.2.1: Realizability

A hypothesis space $\mathcal{H}$ is realizable with respect to a loss ℓ and data distribution $\mathcal{D}$ if $\exists h^* \in \mathcal{H}$ such that $R(h^*) = 0$.

In words, realizability says the target concept itself is expressible inside $\mathcal{H}$, with no approximation error to worry about — the only obstacle between the learner and a perfect predictor is a finite sample. It trivially follows from this definition that $\min_{h \in \mathcal{H}} \widehat{R}_S(h) = 0$ with probability 1 for a sample set S collected i.i.d. from $\mathcal{D}$. Hence, under the realizability assumption, all ERM solutions $h_S \in \arg\min_{h \in \mathcal{H}} \widehat{R}_S(h)$ give zero error.

Definition 6.2.2: Representativeness

A training set S is called $\epsilon-$ representative if

$$\forall h \in \mathcal{H}, |R(h) - \widehat{R}_S(h)| \leq \epsilon.$$

In words, on an ϵ-representative sample, training error is a faithful proxy for true error, simultaneously for *every* hypothesis in $\mathcal{H}$ — not just the one a particular algorithm happens to output.

Lemma 6.2.3

Assume that a training set S is $\frac{\epsilon}{2}$-representative, then any ERM solution $h_S \in \arg\min_{h\in\mathcal{H}} \widehat{R}_S(h)$ satisfies

$$R(h_S) \leq \min_{h\in\mathcal{H}} R(h) + \epsilon.$$

Proof. For every $h \in \mathcal{H}$, $R(h_S) \leq \widehat{R}_S(h_S) + \frac{\epsilon}{2} \leq \widehat{R}_S(h) + \frac{\epsilon}{2} \leq R(h) + \frac{\epsilon}{2} + \frac{\epsilon}{2} = R(h) + \epsilon$, where the first and last steps use $\frac{\epsilon}{2}$-representativeness and the middle step uses that h_S minimizes $\widehat{R}_S$. Taking the minimum over $h \in \mathcal{H}$ on the right gives the claim □

In words: on a representative sample, ERM cannot be fooled — the hypothesis it picks by minimizing *empirical* risk is also near-optimal in *true* risk, because empirical and true risk cannot disagree by more than $\epsilon/2$ for any hypothesis, including both h_S and the true best one.

Definition 6.2.4: Uniform Convergence

A hypothesis class $\mathcal{H}$ has the **uniform convergence property** if there exists a function $m_{\mathcal{H}}^{\mathrm{UC}} : (0,1)^2 \to \mathbb{N}$ such that for every $\epsilon, \delta \in (0,1)$ and every distribution $\mathcal{D}$, any sampled data set $S = \{(x_i, y_i) \overset{i.i.d.}{\sim} D : i = 1, \ldots, m\}$ with $m \geq m_{\mathcal{H}}^{\mathrm{UC}}(\epsilon, \delta)$ is ϵ-representative with probability at least $1 - \delta$.

In words, $\mathcal{H}$ has uniform convergence if, given enough samples, a random data set is (with high probability) ϵ-representative — so that empirical risk minimization is safe to use on it, no matter which distribution $\mathcal{D}$ generated the data.

Definition 6.2.5: Agnostic PAC Learnability

A hypothesis class $\mathcal{H}$ is **agnostic PAC learnable** if there exist a function $m_{\mathcal{H}} : (0,1)^2 \to \mathbb{N}$ and a **learning algorithm** A such that for every $\epsilon, \delta \in (0,1)$ and every distribution $\mathcal{D}$, running the learning algorithm on data set $S = \{(x_i, y_i) \overset{i.i.d.}{\sim} D : i = 1, \ldots, m\}$ with $m \geq m_{\mathcal{H}}(\epsilon, \delta)$ satisfies

$$P\big(\{S \sim \mathcal{D}^m : R(A(S)) \leq \min_{h'\in\mathcal{H}} R(h') + \epsilon\}\big) \geq 1 - \delta.$$

In words, an agnostic PAC learner is guaranteed, given enough data, to output a hypothesis nearly as good as the best one $\mathcal{H}$ has to offer — for *any* data distribution $\mathcal{D}$, and without assuming $\mathcal{H}$ contains a perfect predictor (hence "agnostic").

Corollary 6.2.6

If $|\mathcal{H}| < \infty$ (finite hypothesis class), then it has the uniform convergence property with sample complexity

$$m_{\mathcal{H}}^{\mathrm{UC}}(\epsilon, \delta) \leq \left\lceil \frac{\log(2|\mathcal{H}|/\delta)}{2\epsilon^2} \right\rceil.$$

and is agnostically PAC learnable using the ERM algorithm with sample complexity

$$m_{\mathcal{H}}(\epsilon, \delta) \leq m_{\mathcal{H}}^{\mathrm{UC}}(\epsilon/2, \delta) \leq \left\lceil \frac{2\log(2|\mathcal{H}|/\delta)}{\epsilon^2} \right\rceil.$$

Proof. Definition 6.2.2 asks for a *two-sided* bound, so run the proof of Theorem 6.1.3 over the $2|\mathcal{H}|$ bad events $\{R(h_i) - \widehat{R}_S(h_i) > \epsilon\}$ and $\{\widehat{R}_S(h_i) - R(h_i) > \epsilon\}$, $i = 1, \ldots, |\mathcal{H}|$. The union bound and Hoeffding's inequality (Theorem 5.6.5, applied to each tail separately) give

$$P\Big(\exists h \in \mathcal{H} : |R(h) - \widehat{R}_S(h)| > \epsilon\Big) \leq 2|\mathcal{H}|\, e^{-2m\epsilon^2}.$$

Requiring the right-hand side to be at most δ and solving for m yields $m \geq \log(2|\mathcal{H}|/\delta)/(2\epsilon^2)$, i.e. any such m makes S ϵ-representative with probability at least $1 - \delta$: this is the first claim. For the second, Lemma 6.2.3 turns $\epsilon/2$-representativeness into the agnostic PAC guarantee for ERM, so it suffices to take $m \geq m_{\mathcal{H}}^{\mathrm{UC}}(\epsilon/2, \delta) = \log(2|\mathcal{H}|/\delta)/(2(\epsilon/2)^2) = 2\log(2|\mathcal{H}|/\delta)/\epsilon^2$ □

In short: finiteness of $\mathcal{H}$ alone is already enough to guarantee learnability, with the required sample size growing only *logarithmically* in $|\mathcal{H}|$ — doubling the hypothesis class costs only one extra bit of sample complexity.

Definition 6.2.7: Bayes Error

Given a data distribution $(x, y) \sim \mathcal{D}$, the **Bayes error** is the risk of the best possible predictor over *all* measurable functions $\mathcal{X} \to \mathcal{Y}$:

$$R_{\mathcal{D}}^{*} := \inf_{h' : \mathcal{X} \to \mathcal{Y}} R(h').$$

The infimum above ranges over all measurable functions, not merely over the best predictor inside some hypothesis class $\mathcal{H}$. A predictor h^* that achieves the Bayes error is called a **Bayes (optimal) predictor**. For binary classification with $\mathcal{Y} = \{0, 1\}$, the Bayes predictor is

$$h^*(x) = \arg\max_{y \in \{0,1\}} P(y \mid x),$$

with Bayes error obtained by averaging the pointwise error over the input distribution,

$$R_{\mathcal{D}}^{*} = \mathbb{E}_x\Big[1 - \max_{y \in \{0,1\}} P(y \mid x)\Big] = \mathbb{E}_x\Big[\min_{y \in \{0,1\}} P(y \mid x)\Big];$$

Proof (of the Bayes-error formula). For any predictor h and any x, the conditional risk under the zero-one loss is

$$P\big(h(x) \neq y \mid x\big) = 1 - P\big(y = h(x) \mid x\big) \geq 1 - \max_{y \in \{0,1\}} P(y \mid x) = \min_{y \in \{0,1\}} P(y \mid x).$$

Taking expectations over x gives $R(h) \geq \mathbb{E}_x[\min_y P(y \mid x)]$ for every h. Conversely, fix a measurable tie-breaking rule and let $h^*(x)$ be the selected maximizer of $P(\cdot \mid x)$; then $P(y = h^*(x) \mid x) = \max_y P(y \mid x)$ at every x, so $R(h^*) = \mathbb{E}_x[\min_y P(y \mid x)]$. Since $R^*_{\mathcal{D}}$ is the infimum of $R(h)$ over all measurable predictors, the displayed identity follows. The same argument for a finite label set gives $R^*_{\mathcal{D}} = \mathbb{E}_x[1 - \max_y P(y \mid x)]$. □

This floor is also called the **noise** level of $\mathcal{D}$, since it is the part of the risk that no predictor — however expressive — can remove. It is zero exactly when the label is a deterministic function of the input.

For any predictor h, the difference $R(h) - R^*_{\mathcal{D}}$ is called the **excess risk** of h: the part of the risk that a better predictor could, in principle, remove.

The Bayes error $R^*_{\mathcal{D}}$ is a property of the distribution $\mathcal{D}$ alone. It should not be confused with $\min_{h \in \mathcal{H}} R(h)$, the best risk achievable *within a fixed, possibly restricted* hypothesis class $\mathcal{H}$ — which we call the **approximation error** of $\mathcal{H}$ below. In general $\min_{h \in \mathcal{H}} R(h) \geq R^*_{\mathcal{D}}$, with equality only when $\mathcal{H}$ happens to contain a Bayes predictor for $\mathcal{D}$.

Definition 6.2.8: "Realizable" PAC learnability

A hypothesis class $\mathcal{H}$ is **PAC learnable** if there exist a function $m_{\mathcal{H}} : (0,1)^2 \to \mathbb{N}$ and a learning algorithm A such that for every $\epsilon, \delta \in (0,1)$ and every distribution $\mathcal{D}$ satisfying the realizability assumption (Definition 6.2.1), running A on $S = \{(x_i, y_i) \overset{i.i.d.}{\sim} \mathcal{D} : i = 1, \ldots, m\}$ with $m \geq m_{\mathcal{H}}(\epsilon, \delta)$ satisfies

$$P\big(\{S \sim \mathcal{D} : R(A(S)) \leq \epsilon\}\big) \geq 1 - \delta.$$

Realizability forces $\min_{h \in \mathcal{H}} R(h) = 0$, and in particular $R^*_{\mathcal{D}} = 0$. Definition 6.2.8 is therefore a *weaker demand on the learner* than Definition 6.2.5: it is only required to succeed on the restricted family of distributions for which some hypothesis in $\mathcal{H}$ is perfect, whereas an agnostic learner must succeed on **every** distribution. Consequently, agnostic PAC learnability trivially implies PAC learnability — simply apply Definition 6.2.5 to a realizable $\mathcal{D}$, where $\min_{h' \in \mathcal{H}} R(h') = 0$ makes the two guarantees read identically.

What is far from trivial is that, for binary classification under the zero-one loss, the converse holds as well: the two notions turn out to be *equivalent*, and both are equivalent to $\mathcal{H}$ having a finite VC dimension. This is the content of the Fundamental Theorem of Statistical Learning (Theorem 6.4.7) below. The equivalence is specific to this setting; for general label spaces and loss functions the realizable and agnostic notions genuinely differ, and the sample complexities differ even in the binary case — $\Theta(1/\epsilon)$ in the realizable case against $\Theta(1/\epsilon^2)$ in the agnostic case, as Theorem 6.4.7 makes precise.

6.3 Bias-Complexity Dilemma (Trade-off)

The result below is in the spirit of (Wolpert, 1996).

Theorem 6.3.1: No free lunch

Let A be any learning algorithm for the task of binary classification with respect to the 0/1 loss over a domain $\mathcal{X}$ and $m < |\mathcal{X}|/2$ be a positive integer representing a training set size. Then, there exists a distribution $\mathcal{D}$ over $\mathcal{X} \times \{0,1\}$ such that:

- There exists a function $f : \mathcal{X} \to \{0,1\}$ with $R(f) = 0$.
- With probability of at least $1/7$ over the choice of $S \sim \mathcal{D}^m$ we have that $R(A(S)) \geq 1/8$.

Proof. Fix any $C \subseteq \mathcal{X}$ with $|C| = 2m$; this is possible since $m < |\mathcal{X}|/2$. There are exactly $T := 2^{2m}$ functions $f_1, \dots, f_T : C \to \{0,1\}$. For each i, let $\mathcal{D}_i$ be the distribution on $\mathcal{X} \times \{0,1\}$ that draws x uniformly from C and sets $y = f_i(x)$ deterministically. By construction $R_{\mathcal{D}_i}(f_i) = 0$, so the first bullet holds for every $\mathcal{D}_i$. It remains to exhibit one i for which A fails.

Step 1: reduce to an average. We show

$$\max_{i \in [T]} \mathbb{E}_{S \sim \mathcal{D}_i^m}\big[R_{\mathcal{D}_i}(A(S))\big] \;\geq\; \tfrac{1}{4}.$$

Enumerate the $k := (2m)^m$ possible input sequences $S_1, \dots, S_k$ drawable from C, and write S_j^i for the sequence S_j labeled by f_i. Since drawing $S \sim \mathcal{D}_i^m$ is exactly drawing one of the S_j uniformly and labeling it by f_i,

$$\mathbb{E}_{S \sim \mathcal{D}_i^m}\big[R_{\mathcal{D}_i}(A(S))\big] = \frac{1}{k}\sum_{j=1}^{k} R_{\mathcal{D}_i}\big(A(S_j^i)\big).$$

A maximum is at least an average, and averages may be exchanged, so

$$\begin{aligned}\max_i \frac{1}{k}\sum_j R_{\mathcal{D}_i}(A(S_j^i)) \;&\geq\; \frac{1}{T}\sum_i \frac{1}{k}\sum_j R_{\mathcal{D}_i}(A(S_j^i)) \\ &= \frac{1}{k}\sum_j \frac{1}{T}\sum_i R_{\mathcal{D}_i}(A(S_j^i)) \;\geq\; \min_j \frac{1}{T}\sum_i R_{\mathcal{D}_i}(A(S_j^i)).\end{aligned}$$

Step 2: bound the inner average for a fixed sequence. Fix j, write $S_j = (x_1, \dots, x_m)$, and let $v_1, \dots, v_p$ be the elements of C that **do not** appear in S_j. Since S_j contains at most m distinct elements and $|C| = 2m$, we have $p \geq m$. For any hypothesis h,

$$R_{\mathcal{D}_i}(h) = \frac{1}{2m}\sum_{x \in C} \mathbb{1}\big(h(x) \neq f_i(x)\big) \;\geq\; \frac{1}{2m}\sum_{r=1}^{p} \mathbb{1}\big(h(v_r) \neq f_i(v_r)\big) \;\geq\; \frac{1}{2p}\sum_{r=1}^{p} \mathbb{1}\big(h(v_r) \neq f_i(v_r)\big),$$

the last step using $p \geq m$. Averaging over i and exchanging the two finite averages,

$$\frac{1}{T}\sum_i R_{\mathcal{D}_i}(A(S_j^i)) \;\geq\; \frac{1}{2}\cdot\frac{1}{p}\sum_{r=1}^{p} \underbrace{\frac{1}{T}\sum_i \mathbb{1}\big(A(S_j^i)(v_r) \neq f_i(v_r)\big)}_{=:\,\Pi_r} \;\geq\; \frac{1}{2}\min_r \Pi_r.$$

Step 3: the unseen points are a coin flip. Fix r and pair up the T functions: partition $[T]$ into $T/2$ pairs (i, i') such that f_i and $f_{i'}$ agree everywhere on C **except** at v_r. Since v_r does not occur in S_j, the two labeled sequences coincide, $S_j^i = S_j^{i'}$, so A returns the *same* hypothesis h on both. But $f_i(v_r) \neq f_{i'}(v_r)$, so h disagrees with exactly one of them at v_r. Each pair therefore contributes exactly one to the sum, giving $\Pi_r = \frac{1}{2}$ for every r. Chaining Steps 1-3 gives $\max_i \mathbb{E}_{S\sim\mathcal{D}_i^m}[R_{\mathcal{D}_i}(A(S))] \geq \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Step 4: from expectation to probability. Let $\mathcal{D} := \mathcal{D}_{i^*}$ for a maximizing i^* and let $\theta := R_{\mathcal{D}}(A(S)) \in [0, 1]$, a random variable of the draw of S, with $\mathbb{E}[\theta] \geq \frac{1}{4}$. Apply Markov's inequality (Theorem 5.6.1) to the non-negative variable $1 - \theta$ with threshold $\frac{7}{8}$:

$$P\big(\theta < \tfrac{1}{8}\big) = P\big(1 - \theta > \tfrac{7}{8}\big) \leq \frac{\mathbb{E}[1-\theta]}{7/8} \leq \frac{3/4}{7/8} = \frac{6}{7},$$

hence $P(\theta \geq \frac{1}{8}) \geq \frac{1}{7}$ □

The proof is worth re-reading for what it does *not* assume: nothing about A beyond it being a function of the sample. Step 3 is the crux — on the at least m points of C the learner never saw, the label is, by construction of the family $\{f_i\}$, an independent fair coin flip that no amount of cleverness can predict.

Corollary 6.3.2

Let $\mathcal{X}$ be an infinite domain ($|\mathcal{X}| = \infty$, e.g. a real-valued feature space) and $\mathcal{H}$ be the set of all functions from $\mathcal{X}$ to $\{0, 1\}$. Then $\mathcal{H}$ is not PAC learnable.

Proof (Shalev-Shwartz book). Assume, by way of contradiction, that the class is learnable. Choose some $\epsilon < 1/8$ and $\delta < 1/7$. By the definition of PAC learnability, there must be some learning algorithm A and an integer $m = m(\epsilon, \delta)$, such that for any data-generating distribution over $\mathcal{X} \times \{0, 1\}$, if for some function $f : \mathcal{X} \to \{0, 1\}$, $R(f) = 0$, then with probability greater than $1 - \delta$ when A is applied to samples S of size m, generated i.i.d. by $\mathcal{D}$, $R(A(S)) \leq \epsilon$. However, applying the No-Free-Lunch theorem, since $|\mathcal{X}| > 2m$, for every learning algorithm (and in particular for the algorithm A), there exists a distribution $\mathcal{D}$ such that with probability greater than $1/7 > \delta$, $R(A(S)) > 1/8 > \epsilon$, which leads to the desired contradiction □

The main conclusion of the theorem and its corollary is that there is no universal learner that works well for all data distributions. There always exists a task for which the learner fails. This is a very important result. It tells us that we need to make assumptions about the data distribution in order to design a good learner. These assumptions will constitute the **inductive bias** of the learner. The no free lunch theorem tells us only that we need to induce a degree of bias into the learner. However, it does not address its consequences. Inducing too much bias limits the ability of the learner to explain the training observations. Inducing too little bias leads to overfitting. The goal is to find the right balance between the two. This dilemma is known as the **bias-complexity dilemma**.

6.3.1 Decomposing the Excess Risk

Let us describe the bias-complexity dilemma in more formal terms. Let $h_S \leftarrow A(S)$ be the hypothesis a learning algorithm returns on a sample S — for instance an ERM solution $h_S \in \arg\min_{h\in\mathcal{H}} \widehat{R}_S(h)$. Definition 6.2.7 already identified an irreducible floor $R^*_{\mathcal{D}}$ that no predictor can undercut. Subtracting

it, and adding and subtracting the best risk achievable inside $\mathcal{H}$, splits the generalization error into three terms:

$$\underbrace{R(h_S)}_{\text{generalization error}} = \underbrace{R^*_{\mathcal{D}}}_{\text{Bayes error (noise)}} + \underbrace{\min_{h\in\mathcal{H}} R(h) - R^*_{\mathcal{D}}}_{\epsilon_{app}\text{ (approximation error)}} + \underbrace{R(h_S) - \min_{h\in\mathcal{H}} R(h)}_{\epsilon_{est}\text{ (estimation error)}}.$$

Each term is controlled by a different factor, and only the last two are ours to influence:

- The **Bayes error** $R^*_{\mathcal{D}}$ is a property of $\mathcal{D}$ alone (Definition 6.2.7). No choice of $\mathcal{H}$, algorithm, or sample size touches it.
- The **approximation error** ϵ_{app} measures how badly $\mathcal{H}$ fails to contain a Bayes predictor. It is a property of $\mathcal{H}$ and $\mathcal{D}$, and does *not* depend on S: enlarging the class can only reduce it, since $\mathcal{H}' \supset \mathcal{H}$ implies $\min_{h'\in\mathcal{H}'} R(h') \leq \min_{h\in\mathcal{H}} R(h)$.
- The **estimation error** ϵ_{est} measures how much worse the hypothesis we actually found is than the best one available in $\mathcal{H}$. This is the only term the generalization bounds of this chapter control: representativeness (Definition 6.2.2) with parameter $\epsilon/2$ gives $\epsilon_{est} \leq \epsilon$ by Lemma 6.2.3, and Theorem 6.1.3 bounds it by a quantity growing in $|\mathcal{H}|$ and shrinking in m.

Herein lies the dilemma. Enlarging $\mathcal{H}$ drives ϵ_{app} down but ϵ_{est} up; shrinking it does the reverse. Since we only ever observe $\widehat{R}_S(h_S)$, and a richer class drives *that* towards zero regardless of what happens to $R(h_S)$, an unconstrained search inflates ϵ_{est} invisibly — which is exactly overfitting. This is the **bias-complexity dilemma**.

6.3.2 Bias-Variance Decomposition

For the squared loss, the same excess risk admits a second, more classical split. Consider a regression problem with observations $(x, y) \sim \mathcal{D}$, and let $h_S \in \mathcal{H}$ be fitted by minimizing the mean squared error. For a fixed test point x, the expected squared error over both the noisy label y and the training sample S is

$$\begin{aligned}
\mathbb{E}_{S,y|x}\Big[\big(y - h_S(x)\big)^2\Big] &= \mathbb{E}_{S,y|x}\Big[y^2 - 2y h_S(x) + h_S(x)^2\Big] \\
&= \mathbb{E}_{y|x}[y^2] + \mathbb{E}_S[h_S(x)^2] - 2\mathbb{E}_{y|x}[y]\mathbb{E}_S[h_S(x)] \\
&= \mathbb{E}_{y|x}[y]^2 + \mathrm{Var}_{y|x}[y] + \mathbb{E}_S[h_S(x)]^2 + \mathrm{Var}_S[h_S(x)] - 2\mathbb{E}_{y|x}[y]\mathbb{E}_S[h_S(x)] \\
&= \Big(\underbrace{\mathbb{E}_{y|x}[y] - \mathbb{E}_S[h_S(x)]}_{\text{Estimator Bias}}\Big)^2 + \underbrace{\mathrm{Var}_S[h_S(x)]}_{\text{Estimator Variance}} + \underbrace{\mathrm{Var}_{y|x}[y]}_{\text{Label noise variance}}
\end{aligned}$$

The second line uses that y and S are independent given x, and we note that $\mathbb{E}_{S|x}[h_S(x)] = \mathbb{E}_S[h_S(x)]$ and $\mathrm{Var}_{S|x}[h_S(x)] = \mathrm{Var}_S[h_S(x)]$ since S is collected independently of the test point x.

> ***Example :***
> Take the k-nearest neighbours regressor,
>
> $$h_S(x) = \frac{1}{k}\sum_{i=1}^{k} y_{(i)},$$
>
> where $(x_{(1)}, y_{(1)}), \ldots, (x_{(k)}, y_{(k)})$ are the k training points nearest to x. Writing $f^*(x) := \mathbb{E}[y|x]$

and assuming label noise of variance σ^2, the three terms read

$$\mathbb{E}_{S,y|x}\Big[\Big(y - h_S(x)\Big)^2\Big] = \left(f^*(x) - \mathbb{E}_S\left[\frac{1}{k}\sum_{i=1}^{k} f^*(x_{(i)})\right]\right)^2 + \frac{\sigma^2}{k} + \sigma^2.$$

The middle term follows because averaging k independent noisy labels divides the noise variance by k. The two extremes are instructive:

- $k = m$: every training point is a "neighbour", so h_S predicts the global sample mean of the labels regardless of x. The variance term is at its smallest (σ^2/m), but the bias is large, since a constant cannot track how x influences y. The model **underfits**.
- $k = 1$: h_S predicts the label of the single nearest training point to x. The bias term is small (for dense enough data, $f^*(x_{(1)}) \approx f^*(x)$), but the variance is at its largest (σ^2), since the prediction rides on one noisy label. The model **overfits**: it incurs no error on S, while its test error depends entirely on how representative that particular S was.

6.3.3 The Two Decompositions Are the Same Decomposition

The two splits above look unrelated: one is stated in terms of risks of hypotheses, the other in terms of moments of a predicted value at a point. For the squared loss they are in fact two ways of cutting up one and the same quantity. Let $f^*(x) := \mathbb{E}[y|x]$ denote the Bayes predictor for the squared loss, write $\|g\|^2 := \mathbb{E}_x[g(x)^2]$ for the $L_2(\mathcal{D})$ norm on the input space, and let

$$\bar{h}(x) := \mathbb{E}_S[h_S(x)]$$

be the **average predictor**: the hypothesis we would obtain by averaging the outputs of our learning algorithm over all training samples of size m.

Step 1 (excess risk is a squared distance). Conditioning on x and expanding $(y - h(x))^2 = \big((y - f^*(x)) + (f^*(x) - h(x))\big)^2$, the cross term vanishes because $\mathbb{E}[y - f^*(x) \mid x] = 0$, leaving

$$R(h) = \|h - f^*\|^2 + \underbrace{\mathbb{E}_x\big[\mathrm{Var}[y|x]\big]}_{= R^*_{\mathcal{D}}},$$
$$\text{hence} \quad R(h) - R^*_{\mathcal{D}} = \|h - f^*\|^2.$$

So for the squared loss, excess risk over the Bayes error *is* squared distance to the Bayes predictor, and the label-noise term of the bias-variance decomposition is exactly the Bayes error of Definition 6.2.7.

Step 2 (average over samples). Taking $\mathbb{E}_S$ of Step 1 and splitting $h_S - f^* = (h_S - \bar{h}) + (\bar{h} - f^*)$, whose cross term vanishes by the definition of $\bar{h}$,

$$\mathbb{E}_S\big[R(h_S)\big] - R^*_{\mathcal{D}} = \underbrace{\|\bar{h} - f^*\|^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}_x\big[\mathrm{Var}_S[h_S(x)]\big]}_{\text{Variance}},$$

which is the bias-variance decomposition of the previous section, integrated over the test point x.

Step 3 (the identity). The very same left-hand side is, by the approximation-estimation decomposition,

$$\mathbb{E}_S\big[R(h_S)\big] - R^*_{\mathcal{D}} = \epsilon_{app} + \mathbb{E}_S[\epsilon_{est}].$$

Comparing Steps 2 and 3 gives the conceptual bridge we were after:

$$\boxed{\text{Bias}^2 + \text{Variance} \;=\; \epsilon_{app} + \mathbb{E}_S[\epsilon_{est}] \;=\; \mathbb{E}_S[R(h_S)] - R^*_{\mathcal{D}}}$$

Both decompositions carve up the *expected excess risk over the Bayes error*. They agree on the total and on the noise floor; they differ only in where they draw the internal line.

Where the line differs. Suppose $\mathcal{H}$ is convex, so that the average predictor $\bar{h}$ is itself a member of $\mathcal{H}$. Then

$$\text{Bias}^2 = \|\bar{h} - f^*\|^2 \;\geq\; \min_{h\in\mathcal{H}} \|h - f^*\|^2 = \epsilon_{app},$$
$$\text{and consequently} \quad \text{Variance} \;\leq\; \mathbb{E}_S[\epsilon_{est}].$$

In words: the bias term is the approximation error *plus* whatever part of the estimation error is systematic — the offset that survives averaging over samples. The variance term is the remaining, purely fluctuating part. The correspondence between the two vocabularies is therefore

plays the role of	approximation-estimation	bias-variance (squared loss)
irreducible noise	$R^*_{\mathcal{D}}$	$\mathbb{E}_x[\text{Var}[y \mid x]]$
class too restricted	ϵ_{app}	Bias^2 (which is $\geq \epsilon_{app}$)
finite sample	$\mathbb{E}_S[\epsilon_{est}]$	Variance (which is $\leq \mathbb{E}_S[\epsilon_{est}]$)
grows when $\mathcal{H}$ grows	ϵ_{est}	Variance
shrinks when $\mathcal{H}$ grows	ϵ_{app}	Bias^2

Two caveats, both instructive.

- *Convexity matters.* If $\mathcal{H}$ is not convex, $\bar{h}$ need not lie in $\mathcal{H}$, and it can then be strictly closer to f^* than any single member of $\mathcal{H}$ — so $\text{Bias}^2 < \epsilon_{app}$ becomes possible. This is not a pathology to be avoided but a mechanism to be exploited: it is precisely why averaging many predictors trained on resampled data can beat every individual predictor, which is the theoretical content of **bagging** in Chapter 12.
- *The exact identity is specific to the squared loss.* Step 1 relied on the squared loss to turn excess risk into a squared distance, and Step 2 relied on that distance being a variance. For the zero-one loss no analogous *additive* bias-variance decomposition exists — the various proposals in the literature are definition-dependent and do not decompose additively. The approximation-estimation decomposition, by contrast, is pure bookkeeping (add and subtract $\min_{h\in\mathcal{H}} R(h)$ and $R^*_{\mathcal{D}}$) and therefore holds for **any** loss function whatsoever. This is why statistical learning theory is built on the approximation-estimation split rather than on bias-variance, even though the latter is the more familiar of the two.

6.4 Vapnik - Chervonenkis (VC) Dimension

The generalization bound developed above was for a finite hypothesis set. However, in practice we often deal with infinite hypothesis sets. In order to develop generalization bounds for them, we need a measure

of complexity that stays finite even when $|\mathcal{H}| = \infty$. (Vapnik and Chervonenkis, 1971) built such a measure on a remarkable observation: in a binary classification problem, a hypothesis set can report only a finite number of distinct labelings of a finite sample, even though it contains infinitely many hypotheses. The chain of definitions runs restriction → shattering → growth function → VC dimension; let us take them in turn.

Definition 6.4.1: Restriction

Given $S = \{x_1, \ldots, x_m\} \subset \mathcal{X}$, the following set

$$\mathcal{H}_S = \{(h(x_1), \ldots, h(x_m)) : h \in \mathcal{H}\}$$

is called a **restriction** of $\mathcal{H}$ to S.

In words, a restriction is the set of all unique label vectors a hypothesis class $\mathcal{H}$ can generate from a data set S.

Definition 6.4.2: Shattering

$\mathcal{H}$ is said to **shatter** S if $|\mathcal{H}_S| = 2^{|S|}$.

In words, a hypothesis class $\mathcal{H}$ shatters a data set S if the restriction of $\mathcal{H}$ to S is the set of all functions from S to $\{0, 1\}$.

Corollary 6.4.3

If a data set $C \subset \mathcal{X}$ with $2m$ data points (i.e. $|C| = 2m$) is shattered by $\mathcal{H}$, then for every learning algorithm A, there exists a distribution $\mathcal{D}$ over $\mathcal{X} \times \{0, 1\}$, supported on C, such that some $h \in \mathcal{H}$ satisfies $R(h) = 0$, yet $P\big(\{S' \sim \mathcal{D}^m : R(A(S')) \geq 1/8\}\big) \geq 1/7$.

Proof. Run the proof of Theorem 6.3.1 with $\mathcal{X}$ replaced by C itself, which is legitimate because $|C| = 2m > m$. That proof produced a distribution $\mathcal{D} = \mathcal{D}_{i^*}$, supported on C and labeled by one of the 2^{2m} functions $f_{i^*} : C \to \{0, 1\}$, on which A fails as claimed. The only remaining point to check is that the perfect predictor may be taken *inside* $\mathcal{H}$: since $\mathcal{H}$ shatters C, its restriction $\mathcal{H}_C$ is all of $\{0, 1\}^C$ (Definition 6.4.2), so **every** f_i — f_{i^*} in particular — agrees on C with some $h \in \mathcal{H}$, and that h has $R(h) = 0$ because $\mathcal{D}$ is supported on C □

In words, if $\mathcal{H}$ is large enough to shatter a data set of $2m$ on a domain $\mathcal{X}$, then we cannot find a learning algorithm as good as we desire (arbitrarily good) that can be run on data sets of size m or smaller. The intuitive explanation is that due to the No Free Lunch theorem, under such circumstances, the hypothesis space $\mathcal{H}$ is so large that for any $2m$ data points, the true labels of the half that is not used in training can be perfectly explained by a hypothesis inside $\mathcal{H}$. This is a manifestation of Karl Popper's falsifiability principle in machine learning theory. This principle sets a demarcation point between science and other types of knowledge generation. An explanation is scientific if there exists a set of observations that can falsify it.

Definition 6.4.4: Growth function

The growth function $\tau_{\mathcal{H}} : \mathbb{N}^+ \to \mathbb{N}^+$ of $\mathcal{H}$ is defined as

$$\tau_{\mathcal{H}}(m) = \max_{S \subseteq \mathcal{X}, |S|=m} |\mathcal{H}_S|.$$

In words, the growth function counts the maximum number of distinct labelings that $\mathcal{H}$ can report on any data set on domain $\mathcal{X}$ that contains m data points.

Note that $\mathcal{H}$ shatters some $S \subseteq \mathcal{X}$ with $|S| = m$ if and only if $\tau_{\mathcal{H}}(m) = 2^m$.

Definition 6.4.5: VC dimension

The VC dimension of a hypothesis set $\mathcal{H}$ is the size of the largest data set that $\mathcal{H}$ can shatter:

$$d_{VC}(\mathcal{H}) = \sup\{m : \tau_{\mathcal{H}}(m) = 2^m\},$$

where the supremum is ∞ if $\mathcal{H}$ shatters arbitrarily large sets.

Intuitively, $d_{VC}(\mathcal{H})$ is the largest number of points $\mathcal{H}$ can label in *every* conceivable way — a single number summarizing how much a class can "memorize" arbitrary data, and hence how easily it can overfit; it plays exactly the role $\log |\mathcal{H}|$ played for finite classes, but stays finite even when $\mathcal{H}$ itself is infinite. Let us now see what a finite VC dimension provides — first qualitatively, then quantitatively.

Theorem 6.4.6: VC and PAC learnability

If $d_{VC}(\mathcal{H}) = \infty$ then $\mathcal{H}$ is not PAC learnable.

Proof. For any training set size m, the infinite VC dimension allows $\mathcal{H}$ to shatter a data set of size $2m$. Then the result follows from Corollary 6.4.3 □

This theorem is an analytical way of the argumentation presented above after the No Free Lunch theorem. Learning is only possible with some inductive bias. The bias-complexity dilemma stems from this obligation.

Theorem 6.4.7: The Fundamental Theorem of Statistical Learning

Let $\mathcal{H}$ be a class of binary classifiers under the zero-one loss, and assume that $d_{VC}(\mathcal{H}) = d < \infty$. Then, there exist $C_1, C_2 \in \mathbb{R}^+$ such that for all $\epsilon, \delta \in (0, \frac{1}{16})$

- $\mathcal{H}$ has the uniform convergence property with sample complexity

$$C_1 \frac{d + \log(1/\delta)}{\epsilon^2} \leq m_{\mathcal{H}}^{\text{UC}}(\epsilon, \delta) \leq C_2 \frac{d + \log(1/\delta)}{\epsilon^2}$$

- $\mathcal{H}$ is agnostic PAC learnable with sample complexity $C_1 \frac{d+\log(1/\delta)}{\epsilon^2} \leq m_{\mathcal{H}}(\epsilon, \delta) \leq C_2 \frac{d+\log(1/\delta)}{\epsilon^2}$
- $\mathcal{H}$ is PAC learnable with sample complexity (the lower bound for $d \geq 2$)

$$C_1 \frac{d + \log(1/\delta)}{\epsilon} \leq m_{\mathcal{H}}(\epsilon, \delta) \leq C_2 \frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon}$$

In words, the following statements are equal:

- $\mathcal{H}$ has the uniform convergence property.
- Any ERM rule is a successful agnostic PAC learner for $\mathcal{H}$.
- $\mathcal{H}$ is agnostic PAC learnable.
- $\mathcal{H}$ is PAC learnable.
- Any ERM rule is a successful PAC learner for $\mathcal{H}$.
- $\mathcal{H}$ has a finite VC-dimension.

This outcome is very useful because it allows one to achieve all six of these properties by ensuring only one of them.

Proof. We prove the theorem in two parts: the **equivalence of the six statements**, which is where the conceptual content lies and which we establish completely; and the **sample complexity rates**, which rest on tools developed in Chapter 7: chaining for the agnostic upper bound, and the two-point and Assouad methods for the lower bounds.

Part 1: the equivalence. It suffices to close the cycle

$$\begin{aligned} \text{finite } d_{VC} \;&\Rightarrow\; \text{uniform convergence} \;\Rightarrow\; \text{ERM is agnostic PAC} \\ &\Rightarrow\; \text{agnostic PAC learnable} \;\Rightarrow\; \text{PAC learnable} \;\Rightarrow\; \text{finite } d_{VC}, \end{aligned}$$

after which every statement implies every other, and the two ERM statements follow because the implications that produce them exhibit ERM itself as the successful learner.

1. *Finite $d_{VC} \Rightarrow$ uniform convergence.* By Lemma 6.4.8, $\tau_{\mathcal{H}}(m) \leq (em/d)^d$ is polynomial in m, so $\log \tau_{\mathcal{H}}(2m)/m \to 0$. Theorem 6.4.9 below (whose proof uses only Lemma 6.4.8, symmetrization, and Massart's lemma) then bounds the representativeness gap by a quantity tending to 0, which is exactly Definition 6.2.4.

2. *Uniform convergence ⇒ ERM is an agnostic PAC learner.* Immediate from Lemma 6.2.3: take $m \geq m_{\mathcal{H}}^{\mathrm{UC}}(\epsilon/2, \delta)$, so that S is $\epsilon/2$-representative with probability $\geq 1-\delta$, on which event every ERM output satisfies $R(h_S) \leq \min_h R(h) + \epsilon$.
3. *ERM is an agnostic PAC learner ⇒ agnostic PAC learnable.* By definition — a witness algorithm has been exhibited.
4. *Agnostic PAC learnable ⇒ PAC learnable.* Already argued after Definition 6.2.8: a realizable $\mathcal{D}$ has $\min_h R(h) = 0$, so Definition 6.2.5 specializes to Definition 6.2.8.
5. *PAC learnable ⇒ finite d_{VC}.* This is the contrapositive of Theorem 6.4.6, which was proved from Corollary 6.4.3 and hence from the No Free Lunch theorem.

Part 2: the upper bounds. *Agnostic case.* The chaining bound (7.4) of Chapter 7 states that with probability at least $1-\delta$,

$$\sup_{h \in \mathcal{H}} \left| R(h) - \widehat{R}_S(h) \right| \;\leq\; 96\sqrt{\frac{d}{m}} + 3\sqrt{\frac{\log(4/\delta)}{2m}}.$$

Both terms are at most $\epsilon/2$ as soon as $m \geq 192^2\, d/\epsilon^2$ and $m \geq 18\log(4/\delta)/\epsilon^2$, and $\log(4/\delta) \leq 2\log(1/\delta)$ for $\delta \leq \frac{1}{4}$, so $m_{\mathcal{H}}^{\mathrm{UC}}(\epsilon,\delta) \leq C_2 \frac{d+\log(1/\delta)}{\epsilon^2}$ with $C_2 = 192^2 + 36$. Part 1, step 2 turns this into the same rate for $m_{\mathcal{H}}(\epsilon,\delta) \leq m_{\mathcal{H}}^{\mathrm{UC}}(\epsilon/2,\delta)$, at the price of a factor 4 in C_2. (The more elementary route through Theorem 7.3.5, which bounds the Rademacher complexity by $2\sqrt{d\log(m+1)/m}$ using Lemma 6.4.8 and Massart's lemma alone, gives the same rate up to a factor $\log(1/\epsilon)$; chaining is what removes it.)

Realizable case. Let $\mathcal{D}$ be realizable, so that every ERM hypothesis has $\widehat{R}_S(h_S) = 0$, and let $B := \{\exists h \in \mathcal{H} : \widehat{R}_S(h) = 0,\ R(h) > \epsilon\}$. We show that for $m \geq 8/\epsilon$,

$$P(B) \;\leq\; 2\,\tau_{\mathcal{H}}(2m)\, 2^{-\epsilon m/2}. \tag{6.1}$$

Step 1: a ghost sample. Let S' be an independent sample of size m and $B' := \{\exists h \in \mathcal{H} : \widehat{R}_S(h) = 0,\ \widehat{R}_{S'}(h) \geq \epsilon/2\}$. On B, fix a hypothesis h_S witnessing B, chosen as a function of S alone. Given S, $\widehat{R}_{S'}(h_S)$ is the mean of m i.i.d. Bernoulli variables with mean $\mu := R(h_S) > \epsilon$, so Chebyshev's inequality (Theorem 5.6.2) gives

$$P\big(\widehat{R}_{S'}(h_S) < \tfrac{\epsilon}{2} \,\big|\, S\big) \;\leq\; P\big(|\widehat{R}_{S'}(h_S) - \mu| > \tfrac{\mu}{2} \,\big|\, S\big) \;\leq\; \frac{\mu(1-\mu)/m}{\mu^2/4} \;\leq\; \frac{4}{m\epsilon} \;\leq\; \frac{1}{2}.$$

Hence $P(B' \mid S) \geq \frac{1}{2}$ on B, and $P(B') \geq \frac{1}{2}P(B)$.

Step 2: random swaps. As in the proof of Theorem 6.4.9, exchanging the i-th example of S with the i-th example of S', independently for each i with probability $\frac{1}{2}$, leaves the joint law of (S, S') unchanged. So $P(B')$ equals the probability of B' after the swaps, and it suffices to bound the latter conditionally on the $2m$ examples. On these examples $\mathcal{H}$ has at most $\tau_{\mathcal{H}}(2m)$ distinct behaviours. Fix one, and let r be the number of pairs i on which it errs on exactly one of the two examples. If it errs on both examples of some pair, it cannot have zero error on S after any swap. Otherwise, B' requires that all its mistakes land in S' and that there are at least $\epsilon m/2$ of them, so $r \geq \epsilon m/2$, and each of the r pairs must be swapped in the one favourable way, which happens with probability $2^{-r} \leq 2^{-\epsilon m/2}$. A union bound over the behaviours gives $P(B') \leq \tau_{\mathcal{H}}(2m)\, 2^{-\epsilon m/2}$, and Step 1 gives (6.1).

Step 3: solve for m. For $2m \geq d$, Lemma 6.4.8 gives $\log \tau_{\mathcal{H}}(2m) \leq d \log(2em/d)$, and for any $a > 0$ the inequality $\log u \leq u - 1$ at $u = a \cdot 2em/d$ gives $d \log(2em/d) \leq 2eam - d(1 + \log a)$. With $a := \frac{\epsilon \log 2}{8e}$ this reads $d \log(2em/d) \leq \frac{\epsilon m \log 2}{4} + d \log \frac{8}{\epsilon \log 2}$. The right-hand side of (6.1) is therefore at most δ as soon as

$$\frac{\epsilon m \log 2}{4} \;\geq\; d \log \frac{8}{\epsilon \log 2} + \log \frac{2}{\delta}, \quad \text{i.e.} \quad m \;\geq\; \frac{4}{\epsilon \log 2}\Big(d \log \frac{8}{\epsilon \log 2} + \log \frac{2}{\delta}\Big),$$

which also implies $m \geq 8/\epsilon$ and $2m \geq d$. On the complement of B, every ERM hypothesis has $R(h_S) \leq \epsilon$, so $m_{\mathcal{H}}(\epsilon, \delta) \leq C_2 \frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon}$ for $\epsilon, \delta \in (0, \frac{1}{16})$. The $\log(1/\epsilon)$ enters through $\log \tau_{\mathcal{H}}(2m) = O(d \log m)$. Theorem 6.4.14, at the end of this chapter, is the same argument for a finite class, where no ghost sample is needed. The double-sample argument is due to (Blumer et al., 1989); see also Section 28.3 of (Shalev-Shwartz and Ben-David, 2014).

Part 3: the lower bounds. Let $C \subseteq \mathcal{X}$ be a set of size d shattered by $\mathcal{H}$, and let $m < d/2$. Apply Theorem 6.3.1 with the domain taken to be C (legitimate since $m < |C|/2$) and then Corollary 6.4.3's observation that shattering places the perfect predictor inside $\mathcal{H}$: for **every** learning algorithm A there is a realizable distribution $\mathcal{D}$ on $C \times \{0, 1\}$ with

$$P\big(\{S \sim \mathcal{D}^m : R(A(S)) \geq 1/8\}\big) \geq 1/7.$$

Hence for any $\epsilon < 1/8$ and $\delta < 1/7$ no sample size below $d/2$ can suffice, i.e. $m_{\mathcal{H}}(\epsilon, \delta) \geq d/2$: the sample complexity must grow at least linearly in the VC dimension, in the realizable case and therefore also in the agnostic case. The full rates come from letting the labels on the shattered set be ϵ-biased coins rather than deterministic, so that the learner must additionally *estimate* d separate biases. This is carried out in the minimax section of Chapter 7, whose Theorem 7.9.5 and inequality (7.7) give, for $\epsilon, \delta \in (0, \frac{1}{16})$:

- agnostic: $m_{\mathcal{H}}(\epsilon, \delta) \geq \max\Big(\frac{d}{4096\,\epsilon^2}, \frac{\log(1/(4\delta))}{16\,\epsilon^2}\Big)$;
- realizable, $d \geq 2$: $m_{\mathcal{H}}(\epsilon, \delta) \geq \max\Big(\frac{d-1}{16\,\epsilon}, \frac{\log(1/(2\delta))}{4\,\epsilon}\Big)$.

Since $\log(1/(4\delta)) \geq \frac{1}{2}\log(1/\delta)$ for $\delta \leq \frac{1}{16}$, $d - 1 \geq d/2$ for $d \geq 2$, and a maximum is at least the average, both are of the claimed form $C_1 \frac{d + \log(1/\delta)}{\epsilon^2}$, respectively $C_1 \frac{d + \log(1/\delta)}{\epsilon}$. Finally, $m_{\mathcal{H}}(2\epsilon, \delta) \leq m_{\mathcal{H}}^{\mathrm{UC}}(\epsilon, \delta)$ by Part 1, step 2, so the agnostic lower bound transfers to $m_{\mathcal{H}}^{\mathrm{UC}}$ with C_1 divided by 4. The restriction $d \geq 2$ is necessary: for $\mathcal{H} = \{h, 1 - h\}$ a single example identifies the target $\square$

The VC dimension gives a binary characterization of the hypothesis class. One may want to analyze the capacity of a hypothesis class in relation to an arbitrary data set size smaller or larger than its VC dimension. The following lemma, independently due to (Sauer, 1972) and (Shelah, 1972), is instrumental in relating the growth function of a hypothesis class to the sample size.

Lemma 6.4.8: Sauer-Shelah-Perles

If $d_{VC}(\mathcal{H}) = d < \infty$, then for all $m \geq d$, we have $\tau_{\mathcal{H}}(m) \leq \sum_{i=0}^{d} \binom{m}{i}$. In particular, for $m \geq d$, we have $\tau_{\mathcal{H}}(m) \leq \big(\frac{em}{d}\big)^d$.

Proof. The lemma follows from a stronger, purely combinatorial claim that makes no reference to d at all:

$$\textbf{Claim.} \quad |\mathcal{H}_S| \;\le\; \big|\{B \subseteq S : \mathcal{H} \text{ shatters } B\}\big|, \qquad \text{for every finite } S \subseteq \mathcal{X}.$$

Proof of the claim, by induction on $m = |S|$. For $m = 1$, say $S = \{x_1\}$: if $|\mathcal{H}_S| = 1$ the right-hand side is at least 1 (the empty set is shattered by any non-empty class), and if $|\mathcal{H}_S| = 2$ then $\mathcal{H}$ shatters both $\emptyset$ and $\{x_1\}$, so the right-hand side is 2.

For the inductive step, let $S = \{x_1, \dots, x_m\}$ and $S' := \{x_2, \dots, x_m\}$. Split the labelings in $\mathcal{H}_S$ according to whether their restriction to S' occurs with one or with both values of the first coordinate. Formally, set

$$Y_0 := \mathcal{H}_{S'},$$
$$Y_1 := \big\{(y_2, \dots, y_m) : (0, y_2, \dots, y_m) \in \mathcal{H}_S \text{ and } (1, y_2, \dots, y_m) \in \mathcal{H}_S\big\},$$

so that every element of $\mathcal{H}_S$ is obtained either as the unique extension of an element of Y_0, or as one of the two extensions of an element of Y_1; counting gives exactly

$$|\mathcal{H}_S| = |Y_0| + |Y_1|.$$

The induction hypothesis applied to $\mathcal{H}$ on S' bounds the first term: $|Y_0| = |\mathcal{H}_{S'}| \le |\{B \subseteq S' : \mathcal{H} \text{ shatters } B\}|$. For the second, define the subclass

$$\mathcal{H}' := \big\{h \in \mathcal{H} : \exists h' \in \mathcal{H},\ h'(x_1) = 1 - h(x_1) \text{ and } h'(x_i) = h(x_i)\ \forall i \ge 2\big\},$$

i.e. those hypotheses whose labeling of S' is realized with *both* labels of x_1. By construction $Y_1 = \mathcal{H}'_{S'}$, and crucially: **if $\mathcal{H}'$ shatters $B \subseteq S'$ then $\mathcal{H}'$, hence $\mathcal{H}$, shatters $B \cup \{x_1\}$** — because each of the $2^{|B|}$ labelings of B is available in $\mathcal{H}'$ with both possible labels of x_1. Therefore, by the induction hypothesis applied to $\mathcal{H}'$ on S',

$$\begin{aligned} |Y_1| = |\mathcal{H}'_{S'}| &\le \big|\{B \subseteq S' : \mathcal{H}' \text{ shatters } B\}\big| \\ &= \big|\{B \subseteq S' : \mathcal{H}' \text{ shatters } B \cup \{x_1\}\}\big| \\ &\le \big|\{B \subseteq S : x_1 \in B,\ \mathcal{H} \text{ shatters } B\}\big|. \end{aligned}$$

Adding the two bounds counts each shattered subset of S exactly once — those avoiding x_1 in the first term, those containing it in the second — which proves the claim.

From the claim to the lemma. Since $d_{VC}(\mathcal{H}) = d$, no subset of size $> d$ is shattered, so the right-hand side of the claim counts at most all subsets of size at most d:

$$\tau_{\mathcal{H}}(m) = \max_{|S|=m} |\mathcal{H}_S| \le \sum_{i=0}^{d} \binom{m}{i}.$$

For the closed form, assume $m \ge d$ so that $(d/m)^i \ge (d/m)^d$ for every $i \le d$, and apply the binomial theorem:

$$\Big(\frac{d}{m}\Big)^d \sum_{i=0}^{d} \binom{m}{i} \;\le\; \sum_{i=0}^{d} \binom{m}{i} \Big(\frac{d}{m}\Big)^i \;\le\; \sum_{i=0}^{m} \binom{m}{i} \Big(\frac{d}{m}\Big)^i = \Big(1 + \frac{d}{m}\Big)^m \le e^d,$$

the last step by $1 + u \le e^u$. Rearranging gives $\sum_{i=0}^{d} \binom{m}{i} \le (em/d)^d$ □

The lemma is the combinatorial heart of the whole theory: it converts a *binary* statement ("$\mathcal{H}$ cannot shatter $d+1$ points") into a *quantitative* one ("$\mathcal{H}$ produces at most polynomially many labelings"), and it is the polynomial-versus-exponential gap this opens that every subsequent bound exploits.

Theorem 6.4.9: Sample complexity wrt growth function

Let $\mathcal{H}$ be a hypothesis class with growth function $\tau_{\mathcal{H}}$. Then for every distribution $\mathcal{D}$ and every $\delta \in (0,1)$, a sample set $S = \{(x_i, y_i) \overset{i.i.d.}{\sim} \mathcal{D} : i = 1, \dots, m\}$ satisfies

$$P\left(\left\{S \sim \mathcal{D}^m : \sup_{h \in \mathcal{H}} |R(h) - \widehat{R}_S(h)| \le \frac{1}{\delta}\sqrt{\frac{2\log\left(2\,\tau_{\mathcal{H}}(2m)\right)}{m}}\right\}\right) \ge 1 - \delta.$$

Proof. The proof has two steps: bound the *expected* representativeness gap, then convert to a high-probability statement by Markov's inequality.

Step 1 (expectation bound). Introduce a **ghost sample** $S' = \{(x_i', y_i')\}_{i=1}^m \sim \mathcal{D}^m$, independent of S. Since $R(h) = \mathbb{E}_{S'}[\widehat{R}_{S'}(h)]$, Jensen's inequality (an absolute value of an expectation is at most the expectation of the absolute value, and a supremum of expectations is at most the expectation of the supremum) gives

$$\mathbb{E}_S\Big[\sup_h \big|R(h) - \widehat{R}_S(h)\big|\Big] \;\le\; \mathbb{E}_{S,S'}\Big[\sup_h \big|\widehat{R}_{S'}(h) - \widehat{R}_S(h)\big|\Big].$$

Because S and S' are i.i.d. and mutually independent, exchanging the i-th pair of S with the i-th pair of S' leaves the joint law unchanged. Recording the exchanges with independent Rademacher signs $\sigma_i \in \{-1, +1\}$ therefore changes nothing:

$$\begin{aligned}&\mathbb{E}_{S,S'}\Big[\sup_h \Big|\frac{1}{m}\sum_i \big(\ell(y_i', h(x_i')) - \ell(y_i, h(x_i))\big)\Big|\Big] \\ &= \mathbb{E}_{S,S',\sigma}\Big[\sup_h \Big|\frac{1}{m}\sum_i \sigma_i\big(\ell(y_i', h(x_i')) - \ell(y_i, h(x_i))\big)\Big|\Big].\end{aligned}$$

Now **condition on** S **and** S' and look at the inner expectation over σ alone. For each h define the vector $a^{(h)} \in \mathbb{R}^m$ with $a_i^{(h)} := \frac{1}{m}\big(\ell(y_i', h(x_i')) - \ell(y_i, h(x_i))\big)$. Two observations:

- $a^{(h)}$ depends on h only through h's restriction to the $2m$ points $\{x_1, \dots, x_m, x_1', \dots, x_m'\}$, so at most $\tau_{\mathcal{H}}(2m)$ distinct vectors arise (Definition 6.4.4);
- each coordinate lies in $[-1/m, 1/m]$ because the loss lies in $[0,1]$, hence $\|a^{(h)}\|_2 \le \sqrt{m}/m = 1/\sqrt{m}$.

Put $A := \{a^{(h)} : h \in \mathcal{H}\} \cup \{-a^{(h)} : h \in \mathcal{H}\}$, a finite set of at most $2\tau_{\mathcal{H}}(2m)$ vectors of Euclidean norm at most $1/\sqrt{m}$, and note $\sup_h |\langle \sigma, a^{(h)}\rangle| = \max_{a \in A}\langle \sigma, a\rangle$ — this is why the set is symmetrized. Massart's lemma (Lemma 7.3.3, in the un-normalized form $\mathbb{E}_\sigma[\max_{a\in A}\langle\sigma, a\rangle] \le \max_a \|a\|_2 \sqrt{2\log|A|}$) gives

$$\mathbb{E}_\sigma\Big[\max_{a\in A}\langle\sigma,a\rangle\Big] \;\le\; \frac{1}{\sqrt{m}}\sqrt{2\log\big(2\tau_{\mathcal{H}}(2m)\big)}.$$

The bound is uniform in S, S', so it survives taking the outer expectation:

$$\mathbb{E}_S\Big[\sup_h\big|R(h)-\widehat{R}_S(h)\big|\Big] \;\le\; \sqrt{\frac{2\log\big(2\tau_{\mathcal{H}}(2m)\big)}{m}} \;=:\; \bar{\epsilon}_m.$$

Step 2 (Markov). The random variable $\sup_h|R(h)-\widehat{R}_S(h)|$ is non-negative, so Theorem 5.6.1 with threshold $\bar{\epsilon}_m/\delta$ gives $P\big(\sup_h|R(h)-\widehat{R}_S(h)|\ge\bar{\epsilon}_m/\delta\big)\le\delta$, which is the claim □

Two remarks on the shape of this bound. First, it depends on δ as $1/\delta$ rather than $\sqrt{\log(1/\delta)}$, which is the price of using Markov's inequality on the expectation rather than a concentration inequality for the supremum itself; Chapter 7 replaces this step with McDiarmid's inequality and recovers the $\sqrt{\log(1/\delta)}$ dependence. Second, the constants are not optimal — a more careful sub-Gaussian maximal inequality replaces the bound above by $\frac{4+\sqrt{\log\tau_{\mathcal{H}}(2m)}}{\delta\sqrt{2m}}$ — but the essential dependence, $\sqrt{\log\tau_{\mathcal{H}}(2m)/m}$, is the same and is all that the discussion below uses.

This result is consistent with the implications of the fundamental theorem of statistical learning. The bound is useful exactly when $\sqrt{\log\tau_{\mathcal{H}}(2m)}/\sqrt{2m}\to 0$, i.e. when $\log\tau_{\mathcal{H}}(m)/m\to 0$: the growth function may grow, but only *sub-exponentially* in m. By the Sauer-Shelah-Perles lemma this is precisely what a finite VC dimension guarantees, since $\tau_{\mathcal{H}}(m)\le(em/d)^d$ is polynomial in m, so $\log\tau_{\mathcal{H}}(m)=O(d\log m)$. Conversely, if $\mathcal{H}$ shatters arbitrarily large sets then $\tau_{\mathcal{H}}(m)=2^m$ for every m, $\log\tau_{\mathcal{H}}(m)/m=\log 2$ does not vanish, and the bound degenerates — matching Theorem 6.4.6.

6.4.1 Nonuniform Learnability

Definition 6.4.10: Nonuniform Learnability

A hypothesis class $\mathcal{H}$ is **nonuniformly learnable** if there exist a function $m^{NU}_{\mathcal{H}}:(0,1)^2\times\mathcal{H}\to\mathbb{N}$ and a **learning algorithm** A such that for every $\epsilon,\delta\in(0,1)$ and every distribution $\mathcal{D}$, running A on data set $S=\{(x_i,y_i)\overset{i.i.d.}{\sim}D : i=1,\dots,m\}$ satisfies, for every $h\in\mathcal{H}$ and every $m\ge m^{NU}_{\mathcal{H}}(\epsilon,\delta,h)$,

$$P(\{S\sim\mathcal{D}^m : R(A(S))\le R(h)+\epsilon\})\ge 1-\delta.$$

The difference from Definition 6.2.5 is that the required sample size may depend on the hypothesis h we are competing against, rather than being uniform over the whole class — hence the name.

An agnostic PAC learnable hypothesis class $\mathcal{H}$ is also nonuniform learnable because $R(A(S))\le\min_{h'\in\mathcal{H}}R(h')+\epsilon$ results trivially in $R(A(S))\le R(h)+\epsilon,\forall h\in\mathcal{H}$. Hence, the set of non-uniform learnable hypothesis classes is larger than the agnostic PAC learnable hypothesis classes. With this relaxation, our motivation is to increase the model capacity while sustaining its learnability.

Theorem 6.4.11

If $\mathcal{H} = \cup_{i\in\mathbb{N}}\mathcal{H}_i$ such that each $\mathcal{H}_i$ has the uniform convergence property, then $\mathcal{H}$ is nonuniformly learnable.

Proof. Fix the weights $w(i) := 6/(\pi^2 i^2)$, which satisfy $\sum_{i\geq 1} w(i) = 1$ by the Basel identity $\sum_{i\geq 1} i^{-2} = \pi^2/6$, and take A to be the SRM rule h_{SRM} defined below in this section. Theorem 6.4.12, proved next, guarantees that with probability at least $1-\delta$,

$$R(h) \leq \widehat{R}_S(h) + \epsilon_{i(h)}\big(m, w(i(h))\delta\big)$$
$$\text{simultaneously for every } h \in \mathcal{H}.$$

On that event, for any fixed $h \in \mathcal{H}_i$, the definition of h_{SRM} as the minimizer of the right-hand side gives

$$R(h_{SRM}) \leq \widehat{R}_S(h_{SRM}) + \epsilon_{i(h_{SRM})}\big(m, w(i(h_{SRM}))\delta\big) \leq \widehat{R}_S(h) + \epsilon_i\big(m, w(i)\delta\big) \leq R(h) + 2\,\epsilon_i\big(m, w(i)\delta\big),$$

the last step using the uniform convergence of $\mathcal{H}_i$ once more, in the opposite direction. Since $\mathcal{H}_i$ has the uniform convergence property, $\epsilon_i(m, w(i)\delta) \to 0$ as $m \to \infty$ with i, δ fixed, so for every ϵ, δ and every h there is a finite

$$m^{NU}_{\mathcal{H}}(\epsilon, \delta, h) := \min\big\{m : 2\,\epsilon_{i(h)}\big(m, w(i(h))\delta\big) \leq \epsilon\big\}$$

beyond which $R(A(S)) \leq R(h) + \epsilon$ holds with probability at least $1-\delta$. This is exactly Definition 6.4.10 □

Note precisely where nonuniformity enters: the sample size above depends on h through the index $i(h)$ of the subclass containing it. There is no single m that works for all h at once, because the penalty ϵ_i worsens without bound as i grows.

Vice versa also holds. If a hypothesis class $\mathcal{H}$ is nonuniformly learnable, then it can be expressed as a countable union of agnostic PAC learnable subclasses. Next let us see how we can use this result to build a more powerful learning algorithm than ERM. Remember that we have sample complexity guarantees at the hypothesis-subclass level while we need to do our search at the hypothesis level. The following quantity is instrumental in bridging this gap:

$$\epsilon_i(m, \delta) = \min\{\epsilon \in (0,1) : m^{UC}_{\mathcal{H}_i}(\epsilon, \delta) \leq m\}$$

which in effect inverts the complexity function $m^{UC}_{\mathcal{H}_i}$ of hypothesis subclass i. This quantity gives the smallest representativeness level attainable within subclass i from m data points at confidence $1-\delta$. In other words, the best level of representativeness reachable within this hypothesis subclass, i.e., $P(\{S \sim \mathcal{D}^m : |R(h) - \widehat{R}_S(h)| \leq \epsilon_i(m, \delta)\}) \geq 1-\delta$ for the event of collecting a data set S of size m i.i.d. from any data distribution $\mathcal{D}$.

Theorem 6.4.12

Let $w : \mathbb{N} \to [0,1]$ be a function with $\sum_{i=1}^{\infty} w(i) < 1$ and $\mathcal{H} = \cup_{i\in\mathbb{N}}\mathcal{H}_i$ such that each $\mathcal{H}_i$ has the uniform convergence property with $m^{UC}_{\mathcal{H}_i}$. Then for every $\delta \in (0,1)$, every data distribution $\mathcal{D}$, and every $m \in \mathbb{N}$, and every $i \in \mathbb{N}$ and $h \in \mathcal{H}_i$ we have

$$P\left(\left\{S \sim \mathcal{D}^m : |R(h) - \widehat{R}_S(h)| \leq \epsilon_i(m, w(i)\delta)\right\}\right) \geq 1 - \delta$$

where S is a data set of size m sampled i.i.d. from $\mathcal{D}$.

Proof. Fix i and let B_i be the bad event that S fails to be $\epsilon_i(m, w(i)\delta)$-representative for $\mathcal{H}_i$, i.e. $B_i := \{\exists h \in \mathcal{H}_i : |R(h) - \widehat{R}_S(h)| > \epsilon_i(m, w(i)\delta)\}$. By the definition of ϵ_i as the inverse of the sample complexity function $m^{UC}_{\mathcal{H}_i}$, applying Definition 6.2.4 to $\mathcal{H}_i$ with confidence parameter $w(i)\delta$ gives $P(B_i) \leq w(i)\,\delta$. The union bound over the countably many subclasses then gives

$$P\Big(\bigcup_{i\in\mathbb{N}} B_i\Big) \leq \sum_{i\in\mathbb{N}} P(B_i) \leq \delta \sum_{i\in\mathbb{N}} w(i) \;<\; \delta,$$

using the hypothesis $\sum_i w(i) < 1$. On the complement — an event of probability at least $1-\delta$ — no subclass fails, which is the claim □

The proof shows exactly what the weights accomplish: a union bound over infinitely many events is finite only if the individual confidence levels are made summable, and w is the budget allocating a share $w(i)$ of the total failure probability δ to subclass i. Any summable allocation works; different choices of w simply express different prior preferences over the subclasses.

The goal of this theorem is to express the uniform convergence of a composite hypothesis class comprising subclasses with different uniform convergence guarantees. It only rescales the confidence level of the subclasses with an index function on the classes. As we will see next, this index will allow us to search the composite hypothesis space and thereby define a learning rule that is more powerful than ERM. The above result can also be expressed as follows:

$$P\left(\left\{S \sim \mathcal{D}^m : R(h) \leq \widehat{R}_S(h) + \epsilon_i(m, w(i(h)) \cdot \delta)\right\}\right) \geq 1 - \delta,\ \forall h \in \mathcal{H}$$

where $i(h) = \min_{i:h\in\mathcal{H}_i}$ is a function that finds the most sample-efficient class a hypothesis h belongs to. This result prescribes a search rule for an extended set of hypotheses that satisfies PAC learnability. We can then use it to build a new learning algorithm

$$h_{SRM} \in \arg\min_{h\in\mathcal{H}} \left\{\widehat{R}_S(h) + \epsilon_{i(h)}(m, w(i(h)) \cdot \delta)\right\}$$

which we call **Structural Risk Minimization (SRM)**: it minimizes not the empirical risk itself, but the upper bound on $R(h)$ that the previous display certifies. Remember that by Hoeffding's inequality we have $m^{UC}_h(\epsilon, \delta) = \frac{\log(2/\delta)}{2\epsilon^2}$ for a single hypothesis h, which implies $\epsilon_i(m, \delta) = \sqrt{\frac{\log(2/\delta)}{2m}}$. Defining $w'(h) := w(i(h))$ we get

$$h_{SRM} \in \arg\min_{h\in\mathcal{H}} \widehat{R}_S(h) + \sqrt{\frac{-\log w'(h) + \log(2/\delta)}{2m}}$$

where we use the trivial identity that $\log(2/\delta \cdot w'(h)) = -\log w'(h) + \log(2/\delta)$. In effect, SRM applies a preference over the hypothesis subclasses, represented in the learning rule by the weight $w'(h)$. The higher the weight, the stronger the preference. These preferences can be chosen to reflect prior knowledge coming from domain expertise regarding the collected data.

In many real-world problems, we have little domain knowledge to incorporate into the learning process. It would be useful to have a guideline for building machine learning algorithms without any domain knowledge. Remember that the only condition for combining individually agnostic PAC learnable hypothesis classes into a bigger class is to define a series of positive weights that converge to a value less than one. The following inequality provides an instrumental way of building such a series.

Lemma 6.4.13: Kraft's inequality

Let Σ^* be a set of finite-length strings over the alphabet $\Sigma = \{0, 1\}$ that is **prefix-free**, in the sense that no string $s \in \Sigma^*$ of length k coincides with the first k characters of a longer string in Σ^*. Then $\sum_{s\in\Sigma^*} 2^{-|s|} \leq 1$, where $|s|$ denotes the length of s.

Proof. Generate an infinite sequence of i.i.d. fair bits $b_1, b_2, \ldots$, so that $P(b_1 = c_1, \ldots, b_k = c_k) = 2^{-k}$ for any fixed pattern $c \in \{0, 1\}^k$. For each $s \in \Sigma^*$ define the event

$$A_s := \{b_1 = s_1, \ \ldots, \ b_{|s|} = s_{|s|}\},$$
$$P(A_s) = 2^{-|s|}.$$

The events are **pairwise disjoint**: if A_s and $A_{s'}$ both occurred for $s \neq s'$ with $|s| \leq |s'|$, then s would be the first $|s|$ characters of s', contradicting prefix-freeness. Disjointness and the axioms of a probability measure (σ-additivity plus $P(\Omega) = 1$) give

$$\sum_{s\in\Sigma^*} 2^{-|s|} = \sum_{s\in\Sigma^*} P(A_s) = P\Big(\bigcup_{s\in\Sigma^*} A_s\Big) \leq 1. \qquad \square$$

In words: assigning short codewords is a scarce resource. Prefix-freeness is what makes a code uniquely decodable one symbol at a time, and it forces the codeword lengths to behave like an (unnormalized) probability distribution: one cannot assign a short codeword to every symbol, so some symbols must receive long codewords to pay for the short ones. This is exactly why $2^{-|s(h)|}$ below can be reused as the summable weight $w'(h)$ that Theorem 6.4.12 needs.

The lemma is precisely what Theorem 6.4.12 needs: a description language for hypotheses hands us, for free, a summable weight function $w'(h) := 2^{-|s(h)|}$, with no need to design one by hand. $\square$

Hence, whenever we can find an alphabet that uniquely describes all hypotheses in a class with a string $s(h)$, we can plug Kraft's inequality into Theorem 6.4.12 with $w'(h) := 2^{-|s(h)|}$ and get

$$P\Big(\Big\{S \sim \mathcal{D}^m : R(h) \leq \widehat{R}_S(h) + \sqrt{\frac{|s(h)| + \log(2/\delta)}{2m}}\Big\}\Big) \geq 1 - \delta$$

where we use the property $\log(2) < 1$. The right-hand side of this inequality prescribes the learning rule below

$$h_{MDL} \in \arg\min_{h\in\mathcal{H}} \widehat{R}_S(h) + \sqrt{\frac{|s(h)| + \log(2/\delta)}{2m}}.$$

Note that this rule suggests the hypothesis that can be expressed with the minimum number of characters among the ones that fit the data best. Hence, it is referred to as the **Minimum Description Length (MDL)** rule. The MDL rule is yet another instance of the Occam's Razor principle in machine learning, which also has overarching consequences on hypothesis development in scientific investigation. This rule also sets a theoretical foundation for the idea of applying **regularizers** in machine learning.

6.4.2 The Realizable Case: Fast Rates

Theorem 6.1.3 charges a deviation of order $1/\sqrt{m}$ whatever the risk of the hypothesis. Under realizability (Definition 6.2.1) this is pessimistic: a hypothesis with true risk μ survives m examples with zero training error with probability $(1-\mu)^m$, which decays exponentially in $m\mu$ rather than in $m\mu^2$. The multiplicative Chernoff bounds of Chapter 5 (Theorem 5.6.10 and Corollary 5.6.11) turn this observation into a $1/m$ rate. The second statement below is the relative-deviation form of the same argument: it remains valid for ERM hypotheses with small *nonzero* training error and for an arbitrary data distribution.

Theorem 6.4.14: Realizable case: the fast rate

Let $\mathcal{H}$ be a finite hypothesis class for binary classification under the zero-one loss, realized by $\mathcal{D}$ (Definition 6.2.1), and let h_S be any ERM hypothesis. Then for any $\delta \in (0,1)$, with probability at least $1-\delta$ over $S \sim \mathcal{D}^m$,

$$R(h_S) \;\le\; \frac{\log|\mathcal{H}| + \log(1/\delta)}{m}.$$

Moreover, with probability at least $1-\delta$, simultaneously for all $h \in \mathcal{H}$,

$$R(h) \;\le\; 2\,\widehat{R}_S(h) + \frac{2\log|\mathcal{H}| + 2\log(1/\delta)}{m}.$$

Proof. For the first statement, fix $h \in \mathcal{H}$ with $R(h) = \mu$. Under realizability ERM attains $\widehat{R}_S(h_S) = 0$ (some h^* has zero loss on every sample point, and h_S's training error is minimal). Since the losses $\mathbb{1}(h(x_i) \neq y_i)$ are i.i.d. Bernoulli(μ),

$$P\big(\widehat{R}_S(h) = 0\big) = (1-\mu)^m \le e^{-m\mu},$$

the $z = 0$ case of Theorem 5.6.10, displayed right after its proof. Union-bounding over the bad events $\{\widehat{R}_S(h) = 0\}$ for those h with $R(h) > \epsilon_0$ (for which $e^{-mR(h)} \le e^{-m\epsilon_0}$),

$$P\big(\exists h \in \mathcal{H} : \widehat{R}_S(h) = 0,\; R(h) > \epsilon_0\big) \;\le\; \sum_{h\,:\,R(h)>\epsilon_0} e^{-mR(h)} \;\le\; |\mathcal{H}|\, e^{-m\epsilon_0}.$$

Setting the right-hand side to δ gives $\epsilon_0 = \frac{\log|\mathcal{H}|+\log(1/\delta)}{m}$. On the complement of this event, every zero-training-error hypothesis has $R(h) \le \epsilon_0$ — and h_S, having $\widehat{R}_S(h_S) = 0$, is one of them.

For the second statement, let $\mu := R(h)$, $x := \widehat{R}_S(h)$, and suppose $\mu > 2x + t$. Then the deviation $\epsilon := \mu - x$ satisfies $\epsilon > x + t \ge t$ and, from $x < (\mu - t)/2$, also $\epsilon > \frac{\mu+t}{2}$. Corollary 5.6.11 (lower tail)

therefore applies and gives

$$P\big(\mu > 2x + t\big) \;\le\; P\big(x \le \mu - \tfrac{\mu+t}{2}\big) \;\le\; \exp\Big(-\frac{m(\mu+t)^2/4}{2\mu}\Big) \;\le\; \exp\Big(-\frac{mt}{2}\Big),$$

where the second step uses the deviation $\frac{\mu+t}{2} \le \mu$, i.e. $t \le \mu$ — and if $t > \mu$ the event on the left is empty, since it forces $x < (\mu - t)/2 < 0$ — and the last step uses the AM-GM inequality $(\mu + t)^2 \ge 4\mu t$. Union-bounding over $|\mathcal{H}|$ fixed hypotheses and setting $|\mathcal{H}|\, e^{-mt/2} = \delta$, i.e. $t = \frac{2\log|\mathcal{H}| + 2\log(1/\delta)}{m}$, gives the simultaneous statement □

The contrast with Theorem 6.1.3 could not be sharper: the deviation scale is now $1/m$, not $1/\sqrt{m}$. In the first statement the entire mechanism is the multiplicative Chernoff bound — the exponent $m\mu$ is *linear* in the risk being estimated, which is why a hypothesis of true risk, say, $10/m$ has essentially no chance of surviving m examples with zero training error. The second statement is the *relative-deviation* form: the slack $R(h) - 2\widehat{R}_S(h)$ is $O(1/m)$ whenever the empirical risk itself is small, and the first statement is its special case $\widehat{R}_S(h_S) = 0$ (up to the constant 2). The factor of 2 in front of $\widehat{R}_S(h)$ is the price of turning a lower-tail bound into a statement uniform over all levels of training error. The same mechanism, run over the finitely many behaviours of an infinite class on a sample, gives the realizable rate of the Fundamental Theorem (Part 2 of the proof of Theorem 6.4.7).

Every result in this chapter — the finite-class bound, the VC-dimension-based Fundamental Theorem, the SRM and MDL rules built on top of it — measured the complexity of $\mathcal{H}$ through $|\mathcal{H}|$ or $d_{VC}(\mathcal{H})$: a raw count, or a worst-case combinatorial shattering capacity. Chapter 7 picks up right where the Fundamental Theorem (Theorem 6.4.7) leaves off, and asks what happens if we measure complexity differently — through how well $\mathcal{H}$ can correlate with pure noise (Rademacher complexity), through how finely $\mathcal{H}$ can be approximated by a finite grid (covering numbers), or through how much a data-dependent choice inside $\mathcal{H}$ must diverge from a fixed prior belief (PAC-Bayes bounds) — and shows that all of these, $|\mathcal{H}|$ and $d_{VC}(\mathcal{H})$ included, are instances of one and the same abstract idea.

CHAPTER 7

COMPLEXITY MEASURES AND GENERALIZATION

Chapter 6 built the entire theory of generalization on top of two concrete complexity measures: the raw count $|\mathcal{H}|$ for finite classes, and the VC dimension $d_{VC}(\mathcal{H})$ for infinite ones. Both did the same job: they compressed a hypothesis class down to a single number that a Hoeffding-type concentration inequality, combined with the union bound, could feed on. This chapter makes that role explicit and abstract, then instantiates it three more times.

The general form. Every bound in this course, past and future, has the shape

$$R(h) \le \widehat{R}_S(h) + \underbrace{\text{Complexity}(\mathcal{H}, m)}_{\text{shrinks as } m\to\infty} + \underbrace{\text{Confidence}(\delta, m)}_{\text{shrinks as } m\to\infty}, \quad \forall h \in \mathcal{H}, \text{ w.p. } \ge 1-\delta,$$

i.e. a bound on the **representativeness** gap $\sup_{h\in\mathcal{H}} \left(R(h) - \widehat{R}_S(h)\right)$ from Definition 6.2.2. What is negotiable is only how Complexity$(\mathcal{H}, m)$ is defined. We already have two instances: Complexity $= \sqrt{\log|\mathcal{H}|/(2m)}$ (Theorem 6.1.3) and Complexity $= \sqrt{d_{VC}(\mathcal{H})/m}$ up to logarithmic factors (Theorem 6.4.7). We now add three more: **Rademacher complexity**, **covering numbers**, and **PAC-Bayes bounds**. All five are, in a precise sense we make explicit below, the same quantity viewed through different lenses — and by the end of this chapter, we will have shown algebraically that the very first bound of Chapter 6 (Theorem 6.1.3) is a special case of the PAC-Bayes bound. The second half of the chapter then sharpens the VC rates by chaining, prices norm-constrained combinations, introduces algorithmic stability, and shows by minimax lower bounds that the rates obtained cannot be improved in general.

7.1 A Concentration Tool for Suprema: McDiarmid's Inequality

Hoeffding's inequality (Chapter 5) concentrates a *single* average $\widehat{\mu}_m$ around its mean. Bounding $\sup_{h\in\mathcal{H}} \left(R(h) - \widehat{R}_S(h)\right)$ requires concentrating a *function of the whole sample* — the supremum over (possibly infinitely many) hypotheses — around its mean. The right tool is a generalization, due to (McDiarmid, 1989), of Hoeffding's inequality to arbitrary functions of independent random variables, provided the function does not depend too sensitively on any single one of them.

Theorem 7.1.1: McDiarmid's bounded differences inequality

Let $Z_1, \dots, Z_m$ be independent random variables and let $f : \mathcal{Z}^m \to \mathbb{R}$ satisfy the **bounded differences** property: for every $i \in [m]$ and every $z_1, \dots, z_m, z_i' \in \mathcal{Z}$,

$$\big|f(z_1, \dots, z_i, \dots, z_m) - f(z_1, \dots, z_i', \dots, z_m)\big| \le c_i.$$

Then for any $t > 0$,

$$P\big(|f(Z_1, \dots, Z_m) - \mathbb{E}[f(Z_1, \dots, Z_m)]| \ge t\big) \le 2\exp\Big(\frac{-2t^2}{\sum_{i=1}^m c_i^2}\Big).$$

Proof. The argument is the Chernoff route of Theorem 5.6.5, applied not to the summands themselves — there are none — but to an artificial decomposition of f into m increments, one per input variable. Write $Z_{1:i} := (Z_1, \dots, Z_i)$, $F := f(Z_1, \dots, Z_m)$, and define the **increments**

$$V_i := \mathbb{E}\big[F \mid Z_{1:i}\big] - \mathbb{E}\big[F \mid Z_{1:i-1}\big], \qquad i = 1, \dots, m,$$

with the convention $\mathbb{E}[F \mid Z_{1:0}] := \mathbb{E}[F]$. The sum telescopes exactly:

$$\sum_{i=1}^m V_i = \mathbb{E}[F \mid Z_{1:m}] - \mathbb{E}[F] = F - \mathbb{E}[F].$$

Step 1: each increment has conditional mean zero and conditional range at most c_i. The tower property gives $\mathbb{E}[V_i \mid Z_{1:i-1}] = 0$. For the range, define

$$U_i := \sup_z \mathbb{E}\big[F \mid Z_{1:i-1}, Z_i = z\big] - \mathbb{E}[F \mid Z_{1:i-1}],$$
$$L_i := \inf_z \mathbb{E}\big[F \mid Z_{1:i-1}, Z_i = z\big] - \mathbb{E}[F \mid Z_{1:i-1}],$$

both functions of $Z_{1:i-1}$ alone, so that $L_i \le V_i \le U_i$ by construction. For any two values z, z', independence of the Z's lets us average the bounded-differences hypothesis over the *remaining* variables $Z_{i+1}, \dots, Z_m$:

$$\big|\mathbb{E}[F \mid Z_{1:i-1}, Z_i = z] - \mathbb{E}[F \mid Z_{1:i-1}, Z_i = z']\big| \le \mathbb{E}\big[\,\big|f(\dots, z, \dots) - f(\dots, z', \dots)\big|\,\big] \le c_i,$$

hence $U_i - L_i \le c_i$: conditionally on the past, V_i is a mean-zero random variable confined to an interval of width at most c_i.

Step 2: conditional Hoeffding's lemma, iterated. Lemma 5.6.4 applies verbatim to conditional expectations, so for every $s \in \mathbb{R}$,

$$\mathbb{E}\big[e^{sV_i} \mid Z_{1:i-1}\big] \le e^{s^2 c_i^2/8}.$$

Peeling the increments off one at a time with the tower property,

$$\mathbb{E}\big[e^{s\sum_{i=1}^m V_i}\big] = \mathbb{E}\Big[e^{s\sum_{i<m} V_i}\underbrace{\mathbb{E}\big[e^{sV_m} \mid Z_{1:m-1}\big]}_{\le e^{s^2 c_m^2/8}}\Big] \le \cdots \le \exp\Big(\frac{s^2}{8}\sum_{i=1}^m c_i^2\Big).$$

Step 3: Chernoff. Exactly as in Theorem 5.6.5, Markov's inequality on $e^{s(F-\mathbb{E}[F])}$ gives $P(F-\mathbb{E}[F]\geq t)\leq \exp(-st+\frac{s^2}{8}\sum_i c_i^2)$ for every $s>0$; minimizing the exponent at $s^*=4t/\sum_i c_i^2$ yields $\exp\big(-2t^2/\sum_i c_i^2\big)$. Applying the same argument to $-f$ bounds the other tail identically, and adding the two gives the factor 2 □

Setting $m=1$, or taking $f(z_1,\dots,z_m)=\frac{1}{m}\sum_i z_i$ with $c_i=(b_i-a_i)/m$, recovers Hoeffding's inequality — McDiarmid's inequality is its strict generalization, obtained by replacing "sum of independent variables" with "any function that no single variable can move very far".

For the choice $f(S):=\sup_{h\in\mathcal{H}}\big(R(h)-\widehat{R}_S(h)\big)$ with a loss bounded in $[0,1]$, changing a single training example changes $\widehat{R}_S(h)$ by at most $1/m$ for every h simultaneously, hence f has bounded differences $c_i=1/m$, and McDiarmid's inequality applies with $\sum_i c_i^2=1/m$. This is exactly the role Hoeffding's inequality played for a single hypothesis in Chapter 5 — McDiarmid is its supremum-over-a-class counterpart, and it is what turns a bound on $\mathbb{E}_S[\sup_h(R(h)-\widehat{R}_S(h))]$ into the high-probability statements below.

7.2 The Symmetrization Technique

The remaining question is how to bound $\mathbb{E}_S\big[\sup_{h\in\mathcal{H}}(R(h)-\widehat{R}_S(h))\big]$ itself. Every complexity measure in this chapter is obtained by the same device: introduce an independent **ghost sample** $S'=\{(x_i',y_i')\}_{i=1}^m$, drawn i.i.d. from $\mathcal{D}$ exactly like S, and note that $R(h)=\mathbb{E}_{S'}[\widehat{R}_{S'}(h)]$. Then, since sup of an expectation is at most the expectation of the sup (Jensen),

$$\begin{aligned}\mathbb{E}_S\Big[\sup_{h\in\mathcal{H}}\big(R(h)-\widehat{R}_S(h)\big)\Big]&=\mathbb{E}_S\Big[\sup_{h\in\mathcal{H}}\mathbb{E}_{S'}\big[\widehat{R}_{S'}(h)-\widehat{R}_S(h)\big]\Big]\\&\leq\mathbb{E}_{S,S'}\Big[\sup_{h\in\mathcal{H}}\big(\widehat{R}_{S'}(h)-\widehat{R}_S(h)\big)\Big]\\&=\mathbb{E}_{S,S'}\Big[\sup_{h\in\mathcal{H}}\frac{1}{m}\sum_{i=1}^m\big(\ell(y_i',h(x_i'))-\ell(y_i,h(x_i))\big)\Big].\end{aligned}$$

The last expression is symmetric in the pairing $(x_i,y_i)\leftrightarrow(x_i',y_i')$: since S and S' are i.i.d. and independent of each other, swapping the i-th pair of S with the i-th pair of S' leaves the joint distribution of (S,S') unchanged. Introducing an independent Rademacher variable $\sigma_i\in\{-1,+1\}$ (each sign equally likely) to record whether the i-th pair *was* swapped therefore leaves the expectation unchanged for any choice of signs, hence

$$\begin{aligned}\mathbb{E}_{S,S'}\Big[\sup_h\frac{1}{m}\sum_i\big(\ell(y_i',h(x_i'))-\ell(y_i,h(x_i))\big)\Big]&=\mathbb{E}_{S,S',\sigma}\Big[\sup_h\frac{1}{m}\sum_i\sigma_i\big(\ell(y_i',h(x_i'))-\ell(y_i,h(x_i))\big)\Big]\\&\leq 2\,\mathbb{E}_{S,\sigma}\Big[\sup_h\frac{1}{m}\sum_i\sigma_i\ell(y_i,h(x_i))\Big],\end{aligned}$$

where the last step splits the supremum of a sum into the sum of two suprema (each bounded by the original, by symmetry of $\pm\sigma_i$). This is **symmetrization**, and the quantity on the right is, by definition, twice the (expected) Rademacher complexity of the loss class $\ell\circ\mathcal{H}$.

7.3 Rademacher Complexity

We follow the formulation of (Bartlett and Mendelson, 2002).

Definition 7.3.1: Rademacher complexity

For a set $A \subset \mathbb{R}^m$ and i.i.d. Rademacher signs $\sigma_1, \dots, \sigma_m \in \{-1, +1\}$ ($P(\sigma_i = 1) = P(\sigma_i = -1) = \frac{1}{2}$), define

$$\widehat{\mathfrak{R}}(A) := \mathbb{E}_\sigma\Big[\sup_{a \in A} \frac{1}{m} \sum_{i=1}^{m} \sigma_i a_i\Big].$$

For a hypothesis class $\mathcal{H}$ and a fixed sample $S = \{(x_i, y_i)\}_{i=1}^m$, write $\ell \circ \mathcal{H} \circ S := \big\{(\ell(y_1, h(x_1)), \dots, \ell(y_m, h(x_m))) : h \in \mathcal{H}\big\} \subset \mathbb{R}^m$ for the loss class evaluated on S, and define the **empirical Rademacher complexity** $\widehat{\mathfrak{R}}_S(\ell \circ \mathcal{H}) := \widehat{\mathfrak{R}}(\ell \circ \mathcal{H} \circ S)$ and the **(expected) Rademacher complexity** $\mathfrak{R}_m(\ell \circ \mathcal{H}) := \mathbb{E}_S\big[\widehat{\mathfrak{R}}_S(\ell \circ \mathcal{H})\big]$. For a hypothesis class $\mathcal{H}$, set $\widehat{\mathfrak{R}}_S(\mathcal{H}) := \widehat{\mathfrak{R}}(\mathcal{H}_S)$, where $\mathcal{H}_S$ is the restriction of Definition 6.4.1.

Intuitively, $\widehat{\mathfrak{R}}_S(\ell \circ \mathcal{H})$ measures how well $\mathcal{H}$ can fit **pure random noise**: σ_i is a coin flip drawn independently of the data, so $\sup_h \frac{1}{m}\sum_i \sigma_i \ell(y_i, h(x_i))$ asks how large a spurious correlation the richest hypothesis in $\mathcal{H}$ can manufacture with random labels. A class rich enough to realize any loss vector in $\{0,1\}^m$ (as in the memorization hypothesis of Chapter 2) can match every sign, giving $\widehat{\mathfrak{R}}_S \approx \frac{1}{2}$. A class with a single hypothesis has no supremum to exploit, so $\widehat{\mathfrak{R}}_S = \frac{1}{m}\sum_i \mathbb{E}[\sigma_i]\, a_i = 0$ exactly.

Theorem 7.3.2: Rademacher generalization bound

Let the loss be bounded in $[0, 1]$. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$, simultaneously for all $h \in \mathcal{H}$:

$$R(h) \le \widehat{R}_S(h) + 2\,\widehat{\mathfrak{R}}_S(\ell \circ \mathcal{H}) + 3\sqrt{\frac{\log(2/\delta)}{2m}}.$$

Proof. Write $\Phi(S) := \sup_{h \in \mathcal{H}} \big(R(h) - \widehat{R}_S(h)\big)$, so the claim is a high-probability upper bound on $\Phi(S)$. The proof applies McDiarmid's inequality twice, with symmetrization in between.

Step 1: concentrate Φ around its mean. Replacing one example of S changes $\widehat{R}_S(h)$ by at most $1/m$ for every h at once, hence changes the supremum by at most $1/m$: Φ has bounded differences $c_i = 1/m$, so $\sum_i c_i^2 = 1/m$. The one-sided form of Theorem 7.1.1 at confidence $\delta/2$ gives, with probability at least $1 - \delta/2$,

$$\Phi(S) \;\le\; \mathbb{E}_S[\Phi(S)] + \sqrt{\frac{\log(2/\delta)}{2m}}.$$

Step 2: bound the mean by the expected Rademacher complexity. This is the symmetrization argument of the previous section, verbatim: $\mathbb{E}_S[\Phi(S)] \le 2\,\mathfrak{R}_m(\ell \circ \mathcal{H})$.

Step 3: replace the expected complexity by its empirical counterpart. The map $S \mapsto \widehat{\mathfrak{R}}_S(\ell \circ \mathcal{H})$ also has bounded differences $1/m$: changing one example changes each coordinate of the loss vector by at

most 1, hence changes $\frac{1}{m}\sum_i \sigma_i a_i$ by at most $1/m$ uniformly, and a supremum of functions that all move by at most $1/m$ moves by at most $1/m$. Applying Theorem 7.1.1 once more at confidence $\delta/2$, with probability at least $1-\delta/2$,

$$\mathfrak{R}_m(\ell\circ\mathcal{H}) = \mathbb{E}_S\big[\widehat{\mathfrak{R}}_S(\ell\circ\mathcal{H})\big] \;\le\; \widehat{\mathfrak{R}}_S(\ell\circ\mathcal{H}) + \sqrt{\frac{\log(2/\delta)}{2m}}.$$

Step 4: union bound. Both events hold simultaneously with probability at least $1-\delta$, and on their intersection

$$\sup_h \big(R(h)-\widehat{R}_S(h)\big) \le 2\,\widehat{\mathfrak{R}}_S(\ell\circ\mathcal{H}) + 2\sqrt{\frac{\log(2/\delta)}{2m}} + \sqrt{\frac{\log(2/\delta)}{2m}},$$

where the factor 2 from Step 2 multiplies the deviation of Step 3. Rearranging the supremum gives the claim for every h □

This is the Rademacher-complexity instance of the general form stated at the top of this chapter, with $\text{Complexity}(\mathcal{H},m) = 2\widehat{\mathfrak{R}}_S(\ell\circ\mathcal{H})$.

7.3.1 Massart's Lemma, and Theorem 6.1.3 revisited

Lemma 7.3.3: Massart's lemma

For a finite set $A=\{a^{(1)},\dots,a^{(N)}\}\subset\mathbb{R}^m$ with $\|a^{(j)}\|_2\le B$ for all j,

$$\widehat{\mathfrak{R}}(A) \le \frac{B\sqrt{2\log N}}{m},$$
$$\text{equivalently}\quad \mathbb{E}_\sigma\Big[\max_{j\in[N]}\langle\sigma,a^{(j)}\rangle\Big] \le B\sqrt{2\log N}.$$

Proof. Write $X_j := \langle\sigma,a^{(j)}\rangle = \sum_{i=1}^m \sigma_i a_i^{(j)}$ and assume $N\ge 2$ (for $N=1$ the left side is $\mathbb{E}_\sigma[X_1]=0$ and there is nothing to prove). Fix $s>0$. Because $u\mapsto e^{su}$ is convex and increasing, Jensen's inequality and then the crude bound "a maximum of non-negative terms is at most their sum" give

$$\exp\Big(s\,\mathbb{E}_\sigma\big[\max_j X_j\big]\Big) \;\le\; \mathbb{E}_\sigma\Big[e^{s\max_j X_j}\Big] = \mathbb{E}_\sigma\Big[\max_j e^{sX_j}\Big] \;\le\; \sum_{j=1}^N \mathbb{E}_\sigma\big[e^{sX_j}\big].$$

Each X_j is a sum of m independent, mean-zero terms $\sigma_i a_i^{(j)}$, the i-th confined to the interval $[-|a_i^{(j)}|,+|a_i^{(j)}|]$ of width $2|a_i^{(j)}|$. Hoeffding's lemma (Lemma 5.6.4) applied to each term and independence give

$$\mathbb{E}_\sigma\big[e^{sX_j}\big] = \prod_{i=1}^m \mathbb{E}\big[e^{s\sigma_i a_i^{(j)}}\big] \le \prod_{i=1}^m \exp\Big(\frac{s^2(2a_i^{(j)})^2}{8}\Big) = \exp\Big(\frac{s^2\|a^{(j)}\|_2^2}{2}\Big) \le \exp\Big(\frac{s^2B^2}{2}\Big).$$

Substituting and taking logarithms, $s\,\mathbb{E}_\sigma[\max_j X_j]\le \log N + \frac{s^2B^2}{2}$, i.e.

$$\mathbb{E}_\sigma\big[\max_j X_j\big] \le \frac{\log N}{s} + \frac{sB^2}{2} \quad \text{for every } s > 0.$$

The right-hand side is minimized at $s^* = \sqrt{2\log N}/B$, where its value is $B\sqrt{2\log N}$. Dividing by m turns the left-hand side into $\widehat{\mathfrak{R}}(A)$ □

Note the shape of the result: the dependence on the number of elements is only $\sqrt{\log N}$, which is what makes the union-bound arguments of Chapter 6 so efficient. Note also that the bound is *not* improved by adding a point to A that is far from the others — only the count and the worst-case norm matter.

Apply this to $\ell \circ \mathcal{H} \circ S$ for a **finite** hypothesis class, $N = |\mathcal{H}|$, with loss bounded in $[0,1]$ so $\|a^{(j)}\|_2 \le \sqrt{m}$: this gives $\widehat{\mathfrak{R}}_S(\ell\circ\mathcal{H}) \le \sqrt{2\log|\mathcal{H}|/m}$, and plugging into Theorem 7.3.2 recovers, up to constants, exactly the finite-hypothesis-class bound of Theorem 6.1.3. **The very first generalization bound of Chapter 6 was already an instance of a Rademacher-complexity bound** — we simply proved it directly via Hoeffding and the union bound there, without naming the general machinery.

7.3.2 The Contraction Lemma

Lemma 7.3.4: Contraction

If $\phi_i : \mathbb{R} \to \mathbb{R}$ is ρ-Lipschitz for each i, then $\widehat{\mathfrak{R}}\big(\{(\phi_1(a_1),\dots,\phi_m(a_m)) : a \in A\}\big) \le \rho\,\widehat{\mathfrak{R}}(A)$.

Proof. The whole content is a **one-coordinate** statement, which we prove first and then apply m times:

One-coordinate step. Let $\psi_1,\dots,\psi_{m-1}$ be arbitrary functions and ϕ be ρ-Lipschitz. Then

$$\mathbb{E}_\sigma\Big[\sup_{a\in A}\Big(\sum_{i<m}\sigma_i\psi_i(a_i) + \sigma_m\phi(a_m)\Big)\Big] \;\le\; \mathbb{E}_\sigma\Big[\sup_{a\in A}\Big(\sum_{i<m}\sigma_i\psi_i(a_i) + \sigma_m\,\rho\,a_m\Big)\Big].$$

Granting this, apply it to coordinate m (with $\psi_i := \phi_i$), replacing ϕ_m by $\rho\cdot\mathrm{id}$; then to coordinate $m-1$ of the resulting set, and so on. After m steps every ϕ_i has been replaced by $\rho\cdot\mathrm{id}$, so the left-hand side of the lemma is at most $\widehat{\mathfrak{R}}(\{\rho a : a\in A\}) = \rho\,\widehat{\mathfrak{R}}(A)$, the last equality by homogeneity of the definition. Each step transforms a *different* coordinate, so the factor ρ is collected once in total, not m times.

To prove the one-coordinate step, condition on $\sigma_1,\dots,\sigma_{m-1}$ and abbreviate $u(a) := \sum_{i<m}\sigma_i\psi_i(a_i)$, which does not involve the m-th coordinate's transform. Averaging over the two equally likely values of σ_m,

$$\begin{aligned}
\mathbb{E}_{\sigma_m}\Big[\sup_{a\in A}\big(u(a)+\sigma_m\phi(a_m)\big)\Big] &= \tfrac{1}{2}\sup_{a\in A}\big(u(a)+\phi(a_m)\big)+\tfrac{1}{2}\sup_{a'\in A}\big(u(a')-\phi(a'_m)\big)\\
&= \tfrac{1}{2}\sup_{a,a'\in A}\Big(u(a)+u(a')+\phi(a_m)-\phi(a'_m)\Big)\\
&\le \tfrac{1}{2}\sup_{a,a'\in A}\Big(u(a)+u(a')+\rho\,\big|a_m-a'_m\big|\Big)\\
&= \tfrac{1}{2}\sup_{a,a'\in A}\Big(u(a)+u(a')+\rho\,\big(a_m-a'_m\big)\Big)\\
&= \mathbb{E}_{\sigma_m}\Big[\sup_{a\in A}\big(u(a)+\sigma_m\rho\,a_m\big)\Big],
\end{aligned}$$

where the second line merges the two independent suprema into one over the pair (a, a'); the third applies the ρ-Lipschitz property of ϕ; the fourth drops the absolute value because the expression is symmetric under swapping $a \leftrightarrow a'$, so the supremum is attained at a pair for which $a_m - a'_m \ge 0$ anyway; and the fifth reverses the first two steps with $\rho\, a_m$ in place of $\phi(a_m)$. Taking the expectation over $\sigma_1, \dots, \sigma_{m-1}$ gives the one-coordinate step □

Note where the argument would break for the *signed* variant $\sup_a |\frac{1}{m}\sum_i \sigma_i a_i|$: the second line merges two suprema into one, which is valid for a supremum of a set but not for a supremum of absolute values — this is the source of the extra factor of 2 mentioned below.

This lets us pass, in the definition of $\ell \circ \mathcal{H}$, from the Rademacher complexity of the *loss-composed* class directly to the Rademacher complexity of $\mathcal{H}$ itself, whenever the loss $\ell(\cdot, y)$ is ρ-Lipschitz in its first argument: $\widehat{\mathfrak{R}}_S(\ell \circ \mathcal{H}) \le \rho\,\widehat{\mathfrak{R}}_S(\mathcal{H})$. A word of caution, since this is a place where different textbooks state slightly different constants: the *clean* version above, with no extra factor, holds for the **unsigned** Rademacher complexity (where the supremum in Definition 7.3.1 is *not* wrapped in an absolute value, exactly as we defined it). If instead one works with the signed variant $\sup_a |\frac{1}{m}\sum_i \sigma_i a_i|$, the contraction inequality picks up an extra factor of 2. Precisely: if $0 \in A$ and $\phi_i(0) = 0$ for all i, write $\phi(A) := \{(\phi_1(a_1), \dots, \phi_m(a_m)) : a \in A\}$, which then contains 0. For every realization of the signs both $\sup_{b\in\phi(A)}\langle\sigma, b\rangle$ and $\sup_{b\in\phi(A)}\langle-\sigma, b\rangle$ are ≥ 0 (take $b = 0$), so

$$\sup_{b\in\phi(A)}\big|\langle\sigma, b\rangle\big| = \max\Big(\sup_b\langle\sigma, b\rangle,\ \sup_b\langle-\sigma, b\rangle\Big) \le \sup_b\langle\sigma, b\rangle + \sup_b\langle-\sigma, b\rangle.$$

The two terms have the same expectation, since $-\sigma$ has the same law as σ, and the lemma bounds each by $\rho\, m\, \widehat{\mathfrak{R}}(A)$. Hence

$$\mathbb{E}_\sigma\Big[\sup_{a\in A}\Big|\frac{1}{m}\sum_i \sigma_i\phi_i(a_i)\Big|\Big] \ \le\ 2\rho\,\widehat{\mathfrak{R}}(A). \tag{7.1}$$

Since our Definition 7.3.1 uses the unsigned form throughout, the clean statement of the lemma applies directly throughout this chapter; (7.1) is needed only in Chapter 10.

7.3.3 The Bridge to VC Dimension

Theorem 7.3.5: VC dimension bounds Rademacher complexity

For a hypothesis class $\mathcal{H}$ of binary classifiers with $d_{VC}(\mathcal{H}) = d < \infty$, and the zero-one loss,

$$\widehat{\mathfrak{R}}_S(\mathcal{H}) \le 2\sqrt{\frac{d\log(m+1)}{m}},$$

for every sample S of size m.

Proof. The restriction $\mathcal{H}_S = \{(h(x_1), \dots, h(x_m)) : h \in \mathcal{H}\}$ is a *finite* set (Definition 6.4.1), of size $|\mathcal{H}_S| = \tau_{\mathcal{H}}(m) \le (m+1)^d$ by the Sauer-Shelah-Perles lemma (Lemma 6.4.8, using the simplified polynomial form). Apply Massart's lemma (Lemma 7.3.3) with $N = (m+1)^d$ and $B = \sqrt{m}$ (binary values in $\{0,1\}$, so $\|a^{(j)}\|_2 \le \sqrt{m}$):

$$\widehat{\mathfrak{R}}(\mathcal{H}_S) \le \frac{\sqrt{m}\sqrt{2\log\left((m+1)^d\right)}}{m} = \sqrt{\frac{2d\log(m+1)}{m}} \le 2\sqrt{\frac{d\log(m+1)}{m}},$$

the last step simply using $\sqrt{2} \le 2$, which is where the slack in the stated constant comes from. □

This is the promised unification made concrete: **VC dimension is not a separate theory from Rademacher complexity — it is a specific, combinatorial way of bounding it**, via Sauer-Shelah-Perles turning an infinite hypothesis class into a finite restriction on any given sample, to which Massart's finite-class bound then applies.

7.3.4 A VC-Dimension-Free Example: Linear Predictors

One reason Rademacher complexity is worth the extra abstraction: it can be **finite and small even when** $d_{VC}(\mathcal{H}) = \infty$, provided we constrain the hypothesis class by a *norm* rather than by dimension.

Theorem 7.3.6

Let $\mathcal{H} = \{x \mapsto w^\top x : \|w\|_2 \le B\}$ and suppose $\|x\|_2 \le R$ almost surely under $\mathcal{D}$. Then

$$\widehat{\mathfrak{R}}_S(\mathcal{H}) \le \frac{BR}{\sqrt{m}}.$$

Proof. By definition and linearity, for a fixed realization of the signs,

$$\sup_{\|w\|_2 \le B} \frac{1}{m}\sum_{i=1}^m \sigma_i\, w^\top x_i = \frac{1}{m}\sup_{\|w\|_2 \le B}\left\langle w, \sum_{i=1}^m \sigma_i x_i\right\rangle = \frac{B}{m}\left\|\sum_{i=1}^m \sigma_i x_i\right\|_2,$$

where the second equality is the equality case of the Cauchy-Schwarz inequality: the linear functional $w \mapsto \langle w, v\rangle$ is maximized over the ball $\|w\|_2 \le B$ at $w = B\,v/\|v\|_2$, with value $B\|v\|_2$. Taking $\mathbb{E}_\sigma$ and

applying Jensen's inequality to the concave map $u \mapsto \sqrt{u}$,

$$\widehat{\mathfrak{R}}_S(\mathcal{H}) = \frac{B}{m}\,\mathbb{E}_\sigma\Big[\Big\|\sum_i \sigma_i x_i\Big\|_2\Big] \le \frac{B}{m}\sqrt{\mathbb{E}_\sigma\Big[\Big\|\sum_i \sigma_i x_i\Big\|_2^2\Big]}.$$

Expanding the squared norm and using $\mathbb{E}[\sigma_i\sigma_{i'}] = \mathbb{1}(i = i')$ (the signs are independent and mean-zero, and $\sigma_i^2 = 1$),

$$\mathbb{E}_\sigma\Big[\Big\|\sum_i \sigma_i x_i\Big\|_2^2\Big] = \sum_{i,i'} \mathbb{E}[\sigma_i\sigma_{i'}]\, x_i^\top x_{i'} = \sum_{i=1}^m \|x_i\|_2^2 \le mR^2.$$

Combining, $\widehat{\mathfrak{R}}_S(\mathcal{H}) \le \frac{B}{m}\sqrt{mR^2} = \frac{BR}{\sqrt{m}}$ □

Notice this bound does not depend on the input dimension d at all — only on the norm bound B and the data norm bound R. This matters directly for the kernel methods of Chapter 11, where $\phi(x)$ can live in an infinite-dimensional feature space (for which d_{VC} is typically infinite) while $\|w\|_2$ and $\|\phi(x)\|_2$ remain finite: Rademacher complexity, unlike VC dimension, gives a meaningful, finite generalization guarantee in exactly this regime.

7.4 Covering Numbers

Rademacher complexity measures $\mathcal{H}$'s capacity to correlate with noise. **Covering numbers** measure a more concrete quantity: how many hypotheses it takes to approximate every hypothesis in $\mathcal{H}$ to a given tolerance.

Definition 7.4.1: ϵ-cover, covering number

Let $\mathcal{H}$ be a hypothesis class equipped with a (pseudo-)metric $\mathrm{dist}(\cdot,\cdot)$: a function satisfying axioms M2 and M3 of the metric-space definition of Chapter 3, while M1 is relaxed to the one-directional $\mathrm{dist}(h,h') = 0 \Leftarrow h = h'$. A finite set $\{h_1,\dots,h_N\} \subset \mathcal{H}$ is an **ϵ-cover** of $\mathcal{H}$ if for every $h \in \mathcal{H}$ there exists $j \in [N]$ with $\mathrm{dist}(h,h_j) \le \epsilon$. The **covering number** $N(\epsilon,\mathcal{H},\mathrm{dist})$ is the size of the smallest such cover.

In words, an ϵ-cover is a finite "grid" of representative hypotheses such that every hypothesis in $\mathcal{H}$, however large or infinite the class, is within ϵ of some grid point — turning an infinite class into a finite proxy at the cost of an ϵ-sized approximation error.

The relevant metric for generalization is one under which the risk is Lipschitz: if $\ell(\cdot,y)$ is G-Lipschitz in its first argument for every y, and $\mathrm{dist}(h,h') := \mathbb{E}_{x\sim\mathcal{D}}|h(x) - h'(x)|$, then $|R(h) - R(h')| \le G\,\mathrm{dist}(h,h')$ for all $h,h' \in \mathcal{H}$ — replacing h by the nearest cover element changes the risk by only $G\epsilon$. This particular dist is a genuine pseudo-metric rather than a metric: two distinct hypotheses that differ only on inputs that $\mathcal{D}$ never draws have distance zero, which is exactly why M1 had to be relaxed above — and it is harmless, since such hypotheses also have identical risk.

Theorem 7.4.2: Covering-number generalization bound

Let $\{h_1, \dots, h_{N(\epsilon)}\}$ be an ϵ-cover of $\mathcal{H}$ under dist as above, with loss bounded in $[0, 1]$ and G-Lipschitz in its first argument for every y. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$, simultaneously for all $h \in \mathcal{H}$,

$$R(h) \le \widehat{R}_S(h) + 2G\epsilon + \sqrt{\frac{\log(2N(\epsilon)/\delta)}{2m}}.$$

Proof. Fix $h \in \mathcal{H}$ and choose a cover center h_j with $\mathrm{dist}(h, h_j) \le \epsilon$. The triangle inequality gives

$$\begin{aligned}\left|\widehat{R}_S(h) - R(h)\right| &\le \left|\widehat{R}_S(h) - \widehat{R}_S(h_j)\right| + \left|\widehat{R}_S(h_j) - R(h_j)\right| + \left|R(h_j) - R(h)\right| \\ &\le 2G\epsilon + \left|\widehat{R}_S(h_j) - R(h_j)\right|,\end{aligned}$$

where the last step uses the Lipschitz bound on both the empirical and true risk. The remaining term concerns only the $N(\epsilon)$ **fixed, data-independent** cover centers, so Theorem 6.1.3 applies to the finite class $\{h_1, \dots, h_{N(\epsilon)}\}$ and bounds it, uniformly over j, by $\sqrt{\log(2N(\epsilon)/\delta)/(2m)}$ with probability at least $1 - \delta$ □

This is exactly the finite-hypothesis-class argument of Theorem 6.1.3, applied not to $\mathcal{H}$ itself but to a *finite proxy* for it — the ϵ-cover — with an extra $2G\epsilon$ term paying for the approximation error of using the proxy.

Lemma 7.4.3: Volumetric covering bound

Let $\mathcal{B}_B := \{v \in \mathbb{R}^d : \|v\|_2 \le B\}$ with the Euclidean metric. Then $N(\epsilon, \mathcal{B}_B, \|\cdot\|_2) \le \big(1 + 2B/\epsilon\big)^d$.

Proof. Let $\{v_1, \dots, v_N\} \subseteq \mathcal{B}_B$ be a **maximal** ϵ**-separated** set, i.e. $\|v_j - v_{j'}\|_2 > \epsilon$ for $j \ne j'$ and no further point of $\mathcal{B}_B$ can be added without violating this. Maximality means every $v \in \mathcal{B}_B$ is within ϵ of some v_j — otherwise it could be added — so the set is an ϵ-cover and $N(\epsilon, \mathcal{B}_B, \|\cdot\|_2) \le N$.

To bound N, note the open balls of radius $\epsilon/2$ around the v_j are pairwise disjoint (their centers are more than ϵ apart) and all contained in $\mathcal{B}_{B+\epsilon/2}$. Since the volume of a Euclidean ball of radius r in $\mathbb{R}^d$ is $V_d\, r^d$ for a constant V_d depending only on d, comparing volumes gives

$$N \cdot V_d\Big(\frac{\epsilon}{2}\Big)^d \le V_d\Big(B + \frac{\epsilon}{2}\Big)^d \quad \Longrightarrow \quad N \le \Big(\frac{B + \epsilon/2}{\epsilon/2}\Big)^d = \Big(1 + \frac{2B}{\epsilon}\Big)^d. \qquad \square$$

The exponent d is the entire story: covering numbers grow *exponentially* in the dimension, so $\log N(\epsilon) = O(d \log(1/\epsilon))$ grows only linearly in it — and it is $\log N(\epsilon)$, not $N(\epsilon)$, that enters the bound. This is the same phenomenon as the logarithmic dependence on $|\mathcal{H}|$ in Theorem 6.1.3, and the reason both bounds remain useful in high dimension. □

Balancing $\epsilon \propto 1/\sqrt{m}$ against the second term yields an overall rate of $O\big(\sqrt{(d/m)\log m}\big)$ — the same $\sqrt{d/m}$ rate as the VC/Rademacher bounds above, up to a $\sqrt{\log m}$ factor. That residual log factor

can be removed by not fixing a single resolution ϵ but summing the bound over a geometric sequence of resolutions $\epsilon_k = 2^{-k}$ — a technique called **chaining**. We return to it after the PAC-Bayes bounds, where it removes this factor for VC classes (Corollary 7.6.3).

7.5 PAC-Bayes Bounds

The bounds so far all fix a hypothesis class $\mathcal{H}$ first, and then ask for a guarantee that holds *uniformly*, for every $h \in \mathcal{H}$ at once — this is what makes it valid to then plug in whichever h our learning algorithm happens to output. **PAC-Bayes bounds** (McAllester, 1999) relax this: instead of a single hypothesis, they bound the risk of a whole *distribution* over hypotheses, and let that distribution depend on the data.

Setup. Fix a **prior** distribution q over $\mathcal{H}$, chosen *before* seeing S (it may encode domain knowledge, or simply be uniform/uninformative). After seeing S, we are free to choose any **posterior** distribution ρ over $\mathcal{H}$ — informed by the data in any way whatsoever, including via an algorithm's output — and consider the *randomized* predictor that draws a fresh $h \sim \rho$ at prediction time. Define $R(\rho) := \mathbb{E}_{h\sim\rho}[R(h)]$ and $\widehat{R}_S(\rho) := \mathbb{E}_{h\sim\rho}[\widehat{R}_S(h)]$. The bound below is governed by the **Kullback-Leibler (KL) divergence** between the two,

$$\mathrm{KL}(\rho\|q) := \mathbb{E}_{h\sim\rho}\Big[\log\frac{\rho(h)}{q(h)}\Big],$$

a measure of how distinguishable ρ is from q: it is zero exactly when the two distributions coincide, and grows as ρ places mass where q places little. (Its definition is re-derived and used in the proof, where it appears as the expectation of a log-ratio; only the two displayed properties just stated are needed to read the theorem.)

Theorem 7.5.1: PAC-Bayes bound

Assume the loss is bounded in $[0,1]$. For any prior q fixed before S, any $\beta > 0$, and any $\delta \in (0,1)$, with probability at least $1-\delta$ over $S \sim \mathcal{D}^m$, **simultaneously for every posterior ρ possibly depending on S**:

$$R(\rho) \le \widehat{R}_S(\rho) + \frac{1}{\beta}\,\mathrm{KL}(\rho\|q) + \frac{1}{\beta}\log\frac{1}{\delta} + \frac{\beta}{8m}.$$

Proof. Fix $\beta > 0$ and define, for each hypothesis, the random variable $X(h) := \beta\big(R(h) - \widehat{R}_S(h)\big)$, a function of S.

Step 1: a moment generating function bound for a single, fixed h. Write $\ell_i := \ell(y_i, h(x_i)) \in [0,1]$, independent across i with mean $R(h)$. Then $X(h) = \sum_{i=1}^m \frac{\beta}{m}\big(R(h) - \ell_i\big)$ is a sum of m independent, mean-zero terms, the i-th confined to an interval of width β/m. Hoeffding's lemma (Lemma 5.6.4) and independence give

$$\mathbb{E}_S\big[e^{X(h)}\big] \le \prod_{i=1}^m \exp\Big(\frac{(\beta/m)^2}{8}\Big) = \exp\Big(\frac{\beta^2}{8m}\Big).$$

Step 2: average over the prior, then apply Markov. The bound of Step 1 holds for every h, and q does not depend on S, so the two expectations may be exchanged:

$$\mathbb{E}_S\Big[\mathbb{E}_{h\sim q}\big[e^{X(h)}\big]\Big] = \mathbb{E}_{h\sim q}\Big[\mathbb{E}_S\big[e^{X(h)}\big]\Big] \le e^{\beta^2/(8m)}.$$

The inner quantity $\mathbb{E}_{h\sim q}[e^{X(h)}]$ is a non-negative random variable of S, so Markov's inequality (Theorem 5.6.1) gives: with probability at least $1-\delta$ over S,

$$\mathbb{E}_{h\sim q}\big[e^{X(h)}\big] \;\le\; \frac{1}{\delta}\, e^{\beta^2/(8m)}. \qquad (*)$$

Note carefully that $(*)$ involves **only the prior** q, so it holds on a single event of probability $1-\delta$ that is fixed before any posterior is chosen. This is the entire trick of PAC-Bayes: all the probabilistic work is done under q, and the data-dependent ρ is introduced afterwards, deterministically, on that same event.

Step 3: the Donsker-Varadhan change of measure. For any distribution ρ absolutely continuous with respect to q, and any function X,

$$\mathbb{E}_{h\sim\rho}\big[X(h)\big] \;\le\; \mathrm{KL}(\rho\|q) + \log \mathbb{E}_{h\sim q}\big[e^{X(h)}\big].$$

To see this, define the **tilted** distribution $\tilde{q}(h) := q(h)e^{X(h)}/\mathbb{E}_{h\sim q}[e^{X(h)}]$, which is a valid probability distribution by construction. Expanding the Kullback-Leibler divergence of ρ from $\tilde{q}$,

$$\mathrm{KL}(\rho\|\tilde{q}) = \mathbb{E}_{h\sim\rho}\Big[\log\frac{\rho(h)}{\tilde{q}(h)}\Big] = \underbrace{\mathbb{E}_{h\sim\rho}\Big[\log\frac{\rho(h)}{q(h)}\Big]}_{=\,\mathrm{KL}(\rho\|q)} - \mathbb{E}_{h\sim\rho}\big[X(h)\big] + \log\mathbb{E}_{h\sim q}\big[e^{X(h)}\big],$$

so the claimed inequality is exactly the statement $\mathrm{KL}(\rho\|\tilde{q}) \ge 0$, which holds for any pair of distributions by Jensen's inequality applied to the convex function $u \mapsto -\log u$.

Step 4: combine. On the event of $(*)$, for **every** ρ simultaneously,

$$\beta\big(R(\rho) - \widehat{R}_S(\rho)\big) = \mathbb{E}_{h\sim\rho}\big[X(h)\big] \le \mathrm{KL}(\rho\|q) + \log\Big(\tfrac{1}{\delta}\, e^{\beta^2/(8m)}\Big) = \mathrm{KL}(\rho\|q) + \log\tfrac{1}{\delta} + \frac{\beta^2}{8m},$$

the first equality by linearity of $\mathbb{E}_\rho$ together with the definitions $R(\rho) := \mathbb{E}_\rho[R(h)]$ and $\widehat{R}_S(\rho) := \mathbb{E}_\rho[\widehat{R}_S(h)]$. Dividing by $\beta > 0$ gives the theorem □

The parameter β — called an **inverse temperature** by analogy with statistical physics, where a high temperature flattens a distribution over states and a low one sharpens it — trades off the two error terms and can be optimized; we do this explicitly below. The minimizer of the right-hand side over ρ is the **Gibbs posterior** $\widehat{\rho}_\beta(h) \propto \exp(-\beta\,\widehat{R}_S(h))\,q(h)$, i.e. exactly the tilted distribution $\tilde{q}$ of Step 3 for the choice $X = -\beta\widehat{R}_S$: the prior q, exponentially reweighted towards hypotheses of low empirical risk, with β controlling how sharp the reweighting is. Note that $\widehat{\rho}_\beta$ coincides with the *Bayesian* posterior of Chapter 8 exactly when $\beta = m$; other choices of β are read as different inference temperatures.

7.5.1 Recovering Theorem 6.1.3 as a Special Case

Take $\mathcal{H} = \{h_1, \dots, h_N\}$ finite, the **uniform prior** $q(h_j) = 1/N$, and restrict attention to **Dirac posteriors** $\rho = \delta_h$ that place all their mass on a single hypothesis $h \in \mathcal{H}$ — precisely the deterministic predictors we have used throughout this course. Then $R(\rho) = R(h)$, $\widehat{R}_S(\rho) = \widehat{R}_S(h)$, and

$\mathrm{KL}(\delta_h \| q) = \log(1/q(h)) = \log N$, so Theorem 7.5.1 gives: with probability $\geq 1-\delta$, simultaneously for **every** $h \in \mathcal{H}$,

$$R(h) \leq \widehat{R}_S(h) + \frac{\log N}{\beta} + \frac{1}{\beta}\log\frac{1}{\delta} + \frac{\beta}{8m}, \qquad \forall \beta > 0.$$

Minimizing the right-hand side over β (setting the derivative in β to zero gives $\beta^* = \sqrt{8m(\log N + \log(1/\delta))}$) yields exactly

$$R(h) \leq \widehat{R}_S(h) + \sqrt{\frac{\log N + \log(1/\delta)}{2m}} = \widehat{R}_S(h) + \sqrt{\frac{\log|\mathcal{H}| + \log(1/\delta)}{2m}},$$

which is **precisely Theorem 6.1.3**, recovered as the special case of PAC-Bayes with a uniform prior and Dirac posteriors over a finite $\mathcal{H}$. This closes the loop promised at the start of this chapter: $|\mathcal{H}|$-counting, VC dimension, Rademacher complexity, covering numbers, and PAC-Bayes bounds are not five different theories of generalization — they are one theory, instantiated through five different choices of how to measure "the complexity of $\mathcal{H}$," and each of the first four turns out to be reachable as a special case of the last, more general one.

7.6 Chaining: Sharp Rates for VC Classes

The five complexity measures above complete the abstract framework. The remaining sections refine it in four directions: chaining removes the logarithmic factor left over in the VC bounds (this section); L_1-constrained combinations show that capacity can be priced by a norm on the combination weights (next section); algorithmic stability prices the *algorithm* rather than the class; and minimax lower bounds show that the rates obtained cannot be improved in general.

Recall the residual $\sqrt{\log m}$ factor of the single-resolution covering bound, and the $\sqrt{\log(m+1)}$ factor of Theorem 7.3.5. Both are removed by covering the class at all resolutions $\epsilon_k = 2^{-k}$ simultaneously. Two ingredients are needed: a covering-number estimate for VC classes that does not depend on the sample, and the chaining bound itself. Throughout, for a sample $S = \{x_1, \dots, x_m\}$ we use the empirical pseudo-metrics

$$\mathrm{dist}_{L_1(S)}(g, g') := \frac{1}{m}\sum_{i=1}^{m} |g(x_i) - g'(x_i)|, \qquad \mathrm{dist}_{L_2(S)}(g, g') := \Big(\frac{1}{m}\sum_{i=1}^{m} \big(g(x_i) - g'(x_i)\big)^2\Big)^{1/2},$$

and write $\|g\|_S := \mathrm{dist}_{L_2(S)}(g, 0)$. Covering numbers are those of Definition 7.4.1, with cover elements taken from the class itself.

Lemma 7.6.1: Packing bound for VC classes

Let $\mathcal{G}$ be a class of $\{0,1\}$-valued functions with VC dimension $d \geq 1$, and let P be any distribution on the domain, with $\mathrm{dist}_{L_1(P)}(g, g') := \mathbb{E}_{x\sim P}|g(x) - g'(x)| = P\big(g(x) \neq g'(x)\big)$. Then for every $\epsilon \in (0, 1]$,

$$\log N\big(\epsilon, \mathcal{G}, L_1(P)\big) \;\leq\; 2d \log\frac{4e}{\epsilon}.$$

Proof. Call $g_1, \dots, g_M \in \mathcal{G}$ **ϵ-separated** if $\operatorname{dist}_{L_1(P)}(g_j, g_k) > \epsilon$ for all $j \neq k$. We show that every ϵ-separated family satisfies $\log M \le 2d \log(4e/\epsilon)$. Granting this, a separated family of maximal size exists, and as in the proof of Lemma 7.4.3 it is an ϵ-cover (a function farther than ϵ from all its members could be added to it), which proves the lemma.

Let $M \ge 2$ (for $M = 1$ there is nothing to prove), set $n := \lceil 2 \log M/\epsilon \rceil$, and draw $x_1, \dots, x_n$ i.i.d. from P. For a fixed pair $j < k$, the two functions agree on all n points with probability

$$\big(1 - \operatorname{dist}_{L_1(P)}(g_j, g_k)\big)^n < (1-\epsilon)^n \le e^{-\epsilon n} \le M^{-2}.$$

There are fewer than $M^2/2$ pairs, so by the union bound the probability that *some* pair agrees on all n points is less than $\tfrac{1}{2}$. Hence some realization of the n points — a set of at most n distinct points — separates all M functions, so that M is at most the number of distinct restrictions of $\mathcal{G}$ to that set, i.e. $M \le \tau_{\mathcal{G}}(n)$ (Definition 6.4.4; τ is non-decreasing, so having fewer than n distinct points only helps).

If $n \le d$, then $M \le 2^n \le 2^d$ and $\log M \le d \log 2$, which is below the claim. If $n > d$, Lemma 6.4.8 gives $M \le (en/d)^d$. Write $u := \log M/d$. Since $n \le 2 \log M/\epsilon + 1 = 2du/\epsilon + 1$ and $\epsilon \le 1 \le d$,

$$u \;\le\; \log \frac{e\,n}{d} \;\le\; \log\Big(e\Big(\frac{2u}{\epsilon} + \frac{1}{d}\Big)\Big) \;\le\; 1 + \log\frac{1}{\epsilon} + \log(2u+1).$$

The function $u \mapsto \frac{u}{2} + \log 4 - \log(2u+1)$ has derivative $\frac{1}{2} - \frac{2}{2u+1}$, hence its minimum over $u \ge 0$ at $u = \frac{3}{2}$, where its value is $\frac{3}{4} > 0$; so $\log(2u+1) \le \frac{u}{2} + \log 4$. Substituting, $u \le 1 + \log\frac{1}{\epsilon} + \log 4 + \frac{u}{2}$, i.e. $u \le 2 \log \frac{4e}{\epsilon}$ □

The proof is the probabilistic method in its plainest form: a random sample of $O(\log M/\epsilon)$ points tells M well-separated functions apart, so M cannot exceed the number of labelings a VC class produces on that many points. In particular the bound holds for $L_1(S)$, taking P uniform on the points of S, uniformly over all samples S. The volumetric bound of Lemma 7.4.3 needed the dimension of a parameter space; Lemma 7.6.1 needs only the VC dimension, and it holds for every sample at once. A more delicate argument due to (Haussler, 1995) improves the right-hand side to $1 + \log(d+1) + d\log(2e/\epsilon)$; for our purposes the difference is a constant factor.

For $\{0,1\}$-valued functions $(g(x) - g'(x))^2 = |g(x) - g'(x)|$, so $\operatorname{dist}_{L_2(S)} = \sqrt{\operatorname{dist}_{L_1(S)}}$ and an $L_1(S)$-cover at scale ϵ^2 is an $L_2(S)$-cover at scale ϵ:

$$\log N\big(\epsilon, \mathcal{G}, L_2(S)\big) \;\le\; 2d \log \frac{4e}{\epsilon^2} \qquad (\epsilon \in (0,1]). \tag{7.2}$$

Theorem 7.6.2: Dudley's entropy integral

Let $\mathcal{G}$ be a class of real-valued functions and S a sample of size m with $B := \sup_{g\in\mathcal{G}} \|g\|_S < \infty$. Then

$$\widehat{\mathfrak{R}}_S(\mathcal{G}) \;\le\; \inf_{\epsilon_0 \ge 0} \Big[4\epsilon_0 + \frac{12}{\sqrt{m}} \int_{\epsilon_0}^{\infty} \sqrt{\log N\big(\epsilon, \mathcal{G}, L_2(S)\big)}\, d\epsilon \Big].$$

Proof. Identify each g with its vector $(g(x_1), \dots, g(x_m)) \in \mathbb{R}^m$, and note that for any vector v and any signs, Cauchy-Schwarz gives

$$\frac{1}{m}\langle \sigma, v\rangle \;\le\; \frac{1}{m}\|\sigma\|_2\|v\|_2 = \frac{1}{m}\sqrt{m}\cdot\sqrt{m}\,\|v\|_S = \|v\|_S. \tag{7.3}$$

Step 1: a chain of covers. Let $\epsilon_\ell := 2^{-\ell} B$ for $\ell = 0, 1, 2, \ldots$, let $\mathcal{G}_0 := \{0\}$, and for $\ell \geq 1$ let $\mathcal{G}_\ell \subseteq \mathcal{G}$ be an ϵ_ℓ-cover of minimal size $N_\ell := N(\epsilon_\ell, \mathcal{G}, L_2(S))$. For each g pick $g_\ell \in \mathcal{G}_\ell$ with $\|g - g_\ell\|_S \leq \epsilon_\ell$; for $\ell = 0$ this holds with $g_0 = 0$ by definition of B. For any $L \geq 1$, telescoping gives

$$g = (g - g_L) + \sum_{\ell=1}^{L} (g_\ell - g_{\ell-1}).$$

Step 2: split the supremum. A supremum of a sum is at most the sum of the suprema, so

$$\widehat{\mathfrak{R}}_S(\mathcal{G}) \;\leq\; \mathbb{E}_\sigma\Big[\sup_g \frac{1}{m}\langle \sigma, g - g_L\rangle\Big] + \sum_{\ell=1}^{L} \mathbb{E}_\sigma\Big[\sup_g \frac{1}{m}\langle \sigma, g_\ell - g_{\ell-1}\rangle\Big].$$

By (7.3) the first term is at most ϵ_L. In the ℓ-th term the vector $g_\ell - g_{\ell-1}$ takes at most $N_\ell N_{\ell-1} \leq N_\ell^2$ distinct values (covering numbers do not increase with the scale, so $N_{\ell-1} \leq N_\ell$), each of norm $\|g_\ell - g_{\ell-1}\|_S \leq \|g_\ell - g\|_S + \|g - g_{\ell-1}\|_S \leq \epsilon_\ell + \epsilon_{\ell-1} = 3\epsilon_\ell$, i.e. of Euclidean norm at most $3\epsilon_\ell\sqrt{m}$. Massart's lemma (Lemma 7.3.3) bounds it by

$$\frac{3\epsilon_\ell\sqrt{m}\,\sqrt{2\log N_\ell^2}}{m} \;=\; 6\,\epsilon_\ell\sqrt{\frac{\log N_\ell}{m}} \;=\; \frac{12}{\sqrt{m}}(\epsilon_\ell - \epsilon_{\ell+1})\sqrt{\log N(\epsilon_\ell)} \;\leq\; \frac{12}{\sqrt{m}}\int_{\epsilon_{\ell+1}}^{\epsilon_\ell} \sqrt{\log N(\epsilon)}\, d\epsilon,$$

using $\epsilon_\ell = 2(\epsilon_\ell - \epsilon_{\ell+1})$ and, once more, that $N(\epsilon) \geq N(\epsilon_\ell)$ for $\epsilon \leq \epsilon_\ell$. Summing over $\ell = 1, \ldots, L$,

$$\widehat{\mathfrak{R}}_S(\mathcal{G}) \;\leq\; \epsilon_L + \frac{12}{\sqrt{m}}\int_{\epsilon_{L+1}}^{\infty} \sqrt{\log N(\epsilon)}\, d\epsilon.$$

Step 3: choose L. If $\epsilon_0 = 0$, let $L \to \infty$. If $\epsilon_0 \geq B/4$, the claim holds trivially, since (7.3) gives $\widehat{\mathfrak{R}}_S(\mathcal{G}) \leq B \leq 4\epsilon_0$. Otherwise $\epsilon_2 = B/4 > \epsilon_0$, and we take $L + 1$ to be the largest index with $\epsilon_{L+1} > \epsilon_0$, so that $L \geq 1$, $\epsilon_{L+1} > \epsilon_0$ and $\epsilon_{L+2} \leq \epsilon_0$; then $\epsilon_L = 4\epsilon_{L+2} \leq 4\epsilon_0$ and the integral over $[\epsilon_{L+1}, \infty)$ is at most the integral over $[\epsilon_0, \infty)$ □

The proof is Massart's lemma applied once per scale. A single-scale argument pays $\sqrt{\log N(\epsilon)}$ times the full radius B of the class; chaining pays at each scale only for the *increments* $g_\ell - g_{\ell-1}$, whose radius $3\epsilon_\ell$ shrinks geometrically while their number grows. The resulting series is a Riemann sum of the entropy integral. The bound goes back to (Dudley, 1967); Section 27.2 of (Shalev-Shwartz and Ben-David, 2014) gives a closely related form, and the one above is Theorem 6.25 of (Zhang, 2023).

Corollary 7.6.3

Sharp Rademacher bound for VC classes. Let $\mathcal{G}$ be a class of $\{0, 1\}$-valued functions with VC dimension $d \geq 1$. Then for **every** sample S of size m,

$$\widehat{\mathfrak{R}}_S(\mathcal{G}) \;\leq\; 48\sqrt{\frac{d}{m}}.$$

Proof. Apply Theorem 7.6.2 with $\epsilon_0 = 0$. Two $\{0,1\}$-valued functions are within $L_2(S)$-distance 1 of each other, so $N(\epsilon, \mathcal{G}, L_2(S)) = 1$ for $\epsilon \geq 1$ and the integral runs over $(0,1]$ only. There, by (7.2) and $\sqrt{u+v} \leq \sqrt{u} + \sqrt{v}$,

$$\sqrt{\log N\big(\epsilon, \mathcal{G}, L_2(S)\big)} \;\leq\; \sqrt{2d}\left(\sqrt{\log 4e} + \sqrt{2}\sqrt{\log \tfrac{1}{\epsilon}}\right).$$

The substitution $\epsilon = e^{-s}$, $d\epsilon = -e^{-s}ds$ gives

$$\int_0^1 \sqrt{\log \tfrac{1}{\epsilon}}\, d\epsilon = \int_0^\infty \sqrt{s}\, e^{-s}\, ds = \Gamma\big(\tfrac{3}{2}\big) = \frac{\sqrt{\pi}}{2},$$

so

$$\widehat{\mathfrak{R}}_S(\mathcal{G}) \;\leq\; \frac{12\sqrt{2d}}{\sqrt{m}}\left(\sqrt{\log 4e} + \sqrt{2}\,\frac{\sqrt{\pi}}{2}\right) \;\leq\; 12\sqrt{2}\,(1.545 + 1.254)\sqrt{\frac{d}{m}} \;\leq\; 47.5\sqrt{\frac{d}{m}}. \qquad \square$$

The constant is not optimized; Haussler's sharper covering estimate, for instance, reduces it. $\square$

This removes the $\sqrt{\log(m+1)}$ factor of Theorem 7.3.5: covering a VC class at *every* resolution simultaneously costs only a constant factor more than the single-resolution argument. The zero-one loss $\ell(y, h(x)) = |h(x) - y|$ is 1-Lipschitz in $h(x)$, so the contraction lemma (Lemma 7.3.4) gives $\widehat{\mathfrak{R}}_S(\ell \circ \mathcal{H}) \leq \widehat{\mathfrak{R}}_S(\mathcal{H})$, and feeding Corollary 7.6.3 into Theorem 7.3.2 gives, for a class $\mathcal{H}$ of binary classifiers with $d_{VC}(\mathcal{H}) = d$, with probability at least $1 - \delta$, simultaneously for all h,

$$R(h) \;\leq\; \widehat{R}_S(h) + 96\sqrt{\frac{d}{m}} + 3\sqrt{\frac{\log(2/\delta)}{2m}}.$$

The same argument applied to the loss $1 - \ell$, whose Rademacher complexity equals that of ℓ (replace σ by $-\sigma$), bounds $\widehat{R}_S(h) - R(h)$ identically; with $\delta/2$ for each side,

$$\sup_{h \in \mathcal{H}} \big|R(h) - \widehat{R}_S(h)\big| \;\leq\; 96\sqrt{\frac{d}{m}} + 3\sqrt{\frac{\log(4/\delta)}{2m}} \qquad \text{w.p.} \geq 1 - \delta, \tag{7.4}$$

which is the uniform-convergence upper bound of the Fundamental Theorem (Theorem 6.4.7) without the residual $\log(1/\epsilon)$ factor.

7.7 Sparse Combinations: Maurey's Lemma

The convex hull will appear in Chapter 12's analysis of boosting as the closure of the class the ensemble actually searches. Its Rademacher complexity is that of the base class (Theorem 12.2.6), but a boosting ensemble run for T rounds uses only T members — a *sparse* point of the hull, not an arbitrary one. Two results connect the sparse and dense viewpoints, and both will be consumed by Chapters 10 and 12.

The first result is a lemma of Maurey, published by (Pisier, 1981) and brought to approximation and learning theory by (Jones, 1992) and (Barron, 1993).

Theorem 7.7.1: Maurey's sparse approximation

Fix a sample $x_1, \dots, x_m$ and let Ψ be a family of functions with $\big(\frac{1}{m}\sum_{i=1}^{m}\psi(x_i)^2\big)^{1/2} \le B$ for every $\psi \in \Psi$. Let $f = \sum_{j=1}^{n} w_j\,\psi_j$ be a finite combination with $\psi_j \in \Psi$ and total weight $A := \sum_j |w_j|$. Then for every integer $N \ge 1$ there exist indices $j_1, \dots, j_N$ such that $f_N := \frac{A}{N}\sum_{k=1}^{N} u_{j_k}\psi_{j_k}$ — where $u_j := \operatorname{sign}(w_j)$, so that $u_{j_k}\psi_{j_k}$ is a member of the *symmetrized* class $\Psi \cup (-\Psi)$ — satisfies

$$\frac{1}{m}\sum_{i=1}^{m}\big(f_N(x_i) - f(x_i)\big)^2 \;\le\; \frac{A^2B^2}{N}.$$

Proof. Assume $A > 0$ (otherwise $f = 0$ and $f_N = 0$ works) and let $J_1, \dots, J_N$ be i.i.d. indices with $P(J_k = j) = |w_j|/A$. Define the random function $f_N^{(J)} := \frac{A}{N}\sum_{k=1}^{N} u_{J_k}\psi_{J_k}$. For every fixed x,

$$\mathbb{E}_J\big[f_N^{(J)}(x)\big] = \frac{A}{N}\sum_{k=1}^{N}\mathbb{E}\big[u_{J_k}\psi_{J_k}(x)\big] = A\sum_{j}\frac{|w_j|}{A}\operatorname{sign}(w_j)\psi_j(x) = \sum_{j} w_j\psi_j(x) = f(x),$$

so $f_N^{(J)}$ is an unbiased estimator of f at every point, and an average of N i.i.d. terms $A\,u_{J_k}\psi_{J_k}(x)$. Its variance at each x_i is therefore $\frac{1}{N}\operatorname{Var}\big[A\,u_{J_1}\psi_{J_1}(x_i)\big] \le \frac{A^2}{N}\mathbb{E}\big[\psi_{J_1}(x_i)^2\big]$, and averaging over the sample,

$$\mathbb{E}_J\Big[\frac{1}{m}\sum_{i}\big(f_N^{(J)}(x_i) - f(x_i)\big)^2\Big] \;=\; \frac{1}{m}\sum_{i}\operatorname{Var}_J\big[f_N^{(J)}(x_i)\big] \;\le\; \frac{A^2}{N}\,\mathbb{E}_J\Big[\frac{1}{m}\sum_{i}\psi_{J_1}(x_i)^2\Big] \;\le\; \frac{A^2B^2}{N}.$$

A random variable cannot exceed its mean with probability one, so some realization J satisfies the claimed bound □

In words: **any point of the convex hull ($A = 1$) is, on any finite sample, within $B/\sqrt{N}$ of some N-term average of base functions.** Sparsity costs only $1/\sqrt{N}$ — the same rate at which an i.i.d. average concentrates around its mean, which is exactly what the proof exploits. For a loss $L(f, y)$ that is convex and β-smooth in f (its derivative in f is β-Lipschitz), the excess empirical loss of f_N over f is, on average over J, at most $\frac{\beta}{2}$ times the squared distance above: the first-order term of the Taylor expansion averages to zero because $f_N^{(J)}$ is unbiased. So some N-sparse combination is within $\beta A^2B^2/(2N)$ of any point of the hull in empirical loss. Gradient boosting (Theorem 12.3.1 in Chapter 12) finds such a sparse combination greedily, at the same $O(1/N)$ rate.

The complementary statement is that *constraining the combination weights to an L_1 ball does not increase the Rademacher complexity at all*:

Theorem 7.7.2: Rademacher complexity of L_1-constrained combinations

Let Ψ be a symmetric class of functions ($\psi \in \Psi \Rightarrow -\psi \in \Psi$) and let

$$\mathcal{F}_{A,L_1}(\Psi) := \Big\{ x \mapsto \sum_{j=1}^{k} w_j\, \psi_j(x) \;:\; k \in \mathbb{N},\ \psi_j \in \Psi,\ w_j \in \mathbb{R},\ \sum_j |w_j| \le A \Big\}.$$

Then for every sample S,

$$\widehat{\mathfrak{R}}_S\big(\mathcal{F}_{A,L_1}(\Psi)\big) \;=\; A \cdot \widehat{\mathfrak{R}}_S(\Psi).$$

Proof. "$\ge$": every singleton combination $A \cdot \psi$, $\psi \in \Psi$, lies in $\mathcal{F}_{A,L_1}(\Psi)$, so for every realization of the signs the supremum over $\mathcal{F}_{A,L_1}(\Psi)$ is at least $\sup_{\psi\in\Psi} \frac{A}{m}\sum_i \sigma_i\psi(x_i)$; taking the expectation over σ gives $\widehat{\mathfrak{R}}_S(\mathcal{F}_{A,L_1}(\Psi)) \ge A\,\widehat{\mathfrak{R}}_S(\Psi)$. "$\le$": fix a realization of the signs σ and a combination $g = \sum_j w_j\psi_j$ with $\sum_j |w_j| \le A$. Then

$$\Big|\frac{1}{m}\sum_i \sigma_i g(x_i)\Big| = \Big|\sum_j w_j \frac{1}{m}\sum_i \sigma_i\psi_j(x_i)\Big| \le \sum_j |w_j| \Big|\frac{1}{m}\sum_i \sigma_i\psi_j(x_i)\Big| \le \Big(\sum_j |w_j|\Big) \sup_{\psi\in\Psi} \Big|\frac{1}{m}\sum_i \sigma_i\psi(x_i)\Big|,$$

and since Ψ is symmetric, $\sup_{\psi\in\Psi} \big|\frac{1}{m}\sum_i \sigma_i\psi(x_i)\big| = \sup_{\psi\in\Psi} \frac{1}{m}\sum_i \sigma_i\psi(x_i)$: the absolute value of a negative correlation is attained, with positive sign, by $-\psi \in \Psi$. Taking the supremum over g and then the expectation over σ gives $\widehat{\mathfrak{R}}_S(\mathcal{F}_{A,L_1}(\Psi)) \le A\,\widehat{\mathfrak{R}}_S(\Psi)$ □

Note that the number of terms k and the individual weights never appear in the bound — only the total L_1 mass A does. Since $\mathrm{conv}(\Psi) \subseteq \mathcal{F}_{1,L_1}(\Psi)$, the upper bound in Theorem 12.2.6's "convexification is free" is, for a symmetric base class, the $A = 1$ instance of this statement, and the two proofs are the same argument. What the L_1 form adds is the **norm constraint** reading: capacity is controlled by the *budget* A, not by the number of parameters k — the same moral as Theorem 7.3.6 for L_2-balls, and the same mechanism we will exploit for the output layer of a two-layer network in Chapter 10 and for gradient boosting in Chapter 12.

7.8 Algorithmic Stability

Every measure so far prices the *class* $\mathcal{H}$: the same complexity is charged no matter which algorithm searched it, and no credit is given for the algorithm's actual behavior on the data. **Algorithmic stability** is the complementary theory: it prices the *algorithm*, by asking how much its output can change when a single training example is swapped. We develop it in the distribution-free, expected-risk form initiated by (Bousquet and Elisseeff, 2002).

Throughout this section, a learning algorithm A maps a training set S to a hypothesis $A(S)$, possibly using internal randomness (as stochastic gradient descent does); $\mathbb{E}_A$ denotes the average over that randomness, and is void for a deterministic A; and $\phi(\cdot, z)$ denotes the loss of a hypothesis on example z — the test loss may differ from the training objective, as in regularized ERM.

Definition 7.8.1: Uniform stability

A is ϵ-uniformly stable if for all training sets S, S' that differ in exactly one element,

$$\sup_z \left|\mathbb{E}_A\, \phi(A(S), z) - \mathbb{E}_A\, \phi(A(S'), z)\right| \;\le\; \epsilon.$$

In words: swapping any one training example cannot move any *single* prediction's loss by more than ϵ. Stability is a property of the algorithm's input-output map, not of the size of the class it searches; an algorithm that memorizes S is maximally unstable (swapping a point changes its prediction on that point outright), while an algorithm that averages over data has its outputs anchored by the $m-1$ untouched examples.

Theorem 7.8.2: Stability implies generalization

If A is ϵ-uniformly stable, then

$$\mathbb{E}_{S\sim\mathcal{D}^m}\mathbb{E}_A\big[R(A(S))\big] \;\le\; \mathbb{E}_{S\sim\mathcal{D}^m}\mathbb{E}_A\big[\widehat{R}_S(A(S))\big] + \epsilon.$$

Proof. We write the proof for deterministic A; for a randomized one, replace $\phi(A(S), z)$ by $\mathbb{E}_A\, \phi(A(S), z)$ throughout, which is exactly the quantity Definition 7.8.1 controls. Write $S = (z_1, \dots, z_m)$ and let z be a fresh example. Since $R(A(S)) = \mathbb{E}_z\big[\phi(A(S), z)\big]$, it suffices to compare the two expectations term by term. For each $i \in [m]$, consider the **swap map** T_i on $(m{+}1)$-tuples of examples that exchanges z_i and z,

$$(z_1, \dots, z_{i-1},\ z_i,\ z_{i+1}, \dots, z_m,\ z) \;\mapsto\; (z_1, \dots, z_{i-1},\ z,\ z_{i+1}, \dots, z_m,\ z_i).$$

All $m+1$ coordinates being i.i.d. from $\mathcal{D}$, the law of the tuple is invariant under T_i, so for any function G of the tuple, $\mathbb{E}[G] = \mathbb{E}[G \circ T_i]$. Applying this to $G := \phi(A(S), z)$ (which depends on the first m coordinates only, through A) gives

$$\mathbb{E}_{S,z}\big[\phi(A(S), z)\big] \;=\; \mathbb{E}_{S,z}\big[\phi\big(A(S^{(i)}),\ z_i\big)\big],$$

where $S^{(i)}$ is S with its i-th element replaced by z. Now $S^{(i)}$ and S differ in exactly one element, so by uniform stability $\phi(A(S^{(i)}), z_i) \le \phi(A(S), z_i) + \epsilon$ pointwise. Averaging over i,

$$\mathbb{E}_S\big[R(A(S))\big] = \frac{1}{m}\sum_{i=1}^{m} \mathbb{E}_{S,z}\big[\phi\big(A(S^{(i)}), z_i\big)\big] \;\le\; \frac{1}{m}\sum_{i=1}^{m} \mathbb{E}_{S,z}\big[\phi(A(S), z_i) + \epsilon\big] = \mathbb{E}_S\big[\widehat{R}_S(A(S))\big] + \epsilon. \qquad \square$$

The proof is a pure **ghost-sample swap**: the fresh example z and the training example z_i are statistically exchangeable, so the algorithm's loss on the fresh example can be evaluated as its loss on the replaced example — where stability bridges the two runs. No complexity measure of any class appears anywhere; the entire price of generalization is the one-sample sensitivity ϵ.

A function $\mathcal{J}$ on a convex set Ω is λ-**strongly convex** if for all $w, u \in \Omega$ and $\alpha \in (0, 1)$,

$$\mathcal{J}\big(\alpha w + (1-\alpha)u\big) \;\le\; \alpha\,\mathcal{J}(w) + (1-\alpha)\,\mathcal{J}(u) - \frac{\lambda}{2}\,\alpha(1-\alpha)\|w-u\|_2^2;$$

for $\lambda = 0$ this is convexity. The regularizer $\frac{\lambda}{2}\|w\|_2^2$ is λ-strongly convex (the inequality holds with equality, as expanding the squares shows), and adding a convex function preserves λ-strong convexity. If w^* minimizes a λ-strongly convex $\mathcal{J}$ over Ω, then for every $w \in \Omega$, using $\mathcal{J}(w^*) \le \mathcal{J}(\alpha w + (1-\alpha)w^*)$ in the definition, dividing by α and letting $\alpha \to 0$,

$$\mathcal{J}(w) - \mathcal{J}(w^*) \;\ge\; \frac{\lambda}{2}\|w - w^*\|_2^2. \tag{7.5}$$

In particular a minimizer, when it exists, is unique. □

Theorem 7.8.3: Regularized ERM is uniformly stable

Let Ω be a closed convex set, $\phi(w, z)$ a convex function of $w \in \Omega$ that is G-Lipschitz in w for every z, and $h \ge 0$ a λ-strongly convex regularizer, such that the regularized ERM

$$A(S) \;=\; \arg\min_{w\in\Omega} \mathcal{J}_S(w) := \frac{1}{m}\sum_{i=1}^{m} \phi(w, z_i) + h(w)$$

exists for every S. Then A is $\frac{2G^2}{\lambda m}$-uniformly stable.

Proof. Let S' differ from S in the i-th element only (z_i' instead of z_i), and let $w := A(S)$, $w' := A(S')$. Both $\mathcal{J}_S$ and $\mathcal{J}_{S'}$ are λ-strongly convex, so (7.5) applied to each at its own minimizer gives

$$\mathcal{J}_S(w') - \mathcal{J}_S(w) \;\ge\; \frac{\lambda}{2}\|w' - w\|_2^2, \qquad \mathcal{J}_{S'}(w) - \mathcal{J}_{S'}(w') \;\ge\; \frac{\lambda}{2}\|w' - w\|_2^2.$$

The two objectives differ only in the i-th term, $\mathcal{J}_S - \mathcal{J}_{S'} = \frac{1}{m}\big(\phi(\cdot, z_i) - \phi(\cdot, z_i')\big)$, so adding the two inequalities cancels all remaining terms:

$$\lambda\|w' - w\|_2^2 \;\le\; \frac{\phi(w', z_i) - \phi(w, z_i)}{m} + \frac{\phi(w, z_i') - \phi(w', z_i')}{m} \;\le\; \frac{2G}{m}\|w' - w\|_2,$$

by the Lipschitz property. Hence $\|w - w'\|_2 \le \frac{2G}{\lambda m}$, and for every test example z, again by the Lipschitz property, $\big|\phi(w, z) - \phi(w', z)\big| \le G\,\|w - w'\|_2 \le \frac{2G^2}{\lambda m}$ □

The structural content is that **strong convexity converts a perturbation of the objective that is $\frac{2G}{m}$-Lipschitz into a movement of the minimizer of at most $\frac{2G}{\lambda m}$** — the regularization strength λ is exactly what provides stability, at the price of the approximation error $h(w)$. The argument is that of Section 13.3 of (Shalev-Shwartz and Ben-David, 2014), following Bousquet & Elisseeff. Nothing in it uses finite dimension: it holds verbatim in any Hilbert space, in particular in an RKHS (Chapter 11).

Combining the two theorems gives the standard **expected oracle inequality** for regularized ERM:

$$\mathbb{E}_S\big[R(A(S))\big] \;\le\; \mathbb{E}_S\big[\mathcal{J}_S(A(S))\big] + \frac{2G^2}{\lambda m} \;\le\; \inf_{w\in\Omega}\Big[\mathbb{E}_z\big[\phi(w, z)\big] + h(w)\Big] + \frac{2G^2}{\lambda m},$$

where $R(w) := \mathbb{E}_z[\phi(w, z)]$ and $\widehat{R}_S(w) := \frac{1}{m}\sum_i \phi(w, z_i)$, the first inequality is Theorem 7.8.2 together with $h \ge 0$ (the training objective upper-bounds the training risk), and the second is the optimality of $A(S)$ for $\mathcal{J}_S$ plus $\mathbb{E}_S[\mathcal{J}_S(w)] = \mathbb{E}_z[\phi(w, z)] + h(w)$. Read at the optimal w, the excess expected

risk of regularized ERM over the best regularized population objective is $O(G^2/(\lambda m))$: an $O(1/m)$ rate, in contrast to the $O(1/\sqrt{m})$ of the complexity-based bounds — the price being that the bound is in expectation over S, and is relative to the regularized objective rather than the best-in-class risk.

Two caveats. First, the expectation over S in Theorem 7.8.2 can be upgraded to a high-probability statement. Replacing one example changes $R(A(S)) - \widehat{R}_S(A(S))$ by at most $2\epsilon + M/m$ for a loss with range M, so McDiarmid's inequality applies; this is the route of Bousquet & Elisseeff, and the resulting bound is informative only when $\epsilon = o(1/\sqrt{m})$. Bounds with a nearly optimal dependence on ϵ require a more refined concentration argument, see (Feldman and Vondr{á, 2019). Second, the oracle inequality bounds the excess risk against $\inf_w[R(w) + h(w)]$, in which the regularization bias is paid on the inside; comparing against $\inf_{w\in\Omega} R(w)$ instead requires balancing λ against m and re-optimizing — the same bias-complexity trade-off, now with λ in the role that $\log|\mathcal{H}|$ played in Chapter 6.

7.9 Minimax Lower Bounds

Every bound in this chapter so far is an *upper* bound: it certifies that some algorithm achieves at most a certain rate. The complementary question is whether that rate is *necessary*. The right framework is minimax: fix the sample size m, let the algorithm be chosen *before* the distribution, and ask what the best algorithm can guarantee against the *worst* distribution.

Definition 7.9.1: Minimax risk

For a distribution family $\mathcal{P}$ over examples, a quality measure Q assigning to each pair (hypothesis, distribution) a nonnegative loss, and an algorithm A mapping training sets to hypotheses, the worst-case risk of A is

$$r_m(A, \mathcal{P}, Q) := \sup_{\mathcal{D}\in\mathcal{P}} \mathbb{E}_{S\sim\mathcal{D}^m}\, \mathbb{E}_A\big[Q(A(S), \mathcal{D})\big],$$

where $\mathbb{E}_A$ averages over any internal randomization of A. The **minimax risk** is $r_m(\mathcal{P}, Q) := \inf_A r_m(A, \mathcal{P}, Q)$, the infimum over all algorithms.

A lower bound $r_m(\mathcal{P}, Q) \geq c$ certifies that *no* algorithm — however clever — achieves quality better than c on every $\mathcal{D} \in \mathcal{P}$; matching it with an upper bound identifies r_m up to constants and certifies that the upper-bound proof technique was not merely loose.

The elementary engine of such lower bounds is the **two-point method** (Le Cam's method): to show that m samples do not suffice, exhibit two distributions that are (i) statistically almost indistinguishable from m samples, yet (ii) demand conflicting predictions. Its measure of indistinguishability is the **total variation distance**

$$\mathrm{TV}(P, Q) := \sup_E \big|P(E) - Q(E)\big|,$$

whose single property we need is the change-of-measure bound

$$\big|\mathbb{E}_P[g] - \mathbb{E}_Q[g]\big| \;\leq\; \mathrm{TV}(P, Q) \qquad \text{for every } g \text{ with } 0 \leq g \leq 1, \tag{7.6}$$

which follows from the likelihood-ratio representation of TV. When $P \ll Q$ with likelihood ratio $L :=$

$\frac{dP}{dQ}$, every expectation difference is an expectation under Q:

$$\mathbb{E}_P[g] - \mathbb{E}_Q[g] = \mathbb{E}_Q\big[(L-1)\,g\big] \;\le\; \mathbb{E}_Q\big[(L-1)^+\big],$$

since $g \le 1$ makes $(L-1)\,g \le (L-1)^+$ pointwise (on $\{L \le 1\}$ the left side is ≤ 0). The quantity $\mathbb{E}_Q[(L-1)^+]$ equals $\mathrm{TV}(P,Q)$: the supremum in the definition is attained by the event $E^* = \{L > 1\}$, where $P(E^*) - Q(E^*) = \mathbb{E}_Q[(L-1)\mathbb{1}_{\{L>1\}}] = \mathbb{E}_Q[(L-1)^+]$, and $\mathbb{E}_Q[L-1] = 0$ makes the negative part of $L-1$ equal in magnitude to the positive part, so no event does better in absolute value either.

Lemma 7.9.2: Two-point testing bounds

Let P, Q be two distributions on a common space and W any event.
(i) $P(W) + Q(W^c) \;\ge\; 1 - \mathrm{TV}(P,Q)$.
(ii) For product distributions of m i.i.d. draws, with $\chi^2(P\|Q) := \mathbb{E}_Q\big[\big(\frac{dP}{dQ} - 1\big)^2\big]$ the **chi-squared divergence**,

$$\mathrm{TV}\big(P^{\otimes m}, Q^{\otimes m}\big) \;\le\; \sqrt{\chi^2\big(P^{\otimes m}\|Q^{\otimes m}\big)} \;\le\; \sqrt{2m\,\chi^2(P\|Q)} \qquad \text{whenever } m\,\chi^2(P\|Q) \le 1.$$

(iii) (Bretagnolle-Huber) With $\mathrm{KL}(P\|Q) := \mathbb{E}_P\big[\log\frac{dP}{dQ}\big]$ the Kullback-Leibler divergence,

$$P(W) + Q(W^c) \;\ge\; \tfrac{1}{2}\,e^{-\mathrm{KL}(P\|Q)}, \qquad \mathrm{KL}\big(P^{\otimes m}\big\|Q^{\otimes m}\big) = m\,\mathrm{KL}(P\|Q).$$

Proof. *(i)* By the change-of-measure bound (7.6) with $g := \mathbb{1}_W$, $P(W) \ge Q(W) - \mathrm{TV}(P,Q)$, and $Q(W^c) = 1 - Q(W)$; adding the two gives the claim.

(ii) With $L_m := \prod_{i=1}^m \frac{dP}{dQ}(z_i)$ the likelihood ratio of the product, the representation of TV following (7.6) and Cauchy-Schwarz give

$$\mathrm{TV}\big(P^{\otimes m}, Q^{\otimes m}\big) = \mathbb{E}_{Q^{\otimes m}}\big[(L_m - 1)^+\big] \;\le\; \Big(\mathbb{E}_{Q^{\otimes m}}\big[(L_m-1)^2\big]\Big)^{1/2} = \big(\chi^2(P^{\otimes m}\|Q^{\otimes m})\big)^{1/2}.$$

The chi-squared divergence of products **factorizes**, by independence of the m coordinates and $\mathbb{E}_Q[L] = 1$:

$$1 + \chi^2\big(P^{\otimes m}\big\|Q^{\otimes m}\big) = \mathbb{E}_{Q^{\otimes m}}\big[L_m^2\big] = \prod_{i=1}^m \mathbb{E}_Q\big[L^2\big] = \big(1 + \chi^2(P\|Q)\big)^m \;\le\; e^{m\chi^2(P\|Q)},$$

and $e^x - 1 \le 2x$ on $[0,1]$ gives $\chi^2(P^{\otimes m}\|Q^{\otimes m}) \le 2m\,\chi^2(P\|Q)$.

(iii) Let p, q be densities of P, Q with respect to a common reference measure μ (for instance $\mu = P + Q$). Then $P(W) + Q(W^c) = \int_W p\,d\mu + \int_{W^c} q\,d\mu \ge \int \min(p,q)\,d\mu$. By Cauchy-Schwarz, and since $\int \max(p,q)\,d\mu = \int (p+q)\,d\mu - \int \min(p,q)\,d\mu \le 2$,

$$\Big(\int \sqrt{pq}\,d\mu\Big)^2 = \Big(\int \sqrt{\min(p,q)\max(p,q)}\,d\mu\Big)^2 \;\le\; \int \min(p,q)\,d\mu \cdot \int \max(p,q)\,d\mu \;\le\; 2\int \min(p,q)\,d\mu.$$

By Jensen's inequality for the convex function exp,

$$\begin{aligned}\int \sqrt{pq}\,d\mu \;&\ge\; \int_{\{p>0\}} p\,\exp\Big(-\tfrac{1}{2}\log\frac{p}{q}\Big)d\mu = \mathbb{E}_P\Big[\exp\Big(-\tfrac{1}{2}\log\frac{dP}{dQ}\Big)\Big] \\ &\ge\; \exp\Big(-\tfrac{1}{2}\mathrm{KL}(P\|Q)\Big).\end{aligned}$$

Combining the three displays gives the inequality. For the product, $\log \frac{dP^{\otimes m}}{dQ^{\otimes m}}(z_1,\dots,z_m) = \sum_i \log \frac{dP}{dQ}(z_i)$, whose expectation under $P^{\otimes m}$ is $m\,\mathrm{KL}(P\|Q)$ □

Parts (i) and (ii) control the error probabilities when the two distributions are close in total variation; part (iii) remains informative when they are far apart, where it shows that the error probabilities decay no faster than $e^{-\mathrm{KL}}$. The first gives lower bounds on the expected error, the second on the sample size needed for a given confidence δ.

Theorem 7.9.3: The biased-coin lower bound

Let $\mathcal{X} = \{0\}$ be a one-point input space, $\mathcal{Y} = \{+1,-1\}$, and $\mathcal{H} = \{h_+, h_-\}$ the two constant hypotheses. Let $\mathcal{D}_+$ draw the label $+1$ with probability $\frac{1+\gamma}{2}$ and -1 otherwise, and let $\mathcal{D}_-$ be its mirror image. Let A be any (possibly randomized) algorithm and $\gamma \in (0, \frac{1}{2}]$.
(i) If $m \le \frac{1}{64\gamma^2}$, then

$$\max_{j\in\{+,-\}} \mathbb{E}_{S\sim\mathcal{D}_j^m}\left[R_{\mathcal{D}_j}(A(S)) - R^*_{\mathcal{D}_j}\right] \;\ge\; \frac{\gamma}{4}.$$

(ii) For every $m \ge 1$,

$$\max_{j\in\{+,-\}} P_{S\sim\mathcal{D}_j^m}\left(R_{\mathcal{D}_j}(A(S)) - R^*_{\mathcal{D}_j} \ge \gamma\right) \;\ge\; \tfrac{1}{4}\,e^{-4m\gamma^2}.$$

In words: an algorithm must tell a $(\frac{1}{2}+\frac{\gamma}{2})$-biased coin from a $(\frac{1}{2}-\frac{\gamma}{2})$-biased one, which requires $\Omega(1/\gamma^2)$ flips for constant confidence, and $\Omega(\log(1/\delta)/\gamma^2)$ flips for confidence $1-\delta$.

Proof. Since $\mathcal{X}$ is a single point, *every* hypothesis outputs one of the two labels there — the algorithm's entire decision is binary. Assume first that A is deterministic, and let $W := \{S : A(S) = h_-\}$ be the samples on which it predicts -1. Under $\mathcal{D}_+$ the best hypothesis is h_+ and under $\mathcal{D}_-$ it is h_-; in both cases $R^* = \frac{1-\gamma}{2}$, and a hypothesis that errs under its distribution pays

$$R_{\mathcal{D}_+}(h_-) - R_{\mathcal{D}_+}(h_+) \;=\; R_{\mathcal{D}_-}(h_+) - R_{\mathcal{D}_-}(h_-) \;=\; \gamma.$$

So the excess risk is γ on W under $\mathcal{D}_+$, γ on W^c under $\mathcal{D}_-$, and 0 otherwise.
(i) The expected excess risks are $\gamma\,\mathcal{D}_+^m(W)$ and $\gamma\,\mathcal{D}_-^m(W^c)$, so by Lemma 7.9.2(i),

$$\max_{\pm} \mathbb{E}\left[R_{\mathcal{D}_\pm}(A(S)) - R^*_{\mathcal{D}_\pm}\right] \;=\; \gamma\max\left(\mathcal{D}_+^m(W),\, \mathcal{D}_-^m(W^c)\right) \;\ge\; \frac{\gamma}{2}\left(1 - \mathrm{TV}(\mathcal{D}_+^m, \mathcal{D}_-^m)\right).$$

One label draw has chi-squared divergence

$$\chi^2(\mathcal{D}_+\|\mathcal{D}_-) \;=\; \sum_{y\in\{\pm1\}} \frac{\left(\mathcal{D}_+(y) - \mathcal{D}_-(y)\right)^2}{\mathcal{D}_-(y)} \;=\; \frac{2\gamma^2}{1-\gamma} + \frac{2\gamma^2}{1+\gamma} \;=\; \frac{4\gamma^2}{1-\gamma^2} \;\le\; 6\gamma^2 \qquad (\gamma \le \tfrac{1}{2}),$$

since $\mathcal{D}_+(y) - \mathcal{D}_-(y) = \pm\gamma$. As $m \le \frac{1}{64\gamma^2}$ gives $m\cdot 6\gamma^2 \le \frac{6}{64} \le 1$, Lemma 7.9.2(ii) applies:

$$\mathrm{TV}(\mathcal{D}_+^m, \mathcal{D}_-^m) \;\le\; \sqrt{2m\cdot 6\gamma^2} \;\le\; \sqrt{\tfrac{12}{64}} \;=\; \tfrac{\sqrt{3}}{4} \;<\; \tfrac{1}{2},$$

and the expected excess risk is at least $\frac{\gamma}{2}(1-\frac{1}{2}) = \frac{\gamma}{4}$.

(ii) The two probabilities are $\mathcal{D}_+^m(W)$ and $\mathcal{D}_-^m(W^c)$; their maximum is at least half their sum, hence by Lemma 7.9.2(iii) at least $\frac{1}{4}e^{-m\,\mathrm{KL}(\mathcal{D}_+\|\mathcal{D}_-)}$. Here

$$\mathrm{KL}(\mathcal{D}_+\|\mathcal{D}_-) = \frac{1+\gamma}{2}\log\frac{1+\gamma}{1-\gamma} + \frac{1-\gamma}{2}\log\frac{1-\gamma}{1+\gamma} = \gamma\log\left(1+\frac{2\gamma}{1-\gamma}\right) \;\le\; \frac{2\gamma^2}{1-\gamma} \;\le\; 4\gamma^2,$$

using $\log(1+u) \le u$ and $\gamma \le \frac{1}{2}$.

Finally, a randomized A is a deterministic function of the sample and an independent random variable $\omega \sim \mu$. The joint laws $\mathcal{D}_\pm^m \otimes \mu$ have the same likelihood ratio as $\mathcal{D}_\pm^m$, hence the same total variation and KL divergence, and both arguments apply verbatim to the pair (S, ω) □

Part (i) rests on the factorization $1 + \chi^2(P^{\otimes m}\|Q^{\otimes m}) = (1+\chi^2(P\|Q))^m$ and the single-flip value $\chi^2 = 4\gamma^2/(1-\gamma^2)$: each sample improves distinguishability only by a factor $1+O(\gamma^2)$. The separation γ multiplies the error probability from the outside, while the total variation term only degrades it by a constant factor as long as $m \lesssim 1/\gamma^2$. Part (ii) converts into a sample-size requirement: for $\epsilon \in (0, \frac{1}{4})$ and $\delta \in (0, \frac{1}{4})$, take $\gamma = 2\epsilon$; an algorithm with $P(\text{excess risk} > \epsilon) \le \delta$ under both coins needs $\frac{1}{4}e^{-16m\epsilon^2} \le \delta$, i.e.

$$m \;\ge\; \frac{\log\big(1/(4\delta)\big)}{16\,\epsilon^2}. \tag{7.7}$$

The two-point method compares two distributions. Lower bounds that grow with the complexity of the class need many distributions. The classical tool for this is Fano's inequality, which lower-bounds the *average* error over a finite family through the **mutual information** between the index of the distribution and the sample. For j uniform on $[N]$ and $S \sim \mathcal{D}_j^m$ given j, write $\bar{\mathcal{D}} := \frac{1}{N}\sum_{k=1}^N \mathcal{D}_k^m$ for the law of S, and define

$$I(j,S) := \mathrm{KL}\big(P_{(j,S)} \,\big\|\, P_j \otimes P_S\big) = \frac{1}{N}\sum_{j=1}^N \mathrm{KL}\big(\mathcal{D}_j^m \big\| \bar{\mathcal{D}}\big),$$

the KL divergence of the joint law of (j,S) from the product of its marginals; the second expression follows since on the slice $\{j\}\times\mathcal{Z}^m$ the joint law is $\frac{1}{N}\mathcal{D}_j^m$ and the product law is $\frac{1}{N}\bar{\mathcal{D}}$, so that the likelihood ratio is $d\mathcal{D}_j^m/d\bar{\mathcal{D}}$. $I(j,S) = 0$ exactly when the sample carries no information about j.

Theorem 7.9.4: Fano's inequality, generalized

Let $\mathcal{P} = \{\mathcal{D}_1, \dots, \mathcal{D}_N\}$ be a finite family of distributions, Q a quality measure as in Definition 7.9.1, $\epsilon > 0$, and

$$N_\epsilon := \sup_h \big|\{k : Q(h, \mathcal{D}_k) < \epsilon\}\big|,$$

the largest number of distributions a single hypothesis can serve at quality ϵ. If $N_\epsilon < N$, then every (possibly randomized) algorithm A satisfies

$$\frac{1}{N}\sum_{j=1}^N P_{S\sim\mathcal{D}_j^m}\big(Q(A(S),\mathcal{D}_j) < \epsilon\big) \;\le\; \frac{I(j,S)+\log 2}{\log(N/N_\epsilon)}, \qquad I(j,S) \;\le\; \frac{m}{N^2}\sum_{j,k=1}^N \mathrm{KL}(\mathcal{D}_j\|\mathcal{D}_k).$$

Proof. *Step 1: data processing onto an event.* For distributions $P \ll Q$ and any event F with $0 < P(F) < 1$, with kl the binary KL divergence of Theorem 5.6.10,

$$\mathrm{KL}(P\|Q) \;\geq\; \mathrm{kl}\big(P(F)\,\big\|\,Q(F)\big).$$

Indeed, split $\mathrm{KL}(P\|Q) = P(F)\,\mathbb{E}_P[\log L \mid F] + P(F^c)\,\mathbb{E}_P[\log L \mid F^c]$ with $L := dP/dQ$. By Jensen's inequality for the concave log, $\mathbb{E}_P[\log L \mid F] = -\mathbb{E}_P[\log(1/L) \mid F] \geq -\log \mathbb{E}_P[1/L \mid F] \geq -\log \frac{Q(F)}{P(F)}$, because $\mathbb{E}_P[\mathbb{1}_F/L] = Q(F \cap \{L > 0\}) \leq Q(F)$; likewise on F^c. Adding the two bounds gives $P(F)\log\frac{P(F)}{Q(F)} + P(F^c)\log\frac{P(F^c)}{Q(F^c)}$. (If $P(F) \in \{0,1\}$ the same computation applies with the empty term dropped.)

Step 2: apply it to the success event. Take $P := P_{(j,S)}, Q := P_j \otimes P_S$ and $F := \{(j,S) : Q(A(S), \mathcal{D}_j) < \epsilon\}$. Then $P(F) =: \bar{a}$ is the average success probability on the left of the claim. Under Q, j is uniform and independent of S, and for every fixed S at most N_ϵ indices j satisfy $Q(A(S), \mathcal{D}_j) < \epsilon$, so $Q(F) \leq N_\epsilon/N =: b$. Step 1 gives $I(j,S) \geq \mathrm{kl}(\bar{a}\,\|\,Q(F))$.

Step 3: a lower bound on kl. For $a \in [0,1]$ and $c \in (0,1)$,

$$\mathrm{kl}(a\|c) = a\log\frac{1}{c} + (1-a)\log\frac{1}{1-c} + \big[a\log a + (1-a)\log(1-a)\big] \;\geq\; a\log\frac{1}{c} - \log 2,$$

since the middle term is non-negative and the bracket, a negative entropy of two outcomes, is at least $-\log 2$. With $c = Q(F) \leq b$, $\log\frac{1}{c} \geq \log\frac{N}{N_\epsilon}$; hence $I(j,S) \geq \bar{a}\log(N/N_\epsilon) - \log 2$, which rearranges to the first claim.

Step 4: bound the mutual information. Pointwise, by Jensen's inequality for the convex $-\log$,

$$\log\frac{d\mathcal{D}_j^m}{d\bar{\mathcal{D}}} = -\log\Big(\frac{1}{N}\sum_k \frac{d\mathcal{D}_k^m}{d\mathcal{D}_j^m}\Big) \;\leq\; \frac{1}{N}\sum_k \log\frac{d\mathcal{D}_j^m}{d\mathcal{D}_k^m}.$$

Taking $\mathbb{E}_{\mathcal{D}_j^m}$ and averaging over j gives $I(j,S) \leq \frac{1}{N^2}\sum_{j,k}\mathrm{KL}(\mathcal{D}_j^m\|\mathcal{D}_k^m)$, and $\mathrm{KL}(\mathcal{D}_j^m\|\mathcal{D}_k^m) = m\,\mathrm{KL}(\mathcal{D}_j\|\mathcal{D}_k)$ by Lemma 7.9.2(iii). A randomized A is handled by adjoining its internal randomness to S, which changes no KL divergence □

Fano's inequality turns a lower bound into a *packing* problem: find many distributions that pairwise demand different answers (N_ϵ small) yet are hard to tell apart from m samples ($I(j,S)$ small compared with $\log N$). The inequality goes back to Fano's information theory; versions with approximate recovery, as above, are developed in (Duchi and Wainwright, 2013) and in Chapter 12 of (Zhang, 2023); see also (Tsybakov, 2009). For classification over a shattered set the problem decouples into one binary test per point, and the two-point bounds of Lemma 7.9.2 applied coordinate by coordinate (*Assouad's method*, see Chapter 2 of Tsybakov, 2009) suffice and give simpler constants. The resulting lower bounds are classical; Chapter 28 of (Shalev-Shwartz and Ben-David, 2014) proves the agnostic one with a different construction.

Theorem 7.9.5: Minimax lower bounds for VC classes

Let $\mathcal{H}$ be a class of $\{0,1\}$-valued classifiers with $d_{VC}(\mathcal{H}) = d$, the loss being the zero-one loss, and let A be any (possibly randomized) algorithm and $m \geq 1$.
(i) Agnostic case ($d \geq 1$). There is a distribution $\mathcal{D}$ with

$$P_{S\sim\mathcal{D}^m}\Big(R(A(S)) - \min_{h\in\mathcal{H}} R(h) \;\geq\; \tfrac{1}{64}\min\big(4, \sqrt{d/m}\big)\Big) \;\geq\; \tfrac{1}{7}.$$

(ii) Realizable case ($d \geq 2$). There is a distribution $\mathcal{D}$ realizable by $\mathcal{H}$ with

$$P_{S\sim\mathcal{D}^m}\Big(R(A(S)) \;\geq\; \tfrac{1}{16}\min\big(2, (d-1)/m\big)\Big) \;\geq\; \tfrac{1}{7}.$$

(iii) Realizable case, confidence ($d \geq 2$). For every $\epsilon \in (0, \tfrac{1}{4}]$ there is a distribution $\mathcal{D}$ realizable by $\mathcal{H}$ with

$$P_{S\sim\mathcal{D}^m}\big(R(A(S)) > \epsilon\big) \;\geq\; \tfrac{1}{2}\,e^{-4\epsilon m}.$$

Proof. Assume A deterministic; randomized algorithms are handled as in Theorem 7.9.3. Each part follows the same pattern: a family of distributions indexed by labelings of a shattered set, an average over the family, and the elementary fact that a $[0,1]$-valued random variable F with $\mathbb{E}[F] \geq \frac{1}{4}$ satisfies $P(F \geq \frac{1}{8}) \geq \frac{1}{7}$, because $\mathbb{E}[F] \leq \frac{1}{8} + \frac{7}{8}P(F \geq \frac{1}{8})$.

(i) Let $C = \{x_1, \ldots, x_d\}$ be shattered by $\mathcal{H}$ and $\gamma := \min\big(\frac{1}{2}, \sqrt{d/(64m)}\big)$, so that $m\gamma^2 \leq d/64$. For $\tau \in \{0,1\}^d$ let $\mathcal{D}_\tau$ draw x uniformly from C and, at $x = x_j$, the label τ_j with probability $\frac{1+\gamma}{2}$ and $1-\tau_j$ otherwise. The Bayes predictor $x_j \mapsto \tau_j$ belongs to $\mathcal{H}$ on C (shattering), so $\min_h R_\tau(h) = R^*_{\mathcal{D}_\tau}$, and as in Theorem 7.9.3 every h has excess risk

$$R_\tau(h) - \min_{h'\in\mathcal{H}} R_\tau(h') = \gamma\, F_\tau(h), \qquad F_\tau(h) := \frac{1}{d}\sum_{j=1}^{d} \mathbb{1}\big(h(x_j) \neq \tau_j\big) \in [0,1].$$

Fix j and pair each τ having $\tau_j = 1$ with $\tau' := \tau$ with coordinate j flipped. With $W_j := \{S : A(S)(x_j) = 0\}$, the probabilities of erring at x_j under $\mathcal{D}_\tau$ and $\mathcal{D}_{\tau'}$ are $\mathcal{D}_\tau^m(W_j)$ and $\mathcal{D}_{\tau'}^m(W_j^c)$, whose sum is at least $1 - \mathrm{TV}(\mathcal{D}_\tau^m, \mathcal{D}_{\tau'}^m)$ by Lemma 7.9.2(i). The two distributions differ only in the label law at x_j, which has mass $\frac{1}{d}$, so, as in Theorem 7.9.3,

$$\chi^2(\mathcal{D}_\tau \| \mathcal{D}_{\tau'}) = \frac{1}{d}\cdot\frac{4\gamma^2}{1-\gamma^2} \leq \frac{16\gamma^2}{3d}, \qquad m\,\chi^2 \leq \frac{16}{3}\cdot\frac{1}{64} = \frac{1}{12},$$

and Lemma 7.9.2(ii) gives $\mathrm{TV} \leq \sqrt{2/12} < \frac{1}{2}$. Averaging over the pairs, for τ uniform on $\{0,1\}^d$, the probability of erring at x_j is at least $\frac{1}{2}(1 - \frac{1}{2}) = \frac{1}{4}$, and averaging over j,

$$\mathbb{E}_\tau\, \mathbb{E}_{S\sim\mathcal{D}_\tau^m}\big[F_\tau(A(S))\big] \;\geq\; \tfrac{1}{4}.$$

Some τ attains at least the average; for it $P(F_\tau(A(S)) \geq \frac{1}{8}) \geq \frac{1}{7}$, and on that event the excess risk is at least $\gamma/8 = \frac{1}{64}\min(4, \sqrt{d/m})$.

(ii) Let $C = \{x_0, x_1, \dots, x_{d-1}\}$ be shattered, $p := \min\left(1, \frac{d-1}{2m}\right)$, and for $\tau \in \{0,1\}^{d-1}$ let $\mathcal{D}_\tau$ put mass $1-p$ on x_0 with label 0 and mass $\frac{p}{d-1}$ on each x_j, $j \geq 1$, with label τ_j. By shattering $\mathcal{D}_\tau$ is realizable. Let

$$R'_\tau := \frac{p}{d-1} \sum_{j=1}^{d-1} \mathbb{1}\big(A(S)(x_j) \neq \tau_j\big) \;\leq\; R_\tau(A(S)), \qquad R'_\tau \in [0, p].$$

Fix $j \geq 1$, the input points of the sample, and all coordinates of τ except τ_j. If x_j does not occur in the sample, the sample does not depend on τ_j, so for τ_j uniform $A(S)(x_j) \neq \tau_j$ with probability $\frac{1}{2}$. The point x_j is missed with probability $(1-\frac{p}{d-1})^m \geq 1-\frac{mp}{d-1} \geq \frac{1}{2}$. Hence $\mathbb{E}_\tau \mathbb{E}_S[R'_\tau] \geq \sum_{j\geq 1} \frac{p}{d-1}\cdot\frac{1}{2}\cdot\frac{1}{2} = \frac{p}{4}$; for a τ attaining the average, $F := R'_\tau/p$ has $\mathbb{E}[F] \geq \frac{1}{4}$, so $R_\tau(A(S)) \geq R'_\tau \geq p/8 = \frac{1}{16}\min(2, \frac{d-1}{m})$ with probability at least $\frac{1}{7}$.

(iii) Let $\{x_0, x_1\}$ be shattered and, for $\tau \in \{0,1\}$, let $\mathcal{D}_\tau$ put mass $1-2\epsilon$ on x_0 with label 0 and mass 2ϵ on x_1 with label τ. With probability $(1-2\epsilon)^m \geq e^{-4\epsilon m}$ (by $\log(1-u) \geq -2u$ for $u \in [0, \frac{1}{2}]$) the point x_1 does not occur in S, and then, for τ uniform, $A(S)(x_1) \neq \tau$ with probability $\frac{1}{2}$, which makes $R_\tau(A(S)) \geq 2\epsilon > \epsilon$. Averaged over τ the probability of this is at least $\frac{1}{2}e^{-4\epsilon m}$, and some τ attains the average □

Translated into sample complexities, for $\delta < \frac{1}{7}$: part (i) shows that an agnostic learner with accuracy $\epsilon < \frac{1}{16}$ needs $\sqrt{d/m} \leq 64\epsilon$, i.e. $m \geq \frac{d}{4096\,\epsilon^2}$; part (ii) shows that a realizable learner with accuracy $\epsilon < \frac{1}{8}$ needs $m \geq \frac{d-1}{16\epsilon}$; part (iii) adds $m \geq \frac{\log(1/(2\delta))}{4\epsilon}$ in the realizable case, and (7.7) adds $m \geq \frac{\log(1/(4\delta))}{16\epsilon^2}$ in the agnostic case (put $\mathcal{D}$ on a single point at which two hypotheses of $\mathcal{H}$ disagree). These are the lower bounds of the Fundamental Theorem (Theorem 6.4.7). The restriction $d \geq 2$ in (ii) and (iii) cannot be dropped: if $\mathcal{H} = \{h, 1-h\}$, a single example identifies the target, and the realizable sample complexity is 1 for every ϵ and δ.

What the minimax viewpoint adds to the upper bounds of this chapter is a **certificate of optimality**. The chaining bound (7.4) shows that ERM achieves excess risk $O(\sqrt{d/m})$ on every distribution; Theorem 7.9.5 shows that no algorithm does better in the worst case, and Theorem 7.9.3 that the confidence term $\log(1/\delta)$ is necessary as well. The rates proved in this chapter are therefore properties of the *learning problem*, not artifacts of the proof technique.

7.10 Further Directions

Three lines of work continue the theory of this chapter.

- **Fast rates under low noise.** Apart from the realizable case (Theorem 6.4.14) and the stability-based oracle inequality after Theorem 7.8.3, which pays for its $O(1/m)$ rate with an expectation over S and a comparison against the regularized objective, all bounds above scale as $O(1/\sqrt{m})$. Under a **low-noise (margin) condition** — informally, when $\mathcal{D}$ rarely places points close to the decision boundary — the excess risk of ERM decays faster, interpolating between $1/\sqrt{m}$ and $1/m$ as the condition strengthens. The mechanism is the one behind Theorem 6.4.14: when the variance of the excess loss is controlled by its mean, Bernstein's inequality (Theorem 5.6.9) replaces Hoeffding's, and localized versions of the Rademacher complexity replace the global one. The margin bound for boosting (Theorem 12.2.7) is a related instance.

- **Sharper lower bounds.** Theorem 7.9.5 fixes the rates but not the constants. Exact minimax constants for specific problem families, and lower bounds for estimation problems with continuous parameters, are obtained by sharper packing constructions combined with Fano's inequality and Assouad's method; see (Tsybakov, 2009).
- **Generic chaining.** Dudley's entropy integral (Theorem 7.6.2) is not always tight. Its refinement, generic chaining, characterizes the expected supremum of a Gaussian process up to universal constants (the majorizing-measure theorem), and with it the Rademacher complexity of many classes; see Chapter 12 of (Zhang, 2023).

CHAPTER 8

BAYESIAN LEARNING

8.1 Maximum Likelihood Estimation

Machine learning is the tool of choice especially in real-world problems where there are factors of uncertainty and randomness. Having learned the principled way of accounting for uncertainty via probability theory, next we will see how we can use this theory to build machine learning models. Every machine learning problem has two key components: the data and the model. Let our data set be $S = \{x_1, x_2, \ldots, x_m\}$, where x_i are the observed data points. At the stage of modeling, we draw a hypothesis about how these data could have been generated. Let our hypothesis be that the data are generated as independent samples from a distribution following a parametric probability function $P(x \mid \theta)$, where θ is the parameter of the distribution. Then the joint probability of the random variables representing the occurrence of the data set is given by

$$P(S \mid \theta) = P(x_1, x_2, \ldots, x_m \mid \theta) = \prod_{i=1}^{m} P(x_i \mid \theta).$$

This quantity, viewed as a function of θ for the fixed, observed S, is called the **likelihood** of θ given the data set S. The second equality follows from the assumption that the data points are independent. The maximum likelihood estimation (MLE) is a method of estimating the parameters θ by maximizing the likelihood $P(S \mid \theta)$. This way we aim to find the parameters that are most likely to have generated the data set S, which leads to the optimization problem below:

$$\theta_{MLE} = \arg\max_{\theta} P(S \mid \theta) = \arg\max_{\theta} \prod_{i=1}^{m} P(x_i \mid \theta).$$

Formulating the objective as a product of terms is numerically brittle: if each factor is small, the product of many such terms underflows the floating point system. It is also harder to differentiate a product than a sum. We therefore maximize the **log-likelihood** instead. Since the logarithm is monotonically increasing, its maximizer coincides with that of the likelihood itself:

$$\theta_{MLE} = \arg\max_{\theta} \sum_{i=1}^{m} \log P(x_i \mid \theta).$$

8.1.1 Example: MLE for the Bernoulli distribution

Let $x \in \{0, 1\}$ be the outcome of a coin toss (1 for heads, 0 for tails), modeled as $P(x \mid \theta) := \theta^x(1-\theta)^{1-x}$, where $\theta \in [0, 1]$ is the probability of heads. This is the **Bernoulli distribution**, and θ is its only parameter. Given $S = \{x_1, \ldots, x_m\}$, i.i.d. samples of x, the log-likelihood is

$$\log P(S \mid \theta) = \sum_{i=1}^{m} \log \theta^{x_i}(1-\theta)^{1-x_i} = \Big(\sum_{i=1}^{m} x_i\Big) \log\theta + \Big(m - \sum_{i=1}^{m} x_i\Big) \log(1-\theta).$$

Setting the derivative with respect to θ to zero,

$$\begin{aligned} \frac{\partial}{\partial\theta} \log P(S \mid \theta) = \frac{\sum_{i=1}^{m} x_i}{\theta} - \frac{m - \sum_{i=1}^{m} x_i}{1-\theta} &\triangleq 0 \\ \Rightarrow (1-\theta) \sum_{i=1}^{m} x_i &= \theta\Big(m - \sum_{i=1}^{m} x_i\Big) \\ \Rightarrow \theta_{MLE} &= \frac{1}{m} \sum_{i=1}^{m} x_i. \end{aligned}$$

Unsurprisingly, the MLE of a Bernoulli parameter is simply the observed frequency of heads, i.e. the sample mean of S. This is the estimator we implicitly used throughout Chapter 5 whenever we wrote $\widehat{\mu}_m$ for a sample of Bernoulli random variables.

8.2 Bayesian Learning

The MLE approach commits to a single value of θ and does not represent how uncertain we are about it — an uncertainty that is naturally large when m is small (imagine estimating θ from three coin tosses) and shrinks as m grows. This ansatz, where θ is treated as an unknown but fixed constant, is called **frequentist learning**. **Bayesian learning** instead treats θ itself as a random variable, with a distribution that captures our belief about its likely values before and after seeing data.

Concretely, the modeling assumption — the **generative process** or **data generating process** — becomes

$$\begin{aligned} \theta &\sim p(\theta) \\ x_i \mid \theta &\sim \text{Bernoulli}(\theta), \quad i = 1, \ldots, m. \end{aligned}$$

We start with a **prior distribution** $p(\theta)$ that encodes our belief about θ before seeing any data, and update it to a **posterior distribution** $p(\theta \mid S)$ after observing S, via Bayes' rule:

$$p(\theta \mid S) = \frac{P(S \mid \theta)\, p(\theta)}{P(S)} = \frac{P(S \mid \theta)\, p(\theta)}{\int_0^1 P(S \mid \theta')\, p(\theta')\, d\theta'}.$$

Note the case of the letters, which is not decoration: $P(S \mid \theta)$ and $P(S)$ are **probabilities** of the observed discrete data, while $p(\theta)$ and $p(\theta \mid S)$ are **densities** over the continuous parameter — the latter are not probabilities and can exceed 1. Every mixed expression in this chapter obeys this rule, which is worth checking against as the formulas grow.

The denominator $P(S) = \int_0^1 P(S \mid \theta')p(\theta')\,d\theta'$ is the **evidence**. Two facts about it matter for the rest of this course:

- In almost every real-world model, the evidence is intractable to compute in closed form, forcing us to approximate the posterior. The whole field of Bayesian machine learning is, in large part, about finding good posterior approximations.
- The evidence measures how well the whole model family fits the data, averaged over the prior; it can therefore be used for model selection, by comparing $P(S)$ across candidate models.

The Bernoulli-parameter case is one of the rare instances that admits an exact, closed-form posterior. We choose the prior to be a **Beta distribution**, $\theta \sim \text{Beta}(\alpha, \beta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$ for hyperparameters $\alpha, \beta > 0$, because it is **conjugate** to the Bernoulli likelihood: the posterior comes out in the same family, which makes the update mechanical.

$$\begin{aligned} p(\theta \mid S) &\propto P(S \mid \theta)\, p(\theta) \\ &= \Big[\prod_{i=1}^{m} \theta^{x_i}(1-\theta)^{1-x_i}\Big] \cdot \theta^{\alpha-1}(1-\theta)^{\beta-1} \\ &= \theta^{\sum_{i=1}^m x_i + \alpha - 1}(1-\theta)^{m - \sum_{i=1}^m x_i + \beta - 1}. \end{aligned}$$

This is, up to a normalizing constant, exactly the density of a $\text{Beta}(\alpha', \beta')$ distribution with

$$\alpha' := \alpha + \sum_{i=1}^{m} x_i,$$

$$\beta' := \beta + m - \sum_{i=1}^{m} x_i.$$

Since a probability density integrates to one, the normalizing constant is forced to be the one that makes $\text{Beta}(\alpha', \beta')$ a valid density, so $p(\theta \mid S) = \text{Beta}(\theta \mid \alpha', \beta')$ exactly, with no approximation needed. Note the intuitive interpretation of the hyperparameters: α and β act as **pseudo-counts** of heads and tails observed prior to seeing any real data, and the posterior simply adds the real counts $\sum_i x_i$ and $m - \sum_i x_i$ on top.

For a new observation x_*, the Bayesian model predicts a distribution over outcomes rather than a single value:

$$\begin{aligned} P(x_* = 1 \mid S) &= \int_0^1 P(x_* = 1 \mid \theta)\, p(\theta \mid S)\, d\theta = \int_0^1 \theta \cdot \text{Beta}(\theta \mid \alpha', \beta')\, d\theta \\ &= \mathbb{E}_{\theta \sim p(\theta \mid S)}[\theta] = \frac{\alpha'}{\alpha' + \beta'}, \end{aligned}$$

using the known mean of the Beta distribution. This is the **posterior predictive distribution**. Two of its properties are worth highlighting, both instances of a more general phenomenon in Bayesian models:

- It **averages over all possible values of** θ, weighted by their posterior probability — a form of **model averaging** that accounts for parameter uncertainty, rather than committing to a single point estimate.

- As $m \to \infty$, the pseudo-counts α, β become negligible relative to $\sum_i x_i$ and $m - \sum_i x_i$, so $P(x_* = 1 \mid S) \to \theta_{MLE}$: the Bayesian and frequentist predictions coincide in the large-sample limit, while differing — often substantially — for small m, exactly where accounting for uncertainty matters most.

Given a predictive distribution, there is more than one way to commit to a single prediction. The **Bayes predictor** takes the mode of the predictive distribution — its most probable value, $\arg\max_x P(x_* = x \mid S)$ — and is optimal in the sense of minimizing expected 0/1 loss. The **Gibbs predictor** instead samples a prediction, $\widehat{x} \sim P(x_* \mid S)$. It can never beat the mode: for a binary outcome with predictive probability $p := P(x_* = 1 \mid S)$, its probability of error under the predictive distribution is $p(1-p) + (1-p)p = 2p(1-p)$, against $\min(p, 1-p)$ for the mode — larger unless $p \in \{0, \frac{1}{2}, 1\}$, but never more than twice as large, since $2p(1-p) \leq 2\min(p, 1-p)$.

8.3 Maximum A-Posteriori Estimation

In most real-world applications, the posterior distribution is intractable because of the integral in the evidence term. One common approximation is the **maximum a-posteriori (MAP)** estimate, the mode of the posterior:

$$\theta_{MAP} = \arg\max_\theta p(\theta \mid S) = \arg\max_\theta P(S \mid \theta)p(\theta) = \arg\max_\theta \log P(S \mid \theta) + \log p(\theta).$$

The intractable evidence $P(S)$ drops out since it does not depend on θ, making the objective tractable — this is the entire appeal of MAP over full Bayesian inference. For the Beta-Bernoulli model, since $\text{Beta}(\alpha', \beta')$ has mode $\frac{\alpha'-1}{\alpha'+\beta'-2}$ (for $\alpha', \beta' > 1$),

$$\theta_{MAP} = \frac{\alpha + \sum_{i=1}^m x_i - 1}{\alpha + \beta + m - 2}.$$

Comparing this to $\theta_{MLE} = \frac{1}{m}\sum_i x_i$ makes the role of the prior explicit: MAP behaves like MLE run on an *augmented* data set that includes $\alpha - 1$ extra pseudo-heads and $\beta - 1$ extra pseudo-tails. As $m \to \infty$, $\theta_{MAP} \to \theta_{MLE}$, the same way the full posterior predictive did above.

For a test input, the MAP estimate predicts as a **point estimate**, $x_* \sim P(x_* \mid \theta_{MAP})$, and does not perform model averaging: unlike the fully Bayesian predictive distribution above, it is not a truly Bayesian approach, even though it is derived from the posterior.

8.4 Monte Carlo Integration

Bayesian quantities — evidences, predictive distributions, posterior expectations — are often integrals of the form

$$I = \int f(x)\, p(x)\, dx$$

for some function f and probability density $p(x)$ (with the integral replaced by a sum against a probability mass function $P(x)$ in the discrete case), which is exactly what we needed to evaluate $\mathbb{E}_{\theta \sim p(\theta|S)}[\theta]$

above. When this integral has no closed form (as is typical outside conjugate models like Beta-Bernoulli), and we can draw m samples $x_1, \dots, x_m \sim p(x)$, we can approximate it by the sample average

$$I \approx \frac{1}{m} \sum_{i=1}^{m} f(x_i).$$

This is called **Monte Carlo integration**. By the weak law of large numbers (Chapter 5), this approximation is consistent: it converges to I as $m \to \infty$.

8.5 Generative Models

Consider a supervised learning problem with $(x, y) \sim \mathcal{D}$ for an unknown data distribution $\mathcal{D}$. Take the label y to be discrete and the features x to be continuous, so that — following the convention of Chapter 2 — the joint object is written $p(x, y)$ while its discrete factors are written with a capital P. There are two ways to factorize $p(x, y)$, and each suggests a different modeling strategy:

- $p(x, y) = P(y \mid x)\, p(x)$: model $P(y \mid x)$ directly and treat $p(x)$, the distribution of inputs alone, as a quantity not worth modeling (it enters our formulas only as the distribution we average over — the same "average it out" treatment we gave the label in the marginal $\mathcal{D}_{\mathcal{X}}$ of Chapter 2; when it does appear, it can be approximated by Monte Carlo integration on the training inputs). This is called **discriminative modeling** — the logistic regression model of Chapter 4 is an example.
- $p(x, y) = p(x \mid y)\, P(y)$: model both $P(y)$ and $p(x \mid y)$, i.e. infer the whole data-generating process where a label is chosen first and the corresponding input is generated conditioned on it. This is called **generative modeling**.

8.5.1 Example: a generative Bernoulli classifier

Let us build the simplest possible generative classifier: both the label $y \in \{0, 1\}$ and a single binary feature $x \in \{0, 1\}$ are Bernoulli. We choose

$$y \sim \text{Bernoulli}(\pi),$$
$$x \mid y \sim \text{Bernoulli}(\theta_y),$$

so π is the marginal probability of $y = 1$, and θ_0, θ_1 are the class-conditional probabilities that $x = 1$ given $y = 0$ and $y = 1$ respectively. Given $S = \{(x_i, y_i)\}_{i=1}^{m}$, all three parameters are Bernoulli parameters and each is fit by the MLE formula derived above, applied to the relevant subset of the data:

$$\widehat{\pi} = \frac{1}{m} \sum_{i=1}^{m} \mathbb{1}(y_i{=}1),$$
$$\widehat{\theta}_c = \frac{\sum_{i:y_i=c} x_i}{\sum_{i:y_i=c} 1}, \quad c \in \{0, 1\}.$$

The second formula is exactly θ_{MLE} from before, restricted to the subset of the data with $y_i = c$: fitting a generative model reduces to running the same single-distribution MLE machinery once per class. Given a new x_*, the model predicts

$$P(y{=}1 \mid x_*) = \frac{P(x_* \mid y{=}1)\,P(y{=}1)}{P(x_* \mid y{=}0)P(y{=}0) + P(x_* \mid y{=}1)P(y{=}1)} = \frac{\widehat{\theta}_1^{x_*}(1-\widehat{\theta}_1)^{1-x_*}\,\widehat{\pi}}{\widehat{\theta}_0^{x_*}(1-\widehat{\theta}_0)^{1-x_*}(1-\widehat{\pi}) + \widehat{\theta}_1^{x_*}(1-\widehat{\theta}_1)^{1-x_*}\widehat{\pi}}.$$

This is a complete, if minimal, generative classifier. Real problems have more than one feature, which brings us to Naive Bayes.

8.5.2 Naive Bayes Classifier

Assume now d discrete features $x = (x_1, \ldots, x_d)$, each taking values in a finite alphabet, and a label $y \in \{1, \ldots, C\}$. Modeling the *joint* class-conditional distribution $P(x \mid y{=}c)$ exactly would require a full contingency table with one entry per combination of feature values — exponential in d, and infeasible to estimate from any realistic amount of data. The **naive Bayes assumption** sidesteps this by assuming the features are conditionally independent given the class:

$$P(x \mid y{=}c) = \prod_{j=1}^{d} P(x_j \mid y{=}c).$$

This is "naive" because it is almost always false in the real world (features are typically correlated even within a class) — yet the resulting classifier is often highly effective in practice, and always tractable. For each feature j with alphabet $\{1, \ldots, V_j\}$, we only need to estimate a small **tabular** distribution $P(x_j = v \mid y{=}c)$ per class c and value v, i.e. a $C \times V_j$ table of probabilities. For binary features ($V_j = 2$) this is again exactly the Bernoulli-MLE machinery from before, one Bernoulli parameter $\theta_{j,c}$ per (feature, class) pair:

$$\widehat{\theta}_{j,c} := \widehat{P}(x_j{=}1 \mid y{=}c) = \frac{\sum_{i:y_i=c} x_{ij}}{m_c}, \qquad m_c := \sum_{i:y_i=c} 1,$$

together with the class prior $\widehat{\pi}_c = m_c/m$. Given a query x_*, the classifier predicts

$$\widehat{y} = \arg\max_{c \in [C]} \widehat{\pi}_c \prod_{j=1}^{d} \widehat{\theta}_{j,c}^{x_{*j}} (1-\widehat{\theta}_{j,c})^{1-x_{*j}}.$$

The total number of estimated parameters is $O(Cd)$ — linear in the number of features, in stark contrast to the exponential table size of the unrestricted model. This tabular form generalizes to categorical (non-binary) features and to continuous features (e.g. via a per-class Gaussian $p(x_j \mid y{=}c) = \mathcal{N}(x_j \mid \mu_{j,c}, \sigma_{j,c}^2)$) by the same procedure, replacing the Bernoulli MLE with the MLE of the chosen per-feature family.

PAC analysis of Naive Bayes. Since every one of the $\widehat{\theta}_{j,c}$ (and $\widehat{\pi}_c$) is a Bernoulli-parameter MLE — a sample mean of a $\{0,1\}$-valued quantity — Hoeffding's inequality (Theorem 5.6.5) applies to each of them individually. We have $K = Cd$ such parameters $\widehat{\theta}_{j,c}$. Conditionally on the labels $y_1, \ldots, y_m$, the estimate $\widehat{\theta}_{j,c}$ is an average of m_c i.i.d. $\{0,1\}$-valued variables, so Hoeffding's inequality bounds each of its

two tails by $e^{-2m_c\epsilon^2} \le e^{-2m_{\min}\epsilon^2}$, and the union bound over the $2Cd$ one-sided events — the argument of the uniform Hoeffding bound, Theorem 5.6.6, with sample sizes $m_c \ge m_{\min}$ — gives

$$P\Big(\exists (j,c) \in [d] \times [C] : |\widehat{\theta}_{j,c} - \theta_{j,c}| \ge \epsilon\Big) \le 2\,Cd\,e^{-2m_{\min}\epsilon^2},$$

where $m_{\min} := \min_c m_c$ is the smallest class sample size; the bound holds conditionally on the labels, hence also unconditionally whenever $m_{\min}$ is replaced by a deterministic lower bound on it. Setting the right-hand side to δ and solving for ϵ, with probability at least $1-\delta$, **every** one of the Cd estimated parameters is simultaneously within

$$\epsilon = \sqrt{\frac{\log(2Cd/\delta)}{2m_{\min}}}$$

of its true value. This is a genuine PAC-style guarantee: the sample complexity to achieve a target accuracy ϵ with confidence $1-\delta$ grows only as $O\big(\frac{1}{\epsilon^2}\log(Cd/\delta)\big)$ — logarithmically in the number of features d and classes C, the direct benefit of the naive independence assumption reducing the effective number of parameters from exponential to linear in d. It is, however, a guarantee about *parameter estimation*, not about the risk of the classifier: the classification decision is a nonlinear function of all Cd parameters at once, and errors in individual parameters can compensate one another in the argmax. The next section supplies the missing classifier-level analysis, replacing the Cd parameters by the complexity of the set of decision rules they generate.

8.6 Generalization Analysis of Naive Bayes

The Hoeffding argument of the previous subsection controls the Cd estimated probabilities one at a time. That is a useful statement about *parameter estimation*, but it is not a risk bound for the classifier. The classifier is a nonlinear function of all the parameters at once: two parameter vectors can differ substantially while inducing the same decision rule, and small errors in many parameters can compensate one another. The object whose complexity governs generalization is therefore not the parameter vector but the set of decision rules that the parameters generate.

This section carries out the analysis in three steps. First, we show that the Naive Bayes classifier is a multiclass affine predictor, so its decision rules can be studied as a geometric hypothesis class rather than as a product of independent probability estimates. Second, we introduce the Natarajan dimension, the multiclass analogue of the VC dimension, and prove that the Naive Bayes class has Natarajan dimension at most $(C-1)(d+1)$. Third, we combine the multiclass Sauer–Shelah lemma with symmetrization and Massart's lemma to obtain a uniform convergence bound, from which we read off two guarantees: a risk bound for the Naive Bayes maximum-likelihood estimator, and an oracle inequality for empirical risk minimization over the same hypothesis class. Neither guarantee assumes that the data distribution satisfies the Naive Bayes factorization; the factorization is a property of the model, not an assumption about the world.

8.6.1 Naive Bayes as a Multiclass Affine Predictor

The log-score of class c in the Naive Bayes classifier is

$$\log \pi_c + \sum_{j=1}^{d} \Big[x_j \log \theta_{j,c} + (1 - x_j) \log(1 - \theta_{j,c}) \Big].$$

Writing

$$w_{j,c} := \log \frac{\theta_{j,c}}{1 - \theta_{j,c}}, \qquad b_c := \log \pi_c + \sum_{j=1}^{d} \log(1 - \theta_{j,c}),$$

turns it into the affine function $b_c + \langle w_c, x \rangle = \langle (b_c, w_c), (1, x) \rangle$. The next definition formalizes the resulting hypothesis classes.

Definition 8.6.1: Naive Bayes and affine hypothesis classes

Let $\mathcal{X} = \{0,1\}^d$, $\mathcal{Y} = [C]$, and $\tilde{x} := (1, x) \in \mathbb{R}^{d+1}$. For $v = (v_1, \dots, v_C) \in \mathbb{R}^{C \times (d+1)}$, define $h_v : \mathcal{X} \to \mathcal{Y}$ by

$$h_v(x) := \arg\max_{c \in [C]} \langle v_c, \tilde{x} \rangle,$$

with ties broken in favor of the smallest class index. The class $\mathcal{H}_{\text{aff}} := \{h_v : v \in \mathbb{R}^{C \times (d+1)}\}$ is the class of **multiclass affine predictors**. The **Naive Bayes hypothesis class** $\mathcal{H}_{\text{NB}}$ consists of all h_η of the form

$$h_\eta(x) := \arg\max_{c \in [C]} \Big[\log \pi_c + \sum_{j=1}^{d} \big(x_j \log \theta_{j,c} + (1 - x_j) \log(1 - \theta_{j,c}) \big) \Big],$$

where $\eta = (\pi, \theta)$ ranges over $\pi_c > 0$, $\sum_{c=1}^{C} \pi_c = 1$, and $\theta_{j,c} \in (0,1)$ for all $j \in [d]$ and $c \in [C]$, with the same tie-breaking rule.

The following proposition shows that the reparameterization is not merely a convenient rewriting: it exhausts the affine class.

Proposition 8.6.2

The Naive Bayes and affine classes coincide. $\mathcal{H}_{\text{NB}} = \mathcal{H}_{\text{aff}}$.

Proof. The inclusion $\mathcal{H}_{\text{NB}} \subseteq \mathcal{H}_{\text{aff}}$ follows from the reparameterization above, with $v_c = (b_c, w_c)$.

For the reverse inclusion, fix $v = (v_1, \dots, v_C) \in \mathbb{R}^{C \times (d+1)}$ and write $v_c = (v_{c,0}, v_{c,1}, \dots, v_{c,d})$. Set

$$\theta_{j,c} := \frac{1}{1 + e^{-v_{c,j}}} \in (0,1), \qquad \beta_c := \sum_{j=1}^{d} \log(1 - \theta_{j,c}),$$

and

$$\pi_c := \frac{\exp(v_{c,0} - \beta_c)}{\sum_{c'=1}^{C} \exp(v_{c',0} - \beta_{c'})}.$$

Then $\pi_c > 0$ and $\sum_c \pi_c = 1$, so $\eta = (\pi, \theta)$ is a valid Naive Bayes parameter vector. Its score for class c is

$$\begin{aligned}\log \pi_c + \sum_{j=1}^{d} \big(x_j \log \theta_{j,c} + (1 - x_j) \log(1 - \theta_{j,c})\big) &= \log \pi_c + \beta_c + \sum_{j=1}^{d} x_j v_{c,j} \\ &= v_{c,0} + \sum_{j=1}^{d} x_j v_{c,j} - \log\Big(\sum_{c'=1}^{C} e^{v_{c',0} - \beta_{c'}}\Big) \\ &= \langle v_c, \tilde{x} \rangle - \text{const},\end{aligned}$$

where the constant does not depend on c. Adding the same constant to every class score leaves the argmax and every tie unchanged, so $h_\eta = h_v$. Hence $\mathcal{H}_{\text{aff}} \subseteq \mathcal{H}_{\text{NB}}$, and the two classes are equal. □

Proposition 8.6.2 is the formal content of the familiar observation that binary Naive Bayes is a linear classifier. It also exposes the difference between the number of parameters and the complexity of the decision rules. A Naive Bayes parameter vector has $C - 1 + Cd$ free coordinates, but the induced rule depends only on the differences between the class scores: adding the same vector to every v_c leaves all comparisons unchanged. The space of score differences has dimension $(C - 1)(d + 1)$, which is the quantity measured by the next definition. Parameter counting therefore overstates the capacity of the classifier; what matters is the set of decision rules, not the coordinates used to describe it.

8.6.2 Natarajan Dimension

The VC dimension of Chapter 6 is a binary notion: it asks whether a class can realize all 2^m labelings of an m-point set. For C classes there are C^m labelings, and requiring all of them is far too strong: even the class of all constant classifiers realizes only C of them. Natarajan's definition replaces the Boolean cube $\{0, 1\}^m$ by a Boolean subcube of the C-valued labelings. It chooses two witness labels at each point and requires the class to realize every one of the 2^m ways of switching between them.

Definition 8.6.3: Natarajan shattering and dimension

Let $\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$ with $|\mathcal{Y}| = C \geq 2$. A set $S = \{x_1, \ldots, x_m\} \subseteq \mathcal{X}$ is **N-shattered** by $\mathcal{H}$ if there exist functions $f_1, f_2 : S \to \mathcal{Y}$ with $f_1(x_i) \neq f_2(x_i)$ for every $i \in [m]$ such that for every subset $B \subseteq [m]$ there is $h \in \mathcal{H}$ satisfying

$$h(x_i) = f_1(x_i) \text{ for } i \in B, \qquad h(x_i) = f_2(x_i) \text{ for } i \notin B.$$

The **Natarajan dimension** $d_{\text{N}}(\mathcal{H})$ is the largest cardinality of an N-shattered set, with $d_{\text{N}}(\mathcal{H}) := \infty$ if arbitrarily large sets are N-shattered, and with the convention that the supremum over an empty family is 0.

For $C = 2$, taking $f_1 \equiv 1$ and $f_2 \equiv 0$ on S recovers the usual definition of shattering, so $d_{\text{N}}(\mathcal{H}) = d_{VC}(\mathcal{H})$ for binary classes. The definition is due to (Natarajan, 1989). For $C > 2$, N-shattering is weaker than requiring all C^m labelings, but it is strong enough to force a polynomial growth function (Lemma

8.6.5 below). Unlike a parameter count, d_{N} is a property of the decision rules and is invariant under reparameterization.

Theorem 8.6.4: Natarajan dimension of affine predictors

For $C \geq 2$ and $d \geq 1$,

$$d_{\mathrm{N}}(\mathcal{H}_{\text{aff}}) \leq (C-1)(d+1).$$

Proof. Let $S = \{x_1, \ldots, x_m\} \subseteq \mathcal{X}$ be N-shattered by $\mathcal{H}_{\text{aff}}$, with witness functions $f_1, f_2 : S \to [C]$. For $x \in S$ and $c \in [C]$, let $\Psi(x, c) \in \mathbb{R}^{C\times(d+1)}$ be the matrix whose c-th row is $\tilde{x}$ and whose other rows are zero; then $\langle v, \Psi(x, c)\rangle = \langle v_c, \tilde{x}\rangle$ is the score of class c at x. Define

$$z_i := \Psi(x_i, f_1(x_i)) - \Psi(x_i, f_2(x_i)), \qquad i \in [m].$$

Since $f_1(x_i) \neq f_2(x_i)$, we have $z_i \neq 0$. We claim that $z_1, \ldots, z_m$ are linearly independent.

Suppose otherwise, so that $\sum_{i=1}^m \alpha_i z_i = 0$ for some coefficients $\alpha_i \in \mathbb{R}$, not all zero, and let $B := \{i \in [m] : \alpha_i > 0\}$. By N-shattering, there are hypotheses v_B and v_{B^c} realizing the two complementary labelings: $h_{v_B}(x_i) = f_1(x_i)$ for $i \in B$ and $h_{v_B}(x_i) = f_2(x_i)$ for $i \notin B$, and the opposite assignment for v_{B^c}. If $h_v(x_i) = f_1(x_i)$, then $f_1(x_i)$ is a maximizer of the scores at x_i, so $\langle v, z_i\rangle \geq 0$; if $h_v(x_i) = f_2(x_i)$, then $\langle v, z_i\rangle \leq 0$. Hence

$$0 = \Big\langle v_B, \sum_{i=1}^m \alpha_i z_i \Big\rangle = \sum_{i=1}^m \alpha_i \langle v_B, z_i\rangle \geq 0,$$

where each term is nonnegative because $\alpha_i > 0$ on B and $\alpha_i \leq 0$ outside B. Thus every term vanishes, and $\langle v_B, z_i\rangle = 0$ whenever $\alpha_i \neq 0$. Repeating the computation with B replaced by B^c (the terms are then nonpositive) gives $\langle v_{B^c}, z_i\rangle = 0$ for every i with $\alpha_i \neq 0$.

At such an i, both v_B and v_{B^c} assign equal scores to $f_1(x_i)$ and $f_2(x_i)$. Each hypothesis predicts one of these two labels at x_i, so both labels are tied at the maximum, and the fixed tie-breaking rule returns the same one of them for both hypotheses. But the labelings realized by v_B and v_{B^c} are complementary, so they prescribe opposite labels at every point of S, and in particular at every i with $\alpha_i \neq 0$. This is a contradiction. Hence $z_1, \ldots, z_m$ are linearly independent.

Finally, the sum of the rows of z_i is $\tilde{x}_i - \tilde{x}_i = 0$, so z_i belongs to the subspace

$$\mathcal{U} := \Big\{u \in \mathbb{R}^{C\times(d+1)} : \sum_{c=1}^C u_c = 0\Big\}.$$

The map $u \mapsto \sum_{c=1}^C u_c$ is onto $\mathbb{R}^{d+1}$, so $\dim \mathcal{U} = C(d+1) - (d+1) = (C-1)(d+1)$. Linear independence gives $m \leq \dim \mathcal{U}$. Since this holds for every N-shattered S, the claim follows. □

Together with Proposition 8.6.2, Theorem 8.6.4 gives

$$d_{\mathrm{N}}(\mathcal{H}_{\text{NB}}) = d_{\mathrm{N}}(\mathcal{H}_{\text{aff}}) \leq (C-1)(d+1).$$

This is the multiclass analogue of the VC dimension $d+1$ of affine binary classifiers. The bound is linear in the number of features and in the number of classes; the exponential dependence on d that a full

contingency table would suggest is absent. It is a bound on the *decision rules*, not on the $Cd + C - 1$ parameters of the Naive Bayes parameterization.

8.6.3 Multiclass Uniform Convergence

We now pass from combinatorial capacity to a risk bound. Recall from Definition 6.4.4 that the growth function of a hypothesis class $\mathcal{H}$ is

$$\tau_{\mathcal{H}}(m) := \max_{S \subseteq \mathcal{X},\, |S| = m} |\mathcal{H}_S|, \qquad \mathcal{H}_S := \{(h(x_1), \dots, h(x_m)) : h \in \mathcal{H}\}.$$

Nothing in that definition uses binary labels; it applies verbatim to C-valued classes. The multiclass Sauer–Shelah lemma bounds $\tau_{\mathcal{H}}$ in terms of the Natarajan dimension.

Lemma 8.6.5: Multiclass Sauer–Shelah

Let $\mathcal{H} \subseteq [C]^{\mathcal{X}}$ with $d_{\mathrm{N}}(\mathcal{H}) = d_{\mathrm{N}} < \infty$. Then for every $m \geq 1$,

$$\tau_{\mathcal{H}}(m) \leq \sum_{i=0}^{\min\{d_{\mathrm{N}}, m\}} \binom{m}{i} (C-1)^i.$$

In particular, for $m \geq d_{\mathrm{N}}$,

$$\tau_{\mathcal{H}}(m) \leq \left(\frac{e(C-1)m}{d_{\mathrm{N}}} \right)^{d_{\mathrm{N}}}.$$

Lemma 8.6.5 is the multiclass counterpart of the Sauer–Shelah–Perles lemma (Lemma 6.4.8). The sum has the same shape, with the binary factor 1 replaced by the $C - 1$ alternative labels available each time the Natarajan dimension is charged; for $C = 2$ it reduces exactly to Lemma 6.4.8. The result is due to (Natarajan, 1989); a complete proof appears in (Shalev-Shwartz and Ben-David, 2014), Chapter 29. The closed form follows from the sum by the same binomial estimate as in Lemma 6.4.8, using $m \geq d_{\mathrm{N}}$ and $C - 1 \geq 1$.

Theorem 8.6.6: Multiclass uniform convergence

Let $\mathcal{H} \subseteq [C]^{\mathcal{X}}$, let ℓ be the zero-one loss, and let $\tau_{\mathcal{H}}$ be the growth function. For every distribution $\mathcal{D}$ on $\mathcal{X} \times [C]$ and every $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$,

$$\sup_{h \in \mathcal{H}} \left| R(h) - \widehat{R}_S(h) \right| \leq \sqrt{\frac{16 \log\left(2\tau_{\mathcal{H}}(m)\right) + \log(1/\delta)}{m}}.$$

Proof. Write $\Phi(S) := \sup_{h \in \mathcal{H}} |R(h) - \widehat{R}_S(h)|$.

Step 1: concentration around the mean. Changing one example of S changes $\widehat{R}_S(h)$ by at most $1/m$ for every h simultaneously, hence changes $\Phi(S)$ by at most $1/m$. Thus Φ has bounded differences $c_i = 1/m$ and $\sum_{i=1}^{m} c_i^2 = 1/m$. McDiarmid's inequality (Theorem 7.1.1) gives, with probability at least $1 - \delta$,

$$\Phi(S) \le \mathbb{E}_S[\Phi(S)] + \sqrt{\frac{\log(1/\delta)}{2m}}.$$

Step 2: symmetrization. Let $S' = \{(x_i', y_i')\}_{i=1}^m$ be an independent ghost sample and let $\sigma_1, \dots, \sigma_m$ be independent Rademacher signs. Jensen's inequality and the symmetry of the joint law of (S, S') under independent swaps give

$$\begin{aligned}
\mathbb{E}_S[\Phi(S)] &= \mathbb{E}_S\Big[\sup_{h\in\mathcal{H}} \big|\mathbb{E}_{S'}\big[\widehat{R}_{S'}(h) - \widehat{R}_S(h)\big]\big|\Big] \\
&\le \mathbb{E}_{S,S'}\Big[\sup_{h\in\mathcal{H}} \Big|\frac{1}{m}\sum_{i=1}^m \big(\ell(y_i', h(x_i')) - \ell(y_i, h(x_i))\big)\Big|\Big] \\
&= \mathbb{E}_{S,S',\sigma}\Big[\sup_{h\in\mathcal{H}} \Big|\frac{1}{m}\sum_{i=1}^m \sigma_i\big(\ell(y_i', h(x_i')) - \ell(y_i, h(x_i))\big)\Big|\Big] \\
&\le 2\,\mathbb{E}_{S,\sigma}\Big[\sup_{h\in\mathcal{H}} \Big|\frac{1}{m}\sum_{i=1}^m \sigma_i\,\ell(y_i, h(x_i))\Big|\Big].
\end{aligned}$$

Step 3: Massart's lemma. Condition on S. For each h, let $a^h := (\ell(y_1, h(x_1)), \dots, \ell(y_m, h(x_m))) \in [0,1]^m$. As h ranges over $\mathcal{H}$, the vectors a^h take at most $\tau_{\mathcal{H}}(m)$ distinct values, and each has Euclidean norm at most $\sqrt{m}$. Since the supremum in Step 2 is inside an absolute value, apply Massart's lemma (Lemma 7.3.3) to the symmetrized set $A := \{a^h : h \in \mathcal{H}\} \cup \{-a^h : h \in \mathcal{H}\}$, which has at most $2\tau_{\mathcal{H}}(m)$ elements:

$$\mathbb{E}_\sigma\Big[\sup_{h\in\mathcal{H}} \Big|\frac{1}{m}\sum_{i=1}^m \sigma_i a_i^h\Big|\Big] = \mathbb{E}_\sigma\Big[\max_{a\in A} \frac{1}{m}\sum_{i=1}^m \sigma_i a_i\Big] \le \sqrt{\frac{2\log\big(2\tau_{\mathcal{H}}(m)\big)}{m}}.$$

Hence $\mathbb{E}_S[\Phi(S)] \le 2\sqrt{2\log(2\tau_{\mathcal{H}}(m))/m} = \sqrt{8\log(2\tau_{\mathcal{H}}(m))/m}$.

Step 4: assembling the constants. Combining Steps 1 and 3 and using $(a+b)^2 \le 2a^2 + 2b^2$,

$$\Phi(S) \le \sqrt{\frac{8\log(2\tau_{\mathcal{H}}(m))}{m}} + \sqrt{\frac{\log(1/\delta)}{2m}} \le \sqrt{\frac{16\log(2\tau_{\mathcal{H}}(m)) + \log(1/\delta)}{m}},$$

which is the claim. □

The bound is stated with $\tau_{\mathcal{H}}(m)$ rather than $\tau_{\mathcal{H}}(2m)$. The proof uses the growth function at m because the symmetrization is applied directly to the m-sample; the ghost sample in Theorem 6.4.9 was counted through restrictions on $2m$ points, which is why that earlier bound involved $\tau_{\mathcal{H}}(2m)$. Since $\tau_{\mathcal{H}}$ is nondecreasing, replacing m by $2m$ only weakens the bound.

Corollary 8.6.7

PAC risk bound for the Naive Bayes class. Let $d_{\mathrm{N}} := (C-1)(d+1)$ and $m \geq d_{\mathrm{N}}$. For every distribution $\mathcal{D}$ and every $\delta \in (0,1)$, with probability at least $1-\delta$, simultaneously for every $h \in \mathcal{H}_{\mathrm{NB}}$,

$$R(h) \leq \widehat{R}_S(h) + \sqrt{\frac{16 d_{\mathrm{N}} \log\left(2e(C-1)m/d_{\mathrm{N}}\right) + \log(1/\delta)}{m}}.$$

Proof. By Proposition 8.6.2 and Theorem 8.6.4, $d_{\mathrm{N}}(\mathcal{H}_{\mathrm{NB}}) \leq d_{\mathrm{N}}$. Lemma 8.6.5 with $m \geq d_{\mathrm{N}}$ gives

$$\tau_{\mathcal{H}_{\mathrm{NB}}}(m) \leq \left(\frac{e(C-1)m}{d_{\mathrm{N}}}\right)^{d_{\mathrm{N}}}.$$

Since $d_{\mathrm{N}} \geq 1$,

$$\log\left(2\tau_{\mathcal{H}_{\mathrm{NB}}}(m)\right) \leq \log 2 + d_{\mathrm{N}} \log \frac{e(C-1)m}{d_{\mathrm{N}}} \leq d_{\mathrm{N}} \log \frac{2e(C-1)m}{d_{\mathrm{N}}}.$$

Substituting into Theorem 8.6.6 gives the claim. □

For brevity, denote the deviation term in Corollary 8.6.7 by

$$\epsilon_m(\delta) := \sqrt{\frac{16 d_{\mathrm{N}} \log\left(2e(C-1)m/d_{\mathrm{N}}\right) + \log(1/\delta)}{m}}.$$

The bound is distribution-free: it holds for every $\mathcal{D}$ on $\mathcal{X} \times [C]$, whether or not the conditional independence assumption is true. In the regime of interest the logarithmic factor is negligible, and the deviation scale is

$$\epsilon_m(\delta) = O\left(\sqrt{\frac{(C-1)(d+1)\log\left(Cm/((C-1)(d+1))\right) + \log(1/\delta)}{m}}\right).$$

8.6.4 The Naive Bayes MLE and ERM over the Same Class

We now apply the uniform bound to the two estimators studied in this chapter. The first is the maximum-likelihood estimator. Under the Naive Bayes factorization, the likelihood decouples across classes and features, and its maximizer is given by the per-class frequency counts

$$\widehat{\pi}_c = \frac{m_c}{m}, \quad \widehat{\theta}_{j,c} = \frac{\sum_{i:\, y_i = c} x_{ij}}{m_c}, \quad m_c := \sum_{i:\, y_i = c} 1.$$

This is the estimator analyzed earlier in the chapter. When every count is positive, it lies in the interior of the parameter space and therefore induces a member of $\mathcal{H}_{\mathrm{NB}}$. When some count is zero, the corresponding log-score is infinite; the standard remedy is add-α smoothing,

$$\widetilde{\theta}_{j,c} = \frac{\sum_{i:\, y_i = c} x_{ij} + \alpha}{m_c + 2\alpha}, \qquad \widetilde{\pi}_c = \frac{m_c + \alpha}{m + C\alpha}, \qquad \alpha > 0,$$

which is also what the implementation below uses. The risk bound applies to any data-dependent member of $\mathcal{H}_{\mathrm{NB}}$, so it covers both the interior MLE and its smoothed version.

Corollary 8.6.8

Risk bound for the Naive Bayes MLE. Let $\widehat{\eta}_{\mathrm{MLE}}$ be any maximum-likelihood estimator of the Naive Bayes parameters whose coordinates lie in the interior of the parameter space, or let it be the add-α smoothed estimator. Then, with probability at least $1 - \delta$,

$$R\big(h_{\widehat{\eta}_{\mathrm{MLE}}}\big) \le \widehat{R}_S\big(h_{\widehat{\eta}_{\mathrm{MLE}}}\big) + \epsilon_m(\delta).$$

Proof. The estimator is a data-dependent member of $\mathcal{H}_{\mathrm{NB}}$, and Corollary 8.6.7 is uniform over the class. □

Corollary 8.6.8 is the missing risk bound for the vanilla Naive Bayes estimator. It is a *generalization* guarantee: it compares the true and empirical risks of the estimator actually returned, but it does not compare that estimator with the best hypothesis in $\mathcal{H}_{\mathrm{NB}}$. The latter is the content of the next result.

The second estimator minimizes the empirical zero-one risk over the same class:

$$\widehat{h}_{\mathrm{ERM}} \in \arg\min_{h \in \mathcal{H}_{\mathrm{NB}}} \widehat{R}_S(h).$$

This is a different optimization problem from maximum likelihood. The likelihood objective is a product of per-class and per-feature terms and decomposes into independent one-dimensional estimation problems; the zero-one objective couples all features and all classes through the argmax that defines h_η. In general, $h_{\widehat{\eta}_{\mathrm{MLE}}}$ is not an empirical risk minimizer over $\mathcal{H}_{\mathrm{NB}}$. Both estimators nevertheless live in the same hypothesis class, so the same capacity term controls both.

Theorem 8.6.9: ERM oracle inequality over the Naive Bayes class

Let $\widehat{h}_{\mathrm{ERM}} \in \arg\min_{h \in \mathcal{H}_{\mathrm{NB}}} \widehat{R}_S(h)$. Then, with probability at least $1 - \delta$,

$$R(\widehat{h}_{\mathrm{ERM}}) \le \inf_{h \in \mathcal{H}_{\mathrm{NB}}} R(h) + 2\epsilon_m(\delta).$$

Proof. On the event of Corollary 8.6.7, for every $h \in \mathcal{H}_{\mathrm{NB}}$,

$$R(\widehat{h}_{\mathrm{ERM}}) \le \widehat{R}_S(\widehat{h}_{\mathrm{ERM}}) + \epsilon_m(\delta) \le \widehat{R}_S(h) + \epsilon_m(\delta) \le R(h) + 2\epsilon_m(\delta),$$

where the middle inequality uses the definition of $\widehat{h}_{\mathrm{ERM}}$. Taking the infimum over $h \in \mathcal{H}_{\mathrm{NB}}$ gives the claim. □

If the infimum in Theorem 8.6.9 is not attained, the same argument applies to any ϵ-approximate empirical risk minimizer, at the price of adding ϵ to the right-hand side. The theorem is a *learning* guarantee: it says that ERM over $\mathcal{H}_{\text{NB}}$ is agnostic PAC learnable relative to that class. It does not say that ERM is computationally efficient. Minimizing the zero-one loss over $\mathcal{H}_{\text{NB}}$ is a nonconvex combinatorial problem in general, whereas maximum likelihood is closed-form. This is the precise sense in which the two estimators trade statistical optimality for computational tractability.

The two guarantees are summarized in the following table.

	Naive Bayes MLE	ERM over $\mathcal{H}_{\text{NB}}$
objective	$\sum_i \log P_\eta(x_i, y_i)$	$\sum_i \mathbb{1}\{h_\eta(x_i) \neq y_i\}$
computation	closed-form per-class counts	coupled; nonconvex in general
guarantee	$R \leq \widehat{R}_S + \epsilon_m(\delta)$	$R \leq \inf_{\mathcal{H}_{\text{NB}}} R + 2\epsilon_m(\delta)$
role	the estimator actually used	oracle benchmark and learning guarantee

The ERM guarantee is relative to $\mathcal{H}_{\text{NB}}$, not to the Bayes risk. If the true distribution factorizes as $P(x \mid y) = \prod_j P(x_j \mid y)$, then the Bayes classifier belongs to $\mathcal{H}_{\text{NB}}$, and the infimum in Theorem 8.6.9 equals the Bayes risk; the bound then controls the excess risk over the best possible predictor. If the distribution does not factorize, the infimum may be strictly larger than the Bayes risk, and the difference is the approximation error of the Naive Bayes model. In neither case does the *bound itself* require the factorization: Corollary 8.6.7 and Theorem 8.6.9 are distribution-free, and the factorization enters only in the interpretation of the approximation error.

The parameter-level Hoeffding bound derived earlier and the classifier-level bound of this section are complementary. The former controls the estimation error of each conditional probability and explains why the Naive Bayes estimates are stable; the latter controls the risk of the decision rule and explains why the classifier generalizes. Neither implies the other: parameter errors can compensate in the argmax, and a small risk does not require the individual probabilities to be accurate. What the capacity analysis adds is the correct dependence on the *number of decision rules*, which for Naive Bayes is governed by $(C-1)(d+1)$ rather than by the $Cd + C - 1$ parameters or by the exponentially many entries of a full contingency table.

Two extensions are immediate. For categorical features with V_j values, one-hot encoding replaces d by $\sum_j (V_j - 1)$ and the same bound holds with this effective dimension. For Gaussian Naive Bayes with a shared covariance matrix across classes, the log-score is affine in x, so the analysis applies unchanged; with class-specific covariance matrices the score is quadratic, the relevant Natarajan dimension is of order Cd^2, and the rate changes accordingly. These variants are left as exercises.

8.7 Implementation: Naive Bayes on a Real Data Set

Let us implement the tabular Naive Bayes classifier from scratch and test it on the **Wisconsin Breast Cancer** data set: 569 real tumor samples, each with 30 real-valued measurements (radius, texture, smoothness, ...) of a cell nucleus, labeled malignant or benign. Since our model above is built for *binary* features, we first binarize each measurement at its training-set median — a common, simple discretization strategy that brings a continuous data set into the tabular setting this chapter's math was derived for. Every other step

— the train/test split, the class-conditional MLEs $\widehat{\theta}_{j,c}$, and the prediction rule — is implemented directly from the formulas above, with no black-box model classes.

```
import torch as th
from sklearn.datasets import load_breast_cancer

# The only library-provided piece is the raw data set itself; the
# splitting, estimation, and prediction logic are all implemented below.
data = load_breast_cancer()
X_all = th.tensor(data.data, dtype=th.float64)
y_all = th.tensor(data.target, dtype=th.long)

th.manual_seed(0)
idx = th.randperm(y_all.numel())
n_train = int(0.7 * y_all.numel())
train_idx, test_idx = idx[:n_train], idx[n_train:]

# Binarize each feature at its training-set median.
median = X_all[train_idx].median(dim=0).values
X_train = (X_all[train_idx] > median).double()
X_test = (X_all[test_idx] > median).double()
y_train, y_test = y_all[train_idx], y_all[test_idx]

class BernoulliNaiveBayes:
    def __init__(self, num_classes=2, laplace=1.0):
        self.C = num_classes
        self.alpha = laplace  # Laplace (add-alpha) smoothing pseudo-count

    def learn(self, X, y):
        n, d = X.shape
        self.pi = th.zeros(self.C, dtype=th.float64)
        self.theta = th.zeros(self.C, d, dtype=th.float64)
        for c in range(self.C):
            Xc = X[y == c]
            n_c = Xc.shape[0]
            self.pi[c] = n_c / n
            # theta_{j,c} = (sum_i x_ij + alpha) / (n_c + 2*alpha): the MAP
            # estimate under a Beta(alpha+1, alpha+1) prior derived earlier
            # in this chapter, with alpha=1 recovering add-one smoothing.
            self.theta[c] = (Xc.sum(dim=0) + self.alpha) / (n_c + 2 * self.alpha)

    def predict(self, X):
        # Work in log-space for numerical stability, exactly as motivated
        # by the log-likelihood discussion at the start of this chapter.
        log_theta = th.log(self.theta)
        log_1m_theta = th.log(1 - self.theta)
        log_class_cond = X @ log_theta.T + (1 - X) @ log_1m_theta.T
```

```
        log_prior = th.log(self.pi)
        log_joint = log_class_cond + log_prior
        return th.argmax(log_joint, dim=1)

model = BernoulliNaiveBayes()
model.learn(X_train, y_train)

train_preds = model.predict(X_train)
test_preds = model.predict(X_test)

train_acc = (train_preds == y_train).double().mean().item()
test_acc = (test_preds == y_test).double().mean().item()

print(f"Naive Bayes train accuracy: {train_acc:.3f}")
print(f"Naive Bayes test accuracy:  {test_acc:.3f}")
```

```
Naive Bayes train accuracy: 0.910
Naive Bayes test accuracy:  0.918
```

Despite the crudeness of median binarization (which discards all information about *how far* a measurement is from the median) and the naive independence assumption across all 30 features, the classifier reaches over 90% test accuracy with a training procedure that amounts to frequency counting — no gradient descent, no iterative optimization, just the closed-form MLE/MAP formulas of this chapter applied once per class.

CHAPTER 9

DECISION TREES

Every hypothesis class we have used so far predicts through a fixed algebraic form: a dot product for linear predictors (Chapter 3), a sigmoid or softmax of a dot product for classifiers (Chapter 4), a weighted sum of similarities for kernel methods (Chapter 11). A **decision tree** (Breiman et al., 1984) predicts through a different mechanism entirely: a sequence of "if-then" questions about individual features, organized in a tree, that recursively partitions the feature space $\mathcal{X}$ into regions, each assigned a single output.

9.1 The Hypothesis Class

A decision tree is a rooted tree in which:

- every **internal node** v is labeled with a **splitting rule** $\phi_v : \mathcal{X} \to \{1, \dots, k_v\}$ that routes an input x to one of v's k_v children — for the binary trees we focus on here, $k_v = 2$ and $\phi_v(x) := \mathbb{1}(x_j \le t)$ for some feature index j and threshold t (or $\phi_v(x) := \mathbb{1}(x_j = a)$ for a categorical feature taking the value a),
- every **leaf** u is labeled with a prediction $c_u \in \mathcal{Y}$.

A tree T computes a hypothesis $h_T : \mathcal{X} \to \mathcal{Y}$ by routing x from the root, following $\phi_v(x)$ at each internal node v, until it reaches a leaf u, and outputting $h_T(x) := c_u$. Since each leaf's region is exactly the intersection of the splitting rules on the path from the root, the collection of leaves of T partitions $\mathcal{X}$ into disjoint axis-aligned regions, and h_T is piecewise constant on this partition. The **hypothesis class of decision trees**, $\mathcal{H}_{\text{tree}}$, is the union of h_T over all finite binary trees T built from the allowed splitting rules — an infinite, nonuniform hypothesis class in the sense of Chapter 6, since larger trees induce finer partitions and hence more expressive hypotheses.

Formally, then, a decision tree instantiates the general recursive partitioning scheme

$$h_T(x) = \sum_{u \in \text{leaves}(T)} c_u \cdot \mathbb{1}\big(x \in R_u\big),$$

$$R_u := \bigcap_{v \in \text{path}(u)} \phi_v(x)\text{-consistent half-space at } v,$$

where $\text{path}(u)$ is the sequence of internal nodes from the root to leaf u, and $\{R_u\}_{u \in \text{leaves}(T)}$ partitions $\mathcal{X}$.

9.2 Growing a Tree: A Greedy ERM Surrogate

Given a training set S, ERM over $\mathcal{H}_{\text{tree}}$ would search over all trees for the one minimizing $\widehat{R}_S(h_T)$. This is computationally intractable — even fixing the tree's shape, choosing the best splits jointly is NP-hard. In practice, we build the tree by a **greedy, recursive** procedure:

1. Start with a single leaf holding all of S.
2. At each leaf currently holding a subset $S_v \subseteq S$, consider splitting it into two children by some rule ϕ_v. Among all candidate rules (all features, all thresholds), pick the one that most reduces an **impurity** measure of the label distribution within the node (defined below).
3. Recurse into each child with its corresponding subset of S_v, and repeat, until a stopping criterion is met (e.g. a maximum depth, a minimum leaf size, or zero remaining impurity).
4. Label each final leaf u with the majority class of S_v restricted to that leaf (classification) or the mean label (regression).

The key design choice is the **impurity measure**, which stands in for $\widehat{R}_S(h)$ as a differentiable-in-spirit, locally computable proxy that the greedy procedure can optimize one split at a time. For a node v holding a subset S_v of size $m_v := |S_v|$, with class proportions $\widehat{p}_c := \frac{|\{(x,y)\in S_v : y=c\}|}{m_v}$ for $c \in \{1,\dots,C\}$, two common choices are:

- **Entropy** (of the empirical label distribution at the node): $H(S_v) := -\sum_{c=1}^{C} \widehat{p}_c \log \widehat{p}_c$, borrowed directly from information theory — it is maximized when S_v's labels are uniformly spread across classes (there is the most "uncertainty" about the label at this node) and zero when S_v is pure (a single class). Following the convention of these notes, log here is the natural logarithm, so entropy is measured in nats rather than the more familiar bits; the base only rescales H by a positive constant and therefore never changes which split is selected.
- **Gini index**: $G(S_v) := 1 - \sum_{c=1}^{C} \widehat{p}_c^{\,2} = \sum_{c=1}^{C} \widehat{p}_c(1-\widehat{p}_c)$, the probability that two labels drawn independently (with replacement) from S_v's empirical distribution disagree. It behaves similarly to entropy — zero for a pure node, maximal for a uniform one — but is less costly to compute and does not involve a logarithm.

Given an impurity measure $\text{Imp}(\cdot) \in \{H, G\}$ and a candidate split of S_v into $S_{v,\text{left}}$ and $S_{v,\text{right}}$, the **information gain** of the split is the reduction in impurity, weighted by the resulting subset sizes:

$$\text{Gain}(S_v, \phi_v) := \text{Imp}(S_v) - \frac{|S_{v,\text{left}}|}{m_v}\text{Imp}(S_{v,\text{left}}) - \frac{|S_{v,\text{right}}|}{m_v}\text{Imp}(S_{v,\text{right}}).$$

At each node, the greedy algorithm picks the split ϕ_v maximizing $\text{Gain}(S_v, \phi_v)$ over all candidate features and thresholds. This greedy, node-by-node maximization is what makes decision tree induction tractable: it never revisits an earlier split, and never jointly optimizes two splits together, at the cost of no longer being guaranteed to find the ERM-optimal tree of a given size.

9.3 Overfitting and the Bias-Complexity Dilemma

An unconstrained tree can always drive $\widehat{R}_S(h_T) = 0$ on its training set S: grow it until every leaf holds a single training point (or a set of points sharing one label). This is the tree-induction analogue of the memorization hypothesis from Chapter 2 — it exhibits the same overfitting behavior, with tree size (roughly,

the number of leaves) playing the role that polynomial degree M played in the curve-fitting example, or that $|\mathcal{H}|$ played in the generalization bound of Chapter 6. Two standard remedies, both instances of the Structural Risk Minimization principle from Chapter 6:

- **Early stopping.** Halt the recursive splitting once a stopping criterion is met (maximum depth, minimum node size, or a minimum required Gain), directly capping the tree's complexity during growth.
- **Pruning.** Grow a large tree first, then remove (merge) subtrees whose removal does not substantially hurt performance on a held-out validation set — trading a controlled increase in the empirical risk on the training sample for a reduction in complexity, in the hope of decreasing $R(h_T)$.

Consistent with the fundamental theorem of statistical learning (Chapter 6), the capacity of $\mathcal{H}_{\text{tree}}$ can also be bounded formally.

Proposition 9.3.1

VC dimension of size-limited trees. Let $\mathcal{H}_{\text{tree},L} \subset \mathcal{H}_{\text{tree}}$ be the subclass of binary trees with axis-aligned threshold splits on d real-valued features and at most L leaves, predicting in $\{0,1\}$. Then

$$\tau_{\mathcal{H}_{\text{tree},L}}(m) \;\le\; \underbrace{C_{L-1}}_{\text{tree shapes}} \cdot \underbrace{\big(d(m+1)\big)^{L-1}}_{\text{splits}} \cdot \underbrace{2^{L}}_{\text{leaf labels}},$$

$$\text{hence} \qquad d_{VC}(\mathcal{H}_{\text{tree},L}) = O\big(L\log(Ld)\big),$$

where $C_k \le 4^k$ is the k-th Catalan number, counting the shapes of a binary tree with k internal nodes.

Proof. Fix m points. A tree with at most L leaves has at most $L-1$ internal nodes, and its restriction to the sample is determined by three independent choices:

- the **shape** of the tree, i.e. which binary tree with $L-1$ internal nodes it is: $C_{L-1} \le 4^{L-1}$ possibilities;
- the **splitting rule** at each internal node. A rule $\mathbb{1}(x_j \le t)$ is determined, *as far as its behavior on the sample is concerned*, by the feature index $j \in [d]$ and by the position of t among the m observed values of feature j — at most $m+1$ distinct positions. So at most $d(m+1)$ effective rules per node, and at most $\big(d(m+1)\big)^{L-1}$ jointly;
- the **label** $c_u \in \{0,1\}$ of each of the at most L leaves: at most 2^L possibilities.

Multiplying gives the stated bound on the growth function. Shattering m points requires $2^m \le \tau_{\mathcal{H}_{\text{tree},L}}(m)$, i.e.

$$m\log 2 \;\le\; (L-1)\log 4 + (L-1)\log\big(d(m+1)\big) + L\log 2,$$

so $m = O\big(L\log(d) + L\log m\big)$, which forces $m = O\big(L\log(Ld)\big)$ □

Complexity therefore grows essentially linearly with the number of leaves — up to a logarithmic factor for choosing which feature and threshold to split on at each internal node — a formal restatement of the

intuition that more leaves means more capacity to overfit. The proof is also worth noting for what it does *not* count: the thresholds t range over an uncountable set, yet only $m+1$ of them are distinguishable on a given sample. This is the same "infinite class, finite restriction" phenomenon that Definition 6.4.1 was introduced to capture.

9.4 PAC Analysis of Decision Trees

$\mathcal{H}_{\text{tree}}$ itself is not PAC learnable. Since a tree can grow arbitrarily many leaves, $d_{VC}(\mathcal{H}_{\text{tree}}) = \lim_{L\to\infty} d_{VC}(\mathcal{H}_{\text{tree},L}) = \infty$ (a tree with 2^k leaves can, on real-valued features, always be grown to shatter any k points by isolating each into its own leaf). By Theorem 6.4.6, an infinite VC dimension rules out PAC learnability for the *unconstrained* class: no sample size m guarantees good generalization uniformly over all of $\mathcal{H}_{\text{tree}}$, matching the memorization behavior already noted above. Just as with k-nearest neighbors (Chapter 4) and unconstrained polynomial regression (Chapter 2), the fix is not to abandon the hypothesis class, but to control which part of it we search — exactly the role that early stopping and pruning play informally, and which we now make precise.

Fixed leaf budget: a standard agnostic PAC bound. Fix a leaf budget L and restrict to $\mathcal{H}_{\text{tree},L}$, which has finite VC dimension $d_L := d_{VC}(\mathcal{H}_{\text{tree},L}) = O(L\log(Ld))$. The Fundamental Theorem of Statistical Learning (Theorem 6.4.7) then applies directly: $\mathcal{H}_{\text{tree},L}$ is agnostic PAC learnable, and with probability at least $1-\delta$, every tree $h \in \mathcal{H}_{\text{tree},L}$ satisfies

$$R(h) \le \widehat{R}_S(h) + O\left(\sqrt{\frac{L\log(Ld) + \log(1/\delta)}{m}}\right).$$

This already formalizes the early-stopping heuristic: fixing a maximum depth (hence an upper bound on L) in advance and running ERM (i.e. the greedy growth procedure, as a tractable surrogate for exact ERM) within $\mathcal{H}_{\text{tree},L}$ inherits this generalization guarantee.

Unknown leaf budget: decision trees as an instance of SRM. In practice we do not fix L in advance — the greedy algorithm decides how many leaves to grow from the data itself, effectively searching $\mathcal{H}_{\text{tree}} = \bigcup_{L=1}^{\infty} \mathcal{H}_{\text{tree},L}$. This is precisely the nonuniform learning setting of Theorem 6.4.12: assign each subclass $\mathcal{H}_{\text{tree},L}$ a weight $w(L)$ with $\sum_{L=1}^{\infty} w(L) < 1$ (e.g. $w(L) := 6/(\pi^2 L^2)$, or a prefix-free code on L as in the MDL construction of Lemma 6.4.13), and Theorem 6.4.12 gives, with probability at least $1-\delta$, simultaneously for **every** tree $h \in \mathcal{H}_{\text{tree}}$ of any size $L(h)$,

$$R(h) \le \widehat{R}_S(h) + O\left(\sqrt{\frac{L(h)\log(L(h)d) + \log(1/(w(L(h))\delta))}{m}}\right).$$

The resulting h_{SRM} rule — minimize the right-hand side jointly over tree structure and size — makes explicit what pruning and early stopping approximate: a term that strictly decreases with L (the greedy algorithm's $\widehat{R}_S$, which can always be driven towards 0 by adding leaves) traded off against a term that strictly increases with L (the complexity penalty $\sqrt{L\log(Ld)/m}$). A pruning rule that stops merging leaves once further merges would raise a held-out validation error is, in effect, estimating this trade-off directly from data rather than from the closed-form penalty above — a data-driven proxy for the same SRM objective.

9.5 Illustrative Example

Consider a tiny data set for deciding whether to play outdoor tennis, with two binary features — `Outlook` $\in$ {Sunny, Rainy} and `Wind` $\in$ {Weak, Strong} — and label `Play` $\in$ {Yes, No}:

`Outlook`	`Wind`	`Play`
Sunny	Weak	Yes
Sunny	Weak	Yes
Sunny	Weak	Yes
Sunny	Strong	No
Rainy	Weak	No
Rainy	Weak	No
Rainy	Strong	No
Rainy	Strong	No

Here $m = 8$, with 4 "Yes" and 4 "No", so the root's Gini index is $G(S) = 1 - (\frac{1}{2})^2 - (\frac{1}{2})^2 = 0.5$.

Candidate split on `Outlook`. This sends the 4 Sunny points (3 Yes, 1 No) to the left child and the 4 Rainy points (0 Yes, 4 No) to the right child:

$$G(S_{\text{left}}) = 1 - \left(\tfrac{3}{4}\right)^2 - \left(\tfrac{1}{4}\right)^2 = 0.375,$$
$$G(S_{\text{right}}) = 1 - 1^2 - 0^2 = 0.$$

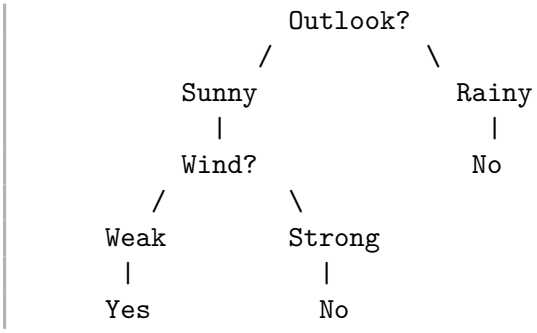
$$\text{Gain}(S, \texttt{Outlook}) = 0.5 - \tfrac{4}{8}(0.375) - \tfrac{4}{8}(0) = 0.5 - 0.1875 = 0.3125.$$

Candidate split on `Wind`. This sends the 5 Weak points (3 Yes, 2 No) to the left child and the 3 Strong points (0 Yes, 3 No) to the right child:

$$G(S_{\text{left}}) = 1 - \left(\tfrac{3}{5}\right)^2 - \left(\tfrac{2}{5}\right)^2 = 0.48,$$
$$G(S_{\text{right}}) = 1 - 0^2 - 1^2 = 0.$$

$$\text{Gain}(S, \texttt{Wind}) = 0.5 - \tfrac{5}{8}(0.48) - \tfrac{3}{8}(0) = 0.5 - 0.3 = 0.2.$$

Since $\text{Gain}(S, \texttt{Outlook}) = 0.3125 > 0.2 = \text{Gain}(S, \texttt{Wind})$, the greedy algorithm splits on `Outlook` first. The right child (Rainy) is already pure ($G = 0$) and becomes a leaf predicting No. The left child (Sunny, 3 Yes/1 No) is not pure, so we recurse: splitting it further on `Wind` sends its single Strong/No point to its own leaf and the remaining 3 Weak/Yes points to another, both pure, terminating the recursion. The resulting tree is



Observe that this tree achieves $\widehat{R}_S(h_T) = 0$ using only 3 leaves out of the $2^2 = 4$ possible input combinations having been observed — the greedy split ordering (largest information gain first) is exactly what let it reach a pure partition using the *fewest* splits, which is the practical mechanism by which impurity-driven greedy growth tends to produce small, and hence better-generalizing, trees compared to growing in an arbitrary order.

9.6 Implementation: Growing a Tree on a Real Data Set

Let us implement the greedy tree-growing algorithm of this chapter from scratch — the Gini impurity, the exhaustive split search, and the recursive growth procedure, all exactly as derived above — and test it on the **Wisconsin Breast Cancer** data set: 569 real tumor samples, each with 30 real-valued measurements of a cell nucleus, labeled malignant or benign. Unlike the Naive Bayes example of Chapter 8, decision trees natively handle real-valued features via threshold splits, so no discretization is needed here.

```
import torch as th
from sklearn.datasets import load_breast_cancer

# The only library-provided piece is the raw data set itself; the
# splitting, growth, and prediction logic are all implemented below.
data = load_breast_cancer()
X_all = th.tensor(data.data, dtype=th.float64)
y_all = th.tensor(data.target, dtype=th.long)

th.manual_seed(0)
idx = th.randperm(y_all.numel())
n_train = int(0.7 * y_all.numel())
train_idx, test_idx = idx[:n_train], idx[n_train:]

X_train, y_train = X_all[train_idx], y_all[train_idx]
X_test, y_test = X_all[test_idx], y_all[test_idx]

def gini(labels, num_classes=2):
    # G(S) = 1 - sum_c p_c^2
    if labels.numel() == 0:
        return th.tensor(0.0)
    counts = th.bincount(labels, minlength=num_classes).double()
    p = counts / labels.numel()
    return 1.0 - th.sum(p ** 2)

def best_split(X, y):
    # Exhaustively search every (feature, threshold) candidate and keep
    # the one with the highest information gain, Gain(S_v, phi_v).
    m, d = X.shape
    parent_impurity = gini(y)
    best_gain, best_feat, best_thresh = 0.0, None, None
    for j in range(d):
```

```
        values = th.unique(X[:, j])
        if values.numel() < 2:
            continue
        # midpoints between consecutive values
        thresholds = (values[:-1] + values[1:]) / 2
        for t in thresholds:
            left_mask = X[:, j] <= t
            n_left = left_mask.sum().item()
            n_right = m - n_left
            if n_left == 0 or n_right == 0:
                continue
            weighted_impurity = (n_left / m) * gini(y[left_mask]) + \
                                (n_right / m) * gini(y[~left_mask])
            gain = (parent_impurity - weighted_impurity).item()
            if gain > best_gain:
                best_gain, best_feat, best_thresh = gain, j, t.item()
    return best_feat, best_thresh, best_gain

class Node:
    def __init__(self, feature=None, threshold=None,
                 left=None, right=None, prediction=None):
        self.feature, self.threshold = feature, threshold
        self.left, self.right = left, right
        self.prediction = prediction  # set only for leaves

def build_tree(X, y, depth=0, max_depth=3, min_leaf_size=10):
    majority = th.bincount(y, minlength=2).argmax().item()
    # Early-stopping criteria: depth cap, minimum leaf size, or purity.
    if depth == max_depth or y.numel() < min_leaf_size or gini(y).item() < 1e-12:
        return Node(prediction=majority)
    feat, thresh, gain = best_split(X, y)
    if feat is None or gain <= 1e-12:
        return Node(prediction=majority)
    left_mask = X[:, feat] <= thresh
    left = build_tree(X[left_mask], y[left_mask],
                      depth + 1, max_depth, min_leaf_size)
    right = build_tree(X[~left_mask], y[~left_mask],
                       depth + 1, max_depth, min_leaf_size)
    return Node(feature=feat, threshold=thresh, left=left, right=right)

def predict_one(node, x):
    while node.prediction is None:  # route x from the root to a leaf
        node = node.left if x[node.feature] <= node.threshold else node.right
    return node.prediction

def predict(tree, X):
    return th.tensor([predict_one(tree, x) for x in X])
```

```
tree = build_tree(X_train, y_train, max_depth=3)

train_preds = predict(tree, X_train)
test_preds = predict(tree, X_test)
train_acc = (train_preds == y_train).double().mean().item()
test_acc = (test_preds == y_test).double().mean().item()

print(f"Decision tree train accuracy: {train_acc:.3f}")
print(f"Decision tree test accuracy:  {test_acc:.3f}")
```

```
Decision tree train accuracy: 0.965
Decision tree test accuracy:  0.918
```

A tree with at most $2^3 = 8$ leaves (`max_depth=3`) already reaches 91.8% test accuracy — close to its 96.5% training accuracy, indicating that the depth cap from Section "PAC Analysis of Decision Trees" above is doing its job of controlling L (and hence $d_{VC}(\mathcal{H}_{\text{tree},L})$) without starving the tree of the capacity it needs to separate the two classes.

Shallow trees are also the standard building blocks of the ensemble methods of Chapter 12: averaging bagged trees gives random forests, and boosting combines depth-one trees (stumps), both as classifiers in AdaBoost and as regression stumps in gradient boosting.

CHAPTER 10

NEURAL NETWORKS

10.1 Neuron (Perceptron)

A neuron, in the original form introduced as the **perceptron** (Rosenblatt, 1958), is the basic unit of a neural network. A neuron identified by index j has the following structure:

$$z = w_j^T x + b_j$$
$$h = g(z).$$

Above,

- x is the input vector,
- w_j is the weight vector, which corresponds to the synapses of the neuron that admit input signals from the previous layer,
- g is the activation function that determines whether the neuron is activated or not,
- b_j is the bias, which is a constant,
- z is the linear activation of the neuron,
- h is the activation output of the neuron.

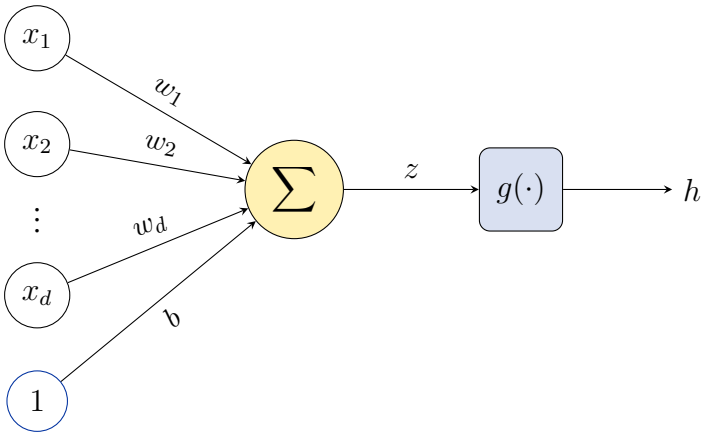


Figure 10.1: A single neuron (perceptron). The inputs $x_1, \ldots, x_d$ are combined through the weights $w_1, \ldots, w_d$ and a bias b into the linear activation $z = w^\top x + b$, which the activation function g maps to the output $h = g(z)$.

The first-generation activation function was the Heaviside step function:

$$g(z) = \begin{cases} 1 & \text{if } z \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

This construction, called a **perceptron**, is not popular nowadays since it is not differentiable. Modern neural networks use differentiable activation functions such as:

- Sigmoid function: $g(z) = \frac{1}{1+e^{-z}}$,
- Hyperbolic tangent function: $g(z) = \tanh(z)$,
- Rectified linear unit (ReLU): $g(z) = \max(0, z)$.

```
import torch as th
import matplotlib.pyplot as plt

# The ReLU, Sigmoid, and Tanh activation functions can be plotted in
# different panels of a figure as below:

x = th.linspace(-10, 10, 100)
x.requires_grad = True  # Enable gradient tracking for x

# Create a figure with two rows and three columns of subplots
fig, axes = plt.subplots(2, 3, figsize=(12, 8))

# Plot the original functions in the top row
# ReLU
y_relu = th.relu(x)
axes[0, 0].plot(x.detach(), y_relu.detach())
axes[0, 0].set_title('ReLU')
axes[0, 0].set_ylim(-1, 10.1)

# Sigmoid
y_sigmoid = th.sigmoid(x)
axes[0, 1].plot(x.detach(), y_sigmoid.detach())
axes[0, 1].set_title('Sigmoid')
axes[0, 1].set_ylim(-0.1, 1.1)

# Tanh
y_tanh = th.tanh(x)
axes[0, 2].plot(x.detach(), y_tanh.detach())
axes[0, 2].set_title('Tanh')
axes[0, 2].set_ylim(-1.1, 1.1)

# Compute gradients
# Compute gradients using the automatic differentiation provided by PyTorch
grad_relu = th.autograd.grad(y_relu.sum(), x)[0].detach()
grad_sigmoid = th.autograd.grad(y_sigmoid.sum(), x)[0].detach()
```

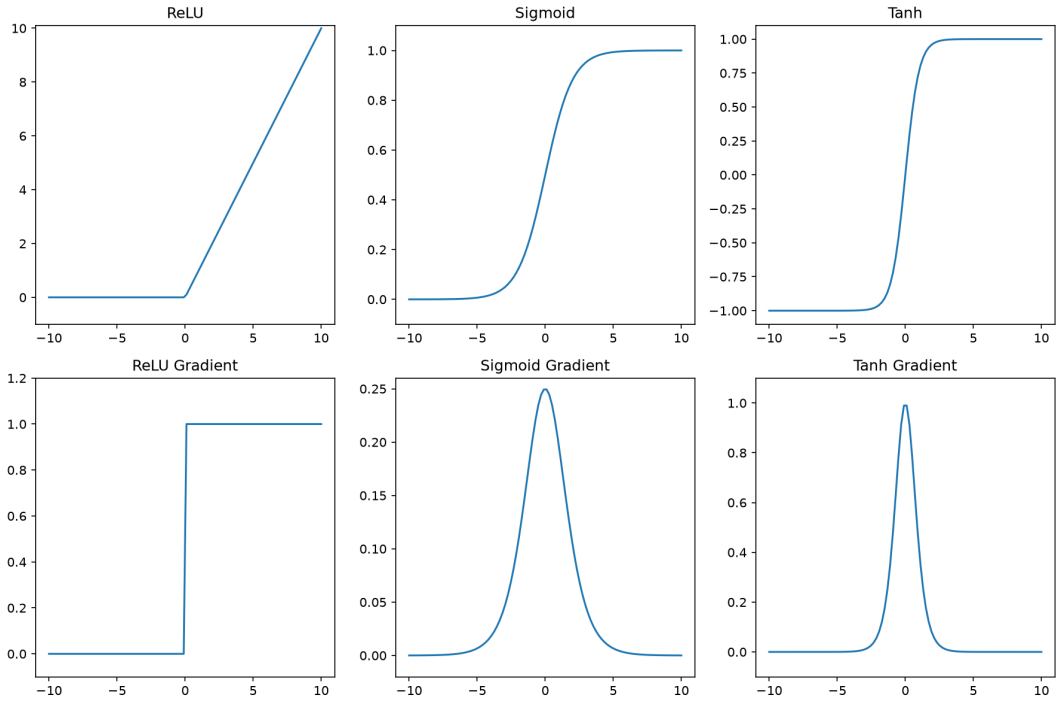


Figure 10.2: The ReLU, sigmoid, and tanh activation functions (top row) and their gradients (bottom row).

```
grad_tanh = th.autograd.grad(y_tanh.sum(), x)[0].detach()

# Plot the gradients in the bottom row
axes[1, 0].plot(x.detach(), grad_relu)
axes[1, 0].set_title('ReLU Gradient')
axes[1, 0].set_ylim(-0.1, 1.2)

axes[1, 1].plot(x.detach(), grad_sigmoid)
axes[1, 1].set_title('Sigmoid Gradient')
axes[1, 1].set_ylim(-0.02, 0.26)  # Adjusted y-axis limits for sigmoid gradient

axes[1, 2].plot(x.detach(), grad_tanh)
axes[1, 2].set_title('Tanh Gradient')
axes[1, 2].set_ylim(-0.1, 1.1)

plt.tight_layout()

plt.show()
```

10.2 Neural Network (Multilayer Perceptron)

A neural network is a collection of neurons. The neurons are organized in layers such that the activation output of the neurons in one layer become the synaptic inputs of the next layer.

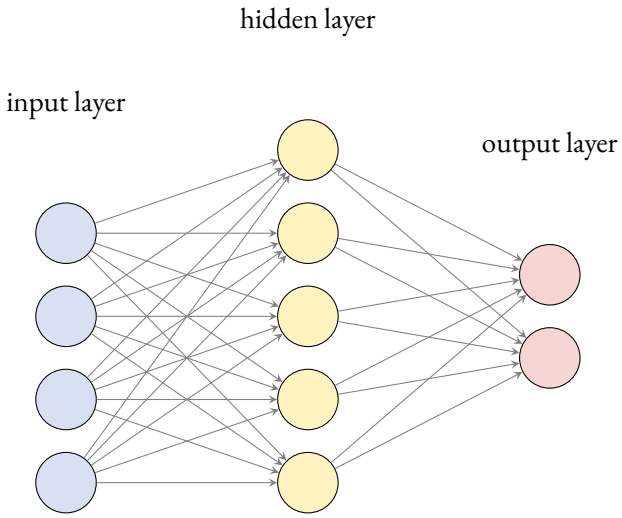


Figure 10.3: A feedforward neural network (multilayer perceptron). Each node is a perceptron unit as in Figure 10.1, organized into layers, with every neuron of one layer feeding into every neuron of the next.

The operation of k neurons sharing the same layer l on the i-th input can be given in the vector form below:

$$z_n^l = W_l^\top h_n^{l-1} + b^l,$$
$$h_n^l = g(z_n^l),$$

where

$$W_l = \begin{bmatrix} w_1^l & w_2^l & \cdots & w_k^l \end{bmatrix}$$

is the matrix of weight vectors and

$$b^l = (b_1^l, \ldots, b_k^l)$$

is the vector of biases. The vector h_n^l is called the **activation map** of layer l for the n-th input. For $l = 0$, the activation map is the input vector x, i.e. $h_n^0 := x_n$. The final pre-activation z_n^H is the output $\widehat{y}_n$ of the neural network.

Assume we have a neural network with two layers. We can express its whole operation as:

$$\widehat{y}_n = W_2^\top g(W_1^\top x_n + b^1) + b^2.$$

Likewise, we can express the operation of a neural network with three layers as:

$$\widehat{y}_n = W_3^\top g(W_2^\top g(W_1^\top x_n + b^1) + b^2) + b^3.$$

10.3 Training a Neural Network: Backpropagation

As with other machine learning models we covered thus far, neural networks are also trained to minimize a loss function. Denote the loss function by $\ell(y, \widehat{y})$ for a prediction $\widehat{y}$ and the true value y. Due to the cascaded application of the nonlinear activation functions, the loss function is not convex. Therefore, we cannot find the optimal solution analytically. Instead, we use gradient descent:

$$w_{ij}^l := w_{ij}^l - \alpha \frac{\partial \ell}{\partial w_{ij}^l},$$

for all synaptic connections $i \to j$ in all layers l. Given a neural network with H layers, define $\widehat{y}_j = z_j^H$ as the output of the j th neuron in the last layer. Also define

$$\delta_j^l = \frac{\partial \ell}{\partial z_j^l}.$$

This expression is called the **error** of the j th neuron in the l th layer. To see why, consider the case where $l = H$ and the loss function is the mean squared error: $\ell(y, \widehat{y}) = \frac{1}{2}\sum_j (y_j - \widehat{y}_j)^2$. Then we have

$$\begin{aligned}
\delta_j^H &= \frac{\partial \frac{1}{2}\sum_j (y_j - \widehat{y}_j)^2}{\partial z_j^H} \\
&= \frac{\partial \frac{1}{2}(y_j - \widehat{y}_j)^2}{\partial \widehat{y}_j} \\
&= \widehat{y}_j - y_j.
\end{aligned}$$

This is the error of the prediction of the neural network on the j output channel. The gradient of the loss with respect to an arbitrary weight is

$$\begin{aligned}
\frac{\partial \ell}{\partial w_{ij}^l} &= \frac{\partial \ell}{\partial z_j^l}\frac{\partial z_j^l}{\partial w_{ij}^l} \\
&= \delta_j^l \frac{\partial z_j^l}{\partial w_{ij}^l} \\
&= \delta_j^l h_i^{l-1}.
\end{aligned}$$

Let us next calculate the error for an intermediate layer l. We have

$$\begin{aligned}
\delta_j^l &= \frac{\partial \ell}{\partial z_j^l} \\
&= \sum_k \frac{\partial \ell}{\partial z_k^{l+1}}\frac{\partial z_k^{l+1}}{\partial z_j^l} \\
&= \sum_k \delta_k^{l+1}\frac{\partial z_k^{l+1}}{\partial z_j^l}
\end{aligned}$$

where k ranges over the neurons in the next layer. Placing the definition of z_k^{l+1} into the above equation, we have

$$\frac{\partial z_k^{l+1}}{\partial z_j^l} = \frac{\partial}{\partial z_j^l}\left(\sum_{j'} w_{j'k}^{l+1} g(z_{j'}^l) + b_k^{l+1}\right)$$
$$= w_{jk}^{l+1} \frac{\partial g(z_j^l)}{\partial z_j^l}.$$

Placing this result into the above expression about the derivative of the error, we get

$$\delta_j^l = \frac{\partial g(z_j^l)}{\partial z_j^l} \sum_k \delta_k^{l+1} w_{jk}^{l+1}$$

This way we obtain a recursive formula for the error of a neuron in an intermediate layer, where the recursion applies in the backward direction of the layers. There are remarkable facts regarding the computational efficiency of this formula that were first observed by Rumelhart et al. in 1986 and made the training of multilayer neural networks feasible:

- It requires only z_j^l, which is already calculated during the forward pass.
- Its computational cost is linear to the number of neurons in the network, i.e. $O(|W|)$ where $|W|$ is the number of weights in the network.

The findings above can be put together into an algorithm and expressed as below.

1. Set $h^0 = x$ (the activation map of the input layer is the input itself).
2. **Forward Pass.** For $l = 1 \to H$ do:
   - Compute and save $z_j^l = \sum_i w_{ij}^l h_i^{l-1} + b_j^l$ for every neuron j in layer l.
   - If $l < H$, compute $h_j^l = g(z_j^l)$. If $l = H$, the network output is $\widehat{y}_j := z_j^H$.
3. **Backward Pass.** For $l = H \to 1$ do:
   - Compute $\delta_j^l = \frac{\partial g(z_j^l)}{\partial z_j^l} \sum_k \delta_k^{l+1} w_{jk}^{l+1}$ for every neuron j in layer l (for $l = H$, use $\delta_j^H = \widehat{y}_j - y_j$ directly, as derived above).
   - Compute $\frac{\partial \ell}{\partial w_{ij}^l} = \delta_j^l h_i^{l-1}$ for every synaptic connection $i \to j$.

This algorithm is called **backpropagation** (Rumelhart et al., 1986). The name refers to the fact that the error term δ_j^l is propagated backward from the output layer to the input layer.

10.3.1 Stochastic Gradient Descent

Above we explained the backpropagation algorithm for a single data point for simplicity. For a data set with m samples, the gradient descent algorithm becomes:

$$w_{ij}^l := w_{ij}^l - \alpha \frac{1}{m} \sum_{i=1}^m \frac{\partial \ell(y_i, \widehat{y}_i)}{\partial w_{ij}^l},$$

This means we need to do a forward pass and a backward pass for all weights and all data points. This is computationally expensive for large data sets. Instead, we can use a technique called **stochastic gradient descent**. In this technique, we use a subset of the data set, called a **mini-batch**, to calculate the gradient. The size of the mini-batch is denoted by b. The stochastic gradient descent algorithm can be expressed as:

$$B \subseteq S, \quad |B| = b, \quad B \text{ drawn uniformly at random}$$
$$w_{ij}^l := w_{ij}^l - \alpha \frac{1}{b} \sum_{i=1}^{b} \frac{\partial \ell(y_i, \widehat{y}_i)}{\partial w_{ij}^l},$$

where the first step draws a mini-batch B uniformly at random from the **training set** S — not from the data distribution $\mathcal{D}$, which we cannot sample from — and the second step performs a gradient descent update using the backpropagation algorithm. The random selection makes the gradient signal a random variable. This is why the algorithm is called stochastic gradient descent. Because B is drawn uniformly from S, this random variable is an unbiased estimate of the full-batch gradient over S, i.e. of the gradient of the empirical risk $\widehat{R}_S$. Hence, under mild conditions (a suitably decaying learning rate α, among others) clarified by (Robbins and Monro, 1951), the algorithm is guaranteed to converge to a stationary point of the loss, just as deterministic gradient descent is. Since the neural network loss is generally non-convex, owing to the cascaded, nonlinear composition of layers described above, neither algorithm is guaranteed to reach the *same* stationary point, or the global optimum; in practice the noise injected by the mini-batch sampling is even considered beneficial, as it helps the current parameter estimate (the *iterate*) escape poor local stationary points that plain gradient descent could get stuck in.

10.3.2 Backpropagation in Matrix Form

The scalar recursion above is what one implements on paper; what one implements on a computer is its **batched matrix** form, which processes all b examples of a mini-batch B in one pass and maps one-to-one onto the code of the next section. Stack the activation maps of the mini-batch as rows,

$$\mathbf{H}^l := \begin{bmatrix} (h_1^l)^\top \\ \vdots \\ (h_b^l)^\top \end{bmatrix} \in \mathbb{R}^{b \times k_l},$$
$$\mathbf{H}^0 := \mathbf{X} \in \mathbb{R}^{b \times d},$$

and collect the layer errors in the same layout, $\square^l := \partial \widehat{R}_B / \partial \mathbf{Z}^l \in \mathbb{R}^{b \times k_l}$, whose (n, j)-th entry is the error δ_j^l of neuron j on example n. Then the entire algorithm is six matrix identities:

$$\textbf{(F1)} \quad \mathbf{Z}^l = \mathbf{H}^{l-1} W_l + \mathbf{1}_b (b^l)^\top, \qquad W_l \in \mathbb{R}^{k_{l-1} \times k_l},$$
$$\textbf{(F2)} \quad \mathbf{H}^l = g(\mathbf{Z}^l) \qquad \text{(elementwise)},$$
$$\textbf{(B1)} \quad \square^H = \tfrac{1}{b}\big(\widehat{\mathbf{Y}} - \mathbf{Y}\big), \qquad \text{(output layer)}$$
$$\textbf{(B2)} \quad \square^{l-1} = \big(\square^l W_l^\top\big) \odot g'(\mathbf{Z}^{l-1}), \qquad \text{(recursion)}$$
$$\textbf{(B3)} \quad \partial \widehat{R}_B / \partial W_l = (\mathbf{H}^{l-1})^\top \square^l,$$
$$\textbf{(B4)} \quad \partial \widehat{R}_B / \partial b^l = (\square^l)^\top \mathbf{1}_b,$$

where $\odot$ is the elementwise (Hadamard) product and $\mathbf{1}_b \in \mathbb{R}^b$ is the all-ones vector. (F1)-(F2) are the forward pass; (B1)-(B4) are the backward pass. Two observations:

- **(B3) is the scalar rule $\partial \ell / \partial w_{ij}^l = \delta_j^l h_i^{l-1}$, summed over the mini-batch** — the matrix product $(\mathbf{H}^{l-1})^\top \square^l$ performs exactly that sum, which is why one gradient step costs one matrix multiplication per layer rather than $|W|$ scalar operations. (The averaging factor $1/b$ that turns the sum into the mini-batch mean has already been folded into $\square^H$ by (B1), so it needs no separate mention here.)
- **(B2) is the scalar recursion $\delta_j^l = g'(z_j^l) \sum_k \delta_k^{l+1} w_{jk}^{l+1}$** — the sum over the next layer's neurons k is the product with $W_l^\top$, and multiplying by g' is the Hadamard product. Every layer therefore needs exactly two quantities from its forward pass: its input $\mathbf{H}^{l-1}$ (for B3) and its pre-activation $\mathbf{Z}^l$ (for g' in B2). This is the entire content of the "cache the forward pass" instruction in the algorithm above.

(B1) for the softmax cross-entropy loss. We derived $\delta_j^H = \widehat{y}_j - y_j$ for the squared loss. For C-class classification the output layer is instead passed through a **softmax**,

$$\widehat{y}_j := \frac{\exp(z_j^H)}{\sum_{c=1}^{C} \exp(z_c^H)},$$

$$\ell(y, \widehat{y}) := -\sum_{c=1}^{C} y_c \log \widehat{y}_c \quad (y \text{ one-hot, i.e. } y_c = \mathbb{1}(c = \text{true class})),$$

and a short calculation gives the *same* form. Writing $\widehat{y}_j = \exp(z_j)/Z$ with $Z := \sum_c \exp(z_c)$,

$$\frac{\partial \widehat{y}_c}{\partial z_j^H} = \widehat{y}_c\big(\mathbb{1}(c{=}j) - \widehat{y}_j\big)$$

$$\Longrightarrow \quad \frac{\partial \ell}{\partial z_j^H} = -\sum_c \frac{y_c}{\widehat{y}_c}\, \widehat{y}_c\big(\mathbb{1}(c{=}j) - \widehat{y}_j\big) = -y_j + \widehat{y}_j \sum_c y_c = \widehat{y}_j - y_j,$$

using $\sum_c y_c = 1$. So $\delta_j^H = \widehat{y}_j - y_j$ holds for the squared loss with a linear output *and* for the cross-entropy loss with a softmax output: in both cases the error that starts the backward recursion is simply **prediction minus target**. This is not a coincidence. It holds for every matched pair of a probability-model output layer (softmax here — one from the standard catalogue of distributions, the *exponential family*, that statistics uses for this purpose) and the loss equal to the negative logarithm of the probability that model assigns to the observed label (called its *negative log-likelihood*). We will meet this pairing again from the modeling side in Chapter 8.

10.4 Implementation: A Multilayer Perceptron From Scratch

We now implement (F1)-(F2) and (B1)-(B4) literally, with no automatic differentiation anywhere: each layer is an object exposing a `forward` map and the corresponding explicitly coded `backward` map, and the optimizer is the four-line stochastic gradient step of the previous section. Two points are worth watching for in the code below.

- **Every `backward` is one of the identities above**, in the same order: `Affine.backward` implements (B3), (B4) and the factor $\square^l W_l^\top$ of (B2); `ReLU.backward` supplies the remaining $\odot\, g'(\mathbf{Z})$ factor; `SoftmaxCrossEntropy.backward` is (B1).
- **The gradients are verified against finite differences.** Since we claim to have differentiated the network by hand, we check that claim numerically: for a random weight w, the central difference $\big(\widehat{R}(w+\epsilon) - \widehat{R}(w-\epsilon)\big)/(2\epsilon)$ must agree with the analytic gradient to within the $O(\epsilon^2)$ truncation error. This **gradient check** is the standard way to debug a hand-written backward pass, and is worth internalizing as a technique in its own right.

The data set is `digits`: 1797 real handwritten digits scanned as 8×8 grayscale images ($d = 64$ features, $C = 10$ classes) — the same task as the MNIST example below, at a scale that trains from scratch in seconds.

```
# A multilayer perceptron implemented from scratch: every forward map, every
# backward map, and every gradient step is written out explicitly as a tensor
# operation. No automatic differentiation is used anywhere in this block --
# not a single tensor below carries requires_grad.

import torch as th
import matplotlib.pyplot as plt
from sklearn.datasets import load_digits

th.manual_seed(0)

# Every tensor below is float64: the finite-difference gradient check needs
# more precision than the float32 that the libraries use by default.

# --- Affine layer: (F1) Z = H W + 1 b^T -----------------------------------

class Affine:
    def __init__(self, k_in, k_out):
        # He initialisation: Var[w] = 2/k_in keeps Var[z^l] ~ Var[z^{l-1}]
        self.W = th.randn(k_in, k_out, dtype=th.float64) * (2.0 / k_in) ** 0.5
        self.b = th.zeros(k_out, dtype=th.float64)

    def forward(self, H):
        self.H = H                          # cached: needed by (B3)
        return H @ self.W + self.b          # (b,k_in)(k_in,k_out) -> (b,k_out)

    def backward(self, Delta):
        # Delta = dR/dZ^l, shape (b,k_out). One line per identity:
        self.dW = self.H.T @ Delta          # (B3)  dR/dW_l = (H^{l-1})^T Delta^l
        self.db = Delta.sum(dim=0)          # (B4)  dR/db^l = (Delta^l)^T 1
        return Delta @ self.W.T             # (B2)  first factor: Delta^l W_l^T

    def step(self, alpha):
        self.W -= alpha * self.dW           # gradient step on the weights
```

```
        self.b -= alpha * self.db        # gradient step on the biases

# --- Activation layer: (F2) H = g(Z), with g the ReLU -----------------------

class ReLU:
    def forward(self, Z):
        self.mask = (Z > 0.0).to(Z.dtype)  # cached: this IS g'(Z)
        return Z * self.mask               # g(z) = max(0,z)

    def backward(self, dH):
        return dH * self.mask            # (B2) second factor: o g'(Z^{l-1})

    def step(self, alpha):
        pass                             # no parameters to update

# --- Loss layer: softmax + cross entropy, differentiated jointly ------------

class SoftmaxCrossEntropy:
    def forward(self, Z, Y):
        Z = Z - Z.max(dim=1, keepdim=True).values     # shift: exp cannot
↪   overflow
        E = Z.exp()
        self.Yhat = E / E.sum(dim=1, keepdim=True)    # softmax probabilities
        self.Y = Y                                    # one-hot targets (see
        ↪   above)
        return -(Y * (self.Yhat + 1e-12).log()).sum() / Z.shape[0]

    def backward(self):
        return (self.Yhat - self.Y) / self.Y.shape[0] # (B1) delta^H = yhat - y

# --- The network: a list of layers, traversed forwards, then backwards ------

class MLP:
    def __init__(self, sizes):
        self.layers = []
        for l in range(len(sizes) - 1):
            self.layers.append(Affine(sizes[l], sizes[l + 1]))
            if l < len(sizes) - 2:                    # no activation on output
                self.layers.append(ReLU())
        self.loss = SoftmaxCrossEntropy()

    def forward(self, X, Y):
        H = X                                         # H^0 := X
        for layer in self.layers:                     # l = 1 -> H
            H = layer.forward(H)
        return self.loss.forward(H, Y)
```

```
    def backward(self):
        Delta = self.loss.backward()                    # delta^H
        for layer in reversed(self.layers):             # l = H -> 1
            Delta = layer.backward(Delta)

    def step(self, alpha):
        for layer in self.layers:
            layer.step(alpha)

    def predict(self, X):
        H = X
        for layer in self.layers:
            H = layer.forward(H)
        return H.argmax(dim=1)

# --- Gradient check: analytic backward pass vs. central finite differences --

net = MLP([5, 4, 3])
Xc = th.randn(6, 5, dtype=th.float64)
Yc = th.eye(3, dtype=th.float64)[th.randint(0, 3, (6,))]
net.forward(Xc, Yc)
net.backward()
W = net.layers[0].W                                     # differentiate w.r.t. W_1
dW_analytic = net.layers[0].dW.clone()
eps, dW_numeric = 1e-6, th.zeros_like(W)
for i in range(W.shape[0]):
    for j in range(W.shape[1]):
        W[i, j] += eps;     Rp = net.forward(Xc, Yc)
        W[i, j] -= 2 * eps; Rm = net.forward(Xc, Yc)
        W[i, j] += eps
        dW_numeric[i, j] = (Rp - Rm) / (2 * eps)
print("max |analytic - numeric| gradient error:",
      f"{(dW_analytic - dW_numeric).abs().max():.2e}")

# --- The pipeline: 8x8 handwritten digits, 1797 samples, 10 classes ---------

digits = load_digits()
X_all = th.tensor(digits.data, dtype=th.float64) / 16.0  # pixel range [0,1]
y_all = th.tensor(digits.target, dtype=th.long)

idx = th.randperm(y_all.numel())
n_tr = int(0.8 * y_all.numel())
tr, te = idx[:n_tr], idx[n_tr:]
mu, sd = X_all[tr].mean(dim=0), X_all[tr].std(dim=0) + 1e-8
X_tr, X_te = (X_all[tr] - mu) / sd, (X_all[te] - mu) / sd
y_tr, y_te = y_all[tr], y_all[te]
Y_tr = th.eye(10, dtype=th.float64)[y_tr]  # one-hot targets: row i of the
↪  identity matrix, a 1 in the true class's slot and 0 elsewhere
```

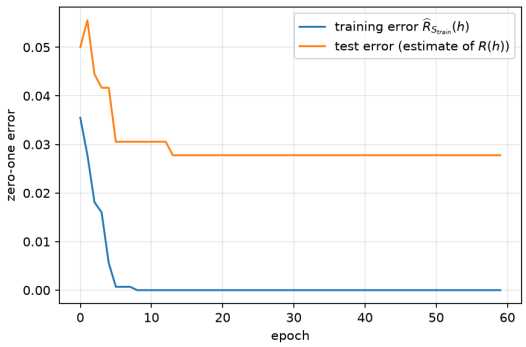


Figure 10.4: Learning curve of the from-scratch multilayer perceptron on the digits data set: training error reaches zero while test error settles near 3%.

```
net = MLP([64, 64, 32, 10])                              # d=64 -> 64 -> 32 -> C=10
alpha, batch_size, epochs = 0.5, 32, 60
curve_tr, curve_te = [], []

for epoch in range(epochs):
    order = th.randperm(n_tr)                            # B ~ Uniform(S), |B| = b
    for start in range(0, n_tr, batch_size):
        B = order[start:start + batch_size]
        net.forward(X_tr[B], Y_tr[B])                    # forward pass  (F1),(F2)
        net.backward()                                   # backward pass (B1)-(B4)
        net.step(alpha)                                  # stochastic gradient step
    curve_tr.append((net.predict(X_tr) != y_tr).double().mean())
    curve_te.append((net.predict(X_te) != y_te).double().mean())

print(f"train error: {curve_tr[-1]:.4f}   test error: {curve_te[-1]:.4f}")

plt.figure(figsize=(6, 4))
plt.plot(curve_tr, label="training error $\\widehat{R}_{S_{train}}(h)$")
plt.plot(curve_te, label="test error (estimate of $R(h)$)")
plt.xlabel("epoch"); plt.ylabel("zero-one error"); plt.legend();
 ↪  plt.grid(alpha=.3)
plt.tight_layout()
plt.show()
```

```
max |analytic - numeric| gradient error: 9.15e-11
train error: 0.0000   test error: 0.0278
```

The analytic and numerical gradients agree to within 10^{-10}, confirming that the four backward identities are implemented correctly. The learning curve shows the training error driven to exactly 0 — the network has $64 \cdot 64 + 64 \cdot 32 + 32 \cdot 10 = 6{,}464$ weights against $m = 1437$ training examples, so it can interpolate the training set outright — while the test error settles near 3%. This is the empirical form

of the puzzle the next section addresses: a hypothesis class whose VC dimension far exceeds m, achieving $\widehat{R}_{S_{train}}(h) = 0$ on its training set, and generalizing anyway.

10.5 PAC Analysis of Neural Networks

A two-layer network with k hidden units and unconstrained real-valued weights has $O(k \cdot (d+1) + k)$ trainable parameters, and its VC dimension grows accordingly — for the sign-thresholded output of a network with piecewise-linear activations (e.g. ReLU), d_{VC} scales at least linearly, and can scale super-linearly, in the number of weights. For any network wide enough to be useful in practice, this VC dimension vastly exceeds the number of training examples m, so the VC-dimension bound of Theorem 6.4.7 is **vacuous**: it provides no guarantee, since its right-hand side exceeds 1. Yet, deep networks with far more parameters than training examples routinely generalize well. Resolving this puzzle needs a complexity measure that does not grow with the number of parameters — exactly the motivation behind Rademacher complexity (Chapter 7). We now derive such a bound for a simplified network, using only the tools already built in Chapter 7.

Simplified setup. Consider a network with a single hidden layer of k units and a scalar output,

$$h_{W,w}(x) = w^\top g(Wx),$$
$$W \in \mathbb{R}^{k \times d},\ w \in \mathbb{R}^k,$$

where g is applied elementwise. We assume:

- g is 1-Lipschitz with $g(0) = 0$ — true for ReLU (and for tanh, but *not* for the sigmoid, whose offset $g(0) = \frac{1}{2}$ would need to be carried through the argument separately);
- every row $w_j^{(1)}$ of W (the incoming weight vector of hidden unit j) satisfies $\|w_j^{(1)}\|_2 \le B_1$;
- the output weights satisfy $\|w\|_2 \le B_2$;
- $\|x\|_2 \le R$ almost surely under $\mathcal{D}$.

Write $\mathcal{H}_{k,B_1,B_2}$ for the resulting hypothesis class. Note that k, B_1, B_2 are hyperparameters we are free to fix in advance — capping them plays exactly the role that capping the leaf budget L played for decision trees in Chapter 9's own PAC analysis.

Theorem 10.5.1: Rademacher complexity of a two-layer network

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_{k,B_1,B_2}) \le \frac{2\sqrt{k}\, B_1 B_2 R}{\sqrt{m}}.$$

Proof. Fix the sample $S = \{x_1, \dots, x_m\}$. Since $h_{W,w}(x) = \sum_{j=1}^k w_j\, g(w_j^{(1)\top} x)$ is linear in w, the supremum over $\|w\|_2 \le B_2$ of a linear functional is attained by Cauchy-Schwarz, for every fixed realization of the Rademacher signs $\sigma_1, \dots, \sigma_m$:

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_{k,B_1,B_2}) = \mathbb{E}_\sigma\Big[\sup_{W,w} \frac{1}{m} \sum_{i=1}^m \sigma_i\, w^\top g(Wx_i)\Big] = \frac{B_2}{m}\, \mathbb{E}_\sigma\Big[\sup_{W} \Big\| \sum_{i=1}^m \sigma_i\, g(Wx_i)\Big\|_2\Big].$$

Write $v_j(W) := \sum_i \sigma_i\, g(w_j^{(1)\top} x_i)$ for the j-th coordinate of the inner vector, where $w_j^{(1)}$ is the j-th row of W. Bounding the Euclidean norm by $\sqrt{k}$ times the sup-norm, and using that the k rows range independently over identical B_1-balls (so the outer supremum and the coordinate-wise maximum commute),

$$\sup_W \big\|v(W)\big\|_2 \le \sqrt{k}\, \sup_W \max_j |v_j(W)| = \sqrt{k}\, \max_j \sup_{\|w_j^{(1)}\|_2 \le B_1} |v_j(W)|$$
$$= \sqrt{k}\, \sup_{\|u\|_2 \le B_1} \Big| \sum_i \sigma_i\, g(u^\top x_i) \Big|,$$

where the last equality holds because every one of the k terms being maximized over is, by construction, the *same* scalar quantity — the row index j only labels which weight vector we call u, the constraint set $\{\|u\|_2 \le B_1\}$ and the data $x_1, \dots, x_m$ being identical for every hidden unit. This collapses a k-fold maximum into a single scalar supremum, over a class with a single B_1-norm-bounded weight vector. The supremum is of an absolute value, so the contraction lemma enters in its signed form (7.1) of Chapter 7, which applies because the set $\{(u^\top x_1, \dots, u^\top x_m) : \|u\|_2 \le B_1\}$ contains 0 and $g(0) = 0$; it costs a factor 2:

$$\mathbb{E}_\sigma\Big[\sup_{\|u\|_2 \le B_1} \Big| \sum_i \sigma_i\, g(u^\top x_i) \Big| \Big] \le 2\, \mathbb{E}_\sigma\Big[\sup_{\|u\|_2 \le B_1} \sum_i \sigma_i\, u^\top x_i \Big] = 2B_1\, \mathbb{E}_\sigma\Big[\Big\| \sum_i \sigma_i x_i \Big\|_2 \Big],$$

the last equality by Cauchy-Schwarz. Finally, by Jensen's inequality and $\mathbb{E}[\sigma_i \sigma_{i'}] = 0$ for $i \ne i'$,

$$\mathbb{E}_\sigma\Big[\Big\| \sum_i \sigma_i x_i \Big\|_2 \Big] \le \sqrt{\mathbb{E}_\sigma\Big[\Big\| \sum_i \sigma_i x_i \Big\|_2^2 \Big]} = \sqrt{\sum_i \|x_i\|_2^2} \le R\sqrt{m}.$$

Assembling the pieces, $\widehat{\mathfrak{R}}_S(\mathcal{H}_{k,B_1,B_2}) \le \frac{B_2}{m} \cdot \sqrt{k} \cdot 2B_1 \cdot R\sqrt{m} = \frac{2\sqrt{k}\, B_1 B_2 R}{\sqrt{m}}$. □

Consequences. Plugging Theorem 10.5.1 into Theorem 7.3.2 (after one more application of the contraction lemma to pass from $\mathcal{H}_{k,B_1,B_2}$ to a Lipschitz loss composed with it) gives, with probability at least $1 - \delta$, simultaneously for every $h \in \mathcal{H}_{k,B_1,B_2}$,

$$R(h) \le \widehat{R}_S(h) + O\Big(\frac{\sqrt{k}\, B_1 B_2 R}{\sqrt{m}} \Big) + O\Big(\sqrt{\frac{\log(1/\delta)}{m}} \Big).$$

Three aspects of this bound are worth noting:

- It **does not depend on the input dimension** d — only on the per-neuron weight-norm bound B_1, the output-norm bound B_2, the data-norm bound R, and the width k. This is precisely the phenomenon flagged in Theorem 7.3.6: capacity control by norm, rather than by counting parameters, is what makes the bound meaningful for models with far more parameters than VC-dimension arguments could ever tolerate.

- It scales with $\sqrt{k}$, not with the number of parameters $k(d+1)$. Width still incurs a cost — but far less than a naive parameter-counting argument (à la Theorem 6.4.7) would suggest. This residual $\sqrt{k}$ is an *artifact of the L_2 constraint* on the output weights: Theorem 10.5.2 below shows that it disappears once the output weights are constrained in the L_1 norm instead.

- The bound gives no credit to gradient descent for finding a good hypothesis — it only certifies that *if* training happens to land on a hypothesis with small weight norms, that hypothesis generalizes well. This matches an empirical regularity of trained networks: gradient descent on overparameterized networks tends to find solutions with small weight norms even without explicit regularization, an implicit-bias phenomenon that is an active research topic — and it is exactly why this norm-based, width-independent view of capacity, rather than the VC-dimension view, is the one that matches practice. We revisit this network in a further-simplified, exactly solvable form — where only the output layer is trained and the hidden layer is fixed — in the Neural Tangent Kernel discussion of Chapter 11.

Theorem 10.5.2: Width-independent bound for an L_1-constrained output layer

In the same setup as Theorem 10.5.1 — g 1-Lipschitz with $g(0) = 0$, $\|w_j^{(1)}\|_2 \le B_1$ per hidden unit, $\|x\|_2 \le R$ — replace the output constraint $\|w\|_2 \le B_2$ by

$$\|w\|_1 := \sum_{j=1}^{k} |w_j| \;\le\; B_2,$$

and write $\mathcal{H}^{(1)}_{k,B_1,B_2}$ for the resulting class. Then for every sample S,

$$\widehat{\mathfrak{R}}_S\big(\mathcal{H}^{(1)}_{k,B_1,B_2}\big) \;\le\; \frac{2B_1B_2R}{\sqrt{m}}.$$

Proof. Fix the signs σ and abbreviate $c(u) := \frac{1}{m}\sum_{i=1}^{m} \sigma_i\, g(u^\top x_i)$ for the correlation of a single hidden unit u. A hypothesis restricted to S is the vector $\big(\sum_j w_j g(w_j^{(1)\top} x_i)\big)_{i=1}^m$, so

$$\sup_{W,w} \frac{1}{m}\sum_{i=1}^{m} \sigma_i h_{W,w}(x_i) \;=\; \sup_{W,w} \sum_{j=1}^{k} w_j\, c\big(w_j^{(1)}\big) \;\le\; \sup_{W,w} \sum_{j=1}^{k} |w_j|\,\big|c\big(w_j^{(1)}\big)\big| \;\le\; \Big(\sum_{j=1}^{k} |w_j|\Big) \sup_{\|u\|_2 \le B_1} |c(u)| \;\le\; B_2 \sup_{\|u\|_2}$$

where the second inequality pulls the total L_1 mass out of the sum, and the third applies the constraint. The point is that the sup over W of the individual correlations $|c(w_j^{(1)})|$ collapses to a single scalar supremum, exactly as in Theorem 10.5.1's proof — the row index j labels identical constraint sets, so maximizing the weighted sum with an L_1 budget puts all mass on the single best hidden unit (a vertex of the L_1 ball). Taking $\mathbb{E}_\sigma$ and applying the same chain as in Theorem 10.5.1's proof (signed contraction (7.1) for the 1-Lipschitz g with $g(0) = 0$, Cauchy-Schwarz, Jensen) gives

$$\widehat{\mathfrak{R}}_S\big(\mathcal{H}^{(1)}_{k,B_1,B_2}\big) \;\le\; B_2\,\mathbb{E}_\sigma\Big[\sup_{\|u\|_2 \le B_1} \big|c(u)\big|\Big] \;\le\; B_2 \cdot \frac{2B_1R\sqrt{m}}{m} \;=\; \frac{2B_1B_2R}{\sqrt{m}}. \qquad \square$$

Bounds of this type, in which the size of the weights rather than the size of the network controls capacity, go back to (Bartlett, 1998). The right-hand side has no dependence on the width k. The comparison between Theorems 10.5.1 and 10.5.2 is the cleanest illustration of **Theorem 7.7.2's moral in action**.

The $\sqrt{k}$ in Theorem 10.5.1 is precisely the cost of the L_2 constraint, measured in L_1 terms: by norm equivalence $\|w\|_1 \le \sqrt{k}\,\|w\|_2$, an L_2-constrained network has total output weight at most $\sqrt{k}\,B_2$, and Theorem 10.5.2 charges exactly that total — so Theorem 10.5.1 is recovered by norm conversion. Conversely, the L_1 form shows that the width is a *bookkeeping device*: the class of L_1-constrained combinations is no richer, in Rademacher terms, than the class of its single hidden units (scaled by the budget B_2), no matter how large k is. The same accounting by total weight underlies the gradient-boosting guarantee of Chapter 12: the price of an ensemble is its total weight, not its member count. Both theorems still constrain each hidden unit separately; the refinement of Golowich, Rakhlin & Shamir (2018) extends the accounting to norms of the hidden weight matrices, and to deeper networks.

Why does SGD generalize? The bounds of this section are *capacity-based*: they certify any low-norm hypothesis the training run happens to land on, and give the training algorithm no credit. A complementary analysis prices the algorithm itself, through **uniform stability** (Definition 7.8.1 and Theorem 7.8.2 of Chapter 7). We carry it out for SGD with single-example updates on a loss $\phi(w, z)$ that is convex in the parameters $w \in \mathbb{R}^p$ — for instance logistic regression (Chapter 4), or a network whose hidden layer is frozen and only the output layer is trained. Two properties of smooth functions are needed. A differentiable $f : \mathbb{R}^p \to \mathbb{R}$ is β-**smooth** if its gradient is β-Lipschitz, $\|\nabla f(u) - \nabla f(v)\|_2 \le \beta\|u - v\|_2$. □

Lemma 10.5.3: Smooth convex functions

Let f be β-smooth. Then for all u, v:
(i) $f(v) \le f(u) + \nabla f(u)^\top (v - u) + \frac{\beta}{2}\|v - u\|_2^2$;
(ii) if f is also convex, $\big(\nabla f(u) - \nabla f(v)\big)^\top (u - v) \ge \frac{1}{\beta}\|\nabla f(u) - \nabla f(v)\|_2^2$;
(iii) if f is also convex and $0 < \alpha \le 2/\beta$, the gradient step $G(u) := u - \alpha\nabla f(u)$ satisfies $\|G(u) - G(v)\|_2 \le \|u - v\|_2$.

Proof. *(i)* By the fundamental theorem of calculus along the segment from u to v, Cauchy-Schwarz and smoothness,

$$f(v)-f(u)-\nabla f(u)^\top(v-u) = \int_0^1 \big(\nabla f(u+t(v-u))-\nabla f(u)\big)^\top (v-u)\,dt \;\le\; \int_0^1 \beta t\,\|v-u\|_2^2\,dt = \frac{\beta}{2}\|v-u\|_2^2.$$

(ii) Fix u and let $\psi(y) := f(y) - \nabla f(u)^\top y$. It is convex and β-smooth with $\nabla\psi(u) = 0$, so u minimizes ψ. Applying (i) to ψ at the point $y - \frac{1}{\beta}\nabla\psi(y)$,

$$\psi(u) \;\le\; \psi\big(y-\tfrac{1}{\beta}\nabla\psi(y)\big) \;\le\; \psi(y)-\tfrac{1}{\beta}\|\nabla\psi(y)\|_2^2+\tfrac{1}{2\beta}\|\nabla\psi(y)\|_2^2 = \psi(y)-\tfrac{1}{2\beta}\|\nabla f(y)-\nabla f(u)\|_2^2.$$

Written out, $f(u) \le f(y) + \nabla f(u)^\top(u - y) - \frac{1}{2\beta}\|\nabla f(y) - \nabla f(u)\|_2^2$. Adding this inequality to the same one with u and y exchanged, and setting $y = v$, gives (ii).

(iii) Expanding the square and using (ii),

$$\|G(u)-G(v)\|_2^2 = \|u-v\|_2^2-2\alpha\big(\nabla f(u)-\nabla f(v)\big)^\top(u-v)+\alpha^2\|\nabla f(u)-\nabla f(v)\|_2^2 \;\le\; \|u-v\|_2^2-\alpha\Big(\frac{2}{\beta}-\alpha\Big)\|\nabla f(u$$

and the last term is non-positive for $\alpha \le 2/\beta$ □

All three facts are standard in convex optimization; see Chapter 2 of (Nesterov, 2004).

Theorem 10.5.4: SGD is uniformly stable

Let $\phi(w, z)$ be convex and β-smooth in w, with $\|\nabla_w \phi(w, z)\|_2 \le G$ for all w, z. Given $S = (z_1, \dots, z_m)$, let SGD start from a fixed w_0, draw i_t uniformly from $[m]$ independently at each step, and update

$$w_{t+1} := w_t - \alpha_t \nabla_w \phi(w_t, z_{i_t}), \qquad 0 < \alpha_t \le 2/\beta, \qquad t = 0, \dots, T-1.$$

The algorithm that returns w_T is ϵ-uniformly stable with

$$\epsilon \;=\; \frac{2G^2}{m} \sum_{t=0}^{T-1} \alpha_t.$$

Proof. Let S' differ from S in position i only, and run SGD on both with the **same** index sequence $i_0, i_1, \dots$; this coupling does not change the law of either run. Let w_t, w'_t be the two iterates and $\delta_t := \|w_t - w'_t\|_2$, so $\delta_0 = 0$. If $i_t \ne i$, both runs apply the gradient step of the *same* function $\phi(\cdot, z_{i_t})$, and Lemma 10.5.3(iii) gives $\delta_{t+1} \le \delta_t$. If $i_t = i$, the triangle inequality and the gradient bound give $\delta_{t+1} \le \delta_t + \alpha_t \|\nabla_w \phi(w_t, z_i)\|_2 + \alpha_t \|\nabla_w \phi(w'_t, z'_i)\|_2 \le \delta_t + 2\alpha_t G$. The second case occurs with probability $\frac{1}{m}$, independently of δ_t, so

$$\mathbb{E}[\delta_{t+1}] \;\le\; \mathbb{E}[\delta_t] + \frac{2\alpha_t G}{m}, \qquad \text{hence} \qquad \mathbb{E}[\delta_T] \le \frac{2G}{m} \sum_{t<T} \alpha_t.$$

The gradient bound makes $\phi(\cdot, z)$ G-Lipschitz, so for every z, $\big|\mathbb{E}\,\phi(w_T, z) - \mathbb{E}\,\phi(w'_T, z)\big| \le G\,\mathbb{E}[\delta_T]$ □

Plugging Theorem 10.5.4 into Theorem 7.8.2 bounds the generalization gap of the *iterate SGD actually returns*, with no complexity term:

$$\mathbb{E}_S \mathbb{E}_{\text{SGD}} \big[R(w_T) - \widehat{R}_S(w_T) \big] \;\le\; \frac{2G^2}{m} \sum_{t<T} \alpha_t.$$

The bound grows with the number of steps: running longer lowers the training loss but raises the stability parameter, an optimization–generalization trade-off expressed entirely through the step sizes. One pass over the data ($T = m$) with constant step $\alpha = c/\sqrt{m}$ gives a gap of at most $2cG^2/\sqrt{m}$. The result is due to (Hardt et al., 2016), who also treat strongly convex losses, where the gap is $O(1/m)$ uniformly in T, and non-convex smooth losses, where it grows polynomially in T. Convexity enters the proof only through Lemma 10.5.3(iii); for a full network trained end to end it fails, and neither the capacity nor the stability analysis is sharp — the two are complementary partial explanations of why over-parameterized networks generalize.

Nearly every neural network used nowadays is trained by minimizing a loss function using stochastic gradient descent with backpropagation. The common practice is to use an automatic differentiation library such as TensorFlow or PyTorch that automates the process of calculating the gradients. These

libraries also provide a variety of neural network architectures and optimization algorithms. Below, we give an example implementation of a neural network using PyTorch on a handwritten digit classification problem. The data set used is among the most famous data sets in machine learning, called MNIST. It contains 70,000 images of handwritten digits. Each image is a 28x28 grayscale image. The goal is to classify the images into 10 classes, one for each digit.

```
# MNIST digits classification with a simple deep neural network,
# this time written the way it is written in practice: with the library's
# own layer, loss and optimizer objects standing in for the code above.

import torch as th
from torch import nn
from torch.utils.data import DataLoader
from torchvision import datasets, transforms

# Load the data from torchvision

# Define the transformations
transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.1307,), (0.3081,))
])

# Download the data
train_ds = datasets.MNIST('data', train=True,
            download=True, transform=transform)
test_ds = datasets.MNIST('data', train=False,
            download=True, transform=transform)

# Create the data loaders
batch_size = 64 # Define the batch size
train_dl = DataLoader(train_ds, batch_size=batch_size, shuffle=True)
test_dl = DataLoader(test_ds, batch_size=batch_size, shuffle=False)

# Define the model
model = nn.Sequential(
    nn.Linear(784, 256),
    nn.ReLU(),
    nn.Linear(256, 512),
    nn.ReLU(),
    nn.Linear(512, 1024),
    nn.ReLU(),
    nn.Linear(1024, 10)
)

# Define the loss function
loss_fn = nn.CrossEntropyLoss()
```

```
# Define the optimizer
optimizer = th.optim.SGD(model.parameters(), lr=1e-4)

# Train the model, tracking the training loss and test accuracy after
# every epoch so that we can plot a proper learning curve instead of a
# wall of per-batch print statements.
num_epochs = 40
train_losses, test_accuracies = [], []

def evaluate(model, test_dl):
    model.eval()
    correct = 0
    for xb, yb in test_dl:
        y_pred = model(xb.view(-1, 784))
        preds = y_pred.argmax(dim=1)
        correct += (preds == yb).float().mean().item()
    model.train()
    return correct / len(test_dl)

for epoch in range(num_epochs):
    epoch_loss = 0.0
    for xb, yb in train_dl:
        # Forward pass
        y_pred = model(xb.view(-1, 784))
        loss = loss_fn(y_pred, yb)

        # Backward pass
        loss.backward()
        optimizer.step()
        optimizer.zero_grad()
        epoch_loss += loss.item()

    train_losses.append(epoch_loss / len(train_dl))
    test_accuracies.append(evaluate(model, test_dl))

print(f"final training loss: {train_losses[-1]:.4f}   "
      f"final test accuracy: {test_accuracies[-1]:.4f}")

fig, axes = plt.subplots(ncols=2, figsize=(11, 4))
axes[0].plot(range(1, num_epochs + 1), train_losses)
axes[0].set_xlabel("epoch"); axes[0].set_ylabel("training loss");
↪  axes[0].grid(alpha=.3)
axes[1].plot(range(1, num_epochs + 1), test_accuracies, color="tab:red")
axes[1].set_xlabel("epoch"); axes[1].set_ylabel("test accuracy");
↪  axes[1].grid(alpha=.3)
plt.tight_layout()
```

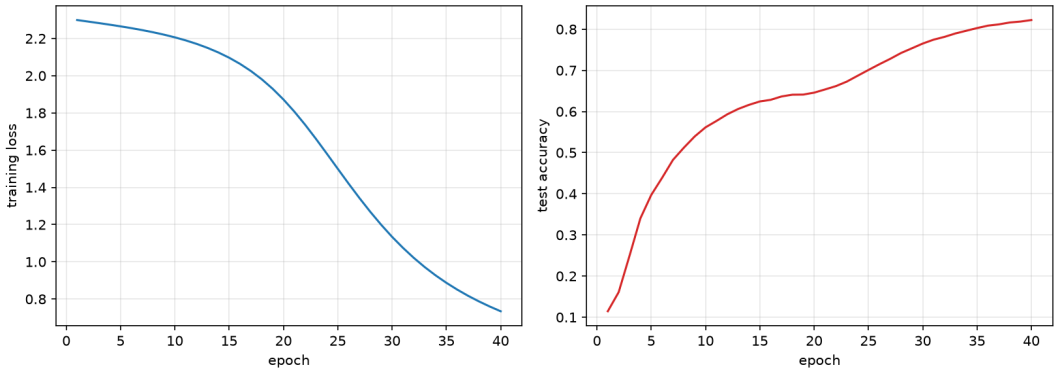


Figure 10.5: Learning curve of the library-based MLP on MNIST: training cross-entropy loss (left) and test accuracy (right) across epochs of plain SGD.

```
plt.show()
```

CHAPTER 11

KERNEL METHODS

11.1 Transforming the input space

Linear predictors have many desirable properties. They:

- can be trained by analytical methods,
- are easy to interpret for humans,
- are theoretically well understood.

However, their expressiveness is limited. For example, they cannot represent the XOR function. The first solution we presented to improve the expressiveness of linear predictors was to pass them through an activation function and build a network of them. The second is **kernel methods**. The core idea is to transform the input space to another space where the problem can be solved by linear methods. For instance, in classification, we expect from the transformed space that there exists a hyperplane that perfectly separates the classes. This way, nonlinearity is handled by the transformation and all the nice properties of linear predictors are preserved.

Consider a 2D feature space in which a data point $x = (x_1, x_2) \in \mathbb{R}^2$ belongs to one of two classes: one class is concentrated around the origin in the shape of a sphere and the other surrounds this sphere in the shape of a ring. The data are not linearly separable in the original space. However, if we transform the data to a new space $\phi(x) = (x_1^2, x_2^2, \sqrt{2}x_1x_2) \in \mathbb{R}^3$, the classes become linearly separable as illustrated below.

```
import torch as th
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

plt.rcParams['text.usetex'] = True # Enable Latex in plots

# Step 1: Generate random samples for two classes
th.manual_seed(0)

# Number of samples for each class
num_samples = 100
```

```
# Class 1:
radius1 = 3
theta1 = 2 * th.pi * th.rand(num_samples)
r1 = th.rand(num_samples).sqrt() * radius1
class1_x = r1 * th.cos(theta1)
class1_y = r1 * th.sin(theta1)

# Class 2
radius2 = 5
theta2 = 2 * th.pi * th.rand(num_samples)
r2 = (-0.5 + th.rand(num_samples)) + radius2
class2_x = r2 * th.cos(theta2)
class2_y = r2 * th.sin(theta2)

# Step 2: Define the transformation function to 3D
def transform_to_3d(x, y):
    x_3d = x**2
    y_3d = y**2
    z_3d = (2 ** 0.5) * x * y
    return x_3d, y_3d, z_3d

# Apply the transformation to both classes
class1_x_3d, class1_y_3d, class1_z_3d = transform_to_3d(class1_x, class1_y)
class2_x_3d, class2_y_3d, class2_z_3d = transform_to_3d(class2_x, class2_y)

# Create a figure with two subplots
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# The classes are not linearly separable in 2D
axes[0].scatter(class1_x, class1_y, label='Class 1', c='b')
axes[0].scatter(class2_x, class2_y, label='Class 2', c='r')
axes[0].set_xlabel('$x_1$', fontsize=14)
axes[0].set_ylabel('$x_2$', fontsize=14)
axes[0].set_title('Original 2D Data')
axes[0].legend(loc='best')

# The classes are linearly separable in 3D after transformation
ax = fig.add_subplot(1, 2, 2, projection='3d', proj_type='ortho')

ax.scatter(class1_x_3d, class1_y_3d, class1_z_3d, label='Class 1', c='b')
ax.scatter(class2_x_3d, class2_y_3d, class2_z_3d, label='Class 2', c='r')
ax.set_xlabel('$x_1^2$', fontsize=14)
ax.set_ylabel('$x_2^2$', fontsize=14)
ax.set_zlabel('$\sqrt{2}  x_1  x_2$')

ax.legend(loc='best')
```

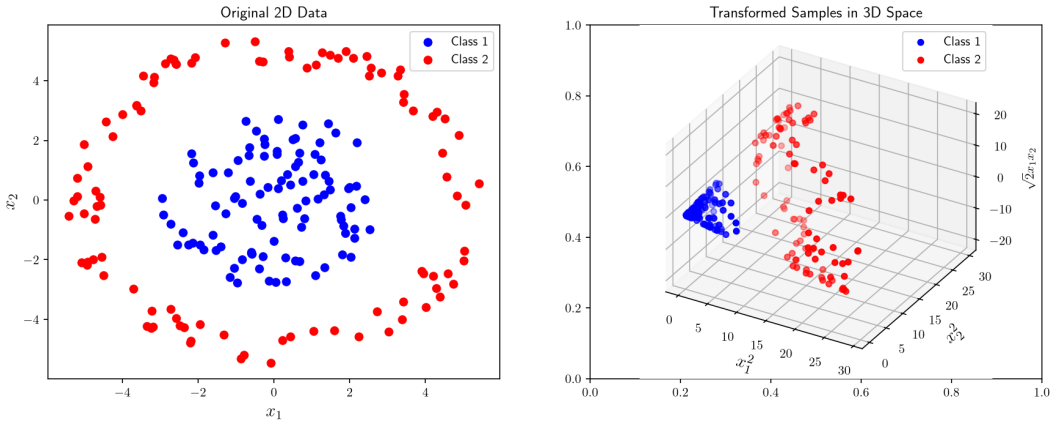


Figure 11.1: Two classes that are not linearly separable in the original 2D input space (left) become linearly separable after the feature transformation $\phi(x) = (x_1^2, x_2^2, \sqrt{2}x_1x_2)$ (right).

```
ax.set_title('Transformed Samples in 3D Space')

plt.show()
```

11.2 Kernel Trick

In real-world applications, it is not as straightforward to find a suitable transformation space as in the example above, one that would make the problem linearly separable. The best one can often do is to transform the data to a space that is as high dimensional as possible. However, this is computationally expensive. There are attractive mathematical tools that allow us to model the **similarity** of a pair of data points in the transformation space without explicitly computing the representations of these data points in the transformation space. This can be achieved by a special family of functions, called **kernel functions** with certain characteristics. Given an input space $\mathcal{X}$, we would like to have a kernel function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ with the following property:

$$k(x_i, x_j) = \phi(x_i)^\top \phi(x_j).$$

In words, we would like to design directly the left hand side of the equation above without going through the computational steps of the right hand side. This is called the **kernel trick**. This is possible for only certain types of functions that have specific properties. The following result, due to (Mercer, 1909), gives a necessary and sufficient condition for a function to be a valid kernel function.

Theorem 11.2.1: Mercer

A symmetric function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ admits a feature map ϕ with $k(x, x') = \phi(x)^\top \phi(x')$ **if and only if** the Gram matrix it generates on an arbitrary data set $\{x_1, \dots, x_m\}$

$$K = \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_m) \\ \vdots & \ddots & \vdots \\ k(x_m, x_1) & \cdots & k(x_m, x_m) \end{bmatrix}$$

is positive semi-definite, i.e. $v^\top K v \geq 0$ for all $v \in \mathbb{R}^m$.

Proof (of the "only if" direction). Suppose $k(x, x') = \phi(x)^\top \phi(x')$ for some feature map ϕ. Then for any $x_1, \dots, x_m$ and any $v \in \mathbb{R}^m$,

$$v^\top K v = \sum_{i=1}^{m} \sum_{j=1}^{m} v_i v_j \, \phi(x_i)^\top \phi(x_j) = \Big\| \sum_{i=1}^{m} v_i \, \phi(x_i) \Big\|_2^2 \;\geq\; 0,$$

since the double sum is exactly the squared norm of the single vector $\sum_i v_i \phi(x_i)$. Symmetry of K is immediate from symmetry of the inner product □

The converse — that positive semi-definiteness *suffices*, i.e. that some feature map always exists — is the substantial half of the theorem, and it is proved constructively in the section on reproducing kernel Hilbert spaces below (Theorem 11.3.2), where the feature map is built as $\phi(x) := k(x, \cdot)$ and the feature space as a space of functions on $\mathcal{X}$. Until then, the "only if" direction just proved, together with the composition rules below, is already enough to certify every kernel we actually employ.

The most commonly used kernel functions are:

- **Linear Kernel:** $k(x, y) = x^\top y$
- **Polynomial Kernel:** $k(x, y) = \big(x^\top y + c\big)^d$
- **Radial Basis Function (RBF) Kernel:** $k(x, y) = \exp(-\frac{\|x-y\|^2}{2\sigma^2})$
- **Sigmoid Kernel:** $k(x, y) = \tanh\big(\gamma x^\top y + c\big)$

The first three are valid kernels for all parameter values ($c \geq 0$ for the polynomial kernel); this is proved below. The sigmoid kernel, borrowed from neural networks, is **not** positive semi-definite in general: for $c < 0$, $k(0, 0) = \tanh(c) < 0$, so already the 1×1 Gram matrix of the single point $x = 0$ violates the condition of Theorem 11.2.1.

There also exist kernels that compute the similarity of a pair of graphs, strings, or other complex objects. The nice property of the kernel functions is that one can compute these similarities directly, without needing to define suitable feature spaces for these object types.

One can imagine that it is not straightforward to devise a function that satisfies this condition. However, there are certain rules that allow us to construct new valid kernel functions from existing ones. These rules are called **kernel composition rules**.

Proposition 11.2.2

Kernel composition rules. Let $k_1, k_2, k_3, \ldots$ be valid kernels, $a \in \mathbb{R}^+$ a constant, B a positive semi-definite matrix, and $q : \mathcal{X} \to \mathbb{R}$ any function. Then each of the following is a valid kernel:

- $k(x, x') = k_1(x, x') + k_2(x, x')$
- $k(x, x') = a \cdot k_1(x, x')$
- $k(x, x') = k_1(x, x') \cdot k_2(x, x')$
- $k(x, x') = x^\top B x'$
- $k(x, x') = q(x)\, k_1(x, x')\, q(x')$
- $k(x, x') = \lim_{n\to\infty} k_n(x, x')$, whenever the limit exists for all x, x'.

Proof. By Theorem 11.2.1 it suffices to verify that each rule produces a positive semi-definite Gram matrix. Write K_1, K_2 for the Gram matrices of k_1, k_2 on an arbitrary sample, both positive semi-definite by hypothesis.

Sum. $v^\top (K_1 + K_2) v = v^\top K_1 v + v^\top K_2 v \geq 0$.

Positive scaling. $v^\top (aK_1) v = a\,(v^\top K_1 v) \geq 0$ since $a > 0$.

Product. The Gram matrix of the product kernel is the entrywise (Hadamard) product $K_1 \odot K_2$, and the **Schur product theorem** states that the Hadamard product of two positive semi-definite matrices is positive semi-definite. Directly: write $K_2 = \sum_r \lambda_r u_r u_r^\top$ with $\lambda_r \geq 0$ (spectral decomposition of a symmetric positive semi-definite matrix). Then for any v,

$$v^\top (K_1 \odot K_2) v = \sum_{i,j} v_i v_j (K_1)_{ij} \sum_r \lambda_r u_{r,i} u_{r,j} = \sum_r \lambda_r \underbrace{\big(v \odot u_r\big)^\top K_1 \big(v \odot u_r\big)}_{\geq 0} \geq 0.$$

Bilinear form. Since B is symmetric positive semi-definite, it has a square root $B^{1/2}$ with $B = B^{1/2} B^{1/2}$, so $x^\top B x' = (B^{1/2} x)^\top (B^{1/2} x')$ exhibits the feature map $\phi(x) := B^{1/2} x$ directly.

Rescaling. The Gram matrix is $D K_1 D$ with $D := \operatorname{diag}(q(x_1), \ldots, q(x_m))$, and $v^\top D K_1 D v = (Dv)^\top K_1 (Dv) \geq 0$.

Limits. The Gram matrices converge entrywise, so $v^\top K v = \lim_n v^\top K_n v \geq 0$ □

These rules certify the kernels listed above. The constant kernel $k \equiv 1$ is valid (its Gram matrix $\mathbf{1}\mathbf{1}^\top$ is positive semi-definite), and the polynomial kernel $(x^\top x' + c)^d$ with $c \geq 0$ is built from it and the linear kernel by d applications of the product rule together with the sum rule and positive scaling (expanding the binomial). For the RBF kernel, expand the square in the exponent:

$$\exp\Big(-\frac{\|x - x'\|^2}{2\sigma^2}\Big) = q(x)\, \exp\Big(\frac{x^\top x'}{\sigma^2}\Big)\, q(x'), \qquad q(x) := \exp\Big(-\frac{\|x\|^2}{2\sigma^2}\Big),$$

and $\exp(x^\top x'/\sigma^2) = \lim_{N\to\infty} \sum_{n=0}^{N} \frac{(x^\top x')^n}{\sigma^{2n}\, n!}$ is a limit of sums of positively scaled powers of the linear kernel. The rescaling and limit rules make it a valid kernel.

11.3 The Reproducing Kernel Hilbert Space

Mercer's theorem guarantees that a positive semi-definite kernel *has* a feature map, but says nothing about which one. There are in general many: the polynomial kernel $(x^\top x')^2$ on $\mathbb{R}^2$ is reproduced by $\phi(x) =$

$(x_1^2, x_2^2, \sqrt{2}x_1x_2)$ and equally by any rotation of it. One choice is canonical, and it turns out to be the one that lets us stop talking about ϕ altogether and talk instead about a **space of functions** — which is what a hypothesis class is. That space is the RKHS of k, following the formulation of (Aronszajn, 1950).

Definition 11.3.1: Reproducing kernel Hilbert space

A **reproducing kernel Hilbert space (RKHS)** for a kernel k on $\mathcal{X}$ is a Hilbert space $\mathcal{H}_k$ of functions $f : \mathcal{X} \to \mathbb{R}$, with inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}_k}$ and induced norm $\|f\|_{\mathcal{H}_k} := \sqrt{\langle f, f \rangle_{\mathcal{H}_k}}$, such that

1. $k(x, \cdot) \in \mathcal{H}_k$ for every $x \in \mathcal{X}$, and
2. **(reproducing property)** $\langle f, k(x, \cdot) \rangle_{\mathcal{H}_k} = f(x)$ for every $f \in \mathcal{H}_k$ and every $x \in \mathcal{X}$.

The reproducing property is the whole point: it says that *evaluating* a function at a point is the same operation as taking an inner product against a fixed element of the space. Applying it twice with $f = k(x', \cdot)$ gives

$$\langle k(x, \cdot),\, k(x', \cdot) \rangle_{\mathcal{H}_k} = k(x, x'),$$

so $\phi(x) := k(x, \cdot)$ is a feature map for k — the **canonical feature map** — whose feature space is $\mathcal{H}_k$ itself. Every function in $\mathcal{H}_k$ is simultaneously a hypothesis and a point of the feature space.

Theorem 11.3.2: Moore-Aronszajn

Every positive semi-definite kernel k has exactly one RKHS.

Proof. *Step 1: a pre-Hilbert space.* Start from the linear span of the kernel sections,

$$\mathcal{V} := \Big\{ f = \sum_{i=1}^{n} a_i\, k(x_i, \cdot) \;:\; n \in \mathbb{N},\; a_i \in \mathbb{R},\; x_i \in \mathcal{X} \Big\},$$

and define, for $f = \sum_i a_i k(x_i, \cdot)$ and $g = \sum_j b_j k(x'_j, \cdot)$,

$$\langle f, g \rangle_{\mathcal{V}} := \sum_{i=1}^{n} \sum_{j=1}^{n'} a_i b_j\, k(x_i, x'_j). \tag{11.1}$$

Well defined. The right-hand side of (11.1) refers to the coefficients a_i and points x_i, which are not unique for a given f. But rewriting it two ways,

$$\sum_{i,j} a_i b_j k(x_i, x'_j) = \sum_{j=1}^{n'} b_j\, f(x'_j) = \sum_{i=1}^{n} a_i\, g(x_i),$$

exhibits it once as a quantity depending on f only through its *values*, and once as a quantity depending on g only through its values. Hence it does not depend on either representation.

Inner product. Bilinearity is immediate from (11.1) and symmetry from $k(x,x') = k(x',x)$. For positive semi-definiteness, $\langle f,f\rangle_{\mathcal{V}} = \sum_{i,j} a_i a_j k(x_i,x_j) = a^\top K a \geq 0$, which is exactly the Gram-matrix condition of Theorem 11.2.1 — this is the *only* place the assumption on k is used, and it is used in full.

Reproducing property. Taking $g = k(x,\cdot)$ (i.e. $n' = 1, b_1 = 1, x'_1 = x$) in (11.1) gives $\langle f, k(x,\cdot)\rangle_{\mathcal{V}} = \sum_i a_i k(x_i,x) = f(x)$.

Definiteness. We must still rule out a non-zero f with $\|f\|_{\mathcal{V}} = 0$. The Cauchy-Schwarz inequality holds for any positive semi-definite bilinear form, so combining it with the reproducing property, for every x,

$$|f(x)| = \big|\langle f, k(x,\cdot)\rangle_{\mathcal{V}}\big| \;\leq\; \|f\|_{\mathcal{V}}\,\|k(x,\cdot)\|_{\mathcal{V}} = \|f\|_{\mathcal{V}}\sqrt{k(x,x)}. \tag{11.2}$$

If $\|f\|_{\mathcal{V}} = 0$ then $f(x) = 0$ for every x, i.e. f is the zero *function*. So $\langle\cdot,\cdot\rangle_{\mathcal{V}}$ is a genuine inner product on $\mathcal{V}$, and $(\mathcal{V}, \langle\cdot,\cdot\rangle_{\mathcal{V}})$ is a pre-Hilbert space satisfying both conditions of Definition 11.3.1.

Step 2: completion. By (11.2), a sequence (f_n) that is Cauchy in $\|\cdot\|_{\mathcal{V}}$ is Cauchy pointwise, $|f_n(x) - f_k(x)| \leq \|f_n - f_k\|_{\mathcal{V}}\sqrt{k(x,x)}$, so it has a pointwise limit. Let $\mathcal{H}_k$ be the set of all functions on $\mathcal{X}$ that arise as pointwise limits of Cauchy sequences in $\mathcal{V}$. It contains $\mathcal{V}$ (constant sequences). The construction rests on one observation:

> *If (f_n) is Cauchy in $\mathcal{V}$ and $f_n \to 0$ pointwise, then $\|f_n\|_{\mathcal{V}} \to 0$.* Indeed, a Cauchy sequence is bounded, $\|f_n\|_{\mathcal{V}} \leq C$. Given $\epsilon > 0$ pick N with $\|f_n - f_N\|_{\mathcal{V}} \leq \epsilon$ for $n \geq N$, and write $f_N = \sum_i a_i k(x_i,\cdot)$. Then $\|f_n\|^2_{\mathcal{V}} = \langle f_n, f_n - f_N\rangle_{\mathcal{V}} + \langle f_n, f_N\rangle_{\mathcal{V}} \leq C\epsilon + \big|\sum_i a_i f_n(x_i)\big|$ by Cauchy-Schwarz and the reproducing property, and the finite sum tends to 0. So $\limsup_n \|f_n\|^2_{\mathcal{V}} \leq C\epsilon$ for every ϵ.

For $f, g \in \mathcal{H}_k$ with approximating sequences $(f_n), (g_n)$ define $\langle f,g\rangle_{\mathcal{H}_k} := \lim_n \langle f_n, g_n\rangle_{\mathcal{V}}$. The limit exists, since $|\langle f_n,g_n\rangle - \langle f_k,g_k\rangle| \leq \|f_n - f_k\|\,\|g_n\| + \|f_k\|\,\|g_n - g_k\|$ makes the sequence Cauchy in $\mathbb{R}$. It does not depend on the chosen sequences: if (f'_n) also converges pointwise to f, then $(f_n - f'_n)$ is Cauchy and tends to 0 pointwise, hence to 0 in norm by the observation, and the same estimate shows the limits agree. The form is bilinear, symmetric and non-negative, inheriting these from $\langle\cdot,\cdot\rangle_{\mathcal{V}}$, and it extends $\langle\cdot,\cdot\rangle_{\mathcal{V}}$. It is definite: if $\langle f,f\rangle_{\mathcal{H}_k} = 0$ then $\|f_n\|_{\mathcal{V}} \to 0$, so $f(x) = \lim_n f_n(x) = 0$ for every x by (11.2).

Step 3: reproducing property and completeness. For $f \in \mathcal{H}_k$, $\langle f, k(x,\cdot)\rangle_{\mathcal{H}_k} = \lim_n \langle f_n, k(x,\cdot)\rangle_{\mathcal{V}} = \lim_n f_n(x) = f(x)$. Moreover $\|f - f_n\|^2_{\mathcal{H}_k} = \lim_k \|f_k - f_n\|^2_{\mathcal{V}} \leq \sup_{k\geq n}\|f_k - f_n\|^2_{\mathcal{V}} \to 0$, so $\mathcal{V}$ is dense in $\mathcal{H}_k$. Now let (h_j) be Cauchy in $\mathcal{H}_k$, and choose $f_j \in \mathcal{V}$ with $\|h_j - f_j\|_{\mathcal{H}_k} \leq 1/j$. Then (f_j) is Cauchy in $\mathcal{V}$; its pointwise limit h lies in $\mathcal{H}_k$, with $\|h - f_j\|_{\mathcal{H}_k} \to 0$ by the density argument, hence $\|h - h_j\|_{\mathcal{H}_k} \leq \|h - f_j\|_{\mathcal{H}_k} + 1/j \to 0$. So $\mathcal{H}_k$ is a Hilbert space satisfying Definition 11.3.1.

Step 4: uniqueness. Let $\mathcal{G}$ be any RKHS for k. By the reproducing property and bilinearity, $\mathcal{G}$ contains $\mathcal{V}$ with the same inner product. A Cauchy sequence (f_n) in $\mathcal{V}$ converges in the complete space $\mathcal{G}$, and convergence in $\mathcal{G}$ implies pointwise convergence (by (11.2), valid in any RKHS), so its limit in $\mathcal{G}$ is its pointwise limit. Hence $\mathcal{H}_k \subseteq \mathcal{G}$, with the same inner product, and $\mathcal{H}_k$ is a closed subspace of $\mathcal{G}$, being complete. If it were a proper subspace, the projection theorem would give a non-zero $g \in \mathcal{G}$ orthogonal to $\mathcal{H}_k$, and then $g(x) = \langle g, k(x,\cdot)\rangle_{\mathcal{G}} = 0$ for every x, a contradiction. So $\mathcal{G} = \mathcal{H}_k$ □

Two consequences are worth recording immediately. First, the construction **completes the proof of Mercer's theorem**: the converse direction we deferred in Theorem 11.2.1 is exactly the statement that a

positive semi-definite k admits a feature map, and $\phi(x) := k(x, \cdot)$ into $\mathcal{H}_k$ is one. Second, every element of $\mathcal{H}_k$ is a limit, in norm and pointwise, of finite kernel expansions $\sum_i a_i k(x_i, \cdot)$ — so the hypothesis class generated by a kernel is precisely the closure of the set of weighted similarity scores to a finite set of points. This is the formal content of the informal description of kernel methods as "predicting by weighted similarity to remembered examples".

Proposition 11.3.3

The RKHS norm controls the function. For every $f \in \mathcal{H}_k$ and every $x \in \mathcal{X}$, $|f(x)| \leq \|f\|_{\mathcal{H}_k} \sqrt{k(x,x)}$. In particular, if $k(x,x) \leq R^2$ uniformly, then $\sup_x |f(x)| \leq R\|f\|_{\mathcal{H}_k}$.

Proof. This is (11.2), which used only the reproducing property and Cauchy-Schwarz, both available in $\mathcal{H}_k$ □

Proposition 11.3.3 is the reason $\|\cdot\|_{\mathcal{H}_k}$ is the right complexity measure for kernel methods, and it identifies the hypothesis class of the generalization section below: for a linear predictor $x \mapsto w^\top \phi(x)$ written in the canonical feature map, w *is* a function $f \in \mathcal{H}_k$ and $\|w\|_2$ *is* $\|f\|_{\mathcal{H}_k}$, so

$$\mathcal{H}_B \;=\; \{f \in \mathcal{H}_k : \|f\|_{\mathcal{H}_k} \leq B\}$$

is a class of uniformly bounded functions, whatever the dimension of the feature space happens to be. That is precisely the norm-constrained, dimension-free regime that Theorem 7.3.6 was built for.

What the norm measures. For the RBF kernel the RKHS norm penalizes wiggliness: one can show that $\|f\|^2_{\mathcal{H}_k}$ equals a weighted integral of $|\widehat{f}(\omega)|^2$ against $e^{\sigma^2\|\omega\|^2/2}$ over the frequency domain, so high-frequency components are charged exponentially, the more so the larger σ is (see (Girosi et al., 1995) for this Fourier-domain description of the norm). This makes the bandwidth σ and the regularization coefficient λ two different dials on the same quantity — σ decides *which* functions are expensive, λ decides *how much* we are willing to pay — and explains, before any experiment, why the fits below vary so sharply with σ.

11.3.1 The Representer Theorem

The RKHS is typically infinite-dimensional, so minimizing an objective over all of $\mathcal{H}_k$ looks like an infinite-dimensional optimization problem. It is not: the solution is always a finite kernel expansion on the training inputs, with one coefficient per training example. This is the single result that makes kernel methods computable, due to (Kimeldorf and Wahba, 1971).

Theorem 11.3.4: Representer theorem

Let $S = \{(x_i, y_i)\}_{i=1}^m$, let $\Psi : \mathbb{R}^m \to \mathbb{R}$ be an arbitrary function of the m predicted values, and let pen : $[0, \infty) \to \mathbb{R}$ be a **strictly increasing** regularizer, in the sense of Chapter 3. Then every minimizer f^* of

$$\min_{f \in \mathcal{H}_k} \; \Psi\big(f(x_1), \ldots, f(x_m)\big) + \mathrm{pen}\big(\|f\|_{\mathcal{H}_k}\big)$$

admits a representation

$$f^*(\cdot) = \sum_{i=1}^{m} a_i \, k(x_i, \cdot),$$
$$a \in \mathbb{R}^m.$$

Proof. Let $\mathcal{V}_S := \mathrm{span}\{k(x_1, \cdot), \ldots, k(x_m, \cdot)\} \subseteq \mathcal{H}_k$, a finite-dimensional and therefore closed subspace, and decompose any $f \in \mathcal{H}_k$ orthogonally as

$$f = f_\parallel + f_\perp,$$
$$f_\parallel \in \mathcal{V}_S, \;\; f_\perp \perp \mathcal{V}_S.$$

The data-fit term Ψ *cannot see* $f_\perp$. By the reproducing property, for every training input x_j,

$$f(x_j) = \langle f, k(x_j, \cdot)\rangle_{\mathcal{H}_k} = \langle f_\parallel, k(x_j, \cdot)\rangle_{\mathcal{H}_k} + \underbrace{\langle f_\perp, k(x_j, \cdot)\rangle_{\mathcal{H}_k}}_{=0, \text{ since } k(x_j,\cdot) \in \mathcal{V}_S} = f_\parallel(x_j),$$

so $\Psi(f(x_1), \ldots, f(x_m)) = \Psi(f_\parallel(x_1), \ldots, f_\parallel(x_m))$ exactly.

The regularizer can only be hurt by $f_\perp$. By the Pythagorean identity in a Hilbert space,

$$\|f\|_{\mathcal{H}_k}^2 = \|f_\parallel\|_{\mathcal{H}_k}^2 + \|f_\perp\|_{\mathcal{H}_k}^2 \;\geq\; \|f_\parallel\|_{\mathcal{H}_k}^2,$$

with equality if and only if $f_\perp = 0$. Since pen is strictly increasing, $\mathrm{pen}(\|f\|_{\mathcal{H}_k}) \geq \mathrm{pen}(\|f_\parallel\|_{\mathcal{H}_k})$, strictly so unless $f_\perp = 0$.

Adding the two, the objective at $f_\parallel$ is no larger than at f, and is **strictly** smaller whenever $f_\perp \neq 0$. A minimizer therefore has $f_\perp^* = 0$, i.e. $f^* = f_\parallel^* \in \mathcal{V}_S$, which is the claimed form □

Read the proof again for what it does *not* assume. Nothing about Ψ: it may be built from the squared loss, the hinge loss of an SVM, the logistic loss, or a non-convex loss with many minima; it need not even be continuous, and it need not decompose across examples at all. The theorem is driven entirely by the reproducing property (which makes Ψ blind to $f_\perp$) and by orthogonality (which makes pen charge for $f_\perp$ without any compensating benefit). What *is* essential is that the regularizer be a strictly increasing function of the RKHS norm specifically — this is why kernel methods regularize with $\|f\|_{\mathcal{H}_k}$ and not with something else.

Corollary 11.3.5

Kernel ridge regression. Take $\Psi(u_1, \dots, u_m) := \sum_{i=1}^{m} (y_i - u_i)^2$ and $\mathrm{pen}(t) := \lambda t^2$ with $\lambda > 0$. Then the minimizer over $\mathcal{H}_k$ is $f^*(\cdot) = \sum_i a_i k(x_i, \cdot)$ with

$$a = (K + \lambda I)^{-1} y.$$

Proof. By Theorem 11.3.4 we may restrict the search to $f = \sum_i a_i k(x_i, \cdot)$, at which point the reproducing property and (11.1) evaluate both terms in terms of the Gram matrix alone:

$$f(x_j) = \sum_i a_i k(x_i, x_j) = (Ka)_j,$$
$$\|f\|_{\mathcal{H}_k}^2 = \sum_{i,j} a_i a_j k(x_i, x_j) = a^\top K a.$$

The infinite-dimensional problem has become the m-dimensional one

$$\min_{a \in \mathbb{R}^m} \|y - Ka\|_2^2 + \lambda\, a^\top K a,$$

whose gradient is $-2Ky + 2KKa + 2\lambda Ka = 2K\big[(K + \lambda I)a - y\big]$. Setting it to zero is solved by $(K + \lambda I)a = y$, and $K + \lambda I \succ 0$ for $\lambda > 0$ makes that system uniquely solvable □

This is exactly the solution the dual derivation of the following section arrives at, and exactly what `precompute_a` computes in the implementation below — but obtained without ever mentioning ϕ, Φ or w. The dual derivation has to *discover* that w lies in the span of the training features, by inspecting the stationarity condition of one particular objective; the representer theorem says it in advance, for every objective of that shape. The practical readings are:

- **Cost.** Training is $O(m^3)$ (one linear solve) and prediction $O(m)$ per query, *independent of the dimension of the feature space* — including when that dimension is infinite, as for the RBF kernel. What is traded away is a cost that grows with the sample size instead, which is the precise sense in which kernel methods are "non-parametric": the model that must be stored is the training set itself.
- **Where the guarantee comes from.** The trained predictor satisfies $\|f^*\|_{\mathcal{H}_k}^2 = a^\top K a \le \|y\|_2^2/\lambda$ by comparing the objective at f^* against the candidate $f = 0$, so λ directly caps the RKHS norm — hence, via Proposition 11.3.3 and Theorem 7.3.6, the Rademacher complexity of the class the solution came from. The section on generalization below makes this quantitative.

11.4 Kernel regression

Let us remember the linear methods. It is possible to apply the kernel trick to all of them, i.e. to **kernelize** them. Let us take **ridge regression** as an example. Consider the case where the raw inputs x_i are passed through a feature extraction step $\phi(x_i)$. Call the matrix that contains $\phi(x_i)^\top$ in its rows $\Phi \in \mathbb{R}^{m \times d}$. The ridge regression objective can then be expressed as

$$\begin{aligned}\mathcal{L}(w) &= \sum_{i=1}^{m}(y_i - w^\top\phi(x_i))^2 + \lambda||w||_2^2 \\ &= y^\top y - 2y^\top\Phi w + w^\top\Phi^\top\Phi w + \lambda w^\top w,\end{aligned}$$

where $y \in \mathbb{R}^m$ is the target vector and $w \in \mathbb{R}^d$ is the weight vector. We perform learning by minimizing this objective with respect to the weights:

$$\begin{aligned}\nabla_w\mathcal{L}(w) &= -2\Phi^\top y + 2\Phi^\top\Phi w + 2\lambda w \triangleq 0 \\ &\Rightarrow (\Phi^\top\Phi + \lambda I)w = \Phi^\top y \\ &\Rightarrow w = (\Phi^\top\Phi + \lambda I)^{-1}\Phi^\top y.\end{aligned}$$

This primal solution still requires inverting a $d \times d$ matrix and computing $\phi(\cdot)$ explicitly, both of which can be prohibitive when ϕ maps into a very high (or infinite) dimensional space. Since the gradient stationarity condition $\Phi^\top\Phi w + \lambda w = \Phi^\top y$ implies $w = \frac{1}{\lambda}\Phi^\top(y - \Phi w)$, the optimal w always lies in the span of the training feature vectors $\phi(x_1), \ldots, \phi(x_m)$. We can therefore reparametrize it as

$$w = \Phi^\top a, \qquad a \in \mathbb{R}^m,$$

and search for the optimal a instead. Placing this expression into the objective gives its so-called **dual representation**, which contains the data only through inner products in the transformed space:

$$\mathcal{L}(a) = y^\top y - 2y^\top\Phi\Phi^\top a + a^\top\Phi\Phi^\top\Phi\Phi^\top a + \lambda a^\top\Phi\Phi^\top a.$$

Note that $K := \Phi\Phi^\top$ is the $m \times m$ dimensional **Gram** matrix with $K_{ij} = \phi(x_i)^\top\phi(x_j)$. This inner product can be replaced by any kernel function, generating a Gram matrix with entries $K_{ij} = k(x_i, x_j)$. The kernelized objective then reads

$$\mathcal{L}(a) = y^\top y - 2y^\top Ka + a^\top KKa + \lambda a^\top Ka.$$

Setting the gradient of $\mathcal{L}$ with respect to a to zero and solving for a gives

$$\begin{aligned}\nabla_a\mathcal{L}(a) &= -2Ky + 2KKa + 2\lambda Ka \triangleq 0 \\ &\Rightarrow K\big[(K + \lambda I)a - y\big] = 0 \\ &\Rightarrow a = (K + \lambda I)^{-1}y,\end{aligned}$$

where the last step assumes K is invertible (it always is a valid solution regardless, since any a satisfying $(K + \lambda I)a = y$ also solves $K\big[(K + \lambda I)a - y\big] = 0$).

To reiterate, we first redefined the parameters w by an expression that involves a newly introduced variable a, and then found the optimal value for this variable that minimizes the training loss. The resulting objective does not have any trainable parameters left that are not summarized by the Gram matrix K. Furthermore, the optimal value of a requires generating a Gram matrix using the whole training set, and it appears as an intermediate result in the calculation of the prediction function. We predict the label y_* of a query input x_* by

$$\begin{aligned}f(x_*) &= \phi(x_*)^\top w = \phi(x_*)^\top\Phi^\top a \\ &= k_*^\top(K + \lambda I)^{-1}y,\end{aligned}$$

where

$$k_* = \begin{bmatrix} k(x_*, x_1) \\ k(x_*, x_2) \\ \vdots, \\ k(x_*, x_m) \end{bmatrix}.$$

Because the prediction function requires computations on the whole training set, kernel regression is called a **lazy learner**: unlike the models we have seen so far, it does all of its work at prediction time, not at training time (the k-nearest neighbor approach of Chapter 4 is the other example — "training" there consists only of memorizing the data set). It is also a **non-parametric** method. As seen in the example implementation given below, its "predict" function takes the whole training set as an input in order to calculate the k_* vector.

A major weakness of the kernel methods is that many common kernel functions have tunable hyperparameters. For instance, the RBF kernel has a hyperparameter σ that controls the length scale of the kernel — and, as the RKHS reading above showed, thereby decides how expensive high-frequency components are in $\|f\|_{\mathcal{H}_k}$. The performance of the model is sensitive to the choice of this hyperparameter. The example below performs kernel regression with three different choices of σ, each giving dramatically different outcomes. While $\sigma = 0.1$ gives a reasonable fit, $\sigma = 0.01$ **overfits** — the kernel sections are so narrow that the fit spikes at each training point and returns to zero between them — and $\sigma = 1$ **underfits**, since at that length scale the RKHS charges so much for curvature that the fit cannot follow a full period of the sine.

```
import matplotlib.pyplot as plt
import torch as th

th.manual_seed(6)

# Generate the data
inputs = th.linspace(0, 1, 50, dtype=th.float64).unsqueeze(1)
outputs = th.sin(2 * th.pi * inputs)
labels = outputs + th.randn(inputs.shape, dtype=th.float64) * 0.25

num_samples = inputs.shape[0]
num_train_samples = num_samples // 2

idx = th.randperm(num_samples)
inputs_train = inputs[idx[:num_train_samples]]
labels_train = labels[idx[:num_train_samples]]

class KernelLinearRegression:
    def __init__(self, lambda_reg=0.1, length_scale=0.1):
        self.lambda_reg = lambda_reg
        self.sigma = length_scale

    def compute_rbf_kernel(self, X, Xp):
```

```
        # k(x,x') = exp(-||x-x'||^2 / (2 sigma^2))
        dist = th.cdist(X, Xp)
        return th.exp(-dist**2 / (2 * self.sigma**2))

    def precompute_a(self, inputs_train, labels_train):
        # a = (K + lambda I)^{-1} y, solved as the linear system
        # (K + lambda I) a = y
        K = self.compute_rbf_kernel(inputs_train, inputs_train)
        A = K + self.lambda_reg * th.eye(inputs_train.shape[0], dtype=K.dtype)
        self.a = th.linalg.solve(A, labels_train)

    def predict(self, inputs_train, inputs):
        # h(x) = sum_i a_i k(x_i, x)
        Kp = self.compute_rbf_kernel(inputs_train, inputs)
        return Kp.T @ self.a

# Create three models with different kernel hyperparameters
model1 = KernelLinearRegression(length_scale=0.01)
model1.precompute_a(inputs_train, labels_train)

model2 = KernelLinearRegression(length_scale=0.1)
model2.precompute_a(inputs_train, labels_train)

model3 = KernelLinearRegression(length_scale=1)
model3.precompute_a(inputs_train, labels_train)

# This is a "nonparametric model", so
#   i)  There is no "learn" function. "precompute_a" only
#       computes a value that is reused across different predictions
#   ii) Predict function takes the training data as input

predictions1 = model1.predict(inputs_train, inputs)
predictions2 = model2.predict(inputs_train, inputs)
predictions3 = model3.predict(inputs_train, inputs)

plt.plot(inputs, outputs, 'k-',
         label="True function", linewidth=4)
plt.plot(inputs_train, labels_train, 'bo',
         label="Training data")
plt.plot(inputs, predictions1, 'r--',
         label=r"Predictions ($\sigma=0.01$)", linewidth=3)
plt.plot(inputs, predictions2, 'g--',
         label=r"Predictions ($\sigma=0.1$)", linewidth=3)
plt.plot(inputs, predictions3, 'm--',
         label=r"Predictions ($\sigma=1$)", linewidth=3)
plt.legend(loc="upper right")
plt.show()
```

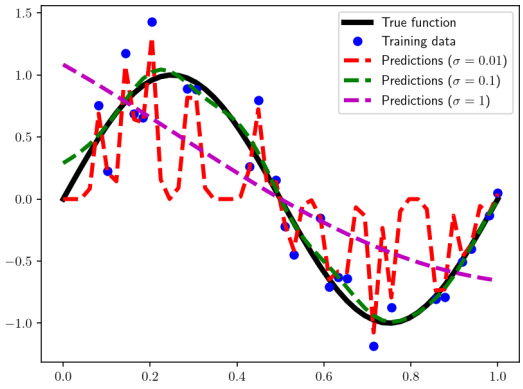


Figure 11.2: Kernel ridge regression fits with an RBF kernel of bandwidth $\sigma \in \{0.01, 0.1, 1\}$, illustrating overfitting, a good fit, and underfitting.

11.4.1 Generalization of Kernel Methods

Kernel methods look, at first glance, like exactly the setting Theorem 7.3.6 warned us about: the feature map $\phi(x)$ can live in an arbitrarily high — even infinite — dimensional space (the RBF kernel above is a standard example), for which d_{VC} is typically infinite, making the VC-dimension bounds of Chapter 6 vacuous. But Theorem 7.3.6 already told us this does not matter: for a norm-constrained linear predictor $x \mapsto w^\top \phi(x)$ with $\|w\|_2 \le B$, the Rademacher complexity is $BR/\sqrt{m}$, where R bounds $\|\phi(x)\|_2$ — with **no dependence on the dimension of $\phi(x)$ at all**. This is precisely the regime kernel methods create by design, and in the canonical feature map of Definition 11.3.1 the constrained class is exactly the RKHS ball $\mathcal{H}_B = \{f \in \mathcal{H}_k : \|f\|_{\mathcal{H}_k} \le B\}$ of Proposition 11.3.3.

We can, in fact, do slightly better and eliminate ϕ from the bound entirely, expressing it purely in terms of the kernel — a small derivation worth doing explicitly, since it shows the kernel trick paying off even inside the generalization theory itself, not only inside the training algorithm. Let $\mathcal{H}_B := \{x \mapsto w^\top \phi(x) : \|w\|_2 \le B\}$. Since $\|\phi(x)\|_2^2 = \phi(x)^\top \phi(x) = k(x,x)$,

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_B) = \frac{B}{m}\,\mathbb{E}_\sigma\Big[\Big\|\sum_{i=1}^m \sigma_i \phi(x_i)\Big\|_2\Big] \le \frac{B}{m}\sqrt{\mathbb{E}_\sigma\Big[\Big\|\sum_i \sigma_i \phi(x_i)\Big\|_2^2\Big]}$$
$$= \frac{B}{m}\sqrt{\sum_{i=1}^m \|\phi(x_i)\|_2^2} = \frac{B}{m}\sqrt{\sum_{i=1}^m k(x_i,x_i)} = \frac{B\sqrt{\mathrm{tr}(K)}}{m},$$

using Jensen's inequality and $\mathbb{E}[\sigma_i \sigma_{i'}] = 0$ for $i \ne i'$ (exactly as in the proof of Theorem 7.3.6), where K is the Gram matrix of Mercer's theorem. This **trace bound** is computable directly from the kernel matrix — never requiring ϕ to be written down — and is always at least as tight as the generic $BR/\sqrt{m}$ bound, since $\mathrm{tr}(K) = \sum_i k(x_i,x_i) \le mR^2$ whenever $k(x,x) \le R^2$ uniformly.

This connects directly to the regularization coefficient λ. Write kernel ridge regression in the averaged form

$$\widehat{f}_S := \arg\min_{f \in \mathcal{H}_k} \mathcal{J}_S(f), \qquad \mathcal{J}_S(f) := \frac{1}{m}\sum_{i=1}^m \big(y_i - f(x_i)\big)^2 + \lambda \|f\|_{\mathcal{H}_k}^2,$$

which is the objective of Corollary 11.3.5 with its λ replaced by $m\lambda$, and assume $|y| \le Y$ and $k(x,x) \le R^2$. Comparing the minimizer with $f = 0$ gives $\lambda \|\widehat{f}_S\|_{\mathcal{H}_k}^2 \le \mathcal{J}_S(\widehat{f}_S) \le \mathcal{J}_S(0) \le Y^2$, so

$$\|\widehat{f}_S\|_{\mathcal{H}_k} \;\le\; \frac{Y}{\sqrt{\lambda}} =: B(\lambda)$$

for **every** sample. The solution therefore always comes from the class $\mathcal{H}_{B(\lambda)}$, which is fixed before the data are seen, so Theorem 7.3.2 applies to it, and the trace bound gives $\widehat{\mathfrak{R}}_S(\mathcal{H}_{B(\lambda)}) \le \frac{YR}{\sqrt{\lambda m}}$. This makes quantitative what Chapter 3 promised only qualitatively when it introduced ridge regression as an instance of Structural Risk Minimization: increasing λ shrinks the class the solution can come from, and with it the complexity term of the generalization bound.

The same regularization also makes the algorithm stable, which gives a faster rate in expectation:

Corollary 11.4.1

Kernel ridge regression is uniformly stable. Under the assumptions above, with the squared loss $\phi(f,(x,y)) = (y - f(x))^2$,

$$\mathbb{E}_S\big[R(\widehat{f}_S)\big] \;\le\; \mathbb{E}_S\big[\widehat{R}_S(\widehat{f}_S)\big] + \frac{4R^2Y^2\big(1 + R/\sqrt{\lambda}\big)^2}{\lambda m}.$$

Proof. Since every minimizer lies in the closed convex ball $\Omega := \{f : \|f\|_{\mathcal{H}_k} \le Y/\sqrt{\lambda}\}$, $\widehat{f}_S$ is also the minimizer of $\mathcal{J}_S$ over Ω. By the reproducing property $f(x) = \langle f, k(x,\cdot)\rangle_{\mathcal{H}_k}$, so the loss is a convex function of f with gradient $-2(y - f(x))\,k(x,\cdot)$, whose norm on Ω is at most $2(Y + RY/\sqrt{\lambda})R =: G$ by Proposition 11.3.3; the loss is therefore G-Lipschitz on Ω. The regularizer $\lambda\|f\|_{\mathcal{H}_k}^2$ is 2λ-strongly convex. Theorem 7.8.3, whose proof holds verbatim in a Hilbert space, makes the algorithm $\frac{2G^2}{2\lambda m}$-uniformly stable, and Theorem 7.8.2 concludes $\square$

The Rademacher route gives a gap of order $1/\sqrt{\lambda m}$ with high probability; the stability route gives order $1/(\lambda m)$ in expectation. Both say that a larger λ produces a smaller gap, and the price is paid in the training error.

How small can the training error, and more importantly the approximation error, become as $\lambda \to 0$, i.e. as the norm budget B grows? For the RBF kernel the answer is: as small as for any continuous predictor.

Theorem 11.4.2: The RBF kernel is universal

Let k be the RBF kernel with bandwidth $\sigma > 0$ on $\mathbb{R}^d$ and $\mathcal{X} \subset \mathbb{R}^d$ compact. For every continuous $g : \mathcal{X} \to \mathbb{R}$ and every $\epsilon > 0$ there is a finite kernel expansion $f = \sum_{i=1}^{n} a_i k(x_i,\cdot)$, $x_i \in \mathbb{R}^d$, with $\sup_{x\in\mathcal{X}} |f(x) - g(x)| \le \epsilon$.

Proof. Let $\mathcal{C}$ be the closure, in the supremum norm on $\mathcal{X}$, of the span of the kernel sections $k(y,\cdot)$, $y \in \mathbb{R}^d$; it is a closed linear subspace of the continuous functions on $\mathcal{X}$, and the claim is that it contains every continuous g. (The kernel and its RKHS are defined on all of $\mathbb{R}^d$, and the centres range over $\mathbb{R}^d$; the

approximation is required only on $\mathcal{X}$.) With $q(x) := e^{-\|x\|^2/(2\sigma^2)}$ as above, $k(y,x) = q(y)\,q(x)\,e^{y^\top x/\sigma^2}$, so $\mathcal{C}$ contains the functions $x \mapsto q(x)e^{y^\top x/\sigma^2}$ for every y.

Step 1: polynomials times q. We show by induction on the degree that $\mathcal{C}$ contains $x \mapsto q(x)\,p(x)\,e^{y^\top x/\sigma^2}$ for every monomial $p(x) = x_1^{\alpha_1}\cdots x_d^{\alpha_d}$ and every y. Degree 0 is the observation above. If it holds for p, consider the difference quotients

$$D_h(x) := q(x)\,p(x)\,\frac{e^{(y+he_j)^\top x/\sigma^2} - e^{y^\top x/\sigma^2}}{h} \;\in\; \mathcal{C}, \qquad h \neq 0,$$

with e_j the j-th unit vector. With $u := hx_j/\sigma^2$, $\left|\frac{e^u-1}{h} - \frac{x_j}{\sigma^2}\right| = \frac{|e^u-1-u|}{|h|} \le \frac{u^2 e^{|u|}}{2|h|} = \frac{|h|\,x_j^2}{2\sigma^4}e^{|hx_j|/\sigma^2}$, which tends to 0 uniformly on the compact $\mathcal{X}$ as $h \to 0$; the remaining factor $q\,p\,e^{y^\top x/\sigma^2}$ is bounded there. So $D_h \to q\,p\,\frac{x_j}{\sigma^2}\,e^{y^\top x/\sigma^2}$ uniformly, and $\mathcal{C}$, being closed, contains the limit. Taking $y = 0$, $\mathcal{C}$ contains $q\,p$ for every polynomial p.

Step 2: Weierstrass. The function g/q is continuous on $\mathcal{X}$, so by the Weierstrass approximation theorem (in its multivariate form) there is a polynomial p with $\sup_{\mathcal{X}} |g/q - p| \le \epsilon$. Since $0 < q \le 1$, $\sup_{\mathcal{X}} |g - q\,p| \le \epsilon$, and $q\,p \in \mathcal{C}$. By the definition of the closure there is a finite kernel expansion within ϵ of $q\,p$, hence within 2ϵ of g; as ϵ was arbitrary, this proves the claim □

Kernels with this property are called **universal**; see (Steinwart, 2001). The linear and polynomial kernels are not: their RKHS consists of linear functions, respectively of polynomials of degree at most d.

Theorem 11.4.2 completes the approximation-estimation decomposition of Chapter 6 for RBF kernel methods. Let the loss be bounded in $[0,1]$ and G-Lipschitz in the prediction, let g be any continuous predictor, and let $\widehat{f}$ minimize the empirical risk over $\mathcal{H}_B$. Given $\epsilon > 0$, Theorem 11.4.2 provides f_ϵ with finite norm $B_\epsilon := \|f_\epsilon\|_{\mathcal{H}_k}$ and $R(f_\epsilon) \le R(g) + G\epsilon$. For every $B \ge B_\epsilon$, the decomposition $R(\widehat{f}) - R(g) = [R(\widehat{f}) - \min_{f\in\mathcal{H}_B} R(f)] + [\min_{f\in\mathcal{H}_B} R(f) - R(g)]$, Lemma 6.2.3, and Theorem 7.3.2 with the contraction lemma and the trace bound give, with probability at least $1-\delta$,

$$R(\widehat{f}) - R(g) \;\le\; \underbrace{O\Big(\frac{GBR}{\sqrt{m}} + \sqrt{\frac{\log(1/\delta)}{m}}\Big)}_{\text{estimation}} + \underbrace{G\epsilon}_{\text{approximation}}.$$

Letting the budget grow with the sample size, $B_m \to \infty$ with $B_m/\sqrt{m} \to 0$, both terms vanish: for every continuous g, the risk of RBF kernel ERM eventually comes within any ϵ of $R(g)$. The bias-complexity dilemma is resolved here by a schedule rather than by a fixed choice — the same role the regularization path $\lambda_m \to 0$ plays in practice.

11.4.2 The Neural Tangent Kernel

Chapter 10 analyzed a two-layer network $h_{W,w}(x) = w^\top g(Wx)$ and bounded its Rademacher complexity by controlling the norms B_1, B_2 of *both* layers — a bound that holds regardless of how W and w are jointly trained. There is an illustrative special case of that same network in which training reduces *exactly* to the kernel methods of this chapter, and it is worth making explicit, since it is the seed of one of the more influential ideas in modern deep learning theory: the **Neural Tangent Kernel (NTK)**.

A simplified case: random features. Suppose we freeze the hidden layer at its random initialization W_0 (drawn once, e.g. with i.i.d. $\mathcal{N}(0,1/d)$ entries per row) and train *only* the output layer w. Then

$\phi(x) := g(W_0 x)$ is a fixed feature map, and $h_{W_0,w}(x) = w^\top \phi(x)$ is exactly a kernel method with kernel $k_{W_0}(x, x') := \phi(x)^\top \phi(x') = \sum_{j=1}^{k} g(w_{0,j}^{(1)\top} x)\, g(w_{0,j}^{(1)\top} x')$ — every tool from this chapter, including the generalization bound just derived, applies to it unchanged. As the width $k \to \infty$, the weak law of large numbers (Theorem 5.6.3, applied to the k i.i.d. terms $g(w_{0,j}^{(1)\top} x)\, g(w_{0,j}^{(1)\top} x')$ as j ranges over hidden units) gives

$$\frac{1}{k} k_{W_0}(x, x') \xrightarrow{P} k_\infty(x, x') := \mathbb{E}_{u \sim \mathcal{N}(0, I/d)} \big[g(u^\top x)\, g(u^\top x') \big],$$

a **deterministic kernel that no longer depends on the random draw of** W_0 at all, only on the architecture (g) and the input distribution of u. This is the "random features" or "infinite-width random network" kernel, and it already illustrates the core NTK phenomenon in the easiest possible case: a wide random network with a trained linear readout *is*, in the limit, exactly kernel regression with a fixed, explicitly computable kernel.

The general case. The full Neural Tangent Kernel result — which trains *all* parameters $\theta = (W, w)$, not just the output layer — is more subtle but built on the same idea. Near the initialization θ_0, a first-order Taylor expansion of the network output in its parameters gives

$$f_\theta(x) \approx f_{\theta_0}(x) + \nabla_\theta f_{\theta_0}(x)^\top (\theta - \theta_0),$$

which is *affine* in θ: exactly the linear-predictor template of this chapter once more, but now with the **tangent feature map** $\phi_{\text{NTK}}(x) := \nabla_\theta f_{\theta_0}(x)$ (one feature per network parameter) and its associated kernel $k_{\text{NTK}}(x, x') := \nabla_\theta f_{\theta_0}(x)^\top \nabla_\theta f_{\theta_0}(x')$. Jacot, Gabriel & Hongler (2018) show two facts that make this linearization exact rather than merely suggestive, in the infinite-width limit and for appropriately scaled initializations: (i) k_{NTK} converges, by the same law-of-large-numbers mechanism as above (now averaging over a growing number of parameters instead of a growing number of hidden units), to a fixed deterministic kernel depending only on the architecture; and (ii) gradient descent moves θ so little from θ_0 relative to the network's width that the linearization above remains accurate for the *entire* training trajectory, not just near initialization. Under these conditions, training an infinitely wide neural network by gradient descent is equivalent to kernel regression (as derived earlier in this chapter) against the fixed kernel k_{NTK}.

Why this matters for generalization. This equivalence resolves, for the infinite-width regime, the very puzzle Chapter 10's PAC analysis raised: an infinitely wide network has infinite VC dimension, yet the NTK correspondence shows it is *simultaneously* a kernel predictor of the exact form Theorem 7.3.6 and the trace bound above already know how to handle — capacity controlled by $\|w\|_{\mathcal{H}}$ (the RKHS norm under k_{NTK}) and by $\operatorname{tr}(K_{\text{NTK}})$, neither of which explodes with the width. This is the precise sense in which Chapter 10's remark that "capacity control by norm, rather than by counting parameters, is what makes the bound meaningful" for neural networks is not a coincidence but a theorem: in the wide-network limit, a neural network trained by gradient descent literally *is* a norm-constrained kernel predictor, and the same Rademacher-complexity machinery developed once, in Chapter 7, governs both.

11.5 Support Vector Machines

A prime application of kernel methods, and the historical reason the representer theorem is stated in the generality it is, is the **Support Vector Machine (SVM)** (Boser et al., 1992). Among many possible hyperplanes that may separate the data points of two classes, SVM chooses the one that maximizes the

smallest distance between a data point and the hyperplane. This distance is called the **margin**. Because of this property, SVM is also called a **maximum margin classifier**. The data points that are closest to the hyperplane are called **support vectors**. The margin-maximizing hyperplane is defined by only these support vectors. In the language of Theorem 11.3.4, the SVM objective is an instance of the representer template with the **hinge** data-fit term $\Psi(u) = \sum_i \max(0, 1 - y_i u_i)$ and $\text{pen}(t) = \lambda t^2$, so its solution is again $f^* = \sum_i a_i k(x_i, \cdot)$ — but the hinge loss, unlike the squared loss, is flat on correctly classified points beyond the margin, which forces $a_i = 0$ for all of them. The support vectors are exactly the indices with $a_i \neq 0$. The other data points do not play any role in prediction. This is a desirable property especially when the SVM is kernelized, as the kernel function is evaluated only on the support vectors instead of the whole training set as was the case for kernel regression above. The figure below illustrates the core idea behind the SVM. We leave the mathematical details out of our scope, since SVMs have been greatly overshadowed by deep learning and Gaussian processes in modern machine learning practice.

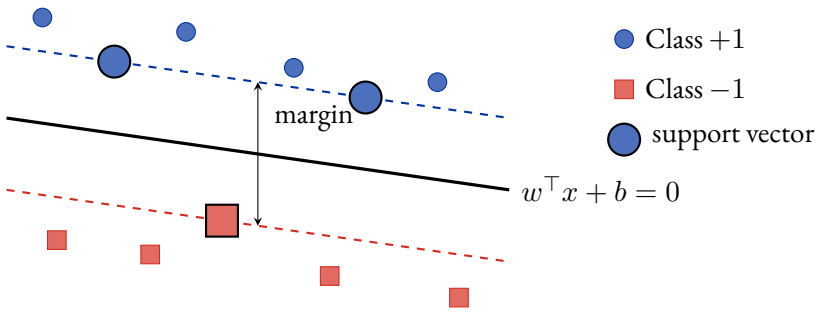


Figure 11.3: The maximum-margin hyperplane of a Support Vector Machine. The decision boundary $w^\top x + b = 0$ is placed to maximize the margin between the two classes; the support vectors (outlined) are the points that lie exactly on the margin boundary and alone determine the solution.

CHAPTER 12

ENSEMBLE METHODS

An **ensemble** aggregates the predictions of several hypotheses $h_1, \dots, h_T$ into a single predictor. The constructions in this chapter differ in *how* the members are produced and in *which* term of the excess-risk decomposition of Chapter 6 they attack:

	members produced	aggregation	attacks
Bagging	in parallel, on resampled data	unweighted average	the **variance** term
Boosting	in sequence, on reweighted data	weighted vote $\sum_t w_t h_t$	the **approximation error** ϵ_{app}
Gradient boosting	in sequence, fit to pseudo-residuals	convex combination of the h_t	the **approximation error** ϵ_{app}, for any smooth convex loss

Bagging takes a low-bias, high-variance base learner and averages its fluctuations away. Boosting takes a high-bias, low-variance base learner — one barely better than a coin flip — and composes many of them into a predictor of much greater capacity; this is the more surprising of the two, and it is where the theory of this chapter is concentrated. Gradient boosting, introduced last, keeps boosting's stagewise structure but replaces the exponential-loss reweighting of AdaBoost by a functional gradient step, which makes the construction work for any smooth convex loss, regression included.

12.1 Bootstrap Aggregation (Bagging)

Bagging (Breiman, 1996) is the technique this section formalizes.

Definition 12.1.1: Bootstrap sample

Given $S = \{(x_i, y_i)\}_{i=1}^m$, a **bootstrap sample** of S is

$$S' := \{(x_{j_k}, y_{j_k}) : j_k \sim \mathcal{U}(\{1, \dots, m\}),\ k = 1, \dots, m\},$$

i.e. m indices drawn uniformly **with replacement**.

Definition 12.1.2: Bagged predictor

Let $S_1, \dots, S_T$ be bootstrap samples of S and $h_{S_t} \leftarrow A(S_t)$. The **bagged predictor** is

$$h_{S_1,\dots,S_T}(x) := \frac{1}{T}\sum_{t=1}^{T} h_{S_t}(x).$$

Two elementary facts explain the benefit of averaging, both stated for the idealized case of T **independent** samples $S_1, \dots, S_T \sim \mathcal{D}^m$; bootstrapping approximates this by reusing one S, at the cost of correlation between the members, which is why the gains below are real but attenuated in practice.

Proposition 12.1.3

Averaging preserves bias and divides variance. Let $h_{S_1}, \dots, h_{S_T}$ be i.i.d. as functions of independent samples $S_1, \dots, S_T \sim \mathcal{D}^m$. Then for every fixed x,

$$\mathbb{E}\left[\tfrac{1}{T}\textstyle\sum_t h_{S_t}(x)\right] = \mathbb{E}_S[h_S(x)],$$
$$\mathrm{Var}\left[\tfrac{1}{T}\textstyle\sum_t h_{S_t}(x)\right] = \tfrac{1}{T}\,\mathrm{Var}_S[h_S(x)].$$

Proof. The first claim is linearity of expectation. For the second, independence makes all cross-covariances vanish, so

$$\mathrm{Var}\left[\frac{1}{T}\sum_{t} h_{S_t}(x)\right] = \frac{1}{T^2}\sum_{t=1}^{T}\mathrm{Var}[h_{S_t}(x)] + \frac{1}{T^2}\sum_{t\neq t'}\underbrace{\mathrm{Cov}[h_{S_t}(x), h_{S_{t'}}(x)]}_{=0} = \frac{1}{T^2}\cdot T\,\mathrm{Var}_S[h_S(x)]. \qquad \square$$

In the vocabulary of Chapter 6's bias-variance decomposition: the Bias^2 term is untouched and the Variance term is divided by T. Letting $T \to \infty$, the weak law of large numbers (Theorem 5.6.3) gives $h_{S_1,\dots,S_T}(x) \to \bar{h}(x) := \mathbb{E}_S[h_S(x)]$, the **average predictor** of Chapter 6. $\square$

Proposition 12.1.4

Bagging never hurts, in expectation. For the squared loss, $R(\bar{h}) \le \mathbb{E}_S[R(h_S)]$, with equality iff $\mathrm{Var}_S[h_S(x)] = 0$ for $\mathcal{D}$-almost every x.

Proof. Chapter 6 established $\mathbb{E}_S[R(h_S)] - R^*_{\mathcal{D}} = \text{Bias}^2 + \text{Variance}$ and $R(\bar{h}) - R^*_{\mathcal{D}} = \text{Bias}^2 = \|\bar{h} - f^*\|^2$. Subtracting, $\mathbb{E}_S[R(h_S)] - R(\bar{h}) = \text{Variance} = \mathbb{E}_x[\mathrm{Var}_S[h_S(x)]] \ge 0$. Equality holds exactly when the integrand vanishes almost everywhere. (Equivalently: this is Jensen's inequality for the convex map $t \mapsto t^2$.) $\square$

Proposition 12.1.4 compares $\bar{h}$ only to the *average* member. It can do better. If $\mathcal{H}$ is **not convex**, then $\bar{h} \notin \mathcal{H}$ in general, and $\bar{h}$ may be strictly closer to f^* than *every* $h \in \mathcal{H}$ — the caveat already flagged in Chapter 6. Decision trees (Chapter 9) are such a class: an average of trees is not a tree. Bagged trees, with an extra randomization over the features considered at each split, are **random forests**, and this non-convexity

is exactly the source of their strength. Bagging a *convex* class such as linear predictors, by contrast, cannot beat its best member, since there $\bar{h} \in \mathcal{H}$ already.

12.2 Boosting

Boosting starts from the opposite premise: the base class is deliberately **too weak** to fit the data, and the ensemble must manufacture capacity rather than stability. Throughout this section $\mathcal{Y} = \{-1, +1\}$, so that the sign of a real-valued score is a prediction and $y\, f(x) > 0$ means "correct".

12.2.1 Weak Learnability

Definition 12.2.1: γ-weak learner

Let $\gamma \in (0, 1/2)$ and let $\mathcal{H}_0 \subseteq \{-1, +1\}^{\mathcal{X}}$ be a **base class**. A **γ-weak learner** for $\mathcal{H}_0$ is a procedure WL that maps a sample S and a probability vector $D \in \Delta_m := \{D \in \mathbb{R}^m_{\geq 0} : \sum_i D_i = 1\}$ to some $h = \mathrm{WL}(S, D) \in \mathcal{H}_0$ whose **D-weighted empirical error** satisfies

$$\widehat{R}_{S,D}(h) := \sum_{i=1}^{m} D_i \, \mathbb{1}\big(h(x_i) \neq y_i\big) \;\leq\; \tfrac{1}{2} - \gamma.$$

The demand is minimal: a coin flip achieves $1/2$ in expectation, so WL must merely beat guessing by a fixed margin γ — but it must do so for **every** reweighting D of the sample, including adversarial ones that concentrate on the points previous rounds got wrong. This uniformity over D is the whole content of the assumption, and it is what boosting consumes.

Definition 12.2.1 is the *empirical* form of weak learnability, stated over reweightings of a fixed sample because that is exactly the interface AdaBoost calls. Its distributional counterpart replaces "D on $[m]$" by "any distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$ realizable by $\mathcal{H}_0$" and asks for $R(h) \leq \frac{1}{2} - \gamma$ with high probability from finitely many samples. The two are linked by the Fundamental Theorem (Theorem 6.4.7): if $d_{VC}(\mathcal{H}_0) < \infty$ then ERM over $\mathcal{H}_0$ is a γ-weak learner in the distributional sense for every $\gamma < 1/2$, given enough samples — so weak learnability is not a weaker *combinatorial* condition than PAC learnability, only a weaker *accuracy* requirement. Corollary 12.2.4 below shows that even that weakening yields no benefit: the two are equivalent.

Definition 12.2.2: Class of T-term linear combinations

For a base class $\mathcal{H}_0$,

$$L(\mathcal{H}_0, T) := \Big\{ x \mapsto \operatorname{sign}\Big(\sum_{t=1}^{T} w_t h_t(x) \Big) \;:\; w \in \mathbb{R}^T,\; h_1, \ldots, h_T \in \mathcal{H}_0 \Big\}.$$

$\mathcal{H}_0 \subseteq L(\mathcal{H}_0, 1) \subseteq L(\mathcal{H}_0, 2) \subseteq \cdots$, so the ensemble class is a **nested family** indexed by T — precisely the setting of nonuniform learnability and SRM (Chapter 6), with the number of rounds T playing the role that the leaf budget played for decision trees in Chapter 9.

Example : decision stumps
$\mathcal{H}_{\text{stump}} := \{x \mapsto s\,\text{sign}(x_j - \theta) : j \in [d],\ \theta \in \mathbb{R},\ s \in \{-1, +1\}\}$. On any m points, a stump's labeling is determined by the feature j (d choices), the position of θ among the m observed values of that feature ($m+1$ choices), and the polarity s (2 choices), so

$$\tau_{\mathcal{H}_{\text{stump}}}(m) \le 2d(m+1).$$

Since $\tau(m) = 2^m$ is needed to shatter, $d_{VC}(\mathcal{H}_{\text{stump}}) = O(\log d)$: a stump is a genuinely weak hypothesis, unable even to represent a two-feature XOR. Yet $L(\mathcal{H}_{\text{stump}}, T)$ grows rich with T, which is what makes stumps the canonical base class for boosting. ■

12.2.2 AdaBoost

AdaBoost (Freund and Schapire, 1997) is the algorithm below.

Input: $S = \{(x_i, y_i)\}_{i=1}^m$ with $y_i \in \{-1, +1\}$; a weak learner WL; a number of rounds T.

1. $D^{(1)} := \left(\frac{1}{m}, \dots, \frac{1}{m}\right)$.
2. For $t = 1, \dots, T$:
   1. $h_t := \text{WL}\left(S, D^{(t)}\right)$
   2. $\epsilon_t := \sum_{i=1}^m D_i^{(t)}\, \mathbb{1}\left(h_t(x_i) \neq y_i\right)$
   3. $w_t := \frac{1}{2}\log\left(\frac{1-\epsilon_t}{\epsilon_t}\right)$
   4. $D_i^{(t+1)} := \dfrac{D_i^{(t)}\, e^{-w_t y_i h_t(x_i)}}{\sum_{j=1}^m D_j^{(t)}\, e^{-w_t y_j h_t(x_j)}}, \qquad i = 1, \dots, m$
3. **Output** $h_S := \text{sign}(f_T) \in L(\mathcal{H}_0, T)$, where $f_T(x) := \sum_{t=1}^T w_t\, h_t(x)$.

Step 2.4 is one line because $y_i h_t(x_i) = +1$ on correct points and -1 on incorrect ones: the update multiplies a correct point's weight by e^{-w_t} and an incorrect point's by e^{+w_t}, then renormalizes. The choice of w_t in step 2.3 is not a guess; it is a minimization, as the proof below shows in one line.

Theorem 12.2.3: AdaBoost drives the training error down exponentially

If $\epsilon_t \le \frac{1}{2} - \gamma$ at every round, then the output of AdaBoost satisfies

$$\widehat{R}_S(h_S) = \frac{1}{m}\sum_{i=1}^m \mathbb{1}\left(h_S(x_i) \neq y_i\right) \ \le\ e^{-2\gamma^2 T}.$$

Proof. Write $f_t := \sum_{p \le t} w_p h_p$, $f_0 := 0$, and define the **exponential loss**

$$Z_t := \frac{1}{m}\sum_{i=1}^m e^{-y_i f_t(x_i)}.$$

Step 1: the exponential loss dominates the zero-one loss. Since $\mathbb{1}(u \le 0) \le e^{-u}$ for all $u \in \mathbb{R}$, and $h_S(x_i) \neq y_i$ exactly when $y_i f_T(x_i) \le 0$,

$$\widehat{R}_S(h_S) \le Z_T.$$

Step 2: the weights are the normalized exponential losses. We claim

$$D_i^{(t)} = e^{-y_i f_{t-1}(x_i)} / \sum_j e^{-y_j f_{t-1}(x_j)}.$$

For $t = 1$ both sides are $1/m$ since $f_0 = 0$. Assuming it at t, step 2.4 gives

$$D_i^{(t+1)} \;\propto\; D_i^{(t)} e^{-w_t y_i h_t(x_i)} \;\propto\; e^{-y_i f_{t-1}(x_i)} e^{-w_t y_i h_t(x_i)} = e^{-y_i f_t(x_i)},$$

and normalizing proves the claim at $t+1$.

Step 3: the per-round progress factor. Using Step 2, and splitting the sum according to whether h_t is correct,

$$\frac{Z_t}{Z_{t-1}} = \frac{\sum_i e^{-y_i f_{t-1}(x_i)}\, e^{-w_t y_i h_t(x_i)}}{\sum_i e^{-y_i f_{t-1}(x_i)}} = \sum_{i=1}^{m} D_i^{(t)} e^{-w_t y_i h_t(x_i)} = \underbrace{(1-\epsilon_t)\, e^{-w_t} + \epsilon_t\, e^{w_t}}_{=:\,\psi(w_t)}.$$

Now $\psi'(w) = -(1-\epsilon_t)e^{-w} + \epsilon_t e^{w} = 0$ gives $e^{2w} = (1-\epsilon_t)/\epsilon_t$, i.e. **exactly the w_t of step 2.3** — this is where that formula comes from, and ψ is convex, so it is a minimum. Substituting $e^{w_t} = \sqrt{(1-\epsilon_t)/\epsilon_t}$,

$$\frac{Z_t}{Z_{t-1}} = (1-\epsilon_t)\sqrt{\tfrac{\epsilon_t}{1-\epsilon_t}} + \epsilon_t\sqrt{\tfrac{1-\epsilon_t}{\epsilon_t}} = 2\sqrt{\epsilon_t(1-\epsilon_t)}.$$

Step 4: telescoping. Writing $\epsilon_t = \tfrac{1}{2} - \gamma_t$ with $\gamma_t \ge \gamma$, we get $2\sqrt{\epsilon_t(1-\epsilon_t)} = \sqrt{1-4\gamma_t^2} \le e^{-2\gamma_t^2} \le e^{-2\gamma^2}$, using $1-u \le e^{-u}$ with $u = 4\gamma_t^2$ under the square root. Since $Z_0 = 1$,

$$\widehat{R}_S(h_S) \le Z_T = \prod_{t=1}^{T} \frac{Z_t}{Z_{t-1}} \le e^{-2\gamma^2 T}. \qquad \square$$

Two readings of the proof are worth keeping. First, AdaBoost is **coordinate-wise greedy minimization of the exponential loss** Z_t: step 2.1 picks the direction h_t and step 2.3 picks the optimal step size w_t along it. Second, the weighting scheme is not an extra heuristic layered on top — Step 2 shows $D^{(t)}$ *is* the exponential loss, normalized, so up-weighting the currently misclassified points is exactly what greedy minimization of Z_t prescribes. $\square$

Corollary 12.2.4

Weak learnability implies strong learnability. Let $\mathcal{H}_0$ admit a γ-weak learner and run AdaBoost for $T \ge \frac{\log m}{2\gamma^2}$ rounds. Then $\widehat{R}_S(h_S) = 0$.

Proof. By Theorem 12.2.3, $\widehat{R}_S(h_S) \le e^{-2\gamma^2 T} \le e^{-\log m} = 1/m$. But $\widehat{R}_S(h_S)$ is a multiple of $1/m$, and the only such value strictly below $1/m$ is 0. $\square$

This is the theorem that made boosting famous: a learner that can only beat a coin flip by γ, on every reweighting of the sample, can be compiled into one that fits the sample perfectly, in $O(\gamma^{-2}\log m)$ rounds. **Weak** and **strong** learnability are the same notion, up to computation. What remains is to price the generalization of the resulting predictor — and this is where two different complexity measures give two genuinely different answers.

12.2.3 What Linear Combinations Cost: Three Accounts

Account 1: the VC dimension of $L(\mathcal{H}_0, T)$ grows with T

Lemma 12.2.5

Let $d_0 := d_{VC}(\mathcal{H}_0) < \infty$. Then

$$\tau_{L(\mathcal{H}_0,T)}(m) \;\le\; \tau_{\mathcal{H}_0}(m)^T \cdot \left(\frac{em}{T}\right)^T \;\le\; \left(\frac{em}{d_0}\right)^{d_0 T}\left(\frac{em}{T}\right)^T,$$
$$\text{hence} \qquad d_{VC}\big(L(\mathcal{H}_0,T)\big) = O\big(d_0 T\log(d_0 T)\big).$$

Proof. Fix $S = \{x_1,\dots,x_m\}$. A hypothesis of $L(\mathcal{H}_0,T)$ is specified by base hypotheses $h_1,\dots,h_T$ and weights w. Its restriction to S depends on the h_t only through their own restrictions to S, and there are at most $\tau_{\mathcal{H}_0}(m)$ of those per slot, hence at most $\tau_{\mathcal{H}_0}(m)^T$ jointly. Having fixed them, map each point to $v_i := (h_1(x_i),\dots,h_T(x_i)) \in \{-1,+1\}^T$; the remaining freedom is the labeling $\operatorname{sign}(w^\top v_i)$, i.e. a homogeneous halfspace in $\mathbb{R}^T$ applied to m fixed points. The class of homogeneous halfspaces in $\mathbb{R}^T$ has VC dimension T, so by Sauer-Shelah-Perles (Lemma 6.4.8) it realizes at most $(em/T)^T$ labelings. Multiplying the two counts gives the first claim, and the second follows from Lemma 6.4.8 applied to $\mathcal{H}_0$. For the VC bound, shattering m points requires $2^m \le \tau_{L(\mathcal{H}_0,T)}(m)$, i.e. $m\log 2 \le (d_0T + T)\log(em)$, which forces $m = O(d_0 T\log(d_0 T))$. □

Feeding $d_{VC}(L(\mathcal{H}_0,T)) = O(d_0T\log(d_0T))$ into the Fundamental Theorem (Theorem 6.4.7) gives, with probability at least $1-\delta$, for every $h \in L(\mathcal{H}_0,T)$,

$$R(h) \;\le\; \widehat{R}_S(h) + O\left(\sqrt{\frac{d_0 T\log(d_0T) + \log(1/\delta)}{m}}\right). \tag{12.1}$$

Combined with Corollary 12.2.4 this already proves boosting works: choose $T \asymp \gamma^{-2}\log m$, get $\widehat{R}_S = 0$, and read off a nonvacuous bound as soon as $m \gtrsim d_0\gamma^{-2}\log^2 m$. It also makes a **prediction**: the bound degrades as $\sqrt{T}$, so running AdaBoost past the point where it fits the sample should start to overfit. Empirically, this prediction fails — AdaBoost's test error typically keeps *improving* for hundreds of rounds after the training error hits zero. The bound is not wrong; it is measuring the wrong quantity.

Account 2: the Rademacher complexity of the convex hull does not grow at all

Assume $\mathcal{H}_0$ is **symmetric** ($h \in \mathcal{H}_0 \Rightarrow -h \in \mathcal{H}_0$; true for stumps, by flipping s). Then we may take every $w_t \geq 0$ without loss of generality, and the **normalized ensemble**

$$\bar{f}_T(x) := \frac{f_T(x)}{\sum_{t=1}^{T} w_t} = \sum_{t=1}^{T} \lambda_t h_t(x), \qquad \lambda_t := \frac{w_t}{\sum_p w_p} \geq 0, \ \sum_t \lambda_t = 1,$$

is a **convex combination** of members of $\mathcal{H}_0$, i.e. $\bar{f}_T \in \operatorname{conv}(\mathcal{H}_0)$, with $\bar{f}_T(x) \in [-1, 1]$. Note $\operatorname{sign}(\bar{f}_T) = \operatorname{sign}(f_T) = h_S$: normalizing changes no prediction, only the scale on which we may measure *how confidently* each prediction was made.

Theorem 12.2.6: Convexification is free

For any $A \subset \mathbb{R}^m$, $\widehat{\mathfrak{R}}(\operatorname{conv}(A)) = \widehat{\mathfrak{R}}(A)$. In particular $\widehat{\mathfrak{R}}_S(\operatorname{conv}(\mathcal{H}_0)) = \widehat{\mathfrak{R}}_S(\mathcal{H}_0)$ for every S, **independently of** T.

Proof. $A \subseteq \operatorname{conv}(A)$ gives "$\geq$". For "$\leq$", fix the signs $\sigma \in \{-1, +1\}^m$ and let $a = \sum_j \lambda_j a^{(j)} \in \operatorname{conv}(A)$ with $\lambda_j \geq 0$, $\sum_j \lambda_j = 1$. By linearity of the inner product,

$$\langle \sigma, a \rangle = \sum_j \lambda_j \langle \sigma, a^{(j)} \rangle \leq \Big(\sum_j \lambda_j \Big) \max_j \langle \sigma, a^{(j)} \rangle \leq \sup_{a' \in A} \langle \sigma, a' \rangle.$$

Taking the supremum over $a \in \operatorname{conv}(A)$ and then $\mathbb{E}_\sigma$ of both sides gives the claim. □

Theorem 12.2.7: Margin bound for boosting

Let $\mathcal{H}_0$ be symmetric with values in $\{-1, +1\}$, fix a **margin threshold** $\nu \in (0, 1]$ and let $\bar{f} \in \operatorname{conv}(\mathcal{H}_0)$ be arbitrary. Then with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$, simultaneously for every such $\bar{f}$,

$$R\big(\operatorname{sign}(\bar{f})\big) \;=\; P_{(x,y)\sim\mathcal{D}}\big(y\,\bar{f}(x) \leq 0\big) \;\leq\; \underbrace{\frac{1}{m}\sum_{i=1}^{m} \mathbb{1}\big(y_i \bar{f}(x_i) \leq \nu\big)}_{\text{fraction of training margins} \leq \nu} \;+\; \frac{2}{\nu}\,\widehat{\mathfrak{R}}_S(\mathcal{H}_0) \;+\; 3\sqrt{\frac{\log(2/\delta)}{2m}}.$$

Proof. Call $y\bar{f}(x) \in [-1, 1]$ the **margin** of the prediction — positive exactly when the weighted vote is correct, and large when it is correct by a wide majority. Define the ν**-margin loss**

$$\phi_\nu(u) := \min\big(1, \max(0, 1 - u/\nu)\big),$$

which is $\frac{1}{\nu}$-Lipschitz, takes values in $[0, 1]$, and satisfies the sandwich $\mathbb{1}(u \leq 0) \leq \phi_\nu(u) \leq \mathbb{1}(u \leq \nu)$.

Apply the Rademacher generalization bound (Theorem 7.3.2) to the loss class $\{(x,y) \mapsto \phi_\nu(y\bar{f}(x)) : \bar{f} \in \mathrm{conv}(\mathcal{H}_0)\}$, whose values lie in $[0,1]$: with probability $\geq 1-\delta$, for all $\bar{f}$,

$$\mathbb{E}_{\mathcal{D}}\big[\phi_\nu(y\bar{f}(x))\big] \leq \frac{1}{m}\sum_i \phi_\nu\big(y_i\bar{f}(x_i)\big) + 2\,\widehat{\mathfrak{R}}_S\big(\phi_\nu \circ \mathrm{conv}(\mathcal{H}_0)\big) + 3\sqrt{\tfrac{\log(2/\delta)}{2m}}.$$

The left side is $\geq P_{\mathcal{D}}\big(y\bar{f}(x) \leq 0\big)$ and the empirical average is $\leq \frac{1}{m}\sum_i \mathbb{1}(y_i\bar{f}(x_i) \leq \nu)$, both by the sandwich. For the middle term, note that the contraction lemma (Lemma 7.3.4) permits a *different* function in each coordinate: take $\phi_i(u) := \phi_\nu(y_i u)$, which is $\frac{1}{\nu}$-Lipschitz for either sign of y_i, so contracting with these removes both the label multiplication and the margin loss in one step. Theorem 12.2.6 then removes the convex hull:

$$\widehat{\mathfrak{R}}_S\big(\phi_\nu \circ \mathrm{conv}(\mathcal{H}_0)\big) \;\leq\; \tfrac{1}{\nu}\,\widehat{\mathfrak{R}}_S\big(\mathrm{conv}(\mathcal{H}_0)\big) \;=\; \tfrac{1}{\nu}\,\widehat{\mathfrak{R}}_S(\mathcal{H}_0). \qquad \square$$

$\square$

The resolution. Compare (12.1) with Theorem 12.2.7, term by term:

	Account 1 (VC of $L(\mathcal{H}_0, T)$)	Account 2 (Rademacher of $\mathrm{conv}(\mathcal{H}_0)$)
fit term	$\widehat{R}_S(h_S)$, hits 0 and then stops moving	$\frac{1}{m}\sum_i \mathbb{1}(y_i\bar{f}_T(x_i) \leq \nu)$, **keeps shrinking**
complexity term	$O(\sqrt{d_0 T \log(d_0 T)/m})$, **grows with** T	$\frac{2}{\nu}\widehat{\mathfrak{R}}_S(\mathcal{H}_0)$, **independent of** T

Account 2 says that additional boosting rounds incur no additional complexity — a convex combination of T members is no richer, in the Rademacher sense, than a single member — while they continue to improve a quantity that the zero-one training error is too coarse to reveal: Theorem 12.2.3's proof shows AdaBoost keeps driving the exponential loss Z_t down after $\widehat{R}_S = 0$, and Z_t is small only when the margins $y_i f_t(x_i)$ are *large*, not merely positive. Boosting past zero training error is therefore not idle: it is converting correct-but-marginal predictions into correct-and-confident ones, which is precisely what the first term of Theorem 12.2.7 rewards. This is the same lesson as Chapter 10's: the complexity measure that grows with the number of parameters is the wrong choice, and a scale-sensitive measure — margins here, weight norms there — is the right one.

The trade-off has not disappeared, it has moved into ν: a larger ν shrinks the complexity term but makes the margin-violation term larger. The bound holds for all ν simultaneously (up to a union bound over a grid of ν's), so one may read it at whichever ν is most favorable for the ensemble actually obtained.

Account 3: the price of an ensemble is its total weight

Account 2 normalized the ensemble to the convex combination $\bar{f}_T = f_T/\sum_t w_t$. The third account prices the *unnormalized* scale. For a symmetric $\mathcal{H}_0$ and $f = \sum_{t=1}^T w_t h_t$, the **total weight** is $\|w\|_1 := \sum_t |w_t|$, and the ensembles of total weight at most A form the class $\mathcal{F}_{A,L_1}(\mathcal{H}_0)$ of Theorem 7.7.2 (Chapter 7), whose empirical Rademacher complexity is exactly

$$\widehat{\mathfrak{R}}_S\big(\mathcal{F}_{A,L_1}(\mathcal{H}_0)\big) = A \cdot \widehat{\mathfrak{R}}_S(\mathcal{H}_0),$$

independently of the number of rounds T. Account 2's "convexification is free" is the case $A = 1$.

- The price of an ensemble is its total weight A, not its member count T — the same accounting as Chapter 10's width-independent Theorem 10.5.2 for L_1-constrained output layers, and the reason an L_1 penalty such as the Lasso of Chapter 3 controls capacity through a norm rather than through the number of parameters.
- For AdaBoost the total weight $\sum_t w_t$ grows with T. Theorem 12.2.7 absorbs this growth by normalizing and measuring margins on the normalized scale; Account 3 says equivalently that the unnormalized f_T belongs to a class whose complexity grows linearly in $\sum_t w_t$.
- Combined with Maurey's lemma (Theorem 7.7.1, Chapter 7), the account explains why stagewise greedy search is reasonable: every point of $\mathcal{F}_{A,L_1}(\mathcal{H}_0)$ is, on the sample, within $AB/\sqrt{N}$ of an N-term average of members of $\mathcal{H}_0$, so a procedure that adds one member per round searches a set that already contains good sparse approximants of every point of the full class. Theorem 12.3.1 below is the algorithmic counterpart of this approximation fact.

12.2.4 Implementation: AdaBoost with Decision Stumps

We implement Definition 12.2.1 and the AdaBoost pseudocode literally: the weak learner is an exhaustive search over $\mathcal{H}_{\text{stump}}$ for the stump of least D-weighted error, and the boosting loop is the four numbered steps. The data set is the **Wisconsin Breast Cancer** set of Chapters 8 and 9 (569 samples, $d = 30$ features), with labels recoded to $\{-1, +1\}$. Alongside the two error curves we track $\min_i y_i \bar{f}_T(x_i)$, the smallest training margin — the quantity Theorem 12.2.7 says should keep improving after the training error stops.

```
# AdaBoost implemented from scratch: the base class is decision stumps, the
# weak learner is an exhaustive weighted-error search over them, and the
# boosting loop is line-for-line the pseudocode above.

import torch as th
import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer

th.manual_seed(0)

data = load_breast_cancer()
X_all = th.tensor(data.data, dtype=th.float64)
y_all = 2 * th.tensor(data.target, dtype=th.float64) - 1   # labels in {-1,+1}
idx = th.randperm(y_all.numel())
n_tr = int(0.7 * y_all.numel())
tr, te = idx[:n_tr], idx[n_tr:]
X, y = X_all[tr], y_all[tr]
X_te, y_te = X_all[te], y_all[te]
m, d = X.shape
order = th.argsort(X, dim=0)                              # sort once, reuse

# --- The base class: decision stumps  h(x) = s * sign(x_j - theta) ----------

def weak_learner(D):
```

```
    """Return the stump of least D-weighted empirical error, by exhaustive
    search over all d features and all m+1 distinct thresholds of each."""
    best = (float('inf'), 0, -float('inf'), 1)          # (err, j, theta, s)
    for j in range(d):
        o = order[:, j]
        xs, ys, Ds = X[o, j], y[o], D[o]
        # Sweep theta from left to right. At theta = -inf the stump predicts
        # +1 everywhere, so its error is the weight of the negative points;
        # each time theta passes a point, that point's prediction flips,
        # changing the error by +D_i if y_i = +1 and by -D_i if y_i = -1.
        errs = Ds[ys == -1].sum() + th.cat([th.zeros(1, dtype=D.dtype),
                                            th.cumsum(Ds * ys, dim=0)])
        thetas = th.cat([th.tensor([-float('inf')], dtype=xs.dtype), xs])
        # A threshold can only fall between two *distinct* values: with ties,
        # x > theta sends all copies of a value to the same side.
        valid = th.ones(m + 1, dtype=th.bool)
        valid[1:m] = xs[:-1] < xs[1:]
        for s in (1, -1):                               # both leaf polarities
            e = errs if s == 1 else 1.0 - errs
            e = th.where(valid, e, th.tensor(float('inf'), dtype=e.dtype))
            k = int(e.argmin())
            if e[k] < best[0]:
                best = (e[k].item(), j, thetas[k].item(), s)
    return best[1:]                                     # (j, theta, s)

def stump_predict(Xq, stump):
    j, theta, s = stump
    return s * th.where(Xq[:, j] > theta, 1.0, -1.0).double()

# --- AdaBoost ----------------------------------------------------------------

T = 200
D = th.full((m,), 1.0 / m, dtype=th.float64)            # step 1: D^(1)
w_sum = 0.0
F, F_te = th.zeros(m, dtype=th.float64), th.zeros(y_te.numel(), dtype=th.float64)
err_tr, err_te, min_margin = [], [], []

for t in range(T):
    stump = weak_learner(D)                             # 2.1  h_t = WL(S,D^(t))
    pred = stump_predict(X, stump)
    eps = D[pred != y].sum()                            # 2.2  epsilon_t
    eps = eps.clamp(1e-10, 1 - 1e-10)                   #      guard log(0)
    w_t = 0.5 * th.log(1.0 / eps - 1.0)                 # 2.3  w_t
    D = D * th.exp(-w_t * y * pred)                     # 2.4  reweight ...
    D = D / D.sum()                                     #      ... and normalise

    w_sum += w_t.item()                                 # bookkeeping only:
```

```
    F += w_t * pred                                    #   f_t on the train set
    F_te += w_t * stump_predict(X_te, stump)           #   f_t on the test set
    err_tr.append((th.sign(F) != y).double().mean().item())
    err_te.append((th.sign(F_te) != y_te).double().mean().item())
    min_margin.append((y * F / w_sum).min().item())    # min_i y_i bar-f_t(x_i)

t0 = next(t for t, e in enumerate(err_tr) if e == 0.0)
print("       T   train err   test err   min margin")
for t in [0, 4, 9, t0, 49, 99, T - 1]:
    print(f"{t+1:8d}    {err_tr[t]:.4f}     {err_te[t]:.4f}
    ↪  {min_margin[t]:+.4f}")
print(f"\ntraining error first reaches 0 at T = {t0+1}")

rounds = th.arange(1, T + 1)
fig, ax = plt.subplots(1, 2, figsize=(11, 4))
ax[0].plot(rounds, err_tr, label=r"training error $\widehat{R}_S(h_S)$ (S =
↪  training set)")
ax[0].plot(rounds, err_te, label=r"test error (estimate of $R(h_S)$)")
ax[0].axvline(t0 + 1, ls=":", c="k")
ax[0].set_xscale("log"); ax[0].set_xlabel("boosting rounds $T$")
ax[0].set_ylabel("zero-one error"); ax[0].legend(); ax[0].grid(alpha=.3)
ax[1].plot(rounds, min_margin, c="C2")
ax[1].axvline(t0 + 1, ls=":", c="k"); ax[1].axhline(0, lw=.5, c="k")
ax[1].set_xscale("log"); ax[1].set_xlabel("boosting rounds $T$")
ax[1].set_ylabel(r"minimum training margin")
ax[1].grid(alpha=.3)
plt.tight_layout(); plt.show()
```

```
       T   train err   test err   min margin
       1    0.0804     0.0819     -1.0000
       5    0.0327     0.0643     -0.3449
      10    0.0151     0.0643     -0.1646
      19    0.0000     0.0526     +0.0225
      50    0.0000     0.0409     +0.0990
     100    0.0000     0.0234     +0.1198
     200    0.0000     0.0175     +0.1441

training error first reaches 0 at T = 19
```

The numbers track the theory. Across the 200 rounds the worst weighted error the stump search ever returns is $\epsilon_t = 0.376$, so the weak-learning parameter realized on this data set is $\gamma = 0.124$. Theorem 12.2.3 then guarantees only $\widehat{R}_S \le e^{-2(0.124)^2 \cdot 19} \approx 0.56$ after 19 rounds, and Corollary 12.2.4's sufficient budget is $\log(398)/(2 \cdot 0.124^2) \approx 193$ rounds — both valid but, as worst-case bounds usually are, far more conservative than what actually happens: the training error is already exactly 0 at $T = 19$. From $T = 19$ onward, the first term of Account 1 is frozen at 0 and its complexity term keeps growing, so (12.1) predicts deterioration; instead the test error *falls further*, from 5.3% to 1.8%. Account 2 explains

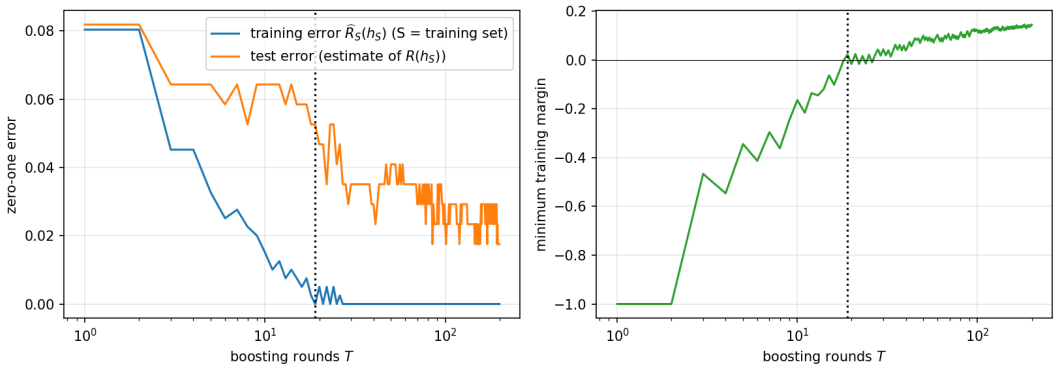


Figure 12.1: AdaBoost's training and test error (left) and minimum training margin (right) as a function of the number of boosting rounds T, on a logarithmic axis; the dotted line marks the round at which training error first reaches zero.

why: over the same rounds the smallest training margin climbs from $+0.023$ to $+0.144$, so the margin-violation term of Theorem 12.2.7 keeps shrinking at any fixed threshold $\nu \lesssim 0.14$, while its complexity term $\frac{2}{\nu}\widehat{\mathfrak{R}}_S(\mathcal{H}_{\text{stump}})$ never moves.

12.3 Gradient Boosting

AdaBoost minimizes the exponential loss by greedy coordinate descent (the first reading of Theorem 12.2.3's proof). **Gradient boosting** — (Friedman, 2001), (Mason et al., 1999) — runs the same greedy scheme on any differentiable loss $L(f, y)$.

Throughout this section $\langle a, b\rangle_S := \frac{1}{m}\sum_{i=1}^m a_i b_i$ and $\|a\|_S := \langle a, a\rangle_S^{1/2}$ for functions or vectors evaluated on the sample, as in Chapter 7, and

$$\widehat{L}(f) := \frac{1}{m}\sum_{i=1}^{m} L\big(f(x_i), y_i\big)$$

is the empirical loss. $\widehat{L}$ depends on f only through the prediction vector $(f(x_1), \dots, f(x_m))$, and its gradient with respect to that vector is $-\frac{1}{m}r$, where

$$r_i := -\frac{\partial L(a, y_i)}{\partial a}\Big|_{a=f(x_i)}, \qquad i = 1, \dots, m,$$

are the **pseudo-residuals**. A gradient step on the prediction vector would produce a vector, not a function; boosting replaces it by a step towards the base hypothesis best aligned with r.

AdaBoost as a special case. For the exponential loss $L(f, y) = e^{-yf}$ the pseudo-residuals are $r_i = y_i e^{-y_i f(x_i)}$, which by Step 2 of the proof of Theorem 12.2.3 are proportional to $y_i D_i^{(t)}$ at $f = f_{t-1}$. For h with values in $\{-1, +1\}$, $y_i h(x_i) = 1 - 2\,\mathbb{1}(h(x_i) \neq y_i)$, so

$$\sum_i r_i h(x_i) \;\propto\; \sum_i D_i^{(t)} y_i h(x_i) = 1 - 2\epsilon_t(h) :$$

the base hypothesis best aligned with the pseudo-residuals is the one of least weighted error, AdaBoost's step 2.1, and Step 3 of the proof showed that w_t is the exact minimizer of the loss along it. For the squared loss $L(f,y)=\frac{1}{2}(f-y)^2$, the pseudo-residuals $r_i=y_i-f(x_i)$ are the ordinary residuals, and boosting fits each new member to what the current ensemble still gets wrong.

The construction. The version we analyze is the conditional-gradient method of (Frank and Wolfe, 1956) run over the convex hull of the base class, in the greedy-approximation form of (Jones, 1992) and (Zhang, 2003). It assumes L convex and β-**smooth** in its first argument: $a\mapsto \partial L(a,y)/\partial a$ is β-Lipschitz for every y. The logistic loss $\log(1+e^{-yf})$ of Chapter 4 has second derivative $\sigma(yf)\sigma(-yf)\le\frac{1}{4}$, so $\beta=\frac{1}{4}$; the squared loss has $\beta=1$; the exponential loss is smooth only on bounded predictions. The base class $\mathcal{H}_0$ consists of real-valued functions with

$$B^2:=\sup_{h\in\mathcal{H}_0}\|h\|_S^2<\infty.$$

Start from $f^{(0)}:=0$ and, for $t=1,\dots,T$, with $\eta_t:=\frac{2}{t+1}$:

1. Compute the pseudo-residuals $r^{(t)}$ at $f^{(t-1)}$.
2. Fit the base learner by least squares, $h_t\in\arg\min_{h\in\mathcal{H}_0}\left\|\beta\eta_t\,h-r^{(t)}\right\|_S^2$.
3. Average, $f^{(t)}:=(1-\eta_t)f^{(t-1)}+\eta_t h_t$.

Since $\eta_1=1$, $f^{(1)}=h_1$, and every $f^{(t)}$ is a convex combination of $h_1,\dots,h_t$: the iterates stay in $\operatorname{conv}(\mathcal{H}_0)$, and the algorithm is a conditional-gradient (Frank-Wolfe) method on the hull. Expanding the square in step 2 and dropping the term $\|r^{(t)}\|_S^2$, which does not depend on h, step 2 minimizes

$$-2\eta_t\langle r^{(t)},h\rangle_S+\beta\eta_t^2\|h\|_S^2,$$

a trade-off between alignment with the pseudo-residuals and the curvature penalty that smoothness charges for a step of length η_t towards h. The target $r^{(t)}/(\beta\eta_t)$ grows as η_t shrinks, but the fit is confined to $\mathcal{H}_0$, so the actual step $\eta_t(h_t-f^{(t-1)})$ shrinks.

Theorem 12.3.1: Gradient boosting converges on the convex hull

Let L be convex and β-smooth in its first argument, let $B^2:=\sup_{h\in\mathcal{H}_0}\|h\|_S^2$, and let each round compute an exact minimizer in step 2. Then for every $T\ge 1$,

$$\widehat{L}\big(f^{(T)}\big)\;\le\;\inf_{f\in\operatorname{conv}(\mathcal{H}_0)}\widehat{L}(f)\;+\;\frac{8\beta B^2}{T+1}.$$

Proof. Fix a competitor $f=\sum_j u_j\psi_j\in\operatorname{conv}(\mathcal{H}_0)$ with $u_j\ge 0$, $\sum_j u_j=1$, $\psi_j\in\mathcal{H}_0$, and let $\Delta_t:=\widehat{L}(f^{(t)})-\widehat{L}(f)$. Fix a round t and abbreviate $\eta:=\eta_t$, $r:=r^{(t)}$, $h:=h_t$, $f_-:=f^{(t-1)}$. The norm $\|\cdot\|_S$ is convex, so $\|f_-\|_S\le B$ (f_- is 0 or a convex combination of members of $\mathcal{H}_0$), and likewise $\|f\|_S\le B$.

Step 1: smoothness. Lemma 10.5.3(i) of Chapter 10, in dimension one, applied to $a\mapsto L(a,y_i)$ between $f_-(x_i)$ and $f^{(t)}(x_i)=f_-(x_i)+\eta\,(h-f_-)(x_i)$ and averaged over i, gives

$$\widehat{L}(f^{(t)})\;\le\;\widehat{L}(f_-)-\eta\langle r,h-f_-\rangle_S+\frac{\beta\eta^2}{2}\|h-f_-\|_S^2\;\le\;\widehat{L}(f_-)-\eta\langle r,h-f_-\rangle_S+\beta\eta^2\big(\|h\|_S^2+B^2\big),$$

using $\|a-b\|_S^2 \le 2\|a\|_S^2 + 2\|b\|_S^2$.

Step 2: least-squares optimality. Comparing h with each ψ_j in step 2 and averaging with the weights u_j,

$$-2\eta\langle r,h\rangle_S + \beta\eta^2\|h\|_S^2 \;\le\; \sum_j u_j\Big(-2\eta\langle r,\psi_j\rangle_S + \beta\eta^2\|\psi_j\|_S^2\Big) \;\le\; -2\eta\langle r,f\rangle_S + \beta\eta^2 B^2.$$

Hence, using also $\|h\|_S \le B$,

$$-\eta\langle r,h\rangle_S + \beta\eta^2\|h\|_S^2 = \tfrac{1}{2}\Big(-2\eta\langle r,h\rangle_S + \beta\eta^2\|h\|_S^2\Big) + \tfrac{\beta\eta^2}{2}\|h\|_S^2 \;\le\; -\eta\langle r,f\rangle_S + \beta\eta^2 B^2,$$

and Step 1 becomes

$$\widehat{L}(f^{(t)}) \;\le\; \widehat{L}(f_-) + \eta\langle r, f_- - f\rangle_S + 2\beta\eta^2 B^2.$$

Step 3: convexity. The first-order condition for the convex $\widehat{L}$, whose gradient is $-\frac{1}{m}r$, reads $\widehat{L}(f) \ge \widehat{L}(f_-) - \langle r, f - f_-\rangle_S$, i.e. $\langle r, f_- - f\rangle_S \le -\Delta_{t-1}$. Therefore

$$\Delta_t \;\le\; (1-\eta_t)\,\Delta_{t-1} + 2\beta\eta_t^2 B^2 = \frac{t-1}{t+1}\,\Delta_{t-1} + \frac{8\beta B^2}{(t+1)^2}.$$

Step 4: the recursion. Multiplying by $t(t+1)$,

$$t(t+1)\,\Delta_t \;\le\; (t-1)t\,\Delta_{t-1} + 8\beta B^2\,\frac{t}{t+1} \;\le\; (t-1)t\,\Delta_{t-1} + 8\beta B^2.$$

Summing over $t = 1,\dots,T$, the left-hand terms telescope and the first one vanishes (its coefficient is $0\cdot 1$), so $T(T+1)\Delta_T \le 8\beta B^2 T$. Since $f \in \operatorname{conv}(\mathcal{H}_0)$ was arbitrary, the claim follows $\square$

Bounds of this type go back to the greedy-approximation analyses of Jones (1992) and Zhang (2003); (Jaggi, 2013) gives the corresponding analysis of the Frank-Wolfe method. The theorem is the algorithmic counterpart of Maurey's lemma (Theorem 7.7.1): the infimum over the *convex hull* is approached, at rate $O(1/T)$, by a T-term combination, found greedily without ever solving an optimization problem over $\operatorname{conv}(\mathcal{H}_0)$.

What the scale B controls. Write $\mathcal{H}_0 = B\cdot\mathcal{H}_1$ for a fixed class $\mathcal{H}_1$ with $\sup_{h\in\mathcal{H}_1}\|h\|_S \le 1$. Then B enters the analysis three times:

- *approximation:* $\operatorname{conv}(B\mathcal{H}_1)$ grows with B, so $\inf_f \widehat{L}(f)$ decreases;
- *optimization:* the convergence term $8\beta B^2/(T+1)$ grows as B^2, so more rounds are needed;
- *complexity:* $\widehat{\mathfrak{R}}_S(\operatorname{conv}(B\mathcal{H}_1)) = B\,\widehat{\mathfrak{R}}_S(\mathcal{H}_1)$ by Theorem 12.2.6, so, through the contraction lemma and Theorem 7.3.2 for a Lipschitz loss such as the logistic one, the generalization gap grows linearly in B — Account 3, with $A = B$.

B is therefore a regularization parameter of the same kind as the number of rounds T in AdaBoost, trading the approximation error against the estimation error.

12.3.1 Implementation: Gradient Boosting with Regression Stumps

We implement the construction on the same split of the Wisconsin Breast Cancer data as the AdaBoost experiment, with the logistic loss ($\beta = \frac{1}{4}$). The base class consists of regression stumps $h(x) = s_L$ if $x_j \le \theta$ and $h(x) = s_R$ otherwise, with leaf values $s_L, s_R \in [-B, B]$, so that $\sup_h \|h\|_S \le B$ as Theorem 12.3.1 requires. Step 2 is solved exactly: for a fixed split the least-squares problem separates over the two leaves, and on each leaf the minimizer of $\sum_i (s - u_i)^2$ over $s \in [-B, B]$ is the leaf mean of the target clipped to $[-B, B]$. Every split is therefore scored in $O(1)$ from prefix sums over the sorted feature values, and the base learner is an exhaustive search over all $O(dm)$ splits. We run the algorithm for $B \in \{1, 4, 16\}$.

```
# Gradient boosting from scratch, as in the pseudocode above: logistic loss,
# base class = regression stumps with leaf values in [-B, B], each round an
# exact least-squares fit of beta*eta_t*h to the pseudo-residuals, and the
# averaging update with eta_t = 2/(t+1) of Theorem 12.3.1.

import torch as th
import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer

th.manual_seed(0)

data = load_breast_cancer()                                  # same split as AdaBoost
X_all = th.tensor(data.data, dtype=th.float64)
y_all = 2 * th.tensor(data.target, dtype=th.float64) - 1    # labels in {-1,+1}
idx = th.randperm(y_all.numel())
n_tr = int(0.7 * y_all.numel())
tr, te = idx[:n_tr], idx[n_tr:]
X, y = X_all[tr], y_all[tr]
X_te, y_te = X_all[te], y_all[te]
m, d = X.shape
order = th.argsort(X, dim=0)                                 # sort once, reuse

BETA = 0.25                                                  # smoothness of the
↪  logistic loss

# --- Base learner: exact least squares over regression stumps ----------------
# A stump h(x) = s_L if x_j <= theta else s_R. For a fixed split, the least-
# squares fit of the target u is separable in the two leaves, and on each leaf
# the box-constrained minimizer of sum (s - u_i)^2 is the leaf mean clipped to
# [-B, B]. The squared error of a leaf with n points, sum q and clipped value s
# is  sum u_i^2 - 2 s q + n s^2,  so every split is scored from prefix sums.

def fit_stump(u, B):
    best = (float('inf'), None)
    tot, sq = u.sum(), u.square().sum()
    for j in range(d):
```

```
        o = order[:, j]
        xs, us = X[o, j], u[o]
        n_l = th.arange(1, m, dtype=th.float64)        # split after point k-1
        q_l = us.cumsum(0)[:-1]                        # left sums
        n_r, q_r = m - n_l, tot - q_l                  # right counts and sums
        s_l = (q_l / n_l).clamp(-B, B)
        s_r = (q_r / n_r).clamp(-B, B)
        sse = sq - 2 * s_l * q_l + n_l * s_l**2 - 2 * s_r * q_r + n_r * s_r**2
        sse[xs[:-1] == xs[1:]] = float('inf')          # cannot split inside ties
        k = int(sse.argmin())
        if sse[k] < best[0]:
            best = (sse[k].item(), (j, xs[k].item(), s_l[k].item(),
↪   s_r[k].item()))
    return best[1]

def stump_predict(Xq, stump):
    j, theta, s_l, s_r = stump
    return th.where(Xq[:, j] <= theta, s_l, s_r).double()

# --- Gradient boosting -------------------------------------------------------

def gradient_boost(B, T):
    F = th.zeros(m, dtype=th.float64)                  # f^(0) = 0 on the train
↪   set
    F_te = th.zeros(y_te.numel(), dtype=th.float64)    # ... and on the test set
    loss_tr, err_tr, err_te = [], [], []
    for t in range(1, T + 1):
        eta = 2.0 / (t + 1)
        r = y * th.sigmoid(-y * F)                     # 1. pseudo-residuals
↪   -dL/df
        stump = fit_stump(r / (BETA * eta), B)         # 2. argmin sum (beta*eta*h
↪   - r)^2
        F = (1 - eta) * F + eta * stump_predict(X, stump)        # 3. averaging
        F_te = (1 - eta) * F_te + eta * stump_predict(X_te, stump)  #  update
        loss_tr.append(th.log1p(th.exp(-y * F)).mean().item())
        err_tr.append((th.sign(F) != y).double().mean().item())
        err_te.append((th.sign(F_te) != y_te).double().mean().item())
    return loss_tr, err_tr, err_te

T = 400
runs = {B: gradient_boost(B, T) for B in (1.0, 4.0, 16.0)}

print("       B       T   train loss   train err   test err")
for B, (loss_tr, err_tr, err_te) in runs.items():
    for t in [10, 100, 400]:
        print(f"{B:8.0f}{t:8d}     {loss_tr[t-1]:.4f}      {err_tr[t-1]:.4f}"
              f"     {err_te[t-1]:.4f}")
```

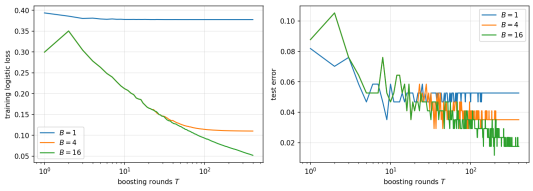


Figure 12.2: Gradient boosting with regression stumps whose leaf values are bounded by B in absolute value: training logistic loss (left) and test error (right) as a function of the number of rounds T, for $B \in \{1, 4, 16\}$, on a logarithmic axis.

```
rounds = th.arange(1, T + 1)
fig, ax = plt.subplots(1, 2, figsize=(11, 4))
for c, (B, (loss_tr, err_tr, err_te)) in enumerate(runs.items()):
    ax[0].plot(rounds, loss_tr, c=f"C{c}", label=f"$B = {B:.0f}$")
    ax[1].plot(rounds, err_te, c=f"C{c}", label=f"$B = {B:.0f}$")
ax[0].set_xscale("log"); ax[0].set_xlabel("boosting rounds $T$")
ax[0].set_ylabel("training logistic loss"); ax[0].legend(); ax[0].grid(alpha=.3)
ax[1].set_xscale("log"); ax[1].set_xlabel("boosting rounds $T$")
ax[1].set_ylabel("test error"); ax[1].legend(); ax[1].grid(alpha=.3)
plt.tight_layout(); plt.show()
```

B	T	train loss	train err	test err
1	10	0.3780	0.0578	0.0585
1	100	0.3777	0.0553	0.0468
1	400	0.3777	0.0553	0.0526
4	10	0.2119	0.0503	0.0468
4	100	0.1142	0.0176	0.0351
4	400	0.1102	0.0176	0.0351
16	10	0.2119	0.0503	0.0468
16	100	0.0931	0.0101	0.0292
16	400	0.0520	0.0075	0.0175

The three runs separate the three roles of B. With $B = 1$ the training loss is flat after about ten rounds at 0.378: Theorem 12.3.1 guarantees a gap of at most $8 \cdot \frac{1}{4} \cdot 1/(T+1) \approx 0.005$ at $T = 400$, so this is (up to that gap) the infimum over $\operatorname{conv}(\mathcal{H}_0)$, and it is high because a convex combination of stumps with values in $[-1, 1]$ cannot exceed 1 in absolute value — the logistic loss of a correct prediction is then at least $\log(1 + e^{-1}) \approx 0.31$. The approximation error dominates, and the test error stays near 5%. With $B = 4$ the hull is larger, the training loss reaches 0.110 and the training error 1.8%; the runs for $B = 4$ and $B = 16$ coincide for the first twenty rounds, until the box $[-4, 4]$ starts to clip a leaf value. (The loss of both rises at $T = 2$: Theorem 12.3.1 bounds the gap to the infimum, not the change from one round to the next.) With $B = 16$ the training loss is still decreasing at $T = 400$ — the guarantee $8 \cdot \frac{1}{4} \cdot 256/401 \approx 1.3$ is vacuous here, so the convergence is faster than the worst case, but visibly slower than for small B — and the test error reaches 1.8%, the value AdaBoost attains at $T = 200$. On this data set the approximation gain of a larger budget outweighs its complexity cost up to $B = 16$; the

complexity term of Account 3 says that this cannot continue indefinitely, since the generalization bound grows linearly in B.

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